6-7

1. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The function f is defined by $f(x) = \int_0^x \left(t^2 - 3t - 10\right) dt$.

(a) Write an equation for the line tangent to the graph of f at $x = 6$.

(b) Find all values of x at which f has a critical point. Determine whether f has a relative minimum, a relative maximum, or neither at each of these values. Justify your answers.

(c) On what open intervals, if any, is the graph of f concave down? Give a reason for your answer.

(d) Let g be the function defined by $g(x) = f(\tan x)$. Find $g'\left(\frac{\pi}{4}\right)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

An unsupported value of -42 does not earn the first point; the minimal supporting work required is $72 - 54 - 60$.

The second point can be earned without presenting an explicit expression for $f'(x)$ by correctly finding $f'(6)$ with supporting work.

Eligibility for third point: The response has earned the second point and used $f(x) = \frac{1}{3}x^3 - \frac{3}{2}x^2 - 10x$ to find $f(6)$. In this case the third point is earned for any form of the equation $y = f(6) + 8(x - 6)$ for the declared value of $f(6)$.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- [ ] $f(6)$
- [ ] $f'(x)$

6-7

- [ ] Equation for tangent line

Solution:

$f(6) = \int_0^6 \left(t^2 - 3t - 10\right)\, dt = \left(\frac{1}{3}t^3 - \frac{3}{2}t^2 - 10t\right)\Big|_0^6$

$= \frac{1}{3} \cdot 6^3 - \frac{3}{2} \cdot 6^2 - 10 \cdot 6 = 72 - 54 - 60 = -42$

$f'(x) = x^2 - 3x - 10$

$f'(6) = 6^2 - 3 \cdot 6 - 10 = 36 - 18 - 10 = 8$

An equation for the tangent line is $y = -42 + 8(x - 6)$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

If any additional critical points are listed or either critical point is missing, the first point is not earned.

If only one of $x = -2$ or $x = 5$ is listed as a critical point and that point is correctly justified as a relative maximum or relative minimum, the response earns one point.

"The slope changes from positive to negative" is not a sufficient justification for the second point. The response must be clear about what slope is changing.

Justifications that appeal only to $f(x)$ changing from increasing to decreasing (or vice versa) are not sufficient for the second point.

A response that imports from part (a) an incorrect $f'(x)$ of the form $ax^2 + bx + c$ (typically, $f'(x) = x^2 - 3x$) may earn the first point for finding the critical points associated with their $f'(x)$. Note: The incorrect $f'(x)$ must be imported from part (a) to be eligible for the first point. Such a response is not eligible for the second point.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Critical points
- [ ] Answers with justification

Solution:

6-7

$f'(x) = x^2 - 3x - 10 = (x-5)(x+2) = 0$ and $x = 5$
$\Rightarrow x = -2$

f has a relative maximum at $x = -2$ because $f'(x)$ changes from positive to negative there.

f has a relative minimum at $x = 5$ because $f'(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that presents $2x - 3 < 0$ (or $2x - 3 = 0$ or $2x - 3 > 0$) earns the first point.

To earn the second point, a response must explain that the graph of f is concave down on $\left(-\infty, \frac{3}{2}\right)$ (or $x < \frac{3}{2}$), because $f''(x) < 0$ on that interval.

The interval may include the right endpoint $x = \frac{3}{2}$.

A response that presents any interval other than $\left(-\infty, \frac{3}{2}\right)$ does not earn the second point.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Find $f''(x)$ and compare to zero
- ☐ Answer with reason

Solution:

$f''(x) = 2x - 3 < 0 \Rightarrow x < \frac{3}{2}$

The graph of f is concave down on the interval $\left(-\infty, \frac{3}{2}\right)$ because $f''(x) < 0$ there.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for $g'(x) = f'(\tan x) \cdot \left(\sec^2 x\right)$.

The answer does not need to be simplified: $\left(\tan^2\left(\frac{\pi}{4}\right) - 3\tan\left(\frac{\pi}{4}\right) - 10\right)\left(\sec^2\left(\frac{\pi}{4}\right)\right)$ or equivalent earns the second point.

6-7

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Chain rule
- ☐ Answer

Solution:

$g'\left(x\right)=f'\left(\tan\ x\right)\cdot\ \left(\sec^2 x\right)=\left(\tan^2 x-3\tan\ x-10\right)\left(\sec^2 x\right)$

$g'\left(\frac{\pi}{4}\right)=\left(\tan^2\left(\frac{\pi}{4}\right)-3\tan\left(\frac{\pi}{4}\right)-10\right)\left(\sec^2\left(\frac{\pi}{4}\right)\right)$
$=\left(1-3-10\right)\left(2\right)=-24$

6-7

2. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	0	2	4	7
$f(x)$	10	7	4	5
$f'(x)$	$\frac{3}{2}$	-8	3	6
$g(x)$	1	2	-3	0
$g'(x)$	5	4	2	8

The functions f and g are twice differentiable. The table shown gives values of the functions and their first derivatives at selected values of x.

(a) Let h be the function defined by $h(x) = f(g(x))$. Find $h'(7)$. Show the work that leads to your answer.

(b) Let k be a differentiable function such that $k'(x) = (f(x))^2 \cdot g(x)$. Is the graph of k concave up or concave down at the point where $x = 4$? Give a reason for your answer.

(c) Let m be the function defined by $m(x) = 5x^3 + \int_0^x f'(t)\,dt$. Find $m(2)$. Show the work that leads to your answer.

(d) Is the function m defined in part (c) increasing, decreasing, or neither at $x = 2$? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for either $h'(x) = f'(g(x)) \cdot g'(x)$ or $h'(7) = f'(g(7)) \cdot g'(7)$.

If the first point is earned, the second point is earned only for an answer of 12 (or equivalent).

If the first point is not earned, the second point can be earned only for a response of either $f'(0) \cdot 8 = 12$ or $\frac{3}{2} \cdot 8$.

A response of 12 with no supporting work does not earn either point.

6-7

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Chain rule
- ☐ Answer

Solution:

$h'(x) = f'(g(x)) \cdot g'(x)$

$h'(7) = f'(g(7)) \cdot g'(7) = f'(0) \cdot 8 = \frac{3}{2} \cdot 8 = 12$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for either $k''(x) = 2f(x) \cdot f'(x) \cdot g(x) + (f(x))^2 \cdot g'(x)$ or $k''(4) = 2f(4) \cdot f'(4) \cdot g(4) + (f(4))^2 \cdot g'(4)$.

The first point is also earned by any of the following incorrect expressions, each of which has a single error in the application of the product rule or the chain rule:

- $2f(x) \cdot g(x) + (f(x))^2 \cdot g'(x)$ or $2f(4) \cdot g(4) + (f(4))^2 \cdot g'(4)$
- $2f'(x) \cdot g(x) + (f(x))^2 \cdot g'(x)$ or $2f'(4) \cdot g(4) + (f(4))^2 \cdot g'(4)$
- $f'(x) \cdot g(x) + (f(x))^2 \cdot g'(x)$ or $f'(4) \cdot g(4) + (f(4))^2 \cdot g'(4)$
- $2f(x) \cdot f'(x) \cdot g'(x)$ or $2f(4) \cdot f'(4) \cdot g'(4)$
- Note: A response that presents one of these expressions cannot earn the second point.

To earn the second point a response must correctly find $k''(4) = -40$ (or equivalent) with supporting work.

The third point is earned for an answer and reason that are consistent with any declared nonzero value of $k''(4)$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

6-7

- [ ] Product or chain rule
- [ ] $k''(4)$
- [ ] Answer with reason

Solution:

$k''(x) = 2f(x) \cdot f'(x) \cdot g(x) + (f(x))^2 \cdot g'(x)$

$k''(4) = 2f(4) \cdot f'(4) \cdot g(4) + (f(4))^2 \cdot g'(4)$
$= 2 \cdot 4 \cdot 3 \cdot (-3) + 4^2 \cdot 2 = -72 + 32 = -40$

The graph of k is concave down at the point where $x = 4$ because $k''(4) < 0$ and k'' is continuous.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned only for an answer of 37 (or equivalent) with supporting work equivalent to
$5 \cdot 8 + (f(2) - f(0))$, $40 + (f(2) - f(0))$, $5 \cdot 8 + (7 - 10)$, $40 + (7 - 10)$.

An answer of 37 with no supporting work does not earn the point.

0	1 ✓

The student response accurately includes the criteria below.

- [ ] Answer with supporting work

Solution:

$m(2) = 5 \cdot 8 + \int_0^2 f'(t)dt = 40 + (f(2) - f(0))$
$= 40 + (7 - 10) = 37$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for considering $m'(x)$, $m'(2)$, or m'. This consideration may appear in a justification statement.

The second point is earned for $m'(2) = 15 \cdot 2^2 + f'(2)$, $m'(2) = 60 + f'(2)$, or $m'(2) = 60 - 8$ but is not earned for an unsupported response of $m'(2) = 52$.

6-7

The third point is earned for an answer and justification consistent with any declared value of $m'(2)$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $m'(x)$
- ☐ $m'(2)$ with supporting work
- ☐ Answer with justification

Solution:

$m'(x) = 15x^2 + f'(x)$

$m'(2) = 15 \cdot 4 + f'(2) = 60 + (-8) = 52$

The graph of m is increasing at $x = 2$ because $m'(2) > 0$.

6-7

3. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

x	0	1	2	3	4	5	6
$f'(x)$	4	3.5	2	0.8	1.7	5.8	7

The function *f* satisfies $f(0) = 20$. The first derivative of *f* satisfies the inequality $0 \leq f'(x) \leq 7$ for all x in the closed interval $[0, 6]$. Selected values of f' are shown in the table above. The function *f* has a continuous second derivative for all real numbers.

(a) Use a midpoint Riemann sum with three subintervals of equal length indicated by the data in the table to approximate the value of $f(6)$.

(b) Determine whether the actual value of $f(6)$ could be 70. Explain your reasoning.

(c) Evaluate $\int_2^4 f''(x)dx$.

(d) Find $\lim_{x \to 0} \frac{f(x) - 20e^x}{0.5f(x) - 10}$.

Part A

1 point is earned for: midpoint sum

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_0^6 f'(x)\, dx \approx 2 \cdot 3.5 + 2 \cdot 0.8 + 2 \cdot 5.8 = 20.2$$

$$f(6) - f(0) = \int_0^6 f'(x)\, dx$$

$$f(6) = f(0) + \int_0^6 f'(x)\, dx \approx 20 + 20.2 = 40.2$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for: midpoint sum

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

6-7

$$\int_0^6 f'(x)\,dx \approx 2 \cdot 3.5 + 2 \cdot 0.8 + 2 \cdot 5.8 = 20.2$$

$$f(6) - f(0) = \int_0^6 f'(x)\,dx$$

$$f(6) = f(0) + \int_0^6 f'(x)\,dx \approx 20 + 20.2 = 40.2$$

Part B

1 point is earned for: integral bound

1 point is earned for: answer with reasoning

Since $f'(x) \leq 7$, $\int_0^6 f'(x)\,dx \leq 6 \cdot 7 = 42$.

$f(6) - f(0) \leq 42 \Rightarrow f(6) \leq 20 + 42 = 62$

Therefore, the actual value of $f(6)$ could not be 70.

0	1	2

The student response earns all of the following points:

1 point is earned for: integral bound

1 point is earned for: answer with reasoning

Since $f'(x) \leq 7$, $\int_0^6 f'(x)\,dx \leq 6 \cdot 7 = 42$.

$f(6) - f(0) \leq 42 \Rightarrow f(6) \leq 20 + 42 = 62$

Therefore, the actual value of $f(6)$ could not be 70.

Part C

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_2^4 f''(x)dx = f'(4) - f'(2) = 1.7 - 2 = -0.3$$

6-7

0	1	2

The student response earns all of the following points:

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_2^4 f''(x)dx = f'(4) - f'(2) = 1.7 - 2 = -0.3$$

Part D

1 point is earned for: L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x\to 0}(f(x) - 20e^x) = 0$$
$$\lim_{x\to 0}(0.5f(x) - 10) = 0$$

Using L'Hospital's Rule,

$$\lim_{x\to 0}\frac{f(x)-20e^x}{0.5f(x)-10} = \lim_{x\to 0}\frac{f'(x)-20e^x}{0.5f'(x)} = \frac{4-20}{0.5(4)} = -8$$

0	1	2

The student response earns all of the following points:

1 point is earned for: L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x\to 0}(f(x) - 20e^x) = 0$$
$$\lim_{x\to 0}(0.5f(x) - 10) = 0$$

Using L'Hospital's Rule,

$$\lim_{x\to 0}\frac{f(x)-20e^x}{0.5f(x)-10} = \lim_{x\to 0}\frac{f'(x)-20e^x}{0.5f'(x)} = \frac{4-20}{0.5(4)} = -8$$

6-7

4. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

x	0	1	2	3	4	5	6
$f'(x)$	4	3.5	2	0.8	1.7	5.8	7

The function f satisfies $f(0) = 20$. The first derivative of f satisfies the inequality $0 \leq f'(x) \leq 7$ for all x in the closed interval $[0,\ 6]$. Selected values of f' are shown in the table above. The function f has a continuous second derivative for all real numbers.

(a) Use a midpoint Riemann sum with three subintervals of equal length indicated by the data in the table to approximate the value of $f(6)$.

(b) Determine whether the actual value of $f(6)$ could be 70. Explain your reasoning.

(c) Evaluate $\int_2^4 f''(x)dx$.

(d) Find $\lim_{x\to 0} \frac{f(x) - 20e^x}{0.5f(x) - 10}$.

Part A

1 point is earned for: midpoint sum

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_0^6 f'(x)\,dx \approx 2 \cdot 3.5 + 2 \cdot 0.8 + 2 \cdot 5.8 = 20.2$$

$$f(6) - f(0) = \int_0^6 f'(x)\,dx$$

$$f(6) = f(0) + \int_0^6 f'(x)\,dx \approx 20 + 20.2 = 40.2$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for: midpoint sum

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

6-7

$$\int_0^6 f'(x)\,dx \approx 2 \cdot 3.5 + 2 \cdot 0.8 + 2 \cdot 5.8 = 20.2$$

$$f(6) - f(0) = \int_0^6 f'(x)\,dx$$

$$f(6) = f(0) + \int_0^6 f'(x)\,dx \approx 20 + 20.2 = 40.2$$

Part B

1 point is earned for: integral bound

1 point is earned for: answer with reasoning

Since $f'(x) \leq 7$, $\displaystyle\int_0^6 f'(x)\,dx \leq 6 \cdot 7 = 42$.

$f(6) - f(0) \leq 42 \Rightarrow f(6) \leq 20 + 42 = 62$

Therefore, the actual value of $f(6)$ could not be 70.

0	1	2

The student response earns all of the following points:

1 point is earned for: integral bound

1 point is earned for: answer with reasoning

Since $f'(x) \leq 7$, $\displaystyle\int_0^6 f'(x)\,dx \leq 6 \cdot 7 = 42$.

$f(6) - f(0) \leq 42 \Rightarrow f(6) \leq 20 + 42 = 62$

Therefore, the actual value of $f(6)$ could not be 70.

Part C

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_2^4 f''(x)dx = f'(4) - f'(2) = 1.7 - 2 = -0.3$$

6-7

0	1	2

The student response earns all of the following points:

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_{2}^{4} f''(x)dx = f'(4) - f'(2) = 1.7 - 2 = -0.3$$

Part D

1 point is earned for: L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x \to 0} (f(x) - 20e^{x}) = 0$$
$$\lim_{x \to 0} (0.5f(x) - 10) = 0$$

Using L'Hospital's Rule,

$$\lim_{x \to 0} \frac{f(x) - 20e^{x}}{0.5f(x) - 10} = \lim_{x \to 0} \frac{f'(x) - 20e^{x}}{0.5f'(x)} = \frac{4 - 20}{0.5(4)} = -8$$

0	1	2

The student response earns all of the following points:

1 point is earned for: L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x \to 0} (f(x) - 20e^{x}) = 0$$
$$\lim_{x \to 0} (0.5f(x) - 10) = 0$$

Using L'Hospital's Rule,

$$\lim_{x \to 0} \frac{f(x) - 20e^{x}}{0.5f(x) - 10} = \lim_{x \to 0} \frac{f'(x) - 20e^{x}}{0.5f'(x)} = \frac{4 - 20}{0.5(4)} = -8$$

6-7

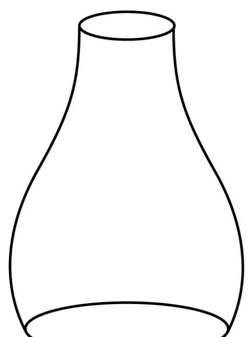
5. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

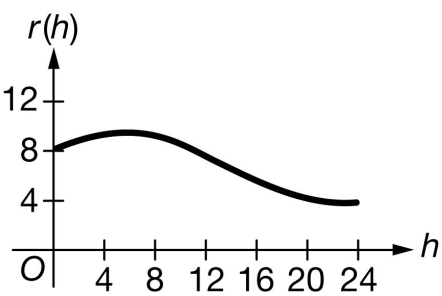
You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.





A 24-centimeter tall vase has cross sections parallel to the base that are circles. Each circular cross section h centimeters above the base has a radius of $r(h) = 8 - 0.1h + 2\sin\left(0.3h^{0.9}\right)$ centimeters for $0 \le h \le 24$. A sketch of the vase and the graph of r are shown above.

(a) Find the area of the region between the graph of r and the h-axis from $h = 0$ to $h = 24$.

(b) Find the volume of the vase.

(c) Evaluate $\int_4^{20} r'(h)\,dh$. Using correct units, interpret the meaning of $\int_4^{20} r'(h)\,dh$ in the context of the problem.

(d) Water is poured into the vase. At the instant when the depth of the water in the vase is 10 centimeters, the depth of the water is increasing at a rate of 0.6 centimeter per second. At that instant, what is the rate of change of the circumference of the surface of the water, in centimeters per second?

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

An unsupported answer of 165.781 (or 165.78) will not earn any points.

6-7

The first point is earned by using $r(h)$ as the integrand in a definite integral. A definite integral with incorrect limits or multiplied by any constant other than 1 will not earn the second point.

Do not read units in this problem.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is generally eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode, area $= 165.509$ (or 165.508).

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Definite integral
- ☐ Answer

Solution:

$$\int_0^{24} r(h)\,dh = 165.780506$$

The area of the region is 165.781 (or 165.78).

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for writing a definite integral (with any limits) with integrand $c(r)^2$, for any nonzero constant c.

Once the first point is earned, it cannot be lost. Any subsequent copy errors in the presence of the correct answer will be considered a calculator restart and will earn the second point.

To earn the second point, a response must earn the first point, show limits from 0 to 24, multiply by π, and present the correct evaluation.

Special case: A response that fails to earn the first point because of a copy error will earn the second point for the correct answer.

A final answer written in terms of π must be correct to three places after the decimal point, rounded or truncated, when converted to a decimal value to earn the second point. (For example, 1252.387π is not correct to three digits after the decimal and therefore does not earn the second point.)

Degree mode: volume $= 3616.746$ (or 3616.745) (See degree mode statement in part (a).)

6-7

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Integrand
- [ ] Answer

Solution:

$$\int_0^{24} \pi(r\,(h))^2\,dh = \pi \cdot 1252.387790 = 3934.492281$$

The volume of the vase is 3934.492 centimeters.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for the value -5.259; no supporting work is required.

The interpretation must clearly indicate that the radius at $h = 20$ is smaller than the radius at $h = 4$. Therefore, “the difference in the radius when $h = 20$ and $h = 4$ …” is not sufficient.

The response, “The difference (or change) in radius from $h = 4$ to $h = 20$ is -5.259 cm” earns both points. However, “The *total* change in radius from $h = 4$ to $h = 20$ is -5.259 cm” earns only the first point.

Degree mode: answer $= -1.481$ (See degree mode statement in part (a).)

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Value
- [ ] Interpretation with units

Solution:

$$\int_4^{20} r'\,(h)\;dh = r\,(20) - r\,(4) = -5.259388$$

6-7

The radius of the vase at $h = 20$ centimeters is 5.259 centimeters less than the radius of the vase at $h = 4$ centimeters.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by using $\frac{dr}{dh} \cdot \frac{dh}{dt}\frac{dr}{dt}$ in an expression for $\frac{dC}{dt}$.

A response may work from $C = 2\pi\left(8 - 0.1h + 2\sin\left(0.3h^{0.9}\right)\right)$. In this case, the first point is earned for $\frac{dC}{dt} = \frac{dC}{dh} \cdot \frac{dh}{dt}$.

The second point is earned for using either the value 0.6 as $\frac{dh}{dt}$ or the value -0.411320 as $\frac{dr}{dh}$ in an expression for $\frac{dC}{dt}$. This point is not earned if either 0.6 or $r'(10)$ is used as $\frac{dr}{dt}$.

The second point can be earned without the first point. For example, a computation or communication error in attempting to use the chain rule for $\frac{dC}{dt}$ does not earn the first point but is eligible for the second point.

A response that uses an incorrect equation for the circumference in terms of r only, such as$C = \pi r^2$, is eligible for the first two points but not the third point.

The third point is earned only for an answer of -1.551 (or -1.55).

A final answer written in terms of π must be correct to three places after the decimal point, rounded or truncated, when converted to a decimal value to earn the third point.

Degree mode: $\text{answer} = -0.349$ (or -0.348) (See degree mode statement in part (a).)

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] $\frac{dr}{dt} = \frac{dr}{dh} \cdot \frac{dh}{dt}$
- [ ] Uses $\left.\frac{dh}{dt}\right|_{h=10} = 0.6$
  – OR –
  Uses $\left.\frac{dr}{dh}\right|_{h=10} = r'(10)$
- [ ] Answer

Solution:

6-7

$$C = 2\pi r \;\Rightarrow\; \frac{dC}{dt} = \frac{d}{dt}(2\pi r) = 2\pi \cdot \left(\frac{dr}{dh} \cdot \frac{dh}{dt}\right)$$

$$\frac{dC}{dt}\Big|_{h=10} = 2\pi \cdot \left(\frac{dr}{dh}\Big|_{h=10} \cdot \frac{dh}{dt}\Big|_{h=10}\right)$$
$$= 2\pi \cdot \left(r'(10) \cdot 0.6\right) = \pi \cdot (-0.493584) = -1.550641$$

The rate of change of the circumference of the surface of the water is -1.551 (or -1.55) centimeters per second.

6-7

6.

t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W, where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t = 0$, the temperature of the water is 55°F. The water is heated for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

(a) Use the data in the table to estimate $W'(12)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of this problem.

(b) Use the data in the table to evaluate $\int_0^{20} W'(t)dt$. Using correct units, interpret the meaning of $\int_0^{20} W'(t)dt$ in the context of this problem.

(c) For $0 \le t \le 20$, the average temperature of the water in the tub is $\frac{1}{20}\int_0^{20} W(t)dt$. Use a left Riemann sum with the four subintervals indicated by the data in the table to approximate $\frac{1}{20}\int_0^{20} W(t)dt$. Does this approximation overestimate or underestimate the average temperature of the water over these 20 minutes? Explain your reasoning.

(d) For $20 \le t \le 25$, the function W that models the water temperature has first derivative given by $W'(t) = 0.4\sqrt{t}\cos(0.06t)$. Based on the model, what is the temperature of the water at time $t = 25$?

Part A

1 point is earned for estimation

1 point is earned for interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

= 1.017 (or 1.016)

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.

0	1	2

The student earns all of the following points:

1 point is earned for estimation

1 point is earned for interpretation with units

6-7

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

= 1.017 (or 1.016)

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.

Part B

1 point is earned for the value

1 point is earned for the interpretation with units

$$\int_0^{20} W'(t)dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.

✓

0	1	2

The student earns all of the following points:

1 point is earned for the value

1 point is earned for the interpretation with units

$$\int_0^{20} W'(t)dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.

Part C

1 point is earned for left Riemann sum

1 point is earned for approximation

1 point is earned for underestimating with reason

6-7

$$
\begin{aligned}
\frac{1}{20}\int_0^{20} W(t)dt &\approx \frac{1}{20}(4\cdot W(0)+5\cdot W(4)+6\cdot W(9)+5\cdot W(15))\\
&=\frac{1}{20}(4\cdot 55.0+5\cdot 57.1+6\cdot 61.8+5\cdot 67.9)\\
&=\frac{1}{20}\cdot 1215.8=60.79
\end{aligned}
$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing

0	1	2	3

The student earns all of the following points:

1 point is earned for left Riemann sum

1 point is earned for approximation

1 point is earned for underestimating with reason

$$
\begin{aligned}
\frac{1}{20}\int_0^{20} W(t)dt &\approx \frac{1}{20}(4\cdot W(0)+5\cdot W(4)+6\cdot W(9)+5\cdot W(15))\\
&=\frac{1}{20}(4\cdot 55.0+5\cdot 57.1+6\cdot 61.8+5\cdot 67.9)\\
&=\frac{1}{20}\cdot 1215.8=60.79
\end{aligned}
$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing

Part D

1 point is earned for the integral

1 point is earned for the answer

$$
\begin{aligned}
W(25) &= 71.0+\int_{20}^{25} W'(t)dt\\
&= 71.0+2.043155=73.043
\end{aligned}
$$

6-7

0	1	2 ✓

The student earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$W(25) = 71.0 + \int_{20}^{25} W'(t)dt$$
$$= 71.0 + 2.043155 = 73.043$$

7. Let *f* be the function defined by $f(x) = \int_0^x \left(2t^3 - 15t^2 + 36t\right) dt$. On which of the following intervals is the graph of $y = f(x)$ concave down?

(A) $(-\infty, 0)$ only

(B) $(-\infty, 2)$

(C) $(0, \infty)$

(D) $(2, 3)$ only ✓

(E) $(3, \infty)$ only

8. Let *f* be a differentiable function such that $f(0) = -5$ and $f'(x) \leq 3$ for all x. Of the following, which is not a possible value for *f*(2)?

(A) -10

(B) -5

(C) 0

(D) 1

(E) 2 ✓

6-7

9. $\int_1^4 \frac{1}{3x^2} dx =$

(A) $-\frac{63}{64}$

(B) $\frac{7}{64}$

(C) $\frac{1}{4}$

(D) $\frac{21}{32}$

Answer C

Correct. The definite integral can be evaluated using the power rule for antidifferentiation and the Fundamental Theorem of Calculus.

$$\int_1^4 \frac{1}{3x^2} dx = \frac{1}{3} \int_1^4 x^{-2} dx = \frac{1}{3} \frac{x^{-1}}{-1}\Big|_1^4 = -\frac{1}{12} - \left(-\frac{1}{3}\right) = -\frac{1}{12} + \frac{4}{12} = \frac{3}{12} = \frac{1}{4}$$

10. $\int_2^x \left(3t^2 - 1\right) dt =$

(A) $x^3 - x - 6$

(B) $x^3 - x$

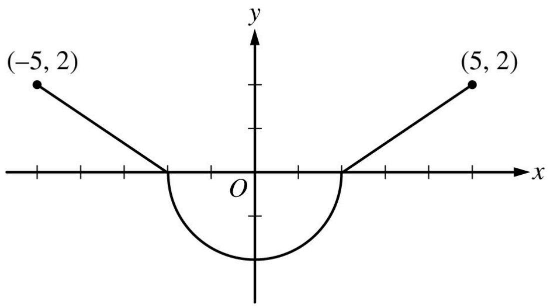
(C) $3x^2 - 12$

(D) $3x^2 - 1$

(E) $6x - 12$

6-7

11.



Graph of f'

The graph of f', the derivative of a function f, consists of two line segments and a semicircle, as shown in the figure above. If $f(2) = 1$, then $f(-5) =$

(A) $2\pi - 2$

(B) $2\pi - 3$

(C) $2\pi - 5$

(D) $6 - 2\pi$

(E) $4 - 2\pi$

12.

If $\sin\left(\frac{1}{x^2+1}\right)$ is an antiderivative for $f(x)$, then $\int_1^2 f(x)\,dx =$

(A) -0.281

(B) -0.102

(C) 0.102

(D) 0.260

(E) 0.282

13.

$\int_1^4 t^{-3/2}\text{dt}$

(A) -1

(B) $-\frac{7}{8}$

(C) $-\frac{1}{2}$

(D) $\frac{1}{2}$

(E) 1

6-7

14. Let $H(x)$ be an antiderivative of $\frac{x^3+\sin x}{x^2+2}$. If $H(5) = \pi$, then $H(2) =$

(A) −9.008

(B) −5.867 ✓

(C) 4.626

(D) 12.150

15. Let $H(x)$ be an antiderivative of $\frac{x^3+\sin x}{x^2+2}$. If $H(5) = \pi$, then $H(2) =$

(A) −9.008

(B) −5.867 ✓

(C) 4.626

(D) 12.150

16. If $g(x) = x^2 - 3x + 4$ and $f(x) = g'(x)$, then $\int_1^3 f(x)dx =$

(A) $-\frac{14}{3}$

(B) -2

(C) 2 ✓

(D) 4

(E) $\frac{14}{3}$

17.

$$f(x) = \begin{cases} x^2 & \text{for } x < 0 \\ -1 & \text{for } x = 0 \\ x & \text{for } x > 0 \end{cases}$$

Let f be the function defined above. What is $\int_{-1}^{1} f(x)dx$?

(A) $\frac{5}{6}$ ✓

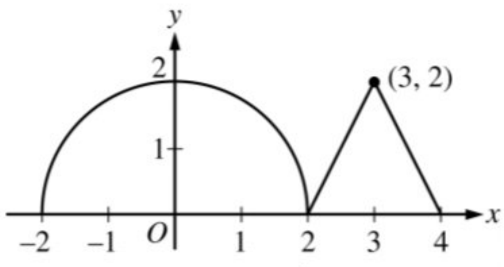
(B) $\frac{2}{3}$

(C) $-\frac{1}{6}$

(D) nonexistent

6-7

18.



Graph of g'

The graph of g', the first derivative of the function g, consists of a semicircle of radius 2 and two line segments, as shown in the figure above.

If $g(0) = 1$, what is $g(3)$?

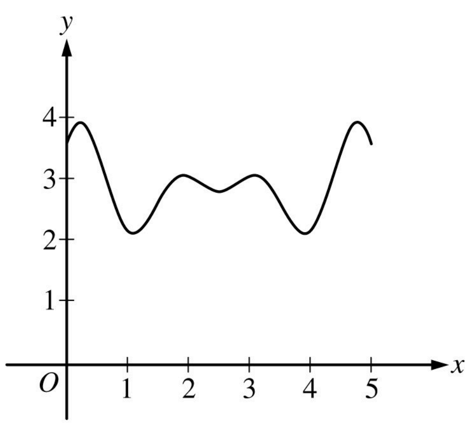
(A) $\pi + 1$

(B) $\pi + 2$ ✓

(C) $2\pi + 1$

(D) $2\pi + 2$

19.



Graph of f'

The graph of f', the derivative of f, is shown in the figure above. If $f(0) = 20$ which of the following could be the value of $f(5)$?

(A) 15

(B) 20

(C) 25

(D) 35 ✓

(E) 40

6-7

20. Let f be a differentiable function such that $f(1) = \pi$ and $f'(x) = \sqrt{x^3 + 6}$. What is the value of $f(5)$?

(A) 11.941

(B) 14.587

(C) 24.672

(D) 27.814

Answer D

This option is correct. By the Fundamental Theorem of Calculus,

$f(5) - f(1) = \int_1^5 f'(x)\,dx$. Therefore $f(5) = f(1) + \int_1^5 \sqrt{x^3+6}\,dx$

$= \pi + 24.672 \approx 27.814.$

21. $\int_{-1}^{3} \left(x^2 - 2x\right) \mathrm{dx} =$

(A) $\frac{-2}{3}$

(B) $\frac{4}{3}$

(C) 8

(D) 12

Answer B

This option is correct. This question involves using the basic power rule

for antidifferentiation and then correctly substituting the endpoints and evaluating:

$\frac{x^3}{3} - x^2\Big|_{-1}^{3} = \left(\frac{27}{3} - 9\right) - \left(-\frac{1}{3} - 1\right) = \frac{4}{3}.$

6-7

22. $$\int_{-2}^{1}\left(8x^3-3x^2\right)dx=$$

(A) -561

(B) -90

(C) -39

(D) 81

Answer C

This option is correct. The question involves using the basic power rule for antidifferentiation and then correctly substituting the endpoints and evaluating:

= (2 - 1) - (32 + 8) = -39.

23. $$\int_{1}^{2}\frac{dx}{2x+1}=$$

(A) $2\ln 2$

(B) $\frac{1}{2}\ln 2$

(C) $2\left(\ln 5-\ln 3\right)$

(D) $\ln 5-\ln 3$

(E) $\frac{1}{2}(\ln 5-\ln 3)$

24. If $\frac{dy}{dt}=-10e^{-t/2}$ and y(0) = 20, what is the value of y(6)?

(A) $20e^{-6}$

(B) $20e^{-3}$

(C) $20e^{-2}$

(D) $10e^{-3}$

(E) $5e^{-3}$

25. If $f'(x)=\cos(x^2)$ and $f(3)=7$, then $f(2)=$

(A) 0.241

(B) 5.831

(C) 6.416

(D) 6.759

(E) 7.241

6-7

26. If $f\,\prime(x) = 3x^2 + 2x$ and $f(2) = 3$, then $f(1) =$

(A) -10

(B) -7

(C) 10

(D) 13

27. If $f'(x) = \sqrt{1+2x^3}$ and f(2) = 0.4 and f(5) =

(A) 29.005

(B) 28.605

(C) 28.205

(D) -28.205

Answer A

This option is correct. By the Fundamental Theorem of Calculus,

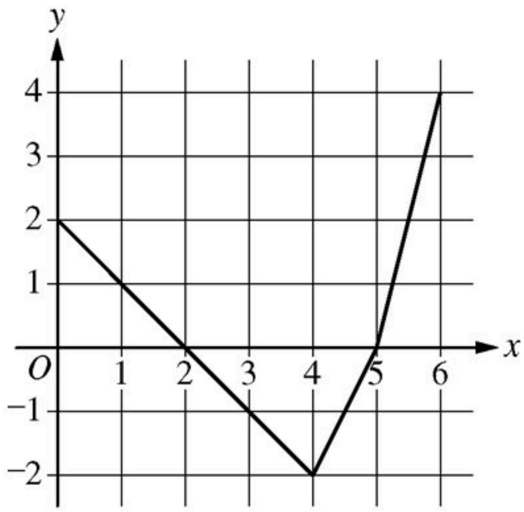
$$f(5) - f(2) = \int_2^5 f'(x)dx$$

Therefore,

$$f(5) = f(2) + \int_2^5 \sqrt{1+2x^3}dx = 0.4 + \int_2^5 \sqrt{1+2x^3}dx = 29.005.$$

6-7

28.



Graph of f

The graph of the function f, shown above, consists of three line segments. If the function g is an antiderivative of f such that $g(2) = 5$, for how many values of c, where $0 \le c \le 6$, does $g(c) = 3$?

(A) zero

(B) one

(C) two

(D) three

Answer D

This option is correct. By the Fundamental Theorem of Calculus and using area and geometry to calculate the definite integral,

$$g(0) = g(2) + \int_2^0 g\prime(x)dx = 5 - \int_0^2 f(x)dx = 5 - 2 = 3.$$

So $c = 0$ is one solution to $g(x) = 3$. Since $g' = f$, the graph of f indicates that g is increasing from $x = 0$ to $x = 2$ since f is positive there.

From $x = 2$ to $x = 4$ the function g decreases by the same amount that it increased from $x = 0$ to $x = 2$ by the symmetry of the regions.

$$g(4) = 5 + \int_2^4 f(x)dx = 5 + (-2) = 3$$

This gives a second solution at $c = 4$. According to the graph of the derivative f, the function g continues to decrease until $x = 5$ and then begins increasing again.

6-7

$$g(5) = 5 + \int_2^5 f(x)dx = 5 + (-3) = 2$$

$$g(6) = 2 + \int_5^6 f(x)dx = 2 + 2 = 4$$

Since g is continuous, the Intermediate Value Theorem guarantees a third solution somewhere on the interval (5, 6). There are no other solutions possible. (The third solution is at $c = 5.5$.)

29.

x	0	1	2	3
$f(x)$	5	2	3	6
$f'(x)$	–3	1	3	4

The derivative of the function f is continuous on the closed interval [0,4]. Values of f and f' for selected values of x are given in the table above. If $\int_0^4 f'(t)dt = 8$, then $f(4) =$

(A) 0
(B) 3
(C) 5
(D) 10
(E) 13

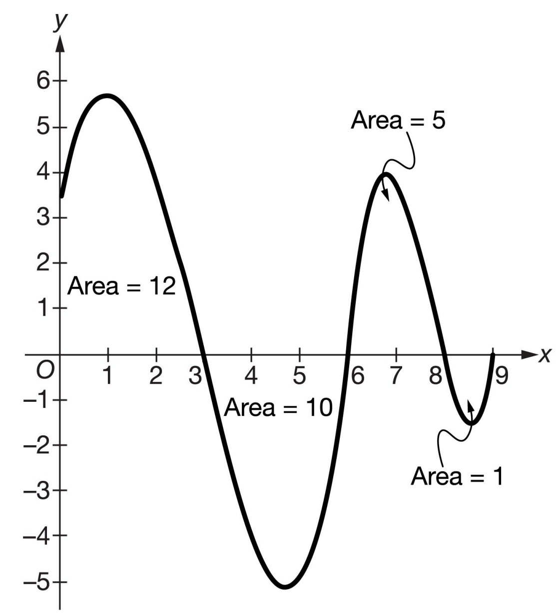
6-7

30. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f'

The figure above shows the graph of f', the derivative of a differentiable function f, on the closed interval $0 \le x \le 9$. The areas of the regions between the graph of f' and the x-axis are labeled in the figure. The function f is defined for all real numbers and satisfies $f(6) = 7$.

Let g be the function defined by $g(x) = x^2 - 1$.

(a) Find the value of $\int_0^9 f'(x)\,dx$.

(b) Given that $f(6) = 7$, write an expression for $f(x)$ that involves an integral. Use this expression to find the absolute minimum value of $f$and the absolute maximum value of $f$on the closed interval $0 \le x \le 9$. Justify your answers.

(c) Find $\int g(x)\,dx$.

(d) Find the value of $\int_2^3 x f'(g(x))\,dx$.

Part A

6-7

The response should include supporting work for the numerical answer of 6.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1

The student response accurately includes a correct value.

Solution:

$$\int_0^9 f'(x)\, dx = 12 - 10 + 5 - 1 = 6$$

Part B

The first point is earned for a correct expression for $f(x)$ OR evidence of correct use of FTC undefined in determining the appropriate $f(x)$ values for consideration.

For the third and fourth points: At most 1 out of 2 points is earned for supporting work that justifies a correct absolute minimum value OR absolute maximum value based on a Candidates Test with a maximum of one arithmetic error.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response

0	1	2	3	4

The student response accurately includes all four of the criteria below.

- ☐ use of Fundamental Theorem of Calculus
- ☐ considers $x = 3,\ x = 6$, and $x = 8$
- ☐ answers for absolute minimum value and absolute maximum value
- ☐ justification

Solution:

$$f(x) = f(6) + \int_6^x f'(t)\, dt$$

The absolute minimum and absolute maximum will occur at a critical point where $f'(x) = 0$

$f\prime(x) = 0 \ \Rightarrow\ x = 3,\ x = 6,\ x = 8$

6-7

The candidates are $x = 0,\ x = 3,\ x = 6,\ x = 8$, and $x = 9$.

x	$f(x)$
0	$f(6)+\int_6^0 f'(t)\,dt = f(6)-\int_0^6 f'(t)\,dt = 7-(12-10)=5$
3	$f(6)+\int_6^3 f'(t)\,dt = f(6)-\int_3^6 f'(t)\,dt = 7-(-10)=17$
6	7
8	$f(6)+\int_6^8 f'(t)\,dt = 7+5=12$
9	$f(6)+\int_6^9 f'(t)\,dt = 7+(5-1)=11$

On the interval $0 \le x \le 9$, the absolute minimum value is $f(0) = 5$ and the absolute maximum value is $f(3) = 17$.

Part C

A response with any errors, including omission of the constant of integration, does not earn the point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1

The student response accurately includes a correct answer.

Solution:

$$\int g(x)\,dx = \int \left(x^2 - 1\right)\,dx = \frac{1}{3}x^3 - x + C$$

Part D

To earn the second point, a response must present a definite integral that correctly handles the substitution of the limits of integration. Correct handling of the reversal of the limits of integration and all calculations are part of the third point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-7

0	1	2	3

The student accurately includes all three of the criteria below.

- ☐ sets $u = x^2 - 1$
- ☐ integral in terms of u
- ☐ answer

Solution:

Let $u = g\left(x\right) = x^2 - 1$.

Then $du = 2x\ dx$ and $\frac{1}{2}\ du = x\ dx$.

$x = 3 \Rightarrow u = 3^2 - 1 = 8$
$x = 2 \Rightarrow u = 2^2 - 1 = 3$

$$\int_2^3 x\,f'\left(g\left(x\right)\right)\ dx = \frac{1}{2}\int_3^8 f'\left(u\right)\ du = \frac{1}{2}(-10+5) = -\frac{5}{2}$$

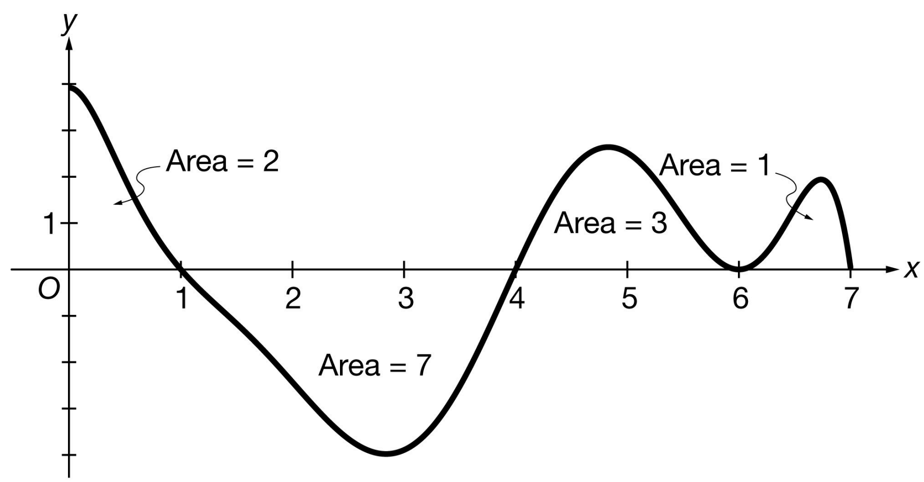
6-7

31. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f'

The figure above shows the graph of f', the derivative of a differentiable function f, on the closed interval $0 \le x \le 7$. The areas of the regions between the graph of f' and the x-axis are labeled in the figure. The function f is defined for all real numbers and satisfies $f(4) = 10$.

Let g be the function defined by $g(x) = 5 - x^2$.

(a) Find the value of $\int_0^7 f'(x)\,dx$.

(b) Given that $f(4) = 10$, write an expression for $f(x)$ that involves an integral. Use this expression to find the absolute minimum value of $f$and the absolute maximum value of $f$on the closed interval $0 \le x \le 7$. Justify your answers.

(c) Find $\int g(x)\,dx$.

(d) Find the value of $\int_1^2 x f'(g(x))\,dx$.

Part A

The response should include supporting work for the numerical answer of -1.

6-7

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1

The student response accurately includes a correct value.

Solution:

$$\int_0^7 f'(x)\,dx = 2 - 7 + 3 + 1 = -1$$

Part B

The first point is earned for a correct expression for $f(x)$ OR evidence of correct use of FTC in determining the appropriate $f(x)$ values for consideration.

For the second point: The response may also consider $x = 6$, though this is not required in order to justify an absolute minimum value or an absolute maximum value on the interval.

For the third and fourth points: At most 1 out of 2 points is earned for supporting work that justifies a correct absolute minimum value OR absolute maximum value based on a Candidates Test with a maximum of one arithmetic error.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4

The student response accurately includes all four of the criteria below.

- ☐ use of Fundamental Theorem of Calculus
- ☐ considers $x = 1$ and $x = 4$
- ☐ answers for absolute minimum value and absolute maximum value
- ☐ justification

Solution:

$$f(x) = f(4) + \int_4^x f'(t)\,dt$$

The absolute minimum and absolute maximum will occur at a critical point where $f'(x) = 0$ or at an endpoint.

6-7

$f\prime\,(x)=0\ \Rightarrow\ x=1,\ x=4,\ x=6$

The critical point at $x=6$ is neither the location of an absolute minimum nor an absolute maximum because f' does not change sign at $x=6$. Thus, the only candidates are $x=0,\ x=1,\ x=4$, and $x=7$.

x	$f(x)$
0	$f(4)+\int_4^0 f'(t)\,dt=f(4)-\int_0^4 f'(t)\,dt$ $=10-(2-7)=15$
1	$f(4)+\int_4^1 f'(t)\,dt=f(4)-\int_1^4 f'(t)\,dt$ $=10-(-7)=17$
4	10
7	$f(4)+\int_4^7 f'(t)\,dt=10+(3+1)=14$

On the interval $0\le x\le 7$, the absolute minimum value is $f\,(4)=10$ and the absolute maximum value is $f\,(1)=17$.

Part C

A response with any errors, including omission of the constant of integration, does not earn the point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1

The student response accurately includes a correct answer.

Solution:

$$\int g\,(x)\;dx=\int\left(5-x^2\right)\;dx=5x-\tfrac{1}{3}x^3+C$$

Part D

To earn the second point, a response must present a definite integral that correctly handles the substitution of the limits of integration. Correct handling of the reversal of the limits of integration and all calculations are part of the third point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-7

0	1	2	3

The student accurately includes all three of the criteria below.

- [ ] sets $u = 5 - x^2$
- [ ] integral in terms of u
- [ ] answer

Solution:

Let $u = g(x) = 5 - x^2$.

Then $du = -2x\ dx$ and $-\frac{1}{2}\ du = x\ dx$.

$x = 2 \Rightarrow u = 5 - 2^2 = 1$
$x = 1 \Rightarrow u = 5 - 1^2 = 4$

$$\int_1^2 x f'(g(x))\ dx = -\frac{1}{2}\int_4^1 f'(u)\ du = \frac{1}{2}\int_1^4 f'(u)\ du = \frac{1}{2}(-7) = -\frac{7}{2}$$

6-7

32. For what value of b does the integral $\int_1^b x^2\, dx$ equal $\lim_{n\to\infty} \sum_{k=1}^{n} \left(1+\frac{2k}{n}\right)^2 \frac{2}{n}$?

(A) $b = 2$ only

(B) $b = 3$ only ✓

(C) b could be any real number.

(D) There is no such value of b.

Answer B

Correct. The sum can be interpreted as a right Riemann sum in the form $\sum_{k=1}^{n} f(1+k\Delta x)\Delta x$, where $f(x) = x^2$ and $\Delta x = \frac{2}{n}$. The value of Δx corresponds to an interval of length 2. The sum starts with the right endpoint $1 + \Delta x$ and ends with the right endpoint $1 + n\Delta x = 1 + 2 = 3$, so the Riemann sum is over the interval $[1, 3]$. The limit of the Riemann sum is the definite integral $\int_1^3 f(x)dx$. There could not be another value of b for which $\int_1^b x^2dx$ has the same value as $\int_1^3 x^2dx$ since $I(b) = \int_1^b x^2dx$ is a strictly increasing function of b. Therefore, $b = 3$ is the only choice.

6-7

33.

x	-1	1
$f(x)$	10	0
$f'(x)$	-1	-9
$f''(x)$	23	-13
$f'''(x)$	-42	6

The function f has derivatives of all orders for all real numbers. The table shown gives values of f and its first three derivatives at selected values of x. What is the value of $\int_{-1}^{1} f''(x)\,dx$?

(A) -36
(B) -10
(C) -8 ✓
(D) 48

Answer C

Correct. By the Fundamental Theorem of Calculus, $\int_{-1}^{1} f''(x)\,dx = f'(1) - f'(-1) = -9 - (-1) = -8$.

34. Let f be a differentiable function such that $f(0) = 5.420$ and $f'(x) = \sqrt{\sin^2 x + x}$. What is the value of $f(2\pi)$?

(A) 7.927
(B) 11.449
(C) 13.295
(D) 16.869 ✓

Answer D

Correct. By the Fundamental Theorem of Calculus, $\int_0^{2\pi} f'(x)\,dx = f(2\pi) - f(0)$. Therefore,

$f(2\pi) = f(0) + \int_0^{2\pi} \sqrt{\sin^2 x + x}\,dx$. The calculator is used to compute the value for the definite integral.

6-7

35. Let g be a differentiable function such that $g(4) = 0.325$ and $g'(x) = \frac{1}{x}e^{-x}\left(\cos\left(\frac{x}{100}\right)\right)$. What is the value of $g(1)$?

(A) 0.109

(B) 0.216

(C) 0.541

(D) 0.688

Answer A

Correct. By the Fundamental Theorem of Calculus, $\int_1^4 g'(x)\,dx = g(4) - g(1)$.

Therefore, $g(1) = g(4) - \int_1^4 g'(x)\,dx = 0.325 - \int_1^4 g'(x)dx = 0.109$.

The calculator is used to do the numerical integration.

36.

x	0	2	4	6
$f(x)$	-22	-6	2	2
$f'(x)$	10	6	2	-2

Selected values of the twice-differentiable function f and its derivative f' are given in the table above. What is the value of $\int_0^6 f'(x)\,dx$?

(A) -12

(B) 12

(C) 24

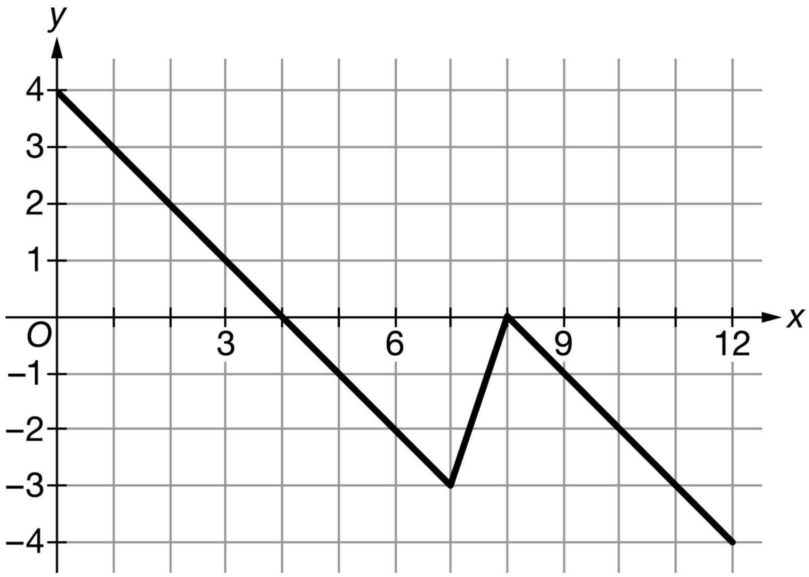
(D) 36

Answer C

Correct. By the Fundamental Theorem of Calculus, $\int_0^6 f'(x)\,dx = f(6) - f(0) = 2 - (-22) = 24$.

6-7

37.



Graph of f

The graph of the piecewise linear function f is shown above.

What is the value of $\int_0^{12} f'(x)\,dx$?

(A) -8

(B) -6

(C) 0

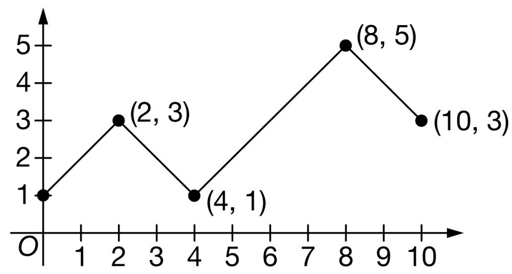
(D) 22

Answer A

Correct. By the Fundamental Theorem of Calculus, $\int_0^{12} f'(x)dx = f(12) - f(0) = (-4) - 4 = -8$, where the values of f at $x = 0$ and $x = 12$ are obtained from the graph.

6-7

38.



Graph of g

The graph of the function g on the closed interval $[0, 10]$ consists of four line segments, as shown above. Let f be the function defined by $f(x) = \int_4^{x^2} g(t)\,dt$. What is the value of $f'(3)$?

(A) -6

(B) 4

(C) 12

(D) 24

Answer D

Correct. The second formulation of the Fundamental Theorem of Calculus along with the chain rule results in the following.

$$f'(x) = \tfrac{d}{dx}\left(\int_4^{x^2} g(t)\,dt\right) = g\left(x^2\right) \cdot \tfrac{d}{dx}\left(x^2\right) = g\left(x^2\right) \cdot 2x$$

$$f'(3) = g(9) \cdot 6 = 4 \cdot 6 = 24$$

The graph of g is used to find the value of $g(9)$.

6-7

39. Let f be a differentiable function such that $f(1) = \frac{\pi}{2}$ and $f'(x) = 3\arctan\left(x^2 - 3x + 2\right)$. What is the value of $f(3)$?

(A) 0.243

(B) 1.328

(C) 2.899

(D) 3.321

Answer C

Correct. By the Fundamental Theorem of Calculus, $\int_1^3 f'(x)\,dx = f(3) - f(1)$. Therefore, $f(3) = f(1) + \int_1^3 f'(x)\,dx = \frac{\pi}{2} + \int_1^3 3\arctan\left(x^2 - 3x + 2\right)dx = 2.899$. The calculator is used to do the numerical integration.

40. Which of the following is an antiderivative of $f(x) = \cos\left(x^2 - 5\right)$?

(A) $\sin\left(x^5 - 5\right)$

(B) $\frac{\sin\left(x^2-5\right)}{2x}$

(C) $\int_{\pi}^{x^2} \cos(t - 5)\,dt$

(D) $\int_{\pi}^{x} \cos\left(t^2 - 5\right)dt$

Answer D

Correct. The function $F(x) = \int_{\pi}^{x} f(t)\,dt$ is an antiderivative of f since $F'(x) = \frac{d}{dx}\int_{\pi}^{x} f(t)\,dt = f(x)$ by the Fundamental Theorem of Calculus.

6-7

41. Let f be a differentiable function such that $f(3) = 2.345$ and $f'(x) = \ln\left(x^2 + 1\right)$. What is the value of $f(5)$?

(A) 3.301

(B) 5.631

(C) 6.950

(D) 7.976

Answer D

Correct. By the Fundamental Theorem of Calculus, $\int_3^5 f'(x)\,dx = f(5) - f(3)$. Therefore, $f(5) = f(3) + \int_3^5 \ln\left(x^2 + 1\right)dx$. The calculator is used to compute the value for the definite integral.

42. Let g be a differentiable function such that $g(10) = 2e$ and $g'(x) = 5e^{-\sqrt{x}}$. What is the value of $g(2)$?

(A) 1.329

(B) 4.107

(C) 6.441

(D) 9.544

Answer A

Correct. By the Fundamental Theorem of Calculus, $\int_2^{10} g'(x)\,dx = g(10) - g(2)$. Therefore, $g(2) = g(10) - \int_2^{10} g'(x)\,dx = 2e - \int_2^{10} 5e^{-\sqrt{x}}dx = 1.329$. The calculator is used to do the numerical integration.

6-7

43.

x	0	1	2	3
$f(x)$	4	9	12	10
$f'(x)$	5	4	1	-6

Selected values of the twice-differentiable function f and its derivative f' are given in the table above. What is the value of $\int_0^3 f'(x)\,dx$?

(A) -11

(B) -1

(C) 6

(D) 10

Answer C

Correct. By the Fundamental Theorem of Calculus, $\int_0^3 f'(x)\,dx = f(3) - f(0) = 10 - 4 = 6$.

44. Let f be a differentiable function such that $f(1) = 2$ and $f'(x) = \sqrt{x^2 + 2\cos x + 3}$. What is the value of $f(4)$?

(A) -6.790

(B) 8.790

(C) 10.790

(D) 12.996

Answer C

Correct. By the Fundamental Theorem of Calculus, $\int_1^4 f'(x)\,dx = f(4) - f(1)$. Therefore, $f(4) = f(1) + \int_1^4 f'(x)\,dx = 2 + \int_1^4 \sqrt{x^2 + 2\cos x + 3}\,dx = 2 + 8.790 = 10.790$. The calculator is used to do the numerical integration.

6-7

45. Which of the following is an antiderivative of $f(x)=\tan\left(e^x+\frac{\pi}{2}\right)$ on $0\leq x\leq\ln\left(\frac{\pi}{2}\right)$?

(A) $\sec^2\left(e^x+\frac{\pi}{2}\right)$

(B) $\frac{\ln\left|\sec\left(e^x+\frac{\pi}{2}\right)\right|}{e^x}$

(C) $\int_0^{e^x}\tan\left(t+\frac{\pi}{2}\right)dt$

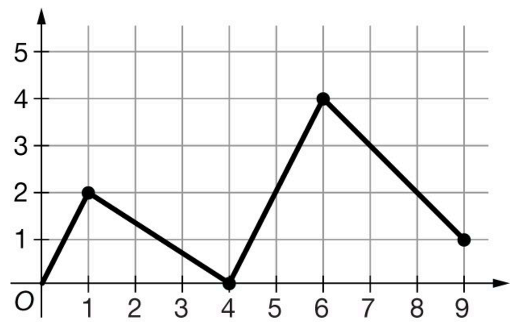
(D) $\int_0^{x}\tan\left(e^t+\frac{\pi}{2}\right)dt$

Answer D

Correct. The function $F(x)=\int_0^x f(t)\,dt$ is an antiderivative of f, since $F'(x)=\frac{d}{dx}\int_0^x f(t)\,dt=f(x)$ by the Fundamental Theorem of Calculus.

6-7

46.



Graph of g

The graph of the function g on the closed interval $[0, 9]$ consists of four line segments, as shown above. Let f be the function defined by $f(x) = \int_4^x g(t)\,dt$. What is the value of $f(8)$?

(A) -1

(B) 2

(C) 10

(D) 14

Answer C

Correct. $f(8) = \int_4^8 g(t)\,dt$

This definite integral is evaluated by computing the area between the graph of g and the x-axis between $x = 4$ and $x = 8$.

6-7

47.

x	$f(x)$	$f'(x)$	$f''(x)$
2	-2	1	3
7	4	5	2

The table shown gives selected values of a twice-differentiable function f and its first two derivatives. What is the value of $\int_2^7 f'(t)\,dt$?

(A) 6

(B) 5

(C) 4

(D) 2

Answer A

Correct. By the Fundamental Theorem of Calculus, $\int_2^7 f'(x)dx = f(7) - f(2) = 4 - (-2) = 6.$

6-7

48. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	6	18
$M(t)$ (coins per day)	40	54	46

A class collected 1-cent coins to raise money for classroom supplies. Students in the class deposited the coins into a container where all the coins were kept until the end of the collection period. For time t days, $0 \leq t \leq 18$, the rate at which coins are being added to the container, in coins per day, is modeled by a twice-differentiable function M. Values of $M(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $M'(6)$ using the average rate of change of M over the interval $0 \leq t \leq 18$.

(b) Explain the meaning of $M'(6)$ in the context of the problem.

(c) The function $C(t)$ models the number of coins in the container at time t days, where $C'(t) = M(t)$. It is known that $C(6) = 300$. Write an equation for the line tangent to the graph of $y = C(t)$ at $t = 6$.

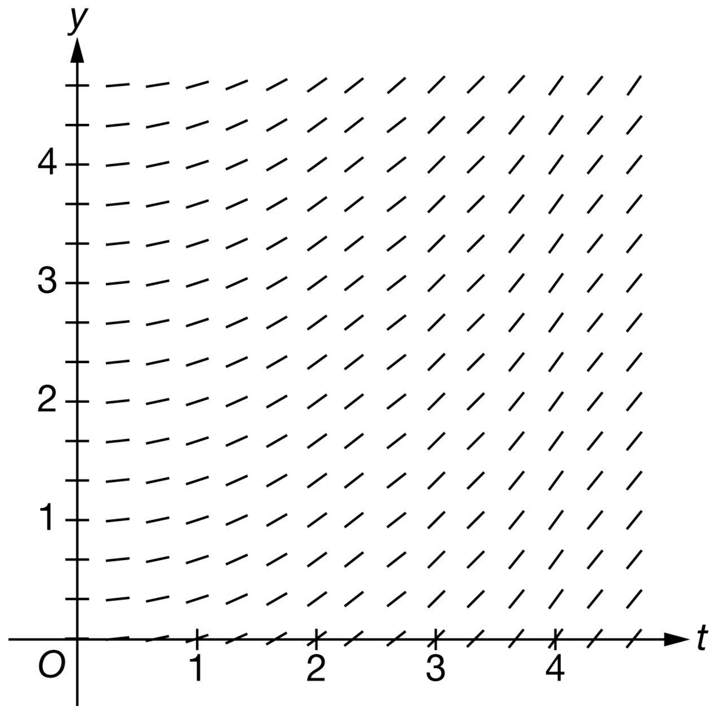
(d) Find the value of $\int_0^6 M'(3t)\,dt$. Show the calculations that lead to your answer.

(e) Use a trapezoidal sum with the two subintervals indicated by the table to approximate $\int_0^{18} M(t)\,dt$. Show the calculations that lead to your answer.

(f) A second class is also collecting 1-cent coins. For $0 \leq t \leq 18$, the rate at which coins are being collected by the second class can be modeled by the differentiable function B, where B is a function of t. Assume the weight of each coin is 3 grams. Write an expression in terms of M and B for the rate of change, in grams per day, of the total weight of the coins collected by both classes at $t = 7$ days.

6-7

(g) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = 0.3y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

Answers without supporting work do not earn this point. Supporting work must be at least a difference and a quotient.

Response does not need to be simplified, but any simplification must be done correctly.

If a decimal is presented, it must be correct to three decimal places. We do not read anything past the third decimal place.

The second point is earned for the correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols.

The response will not be penalized for any verbal attempt to discuss the answer after the correct presentation of the approximation.

The minimum response needed to earn both points is: $\frac{46-40}{18}$ coins per day per day.

6-7

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$M'(6) \approx \frac{M(18)-M(0)}{18-0} = \frac{46-40}{18} = \frac{1}{3}$ coin per day per day

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This part is intended to be independent from part (a), so we do not want or need to see any reference to the value obtained in (a). If that value is referenced, conclusion must be consistent.

Three components of a correct answer:

· Indicating a rate of a rate (look for the word rate to be used correctly twice)

· Including the context of coin collection

· Referring to the specific point in time

Units will not be considered for this point. They do not need to be present, and if they are, they do not need to be correct.

Any incorrect statements make the response ineligible for the point. For example:

· $M'(6)$ is the average (or an approximation of, or an estimate of) rate of change of . . . .

· $M'(6)$ is the acceleration (or rate of change of velocity) of . . . .

“Increasing” or “decreasing” may be used in place of “changing” as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Beware of phrases such as “decreasing at a negative rate.”

· Example: $M'(6)$ is the rate of increase of the rate of change of the number of coins at 6 days. (earns the point)

· Example: $M'(6)$ means the number of coins per day is increasing by 6 coins/day/day. (does not earn the point)

Acceptable references to time include (but are not limited to):

· On day 6

6-7

· At $t = 6$ (with correct units of time, without correct units of time, or even with incorrect units of time)

· After $t = 6$ (with correct units of time, without correct units of time, or even with incorrect units of time)

If the response clearly indicates an interval of time (e.g. “from day 0 to day 6”) the response will not earn the point.

Examples that earn the point include:

· $M'(6)$ represents the rate at which the rate of change of the number of coins is changing on day 6.

· $M'(6)$ is the rate of change in the rate of change of the number of coins at 6 days.

· $M'(6)$ represents the rate at which the rate of change of the number of coins is increasing at $t = 6$.

· $M'(6)$ equals the answer from part (a) means the rate of change in the number of coins per day is increasing at a rate of (answer from part (a)) at $t = 6$.

· $M'(6)$ represents the rate at which the rate of change of the number of coins is decreasing on the sixth day.

Examples that do not earn the point include:

· $M'(6)$ represents the rate at which the number of coins is changing on day 6.

· $M'(6)$ is the rate of change in the number of coins per day at $t = 6$ days

· The rate of change in the number of coins per day is increasing at $t = 6$.

· The rate of change in the rate of change of the number of coins per day is increasing at $t = 6$.

· $M'(6) = -\frac{1}{3}$ means that the rate of change in the number of coins is decreasing at a rate of $-\frac{1}{3}$ at $t = 6$. (double negative).

0	1

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$M'(6)$ is the rate at which the rate of change of the number of 1-cent coins in the container is changing, in coins per day per day, on day 6.

Part C

6-7

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned with a correct tangent line equation in any form. Equivalent forms include (but are not limited to):

· $y = 54\,(t-6)+300$

· $y-300 = 54\,(t-6)$

· $y = 54t-24$

The student may write x in place of t and still earn the point.

No work is required to earn the point.

We will accept $C\ (t)$ in lieu of y in the equation of the tangent line.

0	1

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$y = C'\ (6)\,(t-6)+C\ (6)$

$y = 54\,(t-6)+300$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution.

Two things are required to award this point:

· Coefficient of $\frac{1}{3}$ correctly applied to an integral or an antiderivative

· Evidence that the correct limits of 0 to 18 will be or have been applied

Examples that will earn the first point (these are just a few):

· $\frac{1}{3}\int_0^{18} M'\ (u)\ du$

6-7

- $\frac{1}{3}M\ (u)\big|_0^{18}$ or $\frac{1}{3}M\ (3t)\big|_0^{6}$

Responses that include misstatements such as using t-values for limits with u as the variable of integration in an intermediate step which is later corrected may still earn this point. Intermediate errors in bounds that are subsequently corrected will disqualify the student from earning the third point. If the bounds are never appropriately handled, the response cannot earn the first point. Scratch work that is not connected to the solution will not be read.

Examples:

- $\int_0^6 M'\ (3t)\ dt = \frac{1}{3}\int_0^6 M'\ (u)\ du = \frac{1}{3}M\ (3t)\big|_0^6$ earns the first point, but will not earn the third point.
- $\int_0^6 M'\ (3t)\ dt = \frac{1}{3}\int_0^6 M'\ (u)\ du = \frac{1}{3}[M\ (6) - M\ (0)]$ does not earn the first point.

The second point is for the Fundamental Theorem of Calculus, demonstrating the knowledge that $\int_a^b M'\ (u)\ du = M\ (u)\big|_a^b = M\ (b) - M\ (a)$

This point can be earned with either $kM\ (3t)\ |_a^b$, $kM\ (t)\ |_a^b$, $k\,(M\ (b) - M\ (a))$ where k is a nonzero constant [not necessarily correct], $a = 0$, and $b = 6$ or $b = 18$.

A response can earn this point without earning the first point.

$\frac{1}{3}[M\ (18) - M\ (0)]$ earns the first two points.

The third point is for the correct numerical value. Our answer only.

Values from the table must be used (i.e., $\frac{1}{3}[M\ (18) - M\ (0)]$ is not sufficient to earn the third point).

If the response includes any linkage error, (i.e., the use of equal signs to equate two expressions that are not equal), the third point will not be earned. If an indefinite integral is linked with a definite integral the response will not be penalized.

- $\int_0^6 M'\ (3t)\ dt = \frac{1}{3}\int M'\ (u)\ du = \frac{1}{3}M\ (3t)\big|_0^6 = \ldots$ is eligible for the third point.

Eligibility: Must have earned the first two points.

The minimum response needed to earn all three points is: $\frac{1}{3}(46 - 40)$.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

6-7

- [ ] Substitution
- [ ] Fundamental Theorem of Calculus
- [ ] Answer

Solution:

Let $u = 3t$. Then $du = 3\ dt$.

When $t = 0$, $u = 0$.

When $t = 6$, $u = 18$.

$$\int_0^6 M'(3t)\ dt = \frac{1}{3}\int_0^{18} M'(u)\ du$$

$$= \frac{1}{3}(M(18) - M(0))$$

$$= \frac{1}{3}(46 - 40) = 2$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the form of a trapezoidal sum.

Point is earned for the setup of the trapezoidal sum. Numerical values do not need to be pulled from the table for this point, but Δt values must be presented. Responses that contain one incorrect numerical value in the trapezoidal sum setup are still eligible to earn the first point, but will not earn the second.

Response may present a left Riemann sum and a right Riemann sum, then average the two. Again, one incorrect numerical value will not disqualify the response from earning this first point.

Response may compute trapezoids as areas of rectangles and triangles.

Trapezoidal areas may be given in a graph. In order to earn this point, the response must clearly show the computations leading to the values presented in the graph, and must show that these areas are added to obtain the final approximation value.

A response using subtraction instead of addition will not earn the point (e.g., $\frac{1}{2}[6(M(0) - M(6)) + 12(M(6) - M(18))]$ earns no points).

Response utilizing "Trapezoidal Rule" will not earn this point, unless this represents only one incorrect numerical values.

$\cdot \frac{1}{2}(6)[M(0) + 2M(6) + M(18)]$ is equivalent to $\frac{1}{2}[6(M(0) + M(6)) + 6(M(6) + M(18))]$, thus has only one incorrect numerical value (the second interval width is 6 instead of 12), so can earn the point.

6-7

- $\frac{1}{2}(9)\left[M\ (0) + 2M\ (6) + M\ (18)\right]$ has two incorrect numerical values (both interval widths are incorrect) so will not earn the point.

The second point is for the correct numerical result.

Our answer only. Response does not need to be simplified, but if it is, it must be simplified correctly.

The answer must be accompanied with supporting work. Supporting work may appear in a graph, with clearly labeled areas. It is possible to earn the second point without the first if the supporting work is not sufficient to earn the first point.

Pictures without clear area computations and a clear summation of quantities can earn at most the second point.

Special case: A positive constant multiple of a complete and correctly expressed trapezoidal sum (with the correct values pulled from the table) will earn only the first point:

- $k\left[\left(\frac{40+54}{2}\right) \cdot 6 + \left(\frac{54+46}{2} \cdot 12\right)\right]$ for $k > 0$ earns the first but not the second point.

The minimum response needed to earn both points: $\left(\frac{94}{2}\right) \cdot 6 + \left(\frac{100}{2}\right) \cdot 12$

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Form of trapezoidal sum
- ☐ Answer

Solution:

$$\int_0^{18} M\ (t)\ dt \approx \left(\frac{M\,(0)+M\,(6)}{2}\right)(6-0) + \left(\frac{M\,(6)+M\,(18)}{2}\right)(18-6)$$

$$= \left(\frac{40+54}{2}\right) \cdot 6 + \left(\frac{54+46}{2}\right) \cdot 12 = 882$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a correct constant multiple or correct sum.

Response must contain the constant 3 as a factor of the expression, and/or the correct sum $B\ (7) + M\ (7)$.

If the sum is incorrect, then the constant factor of 3 must be applied either to a sum that involves only $B\ (t)$ and $M\ (t)$, $B'\ (t)$ and $M'\ (t)$, or integrals of $B\ (t)$ and $M\ (t)$, or to one of the correct terms of the sum $B\ (7)$ and

6-7

$M\ (7)$, with no additional incorrect terms.

Examples that earn the point include:

$\cdot\, 3\,[M\ (t) + B\ (t)]$

$\cdot\, 3\displaystyle\int [M\ (t) + B\ (t)]\ dt$

$\cdot\, 3\,[M' + B']$

$\cdot\, 3B\ (7)$

Examples that do not earn the point include:

$\cdot\, 3\,[M\ (t) + B'\ (t)]$

$\cdot\, 3\displaystyle\int M\ (t)\ dt$

$\cdot\, 3M' + B'$

$\cdot\, 3B\ (7) + t$

Special case: $3M\ (7) + B\ (7)$ is an error in parentheses, so will earn the first point but is not eligible to earn the second point.

If the constant multiple is missing or incorrect, the sum must be correct. $k\,[B\ (7) + M\ (7)]$ for any nonzero constant k will earn the first point.

The second point is for the correct expression.

$3B\ (7) + 3M\ (7)$ (or an equivalent expression) earns both points.

Inclusion of units (correct or incorrect) will not affect the score.

A completely correct expression with one additional constant term added/subtracted would earn just the first point. For example, $3B\ (7) + 3M\ (7) + 3\,(2)$ earns only the first point.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Constant multiple or sum
- ☐ Expression

Solution:

6-7

An expression for the rate of change, in grams per day, of the total weight of the coins collected by both classes at time $t = 7$ days is $3B\,(7) + 3M\,(7)$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for a correct statement identifying specific inconsistencies between the given differential equation and the given slope field. There are many such correct statements, such as:

· The response references a specific point on the y-axis (such as $(0,\ 4)$), and indicates that the given slope field shows a slope of zero, yet the differential equation would yield a positive slope.

· The response references a pattern of slopes, and explains why that pattern does not fit our differential equation (e.g. "the slope field shows slopes that change as t changes, but our differential equation depends only on y, so slopes should remain constant for a given value of y as t changes.")

Identifying a mismatch between the differential equation and the slope field at one point is sufficient to earn the point.

Response is not penalized for referencing "x" instead of "t".

If the response contains any incorrect statement, the response will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point, nor will responses that present information beyond what we are given. For example:

· Responses that specify that the slope field shows a particular nonzero slope at a particular point will not earn the point, as this cannot actually be determined visually (i.e., the response should not indicate that at $(1,\ 2)$ the slope shown on the slope field is 1)

· Responses that state a specific slope value on the slope field for a general value of x or y that are not correct for all points with that x or y value do not earn the point. (e.g. "the slope field shows a slope of 0 when $y = 4$" is an incorrect statement)

A response may involve a solution to the differential equation, with an argument based on the resulting function. If done correctly, and presented with a correct argument, this will earn the point.

0	1

The student response accurately includes the criteria below.

☐ Explanation

Solution:

6-7

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

6-7

49. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	4	10	18
$M(t)$ (number of 1-cent coins)	0	200	540	900

A class collected 1-cent coins to raise money for classroom supplies. Students in the class deposited the coins into a container where all the coins were kept until the end of the collection period. For time t days, $0 \leq t \leq 18$, the total number of coins in the container is modeled by an increasing, twice-differentiable function M. Values of $M(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $M'(10)$ using the average rate of change of M over the interval $4 \leq t \leq 18$.

(b) Explain the meaning of $M'(10)$ in the context of the problem.

(c) It is known that $M'(4) = 48$. Write an equation for the line tangent to the graph of $y = M(t)$ at $t = 4$.

(d) Use a right Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{18} M(t)\,dt$. Show the computations that lead to your answer.

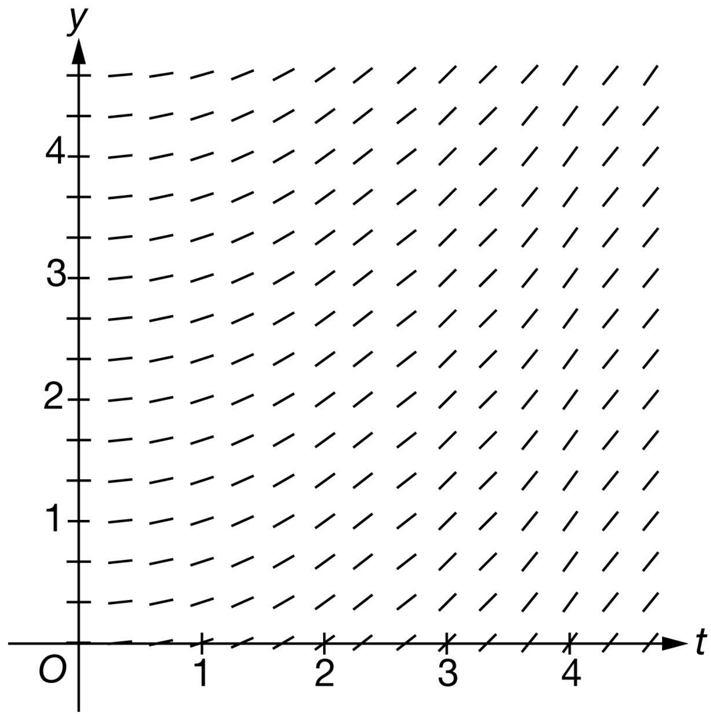
(e) Is the approximation from part (d) an overestimate or an underestimate for $\int_0^{18} M(t)\,dt$? Give a reason for your answer.

(f) Find the value of $\int_0^{6} M'(3t)\,dt$. Show the computations that lead to your answer.

(g) For $0 \leq t \leq 18$, the total number of coins collected can also be modeled by the differentiable function N, where N is a function of t. Assume the weight of each coin is 2.5 grams. Write an expression for the rate of change, in grams per day, of the total weight of the coins at $t = 9$ days.

6-7

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = 0.4y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work. The supporting work must be at least a difference and a quotient. Correct values must be taken from the table to earn the point. The difference quotient does not need to be simplified.

Examples that earn the first point include:

- $\frac{900-200}{18-4}$
- $\frac{M(18)-M(4)}{14} = \frac{700}{14}$
- $\frac{M(18)-M(4)}{14} = 50$

Examples that do not earn the first point include:

- $\frac{M(18)-M(4)}{14}$
- $\frac{M(18)-M(4)}{18-4}$

Minimal responses:

6-7

· $\frac{900-200}{14}$ or $\frac{700}{18-4}$ earns first point

· Neither $\frac{700}{14}$ nor 50, with no supporting work, will earn this point

The second point is for correct units

The correct units must be attached to a numerical value: either correct or incorrect.

The units may be written using words or a combination of words and symbols. For example:

· Coins per day

· Pennies per day

· Cents per day

· Coins/day

The response will not be penalized for any verbal attempt to discuss the answer after the correct answer has been presented.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Approximation
- [ ] Units

Solution:

$$M'(10) \approx \frac{M(18)-M(4)}{18-4} = \frac{900-200}{14} = 50 \text{ coins per day}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

· Indicating a rate of change

· Including the context of coin collection

· Referring to the specific point in time, $t = 10$

6-7

Students are not penalized for incorrect units in this part. (Whether correct or incorrect, units have no effect on the score in this part.)

Responses may use “changing” or “increasing” as a modifier as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Example: “The number of coins is increasing at a rate of 50 coins per day...” is okay.

Acceptable references to a specific point in time include (but are not limited to):

· On day 10 and after 10 days

· At $t = 10$

· At $t = 10$ weeks or any incorrect unit of time

If the response clearly indicates an interval of time (e.g., “from day 10 on,” “during the first 10 days,” or “for $t = 0$ to $t = 10$ ”), the point is not earned.

Examples that earn the point include:

· $M'(10)$ is the rate of change of the number of coins per day at time $t = 10$

· The rate of change of number of coins is 50 coins per day on day 10

· $M'(10)$ is the rate of change in the number of coins at time $t = 10$ in coins per *month*

· 50 is the rate of increase in the number of coins per day on day 10

Examples that do not earn the point include:

· $M'(10)$ is the average rate of change of the total number of coins per day on day 10

· $M'(10)$ is the total number of coins per day per day on day 10

· The rate of change is 50 coins per day (Missing “at $t = 10$ ”)

· $M'(10)$ is the slope of the tangent line at $t = 10$

· Rate of change of M at time $t = 10$

· Coins per day increases on day 10

0	1 ✓

The student response accurately includes the criteria below.

☐ Explanation

6-7

Solution:

$M'(10)$ is the rate at which the number of 1 -cent coins in the container is changing, in coins per day, on day 10 .

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. The use of x in place of t is acceptable. Some examples include

· $y-200=48(t-4)$, $y-200=48(x-4)$

· $y=48(t-4)+200$, $y=48(x-4)+200$

· $y=48t+8$, $y=48x+8$

The key calculus concept being assessed in this part of the problem is that the value of $M'(4)$ is the slope of the tangent line. In that spirit, we are accepting both $y=\ldots$ and $M(t)=\ldots$, but not $M(4)=\ldots$

· $M(t)=48(t-4)+200$ earns the point

· However, $M(4)=48(t-4)+200$ does not earn the point

0	1

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$y=M'(4)(t-4)+M(4)$

$y=48(t-4)+200$ 3

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

· A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example below).

6-7

· Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns. Implicit responses are eligible for both points.

A response presenting a left Riemann sum (instead of right), will not earn any points for part (d).

The second point is for the correct numerical result.

· In order to earn the second point, there must be supporting calculations.

· The numerical values for $M(4)$, $M(10)$, and $M(18)$ must be pulled from the table.

· Any simplification shown must be correct.

Use the following as an illustration to score both points. (In most cases, the expression on the left side represents the entirety of relevant work shown for the response.)

· A response of $200 \cdot (4-0) + 540 \cdot (10-4) + 900 \cdot (18-10)$ earns both points.

· A response of $4M(4) + 6M(10) + 8M(18)$ earns the first point. Eligible for second point if more work is on the page.

· A response of $4M(4) + 6M(10) + 8M(18) = 11,240$ earns both points.

· A response of $4 \cdot 200 + 6 \cdot 540 + 8 \cdot 900$ earns both points.

· A response of $4 \cdot 200 + 10 \cdot 540 + 8 \cdot 900$ contains one numerical error and earns the first point, but is not eligible for the second point.

· A response of $4 \cdot 200 + 6 \cdot 540 + 8 \cdot 900 = 10,240$ earns the first point for left side of equation but does not earn second point due to incorrect simplification.

· A response of $800 + 3240 + 7200$ or $800 + 3240 + 7200 = 11,240$ does not earn the first point because we cannot see the sum of 3 products. However, this is the correct sum, so we see an answer with supporting work. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$4 \cdot 200 + 6 \cdot 540 + 8 \cdot 900$

0	1	2

The student response accurately includes **both** of the criteria below.

6-7

- ☐ Right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{18} M\ (t)\ dt \approx M\ (4)\ (4-0) + M\ (10)\ (10-4) + M\ (18)\ (18-10)$$

$$= 200\ \cdot\ 4 + 540\ \cdot\ 6 + 900\ \cdot\ 8 = 11{,}240$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned when the response states that the approximation is an overestimate because M is increasing.

The response does not need an explicit reference to the approximation in part (d), nor does there need to be an answer given in part (d).

The following statements are sufficient for earning this point:

· Overestimate because M is increasing

· Overestimate because the function is increasing

Note: At this point, there is only one function being discussed in this problem so we will assume “the function” is M.

When a response includes additional information about concavity, interpret that as a lack of understanding about the question, so this will not earn the point. When the response references a graph or the given table rather than the function M, the point is not earned.

Examples that earn the point:

· The right Riemann sum overestimates because the function is increasing.

· The sum is an overestimate because $M\ (t)$ is increasing.

· The sum is an overestimate because $M'\ (t) > 0$.

Examples that do not earn the point:

· The sum is an overestimate because $M'\ (t)$ is increasing. ($M'\ (t)$ increasing is not known.)

· The right Riemann sum overestimates because the graph is increasing.

· The right Riemann sum overestimates because the function is increasing and concave up.

6-7

0	1

The student response accurately includes the criteria below.

- [ ] Answer with reason

Solution:

Because M is an increasing function, and this is a right Riemann sum, the answer from part (d) is an overestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are three components of substitution that need to be seen:

· Uses $u = 3t$, which may be done implicitly

· Uses the correct constant of $\frac{1}{3}$

· Uses the correct limits for the substitution (0 and 18). These limits may only be seen when evaluating the antiderivative. These don't need to be seen as limits on a definite integral.

· Either of the expressions below will earn the substitution point without any additional supporting work:
$\frac{1}{3} M(3t)\Big|_0^6$ or $\frac{1}{3} M(u)\Big|_0^{18}$

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

· Limits for t attached to an integral with u as the variable of integration (or vice versa),

· Improper equals signs, such as an indefinite integral equal to a definite integral

· Reversed limits of integration, for example: $\int_{18}^{0} M'(u)\, du$

The second point is for the Fundamental Theorem of Calculus. This point can be earned with any of the following, where k is a nonzero constant and $b = 6$ or $b = 18$. No explanation or additional supporting work is necessary.

· $kM(3t)\Big|_0^b$

6-7

$\cdot\, k\left(M(b) - M(0)\right)$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is 300.

Values from the table must be used. $\frac{1}{3}[M(18) - M(0)]$ is not enough to earn the answer point.

The expression $\frac{1}{3}(900 - 0)$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

The response cannot earn this point without also earning the FTC point.

Minimalist response: The expression $\frac{1}{3}(900)$ represents the minimum work that must been seen in a response in order to earn all three points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3\ dt$.

When $t = 0$, $u = 0$.

When $t = 6$, $u = 18$.

$$\int_0^6 M'(3t)\ dt = \frac{1}{3}\int_0^{18} M'(u)\ du$$

$$= \frac{1}{3}(M(18) - M(0))$$

$$= \frac{1}{3}(900 - 0) = 300$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-7

This point is earned for a correct expression at $t = 9$. No supporting work or justification is required.

The correct expression of $2.5M'(9)$ or $2.5N'(9)$ earns this point.

No justifications or explanations are required.

Any incorrect statement such as, $M'(9) = 2.5N'(9)$, does not earn this point.

0	1 ✓

The student response accurately includes the criteria below.

☐ Expression

Solution:

An expression for the rate of change, in grams per day, of the total weight of the coins at $t = 9$ days is $2.5N'(9)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between two things: the differential equation, $\frac{dy}{dt} = 0.4y$, and the given slope field.

The response does not need to explicitly reference both the differential equation and the given slope field., but it must explicitly reference one and implicitly reference the other.

Acceptable responses include:

· Identifying an inconsistency between the differential equation and the slope field at one point , e.g., at the point $(4,\ 0)$, the slope of the segment should be 0, but it is positive.

· Referencing a pattern of slopes, and explaining why that pattern does not fit the given differential equation.

A response that lacks even an implicit reference to the one of two things being compared will not earn the point. For example, merely stating either $\frac{dy}{dt}$ is positive, or that the slope field is positive discretely does not earn the point.

· “The values for y are greater than or equal to zero so $\frac{dy}{dt} \geq 0$ ” lacks any reference to the slope field and will not earn the point.

· “All the slopes in the slope field are positive” lacks any reference to the differential equation and will not earn the point.

6-7

Vague answers which have the potential to be incorrect will not earn the point. Some examples are:

· "The field is increasing"

· Responses that state a specific slope value on the slope field for a general value of x or y that is not correct for all points with that x or y value do not earn the point. (e.g. "the slope field shows a slope of 0 when $y = 4$ " is an incorrect statement)

A response may correctly solve or attempt to solve the differential equation and then provide a correct argument without reference to their solution. Such a response would earn the point.

A response might actually solve the differential equation and provide a correct argument on the resulting function.

0	1

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

6-7

50. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	6	13	18
$M(t)$ (coins per day)	42	38	35	30

A class collected 1-cent coins to raise money for classroom supplies. Students in the class deposited the coins into a container where all the coins were kept until the end of the collection period. For time t days, $0 \leq t \leq 18$, the rate at which coins are being added to the container is modeled by a decreasing, twice-differentiable function M. Values of $M(t)$ at various times t are shown in the table above.

(a) Use a right Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{18} M(t)\,dt$. Show the computations that lead to your answer.

(b) Is the approximation from part (a) an overestimate or an underestimate for $\int_0^{18} M(t)\,dt$? Give a reason for your answer.

(c) Using correct units, approximate $M'(13)$ using the average rate of change of M over the interval $6 \leq t \leq 18$.

(d) Explain the meaning of $M'(13)$ in the context of the problem.

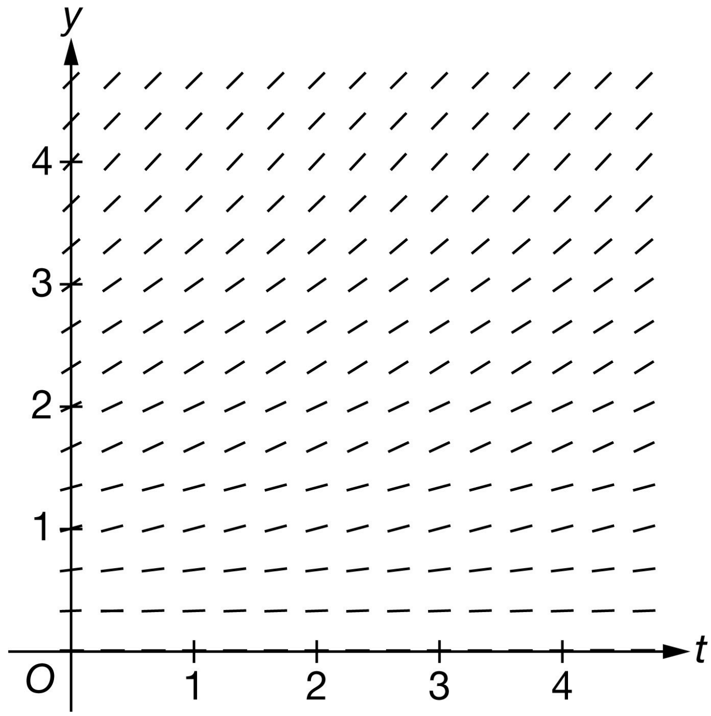
(e) It is known that $M'(6) = -0.5$. Write an equation for the line tangent to the graph of $y = M(t)$ at $t = 6$.

(f) Find the value of $\int_0^{18} M'\left(\frac{t}{3}\right)\,dt$. Show the computations that lead to your answer.

(g) A second class is also collecting 1-cent coins. For $0 \leq t \leq 18$, the total number of coins collected by the second class can be modeled by the decreasing and differentiable function G, where G is a function of t. Assume the weight of each coin is 3 grams. Write an expression for the rate of change, in grams per day, of the total weight of the coins for the second class at $t = 8$ days.

6-7

(h) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = G'(t)$ for the function G defined in part (g).



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example below).

Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns, like so.

38	6	228
35	7	245
30	5	+150
		623

or

6-7

$(6-0)(38)$	$=228$
$(13-6)(35)$	$=245$
$(18-13)(30)$	$=150$
	623

Implicit responses such as those above are eligible for both points.

A response presenting a left Riemann sum (instead of right), will not earn any points for part (a).

The second point is for the correct numerical result.

In order to earn the second point, there must be supporting calculations.

The numerical values for $M\,(6)$, $M\,(13)$, and $M\,(18)$ must be pulled from the table

Any simplification shown must be correct.

Use the following as an illustration how to score both points. (In most cases, you see scores for which the expression represents the entirety of relevant work shown for the response.)

· A response of $38 \cdot (6-0) + 35 \cdot (13-6) + 30 \cdot (18-13)$ earns both points.

· A response of $6 \cdot M\,(6) + 7 \cdot M\,(13) + 5 \cdot M\,(18)$ earns the first point. Eligible for second point if more work is on the page.

· A response of $6 \cdot M\,(6) + 7 \cdot M\,(13) + 5 \cdot M\,(18) = 623$ earns both points.

· A response of $6 \cdot 38 + 7 \cdot 35 + 5 \cdot 30$ earns both points.

· A response of $6 \cdot 38 + 7 \cdot 35 + 6 \cdot 30$ contains one numerical error. Earns the first point. Not eligible for the second point.

· A response of $6 \cdot 38 + 7 \cdot 35 + 5 \cdot 30 = 641$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.

· A response of $228 + 245 + 150$ or $228 + 245 + 150 = 623$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we are not seeing an unsupported answer. The second point will be earned.

The response does not need to include "approximately" in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points. $6 \cdot 38 + 7 \cdot 35 + 5 \cdot 30$

0	1	2

6-7

The student response accurately includes **both** of the criteria below.

- ☐ Right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{18} M(t)\,dt \approx M(6)(6-0) + M(13)(13-6) + M(18)(18-13)$$

$$= 38 \cdot 6 + 35 \cdot 7 + 30 \cdot 5 = 623$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an underestimate because M is decreasing.

The response does not need an explicit reference to part (a) or to a right Riemann sum.

A response that appeals to concavity as a reason will NOT earn this point.

The following examples earn the point:

· Underestimate because M is decreasing

· Underestimate because $M'(t) < 0$

· Underestimate because the graph of M is decreasing (we know they are talking about the given function M)

· Underestimate because the function is decreasing. (There is only one function in the problem at this point.)

The following examples do not earn the point:

· Underestimate because M is concave down/up

· Underestimate because the graph/curve is decreasing. Since no graph/curve is provided, we do not know what graph/curve they are talking about.

· Underestimate because the table [or table values] is/are decreasing. The behavior of the values shown in the table is not the same as the behavior of the function M. The table does not tell us what is happening to M between the given points.

0	1 ✓

6-7

The student response accurately includes the criteria below.

- [ ] Answer with reason

Solution:

Because M is a decreasing function, and this is a right Riemann sum, this is an underestimate.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

We need to see both a difference and a quotient. They don't have to be together.

The expression does not need to be simplified, but values for $M\ (18)$ and $M\ (6)$ must be pulled from the table.

Decimal approximation is not expected, but if presented, then the decimal value must be correct to three decimal places, either rounded or truncated. We do not read anything past the third decimal place.

· If the first three places match -0.666 or -0.667 then the point is earned.

· It does not matter whether any subsequent decimal places match the correct value or not. (For example, the value -0.6672 will earn the point because the first three decimal places match the correct rounded value.)

· Responses with fewer than three decimal places (-0.66, -0.67, -0.7, etc.) will not earn the point

The response does not need to include "approximately" in any way. We will read $=$ and $\approx$ as the same thing.

The second point is for correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols. For example:

· Coins per day per day or Coins/day/day

· Coins per day^2 or Coins/day^2

· Coins per day squared

We do not read work shown in part (d) for this point. The units must appear in part (c).

The letter "C" is not a standard abbreviation for "coins." It will not be interpreted as such.

The response will not be penalized for any attempt to verbally describe their answer after a correct answer has been presented.

6-7

Minimalist response: The following expression and units represent the minimum work that must be seen in a response in order to earn both points. $\frac{30-38}{12}$ coins/day^2

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Approximation
- [ ] Units

Solution:

$$M'(13) \approx \frac{M(18)-M(6)}{18-6} = \frac{30-38}{12} = -\frac{2}{3} \text{ coin per day per day}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

· Indicating a rate of a rate

· Including the context of coin collection

· Explicitly referring to the specified point in time ($t = 13$)

Incorrect or missing units will not be penalized as long as there is a clear connection to the context of the problem.

The words "increasing" and "decreasing" are interpreted as modifiers describing one of the specified rates.

If the numerical value from part (c) is used in this interpretation, then we expect that the choice of "increasing" or "decreasing" is consistent with that value.

The following examples use $-\frac{2}{3}$ as the imported value from part (c). If a different [incorrect] value is imported from part (c), then these statements should be consistent with that value. (Remember that we are reading $=$ and $\approx$ as the same thing.)

· "… increasing at a rate of $-\frac{2}{3}$" is OK [Units not required in part (d)]. (the point is not yet earned with this phrase alone)

· "… decreasing at a rate of $\frac{2}{3}$" is OK [Units not required in part (d)]. (the point is not yet earned with this phrase alone)

6-7

· "... decreasing at a rate of $-\frac{2}{3}$" is not OK and the point is not earned.

Rate of a rate

To earn this point, the response must explicitly discuss the rate of change of a rate of change. For example:

· The rate of change in the rate at which coins are changing

· The rate of change in coins is changing (or decreasing) at a rate of . . .

· The rate at which the rate of change of coins is changing

The words "increasing," "decreasing," or "changing" are not interpreted as rates.

Units are not interpreted as rates. (The phrase "coins per day" is not enough to indicate a rate.)

Slope of tangent line, velocity, and acceleration are not appropriate interpretations for this context.

"Rate of change of M" is not enough without describing what M represents in this context.

Context

A key component of the context for this explanation is "coins."

Explicit reference to $t = 13$

Acceptable reference to specific point in time include (but are not limited to):

· On day 13 (where day could be replaced with any other unit of time)

· At $t = 13$ (with or without units of time)

· After 13 days (where days could be replaced with any other unit of time).

Note: we are choosing to assume that "after" is being used in lieu of "at" unless we have additional evidence to the contrary.

If the response is explicitly indicating an interval of time (e.g., "from day 13 on" or "during the first 13 days"), this point will not be earned.

0	1 ✓

The student response accurately includes the criteria below.

☐ Explanation

Solution:

6-7

$M'(13)$ is the rate at which the rate of change of the number of 1-cent coins in the container is changing, in coins per day per day, on day 13.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. Any of these equations can be written as is without any explanation or additional supporting work. If rewriting of equations is done (transforming to slope-intercept form, perhaps), it must be done correctly.

· $y = -0.5\,(t-6) + 38$

· $y - 38 = -0.5\,(t-6)$

· $y = -0.5t + 41$

The key calculus concept being assessed in this part of the problem is that the value of $M'(6)$ is the slope of the tangent line. In that spirit:

· The response will not be penalized for using x instead of t.

· We are accepting $M\,(t)$ in lieu of y in all forms of the tangent line equation.

0	1

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$y = M'\,(6)\,(t-6) + M\,(6)$

$y = -0.5\,(t-6) + 38$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are three components of substitution that we need to see:

· Uses $u = \frac{t}{3}$, which may be done implicitly

· Uses the correct constant of 3

6-7

· Uses the correct limits for the substitution (0 and 6). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

Either of the expressions $3M\left(\frac{t}{3}\right)\Big|_0^{18}$ or $3M(u)\Big|_0^6$ will earn the substitution point without any additional supporting work.

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

· Limits for t attached to an integral with u as the variable of integration (or vice versa)

· Improper equals signs, such as an indefinite integral equal to a definite integral

· Reversed limits of integration, for example: $\displaystyle\int_6^0 M'(u)\,du$

The second point is for the Fundamental Theorem of Calculus.

This point can be earned with any of the following, where k is a nonzero constant and $b=6$ or $b=18$. No explanation or additional supporting work is necessary.

o $kM\left(\frac{t}{3}\right)\Big|_0^b$

o $k(M(b)-M(0))$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is -12.

Values from the table must be used. $3(M(6)-M(0))$ is not enough to earn the answer point.

The expression $3(38-42)$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

The response cannot earn this point without also earning first two points.

Minimalist response: The expression $3(38-42)$ represents the minimum work that must been seen in a response in order to earn all three points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

6-7

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{3}$. Then $du = \frac{dt}{3}$.

When $t = 0$, $u = 0$.

When $t = 18$, $u = 6$.

$$\int_0^{18} M'\left(\frac{t}{3}\right)\,dt = 3\int_0^6 M'(u)\,du$$

$$= 3\,(M(6) - M(0))$$

$$= 3\,(38 - 42) = -12$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression at $t = 8$ earns the point. It may be written as $3 \cdot G'(8)$ or $3 \cdot \left.\frac{dG}{dt}\right|_{t=8}$.

The expression $3 \cdot G'(t)$ is not sufficient because it does not use $t = 8$.

The response must use G (not M or any other function name) in order to earn the point.

0	1 ✓

The student response accurately includes the criteria below.

- ☐ Expression

Solution:

An expression for the rate of change, in grams per day, of the total weight of the coins for the second class at $t = 8$ days is $3G'(8)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-7

This point is earned for a correct statement identifying the distinction between the differential equation $\frac{dy}{dt} = G'(t)$ and the given slope field.

In order to earn the point, the response must state that the values of the differential equation are negative while the slopes in the given slope field are positive or 0 (nonnegative).

Some ways to say that the values of differential equation are negative:

· G is decreasing

· $G'(t) < 0$

· G has negative slopes

When discussing the slopes in the slope field, the response must explicitly refer to the signs of the slopes.

The following examples could earn the point:

· The slope field has nonnegative slopes.

· The slopes shown are positive or 0.

· The slope field contains positive slopes.

· "The graph" or "The field" could be used in lieu of "The slope field."

The following examples will not earn the point:

· The slope field is increasing/positive.

· The slope field looks like it represents an increasing function.

· The slope field represents a quadratic [or exponential] function.

· "It" (rather than "the slope field")

Statements like "G is a decreasing function, so the signs of the slopes in the slope field should be negative" can earn this point because "should be" is indicating contrast with what is known.

The response will not be penalized for saying that the slopes in the slope field are "always" positive or "all" positive.

0	1 ✓

The student response accurately includes the criteria below.

☐ Explanation

6-7

Solution:

G is a decreasing function, so $\dfrac{dy}{dt} = G'(t) \leq 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are positive.

6-7

51. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	10	35	40
$N(t)$ (coins per minute)	-8	-11	-13	-15

A child has collected 5-cent coins and is preparing them to deposit in the bank by counting them into bags. For time t minutes, $0 \le t \le 40$, the rate of the total number of coins remaining to prepare, in 5-cent coins per minute, is modeled by a decreasing, twice-differentiable function N. Values of $N(t)$ at various times t are shown in the table above.

(a) Use a left Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{40} N(t)\,dt$. Show the calculations that lead to your answer.

(b) Is your answer from part (a) an overestimate or an underestimate for $\int_0^{40} N(t)\,dt$? Give a reason for your answer.

(c) Using correct units, approximate $N'(10)$ using the average rate of change of N over the interval $0 \le t \le 35$.

(d) Explain the meaning of $N'(10)$ in the context of the problem.

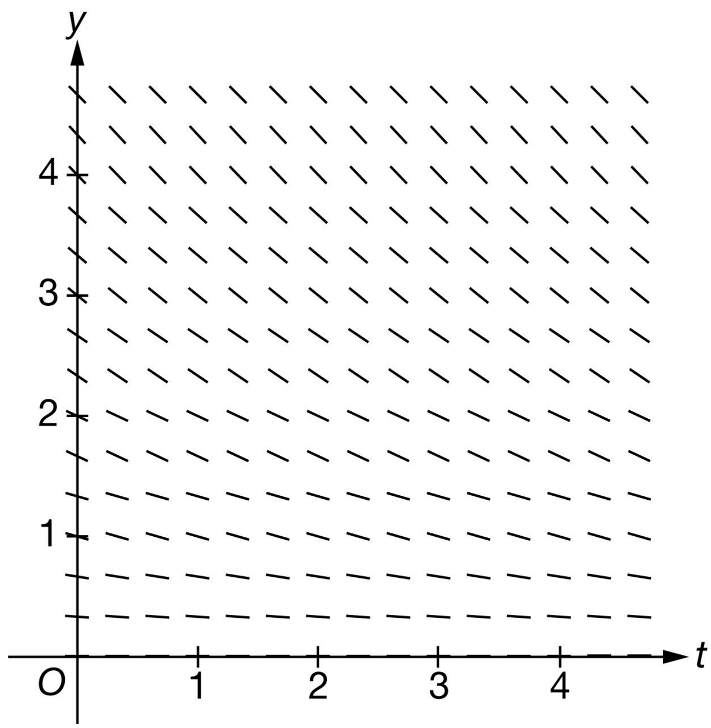
(e) It is known that $N'(35) = -0.2$. Write an equation for the line tangent to the graph of $y = N(t)$ at $t = 35$.

(f) Find the value of $\int_0^{40} N'\left(\frac{t}{4}\right)dt$. Show the calculations that lead to your answer.

(g) A second child is also preparing 5-cent coins for deposit. For $0 \le t \le 40$, the total number of 5-cent coins prepared by the second child for deposit can be modeled by the increasing, differentiable function F, where F is a function of t. Assume the weight of each coin is 5 grams. Write an expression for the rate of change, in grams per minute, of the total weight of the 5-cent coins remaining for the second child at $t = 23$ minutes.

6-7

(h) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = F'(t)$ for the function F defined in part (g).



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a left Riemann sum. The setup must be a sum of three products.

A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example below).

Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns, like so.

10	(-8)	-80
25	(-11)	-275
5	(-13)	$+(-65)$
		-420

or

6-7

$(10-0)(-8)$	$=-80$
$(35-10)(-11)$	$=-275$
$(40-35)(-13)$	$=-65$
	-420

Implicit responses such as those above are eligible for both points.

A response presenting a right Riemann sum (instead of left), will not earn any points for part (a).

The second point is for the correct numerical result.

In order to earn the second point, there must be supporting calculations.

The numerical values for $N(0)$, $N(10)$, and $N(35)$ must be pulled from the table.

Any simplification shown must be correct.

Use the following as an illustration of how to score both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

· A response of $(-8)\cdot(10-0)+(-11)\cdot(35-10)+(-13)\cdot(40-35)$ earns both points.

· A response of $10\cdot N(0)+25\cdot N(10)+5\cdot N(35)$ earns the first point. Eligible for second point if more work is on the page.

· A response of $10\cdot N(0)+25\cdot N(10)+5\cdot N(35)=-420$ earns both points.

· A response of $10\cdot(-8)+25\cdot(-11)+5\cdot(-13)$ earns both points.

· A response of $10\cdot(-8)+35\cdot(-11)+5\cdot(-13)$ contains one numerical error. Earns the first point. Not eligible for the second point.

· A response of $10\cdot(-8)+25\cdot(-11)+5\cdot(-13)=-400$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.

· A response of $(-80)+(-275)+(-65)$ or $(-80)+(-275)+(-65)=-420$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we are not seeing an unsupported answer. The second point will be earned.

The response does not need to include "approximately" in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$10\cdot(-8)+25\cdot(-11)+5\cdot(-13)$

6-7

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Left Riemann sum
- [ ] Answer

Solution:

$$\int_0^{40} N(t)\,dt \approx N(0)(10-0) + N(10)(35-10) + N(35)(40-35)$$

$$= (-8)\cdot 10 + (-11)\cdot 25 + (-13)\cdot 5 = -420$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an overestimate because N is decreasing.

The response does not need an explicit reference to part (a) or to a left Riemann sum.

A response that appeals to concavity as a reason will NOT earn this point.

The following table contains some examples of what WILL or will NOT earn the point.

The following examples earn the point:

· Underestimate because N is decreasing

· Underestimate because $N'(t) < 0$

· Underestimate because the graph of N is decreasing (we know they are talking about the given function N)

· Underestimate because the function is decreasing. (There is only one function in the problem at this point.)

The following examples do not earn the point:

· Underestimate because N is concave down/up

· Underestimate because the graph/curve is decreasing. Since no graph/curve is provided, we do not know what graph/curve they are talking about.

· Underestimate because the table [or table values] is/are decreasing. The behavior of the values shown in the table is not the same as the behavior of the function N. The table does not tell us what is happening to N between the

6-7

given points.

0	1

The student response accurately includes the criteria below.

- [ ] Answer with reason

Solution:

Because N is a decreasing function, and this is a left Riemann sum, this is an overestimate.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

We need to see both a difference and a quotient. They don't have to be together.

The expression does not need to be simplified, but values for $N\ (35)$ and $N\ (0)$ must be pulled from the table.

Decimal approximation is not expected, but if presented, then the decimal value must be correct to three decimal places, either rounded or truncated. We do not read anything past the third decimal place.

· If the first three places match -0.142 or -0.143 then the point is earned.

· It does not matter whether any subsequent decimal places match the correct value or not. (For example, the value -0.1424 will earn the point because the first three decimal places match the correct value.)

· Responses with fewer than 3 decimal places (-0.14, -0.1, etc.) will not earn the point

The response does not need to include "approximately" in any way. We will read $=$ and $\approx$ as the same thing.

The second point is for correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols. For example:

· Coins per minute per minute or Coins/min/min

· Coins per min^2 or Coins/min^2

· Coins per minute squared

We do not read work shown in part (d) for this point. The units must appear in part (c).

6-7

The letter “ C ” is not a standard abbreviation for “Coins.” It will not be interpreted as such.

The response will not be penalized for any attempt to verbally describe their answer after a correct answer has been presented.

Minimalist response: The following expression and units represent the minimum work that must be seen in a response in order to earn both points. $\frac{-13+8}{35}$ coins/min^2 or $\frac{-13-(-8)}{35}$ coins/min^2

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$N'(10) \approx \frac{N(35)-N(0)}{35-0} = \frac{-13-(-8)}{35} = -\frac{1}{7} \text{ coin per minute per minute}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

· Indicating a rate of a rate

· Including the context of coin preparation (coins remaining)

· Explicitly referring to the specified point in time ($t = 10$)

Incorrect or missing units will not be penalized as long as there is a clear connection to the context of the problem.

The words “increasing” and “decreasing” are interpreted as modifiers describing one of the specified rates.

If the numerical value from part (c) is used in this interpretation, then we expect that the choice of “increasing” or “decreasing” is consistent with that value.

The following examples use $\frac{-1}{7}$ as the imported value from part (c). If a different [incorrect] value is imported from part (c), then these statements should be consistent with that value. (Remember that we are reading $=$ and $\approx$ as the same thing.)

6-7

· "… increasing at a rate of $\frac{-1}{7}$ " is OK [Units not required in part (d)] (the point is not earned with this phrase alone)

· "… decreasing at a rate of $\frac{-1}{7}$ " is OK [Units not required in part (d)] (the point is not earned with this phrase alone)

· "… decreasing at a rate of $\frac{-1}{7}$ " is not OK and the point is not earned.

Note: the context is complicated. $N\ (t)$ is describing the rate of change of the remaining unprepared coins. The values in the table indicate that the number of unprepared coins is decreasing more rapidly as time passes. Common sense might tell us that this equates to the idea that the rate at which coins are prepared is increasing as time passes. Explaining the meaning using "coins prepared" rather than "coins remaining" will not prevent the response from earning this point.

Rate of a rate

To earn this point, the response must explicitly discuss the rate of change of a rate of change. For example:

· The rate of change in the rate at which coins are changing

· The rate of change in coins is changing/increasing/decreasing at a rate of . . .

· The rate at which the rate of change of coins is changing

The words "increasing," "decreasing," or "changing" are not interpreted as rates.

Units are not interpreted as rates. (The phrase "coins per minute" is not enough to indicate a rate.)

Slope of tangent line, velocity, and acceleration are not appropriate interpretations for this context.

"Rate of change of N " is not enough without describing what N represents in this context.

Context

A key component of the context for this explanation is "coins."

Any misstatement about coins prepared vs. coins remaining will not prevent the response from earning this point.

Explicit reference to $t = 10$

Acceptable references to a specific point in time include (but are not limited to):

· At minute 10 (where minute could be replaced with any other unit of time)

· At $t = 10$ (with or without units of time)

· After 10 minutes (where minutes could be replaced with any other unit of time).

6-7

Note: we are choosing to assume that “after” is being used in lieu of “at” unless we have additional evidence to the contrary.

If the response is explicitly indicating an interval of time (e.g., “from minute 10 on” or “during the first 10 minutes”), this point will not be earned.

0	1

The student response accurately includes the criteria below.

- [ ] Explanation

Solution:

$N'(10)$ is the rate at which the rate of change in the number of 5 -cent coins remaining is changing, in coins per minute per minute, at minute 10.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. Any of these equations can be written as is without any explanation or additional supporting work. If rewriting of equations is done (transforming to slope-intercept form, perhaps), it must be done correctly.

· $y = -0.2(t - 35) - 13$

· $y + 13 = -0.2(t - 35)$

· $y = -0.2t - 6$

The key calculus concept being assessed in this part of the problem is that the value of $N'(35)$ is the slope of the tangent line. In that spirit:

· The response will not be penalized for using x instead of t.

· We are accepting $N(t)$ in lieu of y in all forms of the tangent line equation.

0	1

The student response accurately includes the criteria below.

- [ ] Tangent line equation

6-7

Solution:

$y = N'(35)(t - 35) + N(35)$

$y = -0.2(t - 35) - 13$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are 3 components of substitution that we need to see:

· Uses $u = \frac{t}{4}$, which may be done implicitly

· Uses the correct constant of 4

· Uses the correct limits for the substitution (0 and 10). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

Either of the expressions $4N\left(\frac{t}{4}\right)\Big|_0^{40}$ or $4N(u)\Big|_0^{10}$ will earn the substitution point without any additional supporting work.

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

· Limits for t attached to an integral with u as the variable of integration (or vice versa)

· Improper equals signs, such as an indefinite integral equal to a definite integral

· Reversed limits of integration, for example: $\int_{10}^{0} N'(u)\, du$

The second point is for the Fundamental Theorem of Calculus.

This point can be earned with any of the following, where k is a nonzero constant and $b = 10$ or $b = 40$. No explanation or additional supporting work is necessary.

· $kN\left(\frac{t}{4}\right)\Big|_0^{b}$

· $k(N(b) - N(0))$

This point can be earned without the substitution point.

The third point is for the correct numerical value, which is -12.

Values from the table must be used. $4(N(10) - N(0))$ is not enough to earn the answer point.

6-7

The expression $4(-11-(-8))$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they do not need to be correct to earn this point.

The response cannot earn this point without also earning first two points.

Minimalist response: The expression $4(-11-(-8))$ represents the minimum work that must been seen in a response in order to earn all three points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{4}$. Then $du = \frac{dt}{4}$.

When $t = 0$, $u = 0$.

When $t = 40$, $u = 10$.

$$\int_0^{40} N'\left(\frac{t}{4}\right) dt = 4\int_0^{10} N'(u)\, du$$

$$= 4(N(10) - N(0))$$

$$= 4(-11-(-8)) = -12$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy for this part: This question was difficult due to the fact that the new function introduced here represents the total number of coins prepared by the second child, but the question asks about the total weight of the coins remaining.

The key calculus concept being assessed in this part of the problem is to understand that the rate of change in the total weight of the coins (either prepared or remaining) at time $t = 23$ could be written as an expression involving

6-7

a constant multiple of $F'(23)$. In that spirit, either of the following expressions will earn the point: $5F'(23)$ or $-5F'(23)$.

Notations such as $5 \cdot \left.\frac{dF}{dt}\right|_{t=23}$ may also be used and earn the point.

The expression must use the time $t = 23$, so $5F'(t)$ is not sufficient to earn the point.

0	1

The student response accurately includes the criteria below.

- [ ] Expression

Solution:

Let $C(0)$ be the total number of coins available to be prepared at time $t = 0$. Let $R(t)$ be a model for the number of coins remaining at time t with $R(t) = C(0) - F(t)$. The rate of change of the number of coins remaining at time t, in grams per minute, is $R'(t) = -F'(t)$. Therefore, the rate of change of the total weight of the 5 -cent coins remaining for the second child at $t = 23$ minutes, in grams per minute, is $-5F'(23)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between the differential equation $\frac{dy}{dt} = F'(t)$ and the given slope field.

In order to earn the point, the response must:

State that the values of the differential equation are positive while the slopes in the given slope field are negative or 0 (non-positive).

Some ways to say that the values of differential equation are positive:

· F is increasing

· $F'(t) > 0$

· F has positive slopes

When discussing the slopes in the slope field, the response must explicitly refer to the signs of the slopes.

The following examples could earn the point:

· The slope field has nonpositive slopes.

6-7

· The slopes shown are negative or 0.

· The slope field contains negative slopes.

· “The graph” or “The field” could be used in lieu of “The slope field.”

The following examples will not earn the point:

· The slope field is decreasing/negative.

· The slope field looks like it represents a decreasing function.

· The slope field represents a quadratic [or exponential] function.

· “It” (rather than “the slope field”)

Statements like “ F is an increasing function, so the signs of the slopes in the slope field should be positive” can earn this point because “should be” is indicating contrast with what is known.

The response will not be penalized for saying that the slopes in the slope field are “always” negative or “all” negative.

0	1

The student response accurately includes the criteria below.

☐ Explanation

Solution:

F is an increasing function, so $\dfrac{dy}{dt} = F'(t) \geq 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are negative.

6-7

52. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	10	40
$N(t)$ (coins per minute)	-10	-16	-8

A child has collected 5-cent coins and is preparing them to deposit in the bank by putting them into bags. For time t minutes, $0 \le t \le 40$, the rate of change in the number of coins remaining to prepare, in coins per minute, is modeled by a twice-differentiable function N. Values of $N(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $N'(10)$ using the average rate of change of N over the interval $0 \le t \le 40$.

(b) Explain the meaning of $N'(10)$ in the context of the problem.

(c) The function $C(t)$ models the number of coins remaining to be put into bags at time t minutes, where $C'(t) = N(t)$. It is known that $C(10) = 400$. Write an equation for the line tangent to the graph of $y = C(t)$ at $t = 10$.

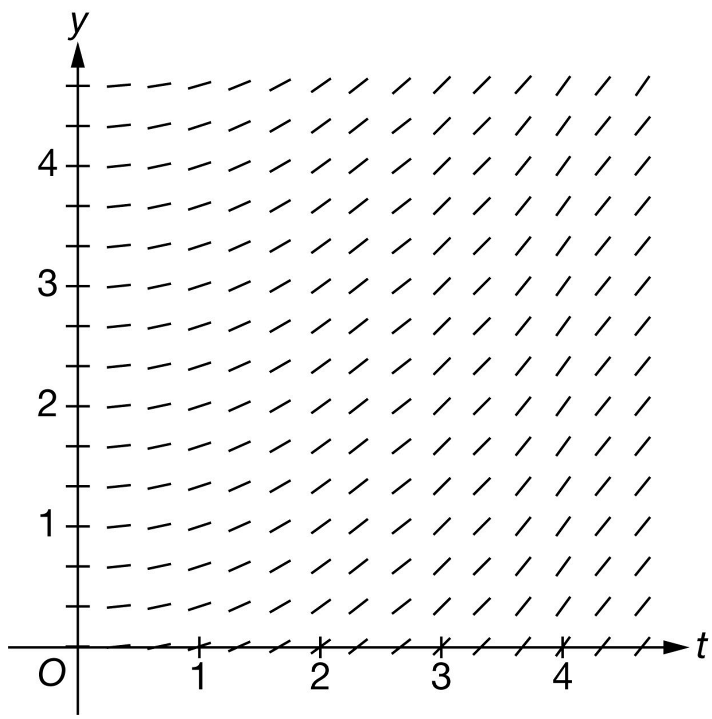
(d) Find the value of $\int_0^{10} N'(4t)\,dt$. Show the calculations that lead to your answer.

(e) Use a trapezoidal sum with the two subintervals indicated by the table to approximate $\int_0^{40} N(t)\,dt$. Show the calculations that lead to your answer.

(f) A second child is preparing another deposit of 5-cent coins. For $0 \le t \le 40$, the rate of change in the number of coins remaining for the second child to prepare, in coins per minute, can be modeled by the differentiable function F, where F is a function of t. Assume the weight of each coin is 5 grams. Write an expression in terms of N and F for the rate of change, in grams per minute, in the combined weight of the 5-cent coins remaining for the two children to prepare at $t = 11$ minutes.

6-7

(g) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = 0.4y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

Answers without supporting work do not earn this point. Supporting work must be at least a difference and a quotient.

Response does not need to be simplified, but any simplification must be done correctly.

If a decimal is presented, it must be correct to 3 decimal places. We do not read anything past the third decimal place.

The second point is earned for the correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols.

The response will not be penalized for any verbal attempt to discuss the answer after the correct presentation of the approximation.

The minimum response needed to earn both points is: $\frac{-8+10}{40}$ coins per minute per minute

6-7

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Approximation
- [ ] Units

Solution:

$$N'(10) \approx \frac{N(40)-N(0)}{40-0} = \frac{-8-(-10)}{40} = \frac{1}{20} \text{ coin per minute per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This part is intended to be independent from part (a), so we do not want or need to see any reference to the value obtained in (a). If that value is referenced, conclusion must be consistent.

Three components of a correct answer:

· Indicating a rate of a rate (look for the word rate to be used correctly twice)

· Including the context of coin collection

· Referring to the specific point in time

Units will not be considered for this point. They do not need to be present, and if they are, they do not need to be correct.

Any incorrect statements make the response ineligible for the point. For example:

· $N'(10)$ is the average (or an approximation of, or an estimate of) rate of change of . . . .

· $N'(10)$ is the acceleration (or rate of change of velocity) of . . . .

"Increasing" or "decreasing" may be used in place of "changing" as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Beware of phrases such as "decreasing at a negative rate."

· Example: $N'(10)$ is the rate of increase of the rate of change of the number of coins at 10 minutes. (earns the point)

· Example: $N'(10)$ means the number of coins per minute is increasing by $\frac{1}{20}$ coins/min/min (does not earn the point)

6-7

Acceptable references to time include (but are not limited to):

· At the tenth minute

· At $t = 10$ (with correct units of time, without correct units of time, or even with incorrect units of time)

· After $t = 10$ (with correct units of time, without correct units of time, or even with incorrect units of time)

If the response clearly indicates an interval of time (e.g. "over the first ten minutes") the response will not earn the point.

Examples that earn the point include:

· $N'(10)$ represents the rate at which the rate of change of the number of coins is changing at minute 10.

· $N'(10)$ is the rate of change in the rate of change of the number of coins at 10 minutes.

· $N'(10)$ represents the rate at which the rate of change of the number of coins is increasing at $t = 10$.

· $N'(10)$ equals (answer from part (a)) means the rate of change in the number of coins per minute is increasing at a rate of (answer from part (a)) at $t = 10$.

· $N'(10)$ represents the rate at which the rate of change of the number of coins is decreasing after 10 10 minutes.

Examples that do not earn the point include:

· $N'(10)$ represents the rate at which the number of coins is changing at minute 10.

· $N'(10)$ is the rate of change in the number of coins per minute at 10 minutes.

· The rate of change in the number of coins per minute is increasing at $t = 10$.

· The rate of change in the rate of change of the number of coins per minute is increasing at $t = 10$.

· $N'(10) = -\frac{1}{20}$ means that the rate of change in the number of coins is decreasing at a rate of $-\frac{1}{20}$ after 10 minutes. (double negative)

✓

0	1

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$N'(10)$ is the rate at which the rate of change of the number of 5-cent coins remaining is changing, in coins per minute per minute, at time $t = 10$ minutes.

6-7

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned with a correct tangent line equation in any form. Equivalent forms include (but are not limited to):

· $y = -16(t - 10) + 400$

· $y - 400 = -16(t - 10)$

· $y = -16t + 560$

The student may write x in place of t and still earn the point.

No work is required to earn the point.

We will accept $C(t)$ in lieu of y in the equation of the tangent line.

0	1

The student response accurately includes the criteria below.

- [ ] Tangent line equation

Solution:

$y = C'(10)(t - 10) + C(10)$

$y = -16(t - 10) + 400$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution.

Two things are required to award this point:

· Coefficient of $\frac{1}{4}$ correctly applied to an integral or an antiderivative

· Evidence that the correct limits of 0 to 40 will be or have been applied

Examples that will earn the first point (these are just a few):

6-7

- $\frac{1}{4}\int_0^{40} N'(u)\, du$

- $\frac{1}{4}N(u)\Big|_0^{40}$ or $\frac{1}{4}N(4t)\Big|_0^{10}$

Responses that include misstatements such as using t-values for limits with u as the variable of integration in an intermediate step which is later corrected may still earn this point. Intermediate errors in bounds that are subsequently corrected will disqualify the student from earning the third point. If the bounds are never appropriately handled, the response cannot earn the first point. Scratch work that is not connected to the solution will not be read.

Examples:

- $\int_0^{10} N'(4t)\, dt = \frac{1}{4}\int_0^{10} N'(u)\, du = \frac{1}{4}N(4t)\Big|_0^{10}$ earns the first point, but will not earn the third point.

- $\int_0^{10} N'(4t)\, dt = \frac{1}{4}\int_0^{10} N'(u)\, du = \frac{1}{4}[N(10) - N(0)]$ does not earn the first point.

The second point is for the Fundamental Theorem of Calculus, demonstrating the knowledge that
$\int_a^b N'(u)\, du = N(u)\big|_a^b = N(b) - N(a)$

This point can be earned with either $kN(4t)\,\big|_a^b$, $kN(t)\,\big|_a^b$, or $k(N(b) - N(a))$ where k is a nonzero constant [not necessarily correct], $a = 0$, and $b = 10$ or $b = 40$.

A response can earn this point without earning the first point.

$\frac{1}{4}[N(40) - N(0)]$ earns the first two points.

The third point is for the correct numerical value. Our answer only.

Values from the table must be used (i.e., $\frac{1}{4}[N(40) - N(0)]$ is not sufficient to earn the third point).

If the response includes any linkage error, (i.e., the use of equal signs to equate two expressions that are not equal), the third point will not be earned. If an indefinite integral is linked with a definite integral the response will not be penalized.

- $\int_0^{10} N'(4t)\, dt = \frac{1}{4}\int N'(u)\, du = \frac{1}{4}N(4t)\Big|_0^{10} = \ldots$ is eligible for the third point.

Eligibility: Must have earned the first two points.

The minimum response needed to earn all three points is: $\frac{1}{4}(-8 + 10)$.

6-7

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 4t$. Then $du = 4\ dt$.

When $t = 0$, $u = 0$.

When $t = 10$, $u = 40$.

$$\int_0^{10} N'(4t)\ dt = \frac{1}{4}\int_0^{40} N'(u)\ du$$

$$= \frac{1}{4}(N(40) - N(0))$$

$$= \frac{1}{4}(-8 - (-10)) = \frac{1}{2}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a trapezoidal sum

Point is earned for the setup of the trapezoidal sum. Numerical values do not need to be pulled from the table for this point, but Δt values must be presented. Responses that contain one incorrect numerical value in the trapezoidal sum setup are still eligible to earn the first point, but will not earn the second.

Response may present a left Riemann sum and a right Riemann sum, then average the two. Again, one incorrect numerical value will not disqualify the response from earning this first point.

Response may compute trapezoids as areas of rectangles and triangles.

Trapezoidal areas may be given in a graph. In order to earn this point, the response must clearly show the computations leading to the values presented in the graph, and must show that these areas are added to obtain the final approximation value.

A response using subtraction instead of addition will not earn the point

6-7

(e.g. $\frac{1}{2}[10\,(N\,(0) - N\,(10)) + 30\,(N\,(10) - N\,(40))]$ earns no points.)

Response utilizing “Trapezoidal Rule” will not earn this point, unless this represents only one incorrect numerical value.

· $\frac{1}{2}(10)\,[N\,(0) + 2N\,(10) + N\,(40)]$ is equivalent to $\frac{1}{2}[10\,(N(0) + N(10)) + 10\,(N(10) + N(40))]$
$\frac{1}{2}[10\,(N\,(0) + N\,(10)) + 10\,(N\,(10) + N\,(40))]$, thus has only one incorrect numerical value (the second interval width is 10 instead of 30), so can earn the point.

· $\frac{1}{2}(20)\,[N\,(0) + 2N\,(10) + N\,(40)]$ has two incorrect numerical values (both interval widths are incorrect) so will not earn the point.

The second point is for the correct numerical result.

Our answer only. Response does not need to be simplified, but if it is, it must be simplified correctly.

The answer must be accompanied with supporting work. Supporting work may appear in a graph, with clearly labeled areas. It is possible to earn the second point without earning the first point, if the supporting work is not sufficient to earn the first point.

Pictures without clear area computations and a clear summation of quantities can earn at most the second point.

Special case: A positive constant multiple of a complete and correctly expressed trapezoidal sum (with the correct values pulled from the table) will earn the first, but not the second point:

· $k\left[\left(\frac{-10-16}{2}\right)\cdot 10 + \left(\frac{-16-8}{2}\cdot 30\right)\right]$ for $k > 0$ earns only the first point.

The minimum response needed to earn both points: $\left(\frac{-26}{2}\right)\cdot 10 + \left(\frac{-24}{2}\right)\cdot 30$

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Form of trapezoidal sum
- ☐ Answer

Solution:

$$\int_0^{40} N\,(t)\,dt \approx \left(\frac{N\,(0)+N\,(10)}{2}\right)(10-0) + \left(\frac{N\,(10)+N\,(40)}{2}\right)(40-10)$$

$$= \left(\frac{-10-16}{2}\right)\cdot 10 + \left(\frac{-16-8}{2}\right)\cdot 30 = -490$$

6-7

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a correct constant multiple or correct sum.

Response must contain the constant 5 as a factor of the expression, and/or the correct sum $N(11) + F(11)$.

If the sum is incorrect, then the constant factor of 5 must be applied either to a sum that involves only $N(t)$ and $F(t)$, $N'(t)$ and $F'(t)$, or integrals of $N(t)$ and $F(t)$ or to one of the correct terms of the sum $N(11)$ or $F(11)$, with no additional incorrect terms.

Examples that earn the point include:

- $5\,[N\,(t) + F\,(t)]$
- $5\displaystyle\int [N\,(t) + F\,(t)]\;dt$
- $5\,[N' + F']$
- $5N\,(11)$

Examples that do not earn the point include:

- $5\,[N\,(t) + F'\,(t)]$
- $5\displaystyle\int N\,(t)\;dt$
- $5N' + F'$
- $5N\,(11) + t$

Special case: $5N\,(11) + F\,(11)$ is an error in parentheses, so will earn the first point but is not eligible to earn the second point.

If the constant multiple is missing or incorrect, the sum must be correct. $k\,[N\,(11) + F\,(11)]$, for any nonzero constant k will earn the first point.

The second point is for the correct expression.

$5\,[N\,(11) + F\,(11)]$ (or an equivalent expression) earns both points.

Inclusion of units (correct or incorrect) will not affect the score.

A completely correct expression with one additional constant term added/subtracted would earn the first point only. For example: $5\,[N\,(11) + F\,(11)] + 5\,(2)$ earns only the first point.

6-7

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Constant multiple or sum
- [ ] Expression

Solution:

The rate of change, in grams per minute, in the combined weight of the 5-cent coins remaining for the two children to prepare at $t = 11$ minutes is $5N\ (11) + 5F\ (11)$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for a correct statement identifying specific inconsistencies between the given differential equation and the given slope field. There are many such correct statements, such as:

· The response references a specific point on the y-axis (such as $(0,\ 4)$), and indicates that the given slope field shows a slope of zero, yet the differential equation would yield a positive slope.

· The response references a pattern of slopes, and explains why that pattern does not fit our differential equation (e.g. "the slope field shows slopes that change as t changes, but our differential equation depends only on y, so slopes should remain constant for a given value of y, as t changes.")

Identifying a mismatch between the differential equation and the slope field at one point is sufficient to earn the point.

Response is not penalized for referencing "x" instead of "t".

If the response contains any incorrect statement, the response will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point, nor will responses that present information beyond what we are given. For example:

· Responses that specify that the slope field shows a particular nonzero slope at a particular point will not earn the point, as this cannot actually be determined visually (i.e., the response should not indicate that at $(1,\ 2)$ the slope shown on the slope field is 1)

· Responses that state a specific slope value on the slope field for a general value of x or y that are not correct for all points with that x or y value do not earn the point. (e.g. "the slope field shows a slope of 0 when $y = 4$" is an incorrect statement)

6-7

A response may involve a solution to the differential equation, with an argument based on the resulting function. If done correctly, and presented with a correct argument, this will earn the point.

0	1

The student response accurately includes the criteria below.

- [ ] Explanation

Solution:

When $y = 0$, $\dfrac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

6-7

53. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	10	20	30	40
$N(t)$ (number of 5-cent coins remaining)	1200	1000	750	400	0

A child has collected 5-cent coins and is preparing them to deposit in the bank by counting them into bags. The child begins with 1200 coins. For time t minutes, $0 \le t \le 40$, the total number of coins remaining to prepare is modeled by a decreasing, twice-differentiable function N. Values of $N(t)$ at various times t are shown in the table above.

(a) Use a midpoint Riemann sum with the two subintervals indicated by the table to approximate $\int_0^{40} N(t)\, dt$. Show the computations that lead to your answer.

(b) Find the value of $\int_0^{10} N'(2t)\, dt$. Show the computations that lead to your answer.

(c) Using correct units, approximate $N'(20)$ using the average rate of change of N over the interval $10 \le t \le 30$.

(d) Explain the meaning of $N'(20)$ in the context of the problem.

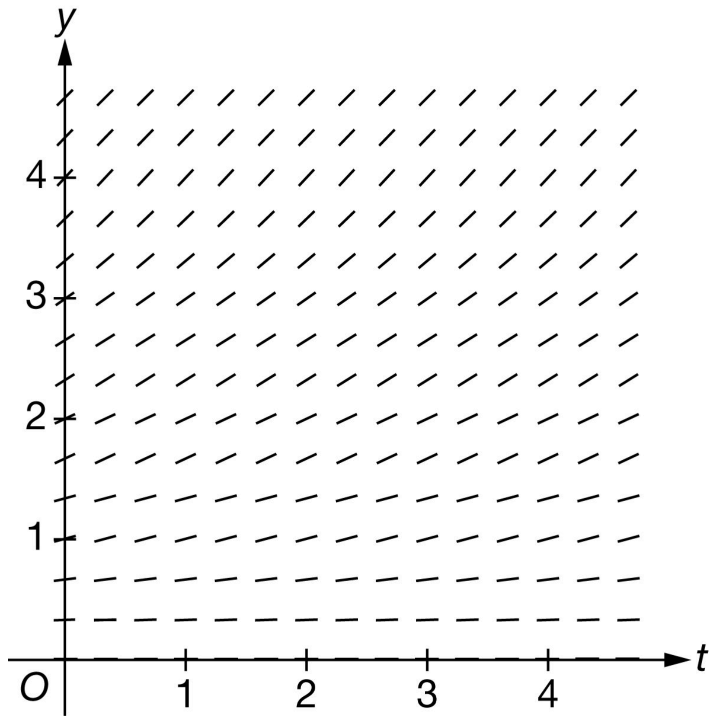
(e) It is known that $N'(10) = -20$. Write an equation for the line tangent to the graph of $y = N(t)$ at $t = 10$.

(f) It is known that $N''(t) < 0$ for $10 \le t \le 20$. If $N(17)$ is approximated using the equation from part (e), would the approximation be an overestimate or underestimate for the actual value of $N(17)$? Give a reason for your answer.

(g) For $0 \le t \le 40$, the total number of 5-cent coins remaining can also be modeled by the differentiable function G, where G is a function of t. Assume the weight of each coin is 6 grams. Write an expression for the rate of change, in grams per minute, of the total weight of the 5-cent coins remaining at $t = 37$ minutes.

6-7

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.4y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Form of midpoint sum): Philosophy: The response should provide clear evidence adding the area of two rectangles whose heights are found using the midpoint of the interval.

Acceptable evidence:

· $20 \cdot (N(10) + N(30))$ or $20 \cdot N(10) + 20 \cdot N(30)$

· $(20 - 0) \cdot 1000 + (40 - 20) \cdot 400$

· $20 \cdot (1000 + 400)$ or $20 \cdot 1000 + 20 \cdot 400$

To earn this point the response can have at most one mistake in the expression or equation. So, expressions of the form $\frac{20}{b} \cdot (N(10) + N(30))$, for any nonzero value of b would earn the point. Note: The value $\frac{20}{b}$ might be distributed across the sum.

Expressions of the form $\frac{a}{b} \cdot (N(10) + N(30))$ for $a \neq 20$ will not earn this point.

Once the first point is earned, it cannot be lost.

Second Point (Answer)

6-7

To earn the point, the answer must be correct. It does not need to be simplified, but if it is, the simplification must be correct.

The answer point cannot be earned unless the first (midpoint sum) point was earned.

Minimal complete answers that earn both points:

· $20 \cdot (1000 + 400)$

· $20 \cdot 1000 + 20 \cdot 400$

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Form of midpoint sum
- [ ] Answer

Solution:

$$\int_0^{40} N(t)\,dt \approx N(10)(20-0) + N(30)(40-20),$$

$$= 1000 \cdot 20 + 400 \cdot 20 = 28{,}000$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Substitution) Philosophy: This point is earned when the response has all three components of the substitution done correctly.

Uses $u = 2t$, perhaps implicitly (any variable could be used for u)

Uses the correct constant of $\frac{1}{2}$

Uses the correct limits for their substitution.

· The limits might not be seen until the final evaluation, and at that point, response may have returned to the variable t.

· Limits for t attached to an integral involving u or vice versa will not earn this substitution point but could still earn the answer point, if the numerical answer is correct.

6-7

Second point: (Fundamental Theorem of Calculus) Philosophy: This point is earned when the FTC has been applied correctly to either $\int N'(2t)\ dt$ or $\int N'(u)\ du$.

The point is earned for any of the following:

· The equation $\int N'(2t)\ dt = N(2t)$ or $\int N'(u)\ du = N(u)$

· $\frac{1}{2}N\ (2t)\Big|_0^{10}$ or $\frac{1}{2}N\ (u)\Big|_0^{20}$

· $N\ (20) - N\ (0)$, $N\ (20)$, or $N\ (0)$

This point can be earned if the substitution point is not.

Note: $\int_a^b N'(u)\ du = N'(b) - N'(a)$ does not earn the point. The FTC point is for the response communicating that the antiderivative of N' is N.

Third Point (Answer)

Must be the correct numerical answer.

Must have earned the second point for the FTC.

Might have not earned the first point for substitution because of an incorrect use of limits, but will still earn the third point for the correct answer.

$N\ (20)$ and $N\ (0)$ must be evaluated; the expression $\frac{1}{2}(N\ (20) - N\ (0))$ by itself does not earn the third point.

The answer may be unsimplified, but if simplified it must be done correctly.

Scoring common responses:

The minimal work required to earn all three points is $\frac{1}{2}(750 - 1200)$ or $\frac{1}{2}(750) - \frac{1}{2}(1200)$.

The following responses would earn the first two points and be eligible for the answer point if the response continues:

· $\frac{1}{2}N\ (2t)\Big|_0^{10}$

· $\frac{1}{2}(N\ (20) - N\ (0))$

Responses of only the following would earn only the Second point:

· $N\ (20) - N\ (0)$

· $N\ (10) - N\ (0)$

6-7

· $\int_0^{10} N'(u)\ du = N(10) - N(0)$

· $N(2t)|_0^{10}$ or $N(u)|_0^{20}$

The following responses would earn no points:

· -225 in the absence of one of the first two points.

· $\int_0^{10} N'(2t)\ dt = N(10) - N(0)$

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Substitution
- [ ] Fundamental Theorem of Calculus
- [ ] Answer

Solution:

Let $u = 2t$. Then $du = 2\ dt$.

When $t = 0$, $u = 0$.

When $t = 10$, $u = 20$.

$$\int_0^{10} N'(2t)\ dt = \frac{1}{2}\int_0^{20} N'(u)\ du$$

$$= \frac{1}{2}(N(20) - N(0))$$

$$= \frac{1}{2}(750 - 1200) = -225$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Approximation)

The response must be a difference and a quotient.

The answer does not need to be simplified, but if it is simplified it must be done correctly.

6-7

The minimal work that must be shown to earn this point is either $\frac{400-1000}{20}$ or $\frac{-600}{30-10}$.

Second point (Units)

The units must be attached to a value (either correct or incorrect).

The response will not be penalized for any verbal attempt to discuss the answer after the correct approximation with units is presented.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$N'(20) \approx \frac{N(30)-N(10)}{30-10} = \frac{400-1000}{20} = -30 \text{ coins per minute}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The interpretation must include three components:

· indicate the rate of change

· the specific context of the number of coins remaining

· the specific point in time, $t = 20$.

Responses do not need to include correct units or any units.

The word "decreasing" may be used in place of the word "changing".

The response is not earned for:

· "the rate of coins remaining at minute 20 " as this does not indicate a change in the rate of deposit.

· "decreasing at a rate of -30 coins per minute"

"the rate of change of N " without specifying that N is the number of coins remaining, does not earn the point.

Acceptable ways to provide the specific point in time:

6-7

· At minute 20 or after 20 minutes.

· At $t = 20$ (missing the unit of time)

· At 20 seconds (incorrect unit of time)

A response that clearly indicates an interval of time (e.g. "during the first 20 minutes" or "from minute 20 on") does not earn the point.

0	1

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$N'(20)$ is the rate at which the number of 5-cent coins remaining is changing, in coins per minute, at time $t = 20$ minutes.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of $N\ (t)$ in place of y earns the point.

The use of x in place of t earns the point.

Tangent lines do not need to be simplified, but any simplification must be done correctly.

Any correct form of the tangent line earns the point. For example:

· $y = 1000 - 20\,(t - 10)$

· $y - 1000 = 20\,(t - 10)$

· $y = -20t + 200 + 1000$

· $y = -20t + 1200$

0	1

The student response accurately includes the criteria below.

6-7

☐ Tangent line equation

Solution:

$y = N'(10)(t - 10) + N(10)$

$y = -20(t - 10) + 1000$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point the response must include $N''(t) < 0$ and overestimate.

" $N(t)$ is concave down" can be used instead of $N''(t) < 0$.

$N'(t)$ by itself is not sufficient and does not earn this point

"Overestimate because it is concave down" is not sufficient by itself because of the use of "it". The response must clearly, directly or indirectly, indicate to what "it" is referring.

If a response makes the correct conclusion and states $N''(t) < 0$ and then goes on and makes an incorrect statement or a statement that cannot be verified will not earn this point.

For example: "$N'(t)$ is concave down, overestimate" does not earn this point.

0	1 ✓

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because $N''(t) < 0$, the tangent line approximation would produce an overestimate.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression of only $6 \cdot G'(37)$ earns this point.

Only expression $6 \cdot N'(37)$ also earns this point.

No justifications or explanations are required.

6-7

Any incorrect statement such as, $G'(37) = 6 \cdot N'(37)$, does not earn this point.

0	1

The student response accurately includes the criteria below.

- [ ] Expression

Solution:

The rate of change, in grams per minute, of the total weight of the 5-cent coins remaining at $t = 37$ minutes is $6G'(37)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Response clearly references what part of the differential equation is different from the slope field and the response needs only to clearly and explicitly reference one of them.

The response does not need to explicitly reference both the differential equation and the given slope field.

The distinction can be made with explicit reference to one of the two things and an implicit reference to the other

For example:

· "$\frac{dy}{dt}$ is negative, not positive". We will infer that positive is referring to the slopes shown in the slope field.

· "This would be correct for a positive derivative, but $\frac{dy}{dt}$ is negative." We will read "This" as "the slope field" in this sentence.

· "All of the slopes in the slope field are positive, not negative." We will infer that the values of $\frac{dy}{dt}$ at points in the first quadrant are negative.

A response that lacks even an implicit reference to either the differential equation or the slope field will not earn the point. For example:

· "$\frac{dy}{dt}$ is negative" lacks any reference to the slope field and so will not earn the point.

· "All the slopes in the graph are positive" lacks any reference to the differential equation, and so will not earn the point.

· Vague answers such as "The slope field has positive points" or "The field is increasing" will not earn the point.

6-7

0	1

The student response accurately includes the criteria below.

- [ ] Explanation

Solution:

When $y > 0$, $\frac{dy}{dt} < 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are positive.

6-7

54. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	2	5	10
$S(t)$ (number of boxes of snacks remaining)	500	340	200	20

A student organization is selling boxes of snacks to raise money for a school trip. The student organization begins with 500 boxes of snacks and sells them over a ten-day period. For time t days, $0 \le t \le 10$, the total number of boxes of snacks remaining to sell is modeled by a decreasing, twice-differentiable function S. Values of $S(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $S'(5)$ using the average rate of change of S over the interval $2 \le t \le 10$.

(b) Explain the meaning of $S'(5)$ in the context of the problem.

(c) It is known that $S'(2) = -55$. Write an equation for the line tangent to the graph of $y = S(t)$ at $t = 2$.

(d) Use a left Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{10} S(t)\,dt$. Show the computations that lead to your answer.

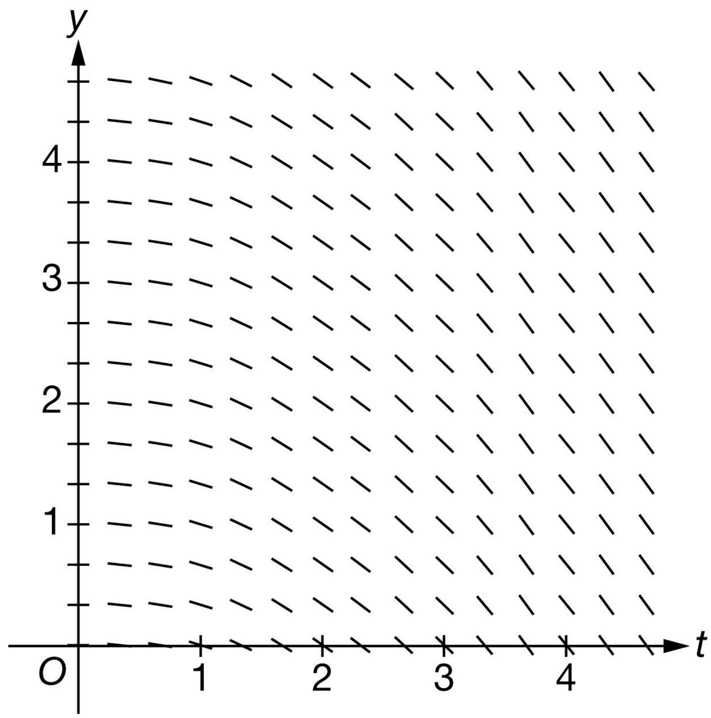
(e) Is the approximation from part (d) an overestimate or an underestimate for $\int_0^{10} S(t)\,dt$? Give a reason for your answer.

(f) Find the value of $\int_0^{5} S'(2t)\,dt$. Show the computations that lead to your answer.

(g) For $0 \le t \le 10$, the total number of boxes of snacks remaining can also be modeled by the differentiable function M, where M is a function of t. Assume the weight of each box is 7 ounces. Write an expression for the rate of change, in ounces per day, of the total weight of the boxes of snacks remaining at $t = 8$ days.

6-7

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.2y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work. The supporting work must be at least a difference and a quotient. Correct values must be taken from the table to earn the point. The difference quotient does not need to be simplified.

Examples that earn the first point include:

- $\frac{20-340}{10-2}$
- $\frac{S(10)-S(2)}{8} = \frac{-320}{8}$
- $\frac{S(10)-S(2)}{8} = -40$

Examples that do not earn the first point include:

- $\frac{S(10)-S(2)}{8}$
- $\frac{S(10)-S(2)}{10-2}$

Minimal response that earns the point: $\frac{20-340}{10-2}$ or $\frac{20-340}{8}$

6-7

Neither $\frac{-320}{8}$ nor -40, with no supporting work, will earn this point

The second point is for correct units

The correct units must be attached to a numerical value: either correct or incorrect.

The units may be written using words or a combination of words and symbols. For example:

· Boxes per day

· Boxes of snacks per day

· Snacks per day

· Boxes/day

The response will not be penalized for any verbal attempt to discuss the answer after the correct answer has been presented.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Approximation
- [ ] Units

Solution:

$$S'(5) \approx \frac{S(10)-S(2)}{10-2} = \frac{20-340}{8} = -40$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

· Indicating a rate of change

· Including the context of boxes of snack collection

· Referring to the specific point in time, $t = 5$

Students are not penalized for incorrect units as a modifier in this part. (Whether correct or incorrect, units have no effect on the score in this part.) Example: “The number of coins is decreasing at a rate of 40 coins per day.”

6-7

Responses may use "changing" or "decreasing" as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Example: "The number of boxes is decreasing at a rate of 40 boxes per day" is okay, but "The number of boxes is decreasing at a rate of -40 boxes per day" would not earn the point.

Acceptable references to a specific point in time include (but are not limited to):

· On day 5 and after 5 days

· At $t = 5$

· At $t = 5$ weeks or any incorrect unit of time

If the response clearly indicates an interval of time (e.g., "from day 5 on," "during the first 5 days," or "for $t = 0$ to $t = 5$ "), the point is not earned.

Examples that earn the point include:

· $S'(5)$ is the rate of change of the number of boxes of snacks per day at time $t = 5$

· The rate of change of number of boxes of snacks per day is -40 boxes of snacks per day on day 5

· $S'(5)$ is the rate of change in the number of boxes of snacks at time $t = 5$ in boxes of snacks per *month*

· 40 is the rate of decrease in the number of boxes of snacks per day on day 5

Examples that do not earn the point include:

· $S'(5)$ is the average rate of change of the total number of boxes of snacks per day on day 5

· $S'(5)$ is the total number of boxes of snacks per day on day 5

· The rate of change is -40 boxes of snacks per day (Missing "at $t = 5$ ")

· $S'(5)$ is the slope of the tangent line at $t = 5$

· Rate of change of S at time $t = 5$

· Increase in number of boxes of snacks at $t = 5$

· Boxes per day increases on day 5

0	1 ✓

The student response accurately includes the criteria below.

☐ Explanation

6-7

Solution:

$S'(5)$ is the rate at which the number of boxes of snacks remaining is changing, in boxes per day, on day 5.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. The use of x in place of t is acceptable.

Some examples include

· $y - 340 = -55(t-2)$, $y - 340 = -55(x-2)$

· $y = -55(t-2) + 340$, $y = -55(x-2) + 340$

· $y = -55t + 450$, $y = -55x + 450$

The key calculus concept being assessed in this part of the problem is that the value of $S'(2)$ is the slope of the tangent line. In that spirit, we are accepting $S(t) = \ldots$ in lieu of $y = \ldots$, but not $S(2) = \ldots$

0	1 ✓

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$y = S'(2)(t-2) + S(2)$

$y = -55(t-2) + 340$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a left Riemann sum. The setup must be a sum of 3 products.

· A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example below).

· Responses with the numerical error described above are not eligible for the second point.

6-7

NOTE: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns. Implicit responses are eligible for both points.

A response presenting a right Riemann sum (instead of left), will not earn any points for part (a).

The second point is for the correct numerical result.

- In order to earn the second point, there must be supporting calculations.
- The numerical values for $S(2)$, $S(5)$, and $S(10)$ must be pulled from the table
- Any simplification shown must be correct.

Use the following as an illustration to score both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

- A response of $500 \cdot (2-0) + 340 \cdot (5-2) + 200 \cdot (10-5)$ earns both points.
- A response of $2 \cdot S(0) + 3 \cdot S(2) + 5 \cdot S(5)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $2 \cdot S(0) + 3 \cdot S(2) + 5 \cdot S(5) = 3020$ earns both points.
- A response of $2 \cdot 500 + 3 \cdot 340 + 5 \cdot 200$ earns both points.
- A response of $2 \cdot 500 + 10 \cdot 340 + 5 \cdot 200$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $2 \cdot 500 + 3 \cdot 340 + 5 \cdot 200 = 3200$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $1000 + 1020 + 1000$ or $1000 + 1020 + 1000 = 3020$ does not earn the first point because we cannot see the sum of 3 products. However, this is the correct sum, so we see an answer with supporting work. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$2 \cdot 500 + 10 \cdot 340 + 5 \cdot 200$

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Left Riemann sum

6-7

☐ Answer

Solution:

$$\int_0^{10} S(t)\ dt \approx S(0)(2-0) + S(2)(5-2) + S(5)(10-5)$$

$$= 500 \cdot 2 + 340 \cdot 3 + 200 \cdot 5 = 3020$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an overestimate because S is decreasing.

The response does not need an explicit reference to the approximation in part (d).

The following statements are sufficient for earning this point:

· Overestimate because S is decreasing

· Overestimate because the function is decreasing
NOTE: At this point, there is only one function being discussed in this problem, so we will assume "the function" is S .

When a response includes additional information about concavity, we interpret that as a lack of understanding about the question, so this will not earn the point. When the response references a graph or the given table rather than the function S , the point is not earned.

Examples that earn the point:

· The left Riemann sum overestimates because the function is decreasing.

· The sum is an overestimate because $S(t)$ is decreasing.

· The sum is an overestimate because $S'(t) < 0$.

Examples that do not earn the point:

· The sum is an overestimate because $S'(t)$ is decreasing. ($S'(t)$ decreasing is not known)

· The left Riemann sum overestimates because the graph is decreasing.

· The left Riemann sum overestimates because the function is decreasing and concave up.

0	1 ✓

6-7

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because S is a decreasing function, and this is a left Riemann sum, the answer from part (d) is an overestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are three components of substitution that need to be seen:

· Uses $u = 2t$, which may be done implicitly

· Uses the correct constant of $\frac{1}{2}$

· Uses the correct limits for the substitution (0 and 10). These limits may only be seen when evaluating the antiderivative. These don't need to see them as limits on a definite integral.

· Either of the expressions below will earn the substitution point without any additional supporting work:
$\frac{1}{2} S\,(2t)\Big|_0^5$ or $\frac{1}{2} S\,(u)\Big|_0^{10}$

Special Handling: (Notation Errors)

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

· Limits for t attached to an integral with u as the variable of integration (or vice versa),

· Improper equals signs, such as an indefinite integral equal to a definite integral

· Reversed limits of integration, for example: $\int_{10}^{0} S'\,(u)\ du$

The second point is for the Fundamental Theorem of Calculus. This point can be earned with any of the following, where k is a nonzero constant and $b = 5$ or $b = 10$. No explanation or additional supporting work is necessary.

· $k\,S\,(2t)\Big|_0^b$

· $k\,(S\,(b) - S\,(0))$

· This point can be earned without earning the substitution point.

6-7

The third point is for the correct numerical value, which is -240.

Values from the table must be used. $\frac{1}{2}[S\ (10) - S\ (0)]$ is not enough to earn the answer point

The expression $\frac{1}{2}(20 - 500)$ close parenthesis is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

The response cannot earn this point without also earning the FTC point.

Minimalist response**:** The expression $\frac{1}{2}(-480)$ represents the minimum work that must be seen in a response in order to earn all three points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t$. Then $du = 2\ dt$.

When $t = 0$, $u = 0$.

When $t = 5$, $u = 10$.

$$\int_0^5 S'\ (2t)\ dt = \frac{1}{2}\int_0^{10} S'\ (u)\ du$$

$$= \frac{1}{2}(S\ (10) - S\ (0))$$

$$= \frac{1}{2}(20 - 500) = -240$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct expression at $t = 8$. No supporting work or justification is required.

The correct expression of $7 \cdot S'\ (8)$ or $7 \cdot M'\ (8)$ earns this point.

6-7

No justifications or explanations are required.

Any incorrect statement such as, $S'(8) = 7 \cdot M'(8)$, does not earn this point.

0	1

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in ounces per day, of the total weight of the boxes of snacks remaining at $t = 8$ days is $7M'(8)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between two things: the differential equation, $\frac{dy}{dt} = -0.2y$, and the given slope field.

The response does not need to explicitly reference both the differential equation and the given slope field, but it must explicitly reference one and implicitly reference the other.

Acceptable responses include

· Identifying an inconsistency between the differential equation and the slope field at one point , e.g., at the point $(4,\ 0)$, the slope of the segment should be 0, but it is negative.

· Referencing a pattern of slopes, and explaining why that pattern does not fit the given differential equation.

A response that lacks even an implicit reference to the one of two things being compared will not earn the point. For example, merely stating either $\frac{dy}{dt}$ is negative, or that the slope field is negative does not earn the point.

· “The values for y are less than or equal to zero so $\frac{dy}{dt} \leq 0$ " lacks any reference to the slope field and will not earn the point.

· “All the slopes in the slope field are negative” lacks any reference to the differential equation and will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point. Some examples are:

· “The field is decreasing”

6-7

· Responses that state a specific slope value on the slope field for a general value of x or y that is not correct for all points with that x or y value do not earn the point. (e.g. "the slope field shows a slope of 0 when $y = 4$ " is an incorrect statement)

A response may correctly solve or attempt to solve the differential equation and then provide a correct argument without reference to their solution. Such a response would earn the point.

A response might actually solve the differential equation and provide a correct argument on the resulting function.

0	1

The student response accurately includes the criteria below.

- [ ] Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

6-7

55. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	5	9
$S(t)$ (boxes per day)	-70	-30	-16

A student organization is selling boxes of snacks to raise money for a school trip. For time t days, $0 \leq t \leq 9$, the rate of change in the number of boxes of snacks remaining to sell is modeled by an increasing, twice-differentiable function S. Values of $S(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $S'(5)$ using the average rate of change of S over the interval $0 \leq t \leq 9$.

(b) Explain the meaning of $S'(5)$ in the context of the problem.

(c) The function $B(t)$ models the number of boxes remaining to sell at time t days, where $B'(t) = S(t)$ .It is known that $B(5) = 240$. Write an equation for the line tangent to the graph of $y = B(t)$ at $t = 5$.

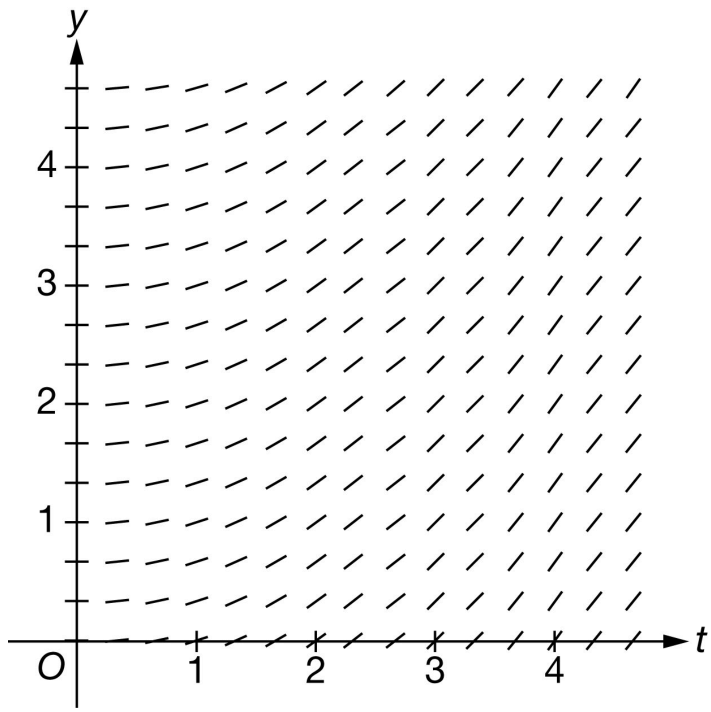
(d) Find the value of $\int_0^{4.5} S'(2t)\,dt$. Show the calculations that lead to your answer.

(e) Use a trapezoidal sum with the two subintervals indicated by the table to approximate $\int_0^9 S(t)\,dt$. Show the calculations that lead to your answer.

(f) Another student organization is also selling boxes of snacks. For $0 \leq t \leq 9$, the rate of change in the number of boxes of snacks remaining for the second student organization to sell is modeled by a decreasing, differentiable function M, where M is a function of t. Assume the weight of each box of snacks is 7 ounces. Write an expression in terms of M and S for the rate of change, in ounces per day, of the combined weight of the boxes of snacks remaining for the two organizations to sell at $t = 8$ days.

6-7

(g) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = 0.25y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

Answers without supporting work do not earn this point. We need to see both a difference and a quotient for the supporting work.

Response does not need to be simplified, but any simplification must be done correctly.

If a decimal is presented, it must be correct to three decimal places. We do not read anything past the third decimal place.

The second point is earned for the correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols.

The response will not be penalized for any verbal attempt to discuss the answer after the correct presentation of the approximation.

The minimum response needed to earn both points is $\frac{-16+70}{9}$ boxes per day per day

6-7

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$S'(5) \approx \frac{S(9)-S(0)}{9-0} = \frac{-16-(-70)}{9} = 6$ boxes per day per day

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This part is intended to be independent from part (a), so we do not want or need to see any reference to the value obtained in (a). If that value is referenced, conclusion must be consistent.

Three components of a correct answer:

· Indicating a rate of a rate (look for the word rate to be used correctly twice)

· Including the context of boxes

· Referring to the specific point in time

Units will not be considered for this point. They do not need to be present, and if they are, they do not need to be correct.

Any incorrect statements make the response ineligible for the point. For example:

· $S'(5)$ is the average (or an approximation of, or an estimate of) rate of change of . . . .

· $S'(5)$ is the acceleration (or rate of change of velocity) of . . .

“Increasing” or “decreasing” may be used in place of “changing” as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Beware of phrases such as “decreasing at a negative rate.”

· Example: $S'(5)$ is the rate of increase of the rate of change of the number of boxes at 5 days. (earns the point)

· Example: $S'(5)$ means the number of boxes per day is increasing by 5 boxes/day/day. (does not earn the point)

Acceptable references to time include (but are not limited to):

· On day 5

6-7

· At $t=5$ (with correct units of time, without correct units of time, or even with incorrect units of time)

· After $t=5$ (with correct units of time, without correct units of time, or even with incorrect units of time)

If the response clearly indicates an interval of time (e.g. "from day 0 to day 5 ") the response will not earn the point.

Examples that earn the point include:

· $S'(5)$ represents the rate at which the rate of change of the number of boxes is changing on day 5.

· $S'(5)$ is the rate of change in the rate of change of the number of boxes at 5 days.

· $S'(5)$ represents the rate at which the rate of change of the number of boxes is increasing at $t=5$.

· " $S'(5) =$ (answer from part (a))" means the rate of change in the number of boxes per day is increasing at a rate of (answer from part (a)) at $t=5$.

· $S'(5)$ represents the rate at which the rate of change of the number of boxes is decreasing on the fifth day.

Examples that do not earn the point include:

· $S'(5)$ represents the rate at which the number of boxes is changing on day 5.

· $S'(5)$ is the rate of change in the number of boxes per day at $t=5$ days

· The rate of change in the number of boxes per day is increasing at $t=5$.

· The rate of change in the rate of change of the number of boxes per day is increasing at $t=5$.

· $S'(5)=-6$ means that the rate of change in the number of boxes is decreasing at a rate of -6 at $t=5$ (double negative).

0	1 ✓

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$S'(5)$ is the rate at which the rate of change in the number of boxes per day is changing, in boxes per day per day, on day 5.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-7

The point is earned with a correct tangent line equation in any form. For example:

$y = -30\,(t - 5) + 240$

$y - 240 = -30\,(t - 5)$

$y = -30t + 390$

The student may write x in place of t and still earn the point.

No work is required to earn the point.

We will accept $B\,(t)$ in lieu of y in the equation of the tangent line.

0	1

The student response accurately includes the criteria below.

- [ ] Tangent line equation

Solution:

$y = B'(5)\,(t - 5) + B(5)$

$y = -30\,(t - 5) + 240$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution.

Two things are required as evidence of correct substitution:

· Coefficient of $\frac{1}{2}$ correctly applied to an integral or an antiderivative

· Evidence that the correct limits of 0 to 9 will be applied

Examples that will earn the first point (these are just a few):

· $\frac{1}{2}\int_0^9 S'\,(u)\,du$

· $\frac{1}{2}S\,(u)\big|_0^9$ or $\frac{1}{2}S\,(2t)\big|_0^{4.5}$

6-7

Responses that include misstatements such as using t-values for limits with u as the variable of integration in an intermediate step which is later corrected may still earn this point. Intermediate errors in bounds that are subsequently corrected will disqualify the student from earning the third point. If the bounds are never appropriately handled, the response cannot earn the first point. Scratch work that is not connected to the solution will not be read.

Examples:

- $\int_0^{4.5} S'(2t)\,dt = \frac{1}{2}\int_0^{4.5} S'(u)\,du = \frac{1}{2}S(2t)\Big|_0^{4.5}$ earns the first point, but will not earn the third point.
- $\int_0^{4.5} S'(2t)\,dt = \frac{1}{2}\int_0^{4.5} S'(u)\,du = \frac{1}{2}[S(4.5) - S(0)]$ does not earn the first point.

The second point is for the Fundamental Theorem of Calculus, demonstrating the knowledge that

$$\int_a^b S'(u)\,du = S(u)\big|_a^b = S(b) - S(a)$$

This point can be earned with either $kS(2t)\big|_a^b$, $kS(t)\big|_a^b$, or $k(S(b) - S(a))$ where k is a nonzero constant [not necessarily correct], $a = 0$, and $b = 4.5$ or $b = 9$.

The response can earn this point without earning the first point.

$\frac{1}{2}[S(9) - S(0)]$ earns the first two points.

The third point is for the correct numerical value. Our answer only.

Values from the table must be used (i.e., $\frac{1}{2}[S(9) - S(0)]$ is not sufficient to earn the third point).

If the response includes any linkage error, (i.e., the use of equal signs to equate two expressions that are not equal), the third point will not be earned. If an indefinite integral is linked with a definite integral the response will not be penalized.

- $\int_0^{4.5} S'(2t)\,dt = \frac{1}{2}\int S'(u)\,du = \frac{1}{2}S(2t)\Big|_0^{4.5} = \ldots$ is eligible for the third point.

Eligibility for the third point: Must have earned the first two points.

The minimum response needed to earn all three points is: $\frac{1}{2}(-16 + 70)$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Substitution

6-7

- [ ] Fundamental Theorem of Calculus
- [ ] Answer

Solution:

Let $u = 2t$. Then $du = 2\ dt$.

When $t = 0$, $u = 0$.

When $t = 4.5$, $u = 9$.

$$\int_0^{4.5} S'\ (2t)\ dt = \tfrac{1}{2} \int_0^{9} S'\ (u)\ du$$

$$= \tfrac{1}{2}(S\ (9) - S\ (0))$$

$$= \tfrac{1}{2}(-16 - (-70)) = \tfrac{1}{2}(54) = 27$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the form of a trapezoidal sum

Point is earned for the setup of the trapezoidal sum, with one numerical error allowed. Numerical values do not need to be pulled from the table for this point, but Δt values must be presented. Responses that contain one incorrect numerical value in the trapezoidal sum setup are still eligible to earn the first point, but will not earn the second.

Response may present a left Riemann sum, a right Riemann sum, then average the two. Again, one incorrect numerical value will not disqualify the response from earning the first point.

Response may compute trapezoids as areas of rectangles and triangles.

Trapezoidal areas may be given in a graph. In order to earn this point, the response must clearly show the computations leading to the values presented in the graph, and must show that these areas are added to obtain the final approximation value.

A response using subtraction instead of addition will not earn the point (e.g. $\tfrac{1}{2}[5\,(S\ (0) - S\ (5)) + 4\,(S\ (5) - S\ (9))]$ earns no points).

Response utilizing “Trapezoidal Rule” will not earn this point, unless this represents only one error in the numerical values.

- $\tfrac{1}{2}(5)\ [S\ (0) + 2S\ (5) + S\ (9)]$ is equivalent to $\tfrac{1}{2}[5\,(S\ (0) + S\ (5)) + 5\,(S\ (5) + S\ (9))]$, has only one incorrect numerical value (the second interval width is 5 instead of 4), so can earn the point.

6-7

· $\frac{1}{2}(4.5)\left[S(0)+2S(5)+S(9)\right]$ has two incorrect numerical values (both interval widths are incorrect) so will not earn the point.

The second point is for the correct numerical result.

Our answer only. Response does not need to be simplified, but if it is, it must be simplified correctly.

The answer must be accompanied with supporting work. Supporting work may appear in a graph, with clearly labeled areas. It is possible to earn the second point without earning the first if the supporting work is not sufficient to earn the first point.

Pictures without clear area computations and a clear summation of quantities can earn at most the second point.

Special case: A positive constant multiple of a complete and correctly expressed trapezoidal sum (with the correct values pulled from the table) will earn the first point but not the second:

· $k\left[\left(\frac{-70-30}{2}\right)\cdot 5+\left(\frac{-30-16}{2}\right)\cdot 4\right]$ for $k>0$ earns just the first point.

The minimum response needed to earn both points is: $\left(\frac{-100}{2}\right)\cdot 5+\left(\frac{-46}{2}\right)\cdot 4$

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Form of trapezoidal sum
- ☐ Answer

Solution:

$$\int_0^9 S(t)\,dt \approx \left(\frac{S(0)+S(5)}{2}\right)(5-0)+\left(\frac{S(5)+S(9)}{2}\right)(9-5)$$

$$=\left(\frac{-70-30}{2}\right)\cdot 5+\left(\frac{-30-16}{2}\right)\cdot 4=-342$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a correct constant multiple or correct sum.

Response must contain the constant 7 as a factor of the expression, and/or the correct sum $M(8)+S(8)$.

If the sum is incorrect, then the constant factor of 7 must be applied either to a sum that involves only $M(t)$ and $S(t)$, $M'(t)$ and $S'(t)$, or integrals of $M(t)$ and $S(t)$, or to one of the correct terms of the sum $M(8)$ or

6-7

$S(8)$, with no additional incorrect terms.

Examples that earn the point include:

$\cdot\, 7[M(t) + S(t)]$

$\cdot\, 7\int [M(t) + S(t)]\, dt$

$\cdot\, 7[M' + S']$

$\cdot\, 7S(8)$

Examples that do not earn the point include:

$\cdot\, 7[M(t) + S'(t)]$

$\cdot\, 7\int M(t)\, dt$

$\cdot\, 7M' + S'$

$\cdot\, 7S(8) + t$

Special case: $7M(8) + S(8)$ is an error in parentheses, so will earn the first point but is not eligible to earn the second point.

If the constant multiple is missing or incorrect, the sum must be correct. $k[M(8) + S(8)]$ for any nonzero constant k will earn the first point.

The second point is for the correct expression.

$7M(8) + 7S(8)$ (or an equivalent expression) earns both points.

Inclusion of units (correct or incorrect) will not affect the score.

A completely correct expression with one additional constant term added/subtracted would earn a just the first point. For example, $7M(8) + 7S(8) + 7(2)$ earns only the first point.

✓

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Constant multiple or sum
- ☐ Expression

Solution:

6-7

The rate of change, in ounces per day, of the combined weight of the boxes of snacks remaining for the two organizations to sell at $t = 8$ days is $7M\ (8) + 7S\ (8)$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for a correct statement identifying specific inconsistencies between the given differential equation and the given slope field. There are many such correct statements, such as:

· The response references a specific point on the y-axis (such as $(0,\ 4)$), and indicates that the given slope field shows a slope of zero, yet the differential equation would yield a positive slope.

· The response references a pattern of slopes, and explains why that pattern does not fit our differential equation (e.g. "the slope field shows slopes that change as t changes, but our differential equation depends only on y, so slopes should remain constant for a given value of y as t changes.")

Identifying a mismatch between the differential equation and the slope field at one point is sufficient to earn the point.

Response is not penalized for referencing " x " instead of " t ".

If the response contains any incorrect statement, the response will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point, nor will responses that present information beyond what we are given. For example:

· Responses that specify that the slope field shows a particular nonzero slope at a particular point will not earn the point, as this cannot actually be determined visually (i.e., the response should not indicate that at $(2,\ 2)$ the slope shown on the slope field is 1)

· Responses that state a specific slope value on the slope field for a general value of x or y that are not correct for all points with that x or y value do not earn the point. (e.g. "the slope field shows a slope of 0 when $y = 4$ " is an incorrect statement)

A response may involve a solution to the differential equation, with an argument based on the resulting function. If done correctly, and presented with a correct argument, this will earn the point.

0	1 ✓

The student response accurately includes the criteria below.

☐ Explanation

Solution:

6-7

When $y=0$, $\dfrac{dy}{dt}=0$. However, in the given slope field, for $t>0$, the slopes of the line segments for $y=0$ are nonzero.

6-7

56. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	3	6	9	12
$S(t)$ (number of boxes of snacks remaining)	600	500	440	240	20

A student organization is selling boxes of snacks to raise money for a school trip. The student organization begins with 600 boxes of snacks and sells them over a twelve-day period. For time t days, $0 \le t \le 12$, the total number of boxes of snacks remaining to sell is modeled by a twice-differentiable function S. Values of $S(t)$ at various times t are shown in the table above.

(a) Use a midpoint Riemann sum with the two subintervals indicated by the table to approximate $\int_0^{12} S(t)\,dt$. Show the computations that lead to your answer.

(b) Find the value of $\int_0^{3} S'(3t)\,dt$. Show the computations that lead to your answer.

(c) Using correct units, approximate $S'(9)$ using the average rate of change of S over the interval $6 \le t \le 12$.

(d) Explain the meaning of $S'(9)$ in the context of the problem.

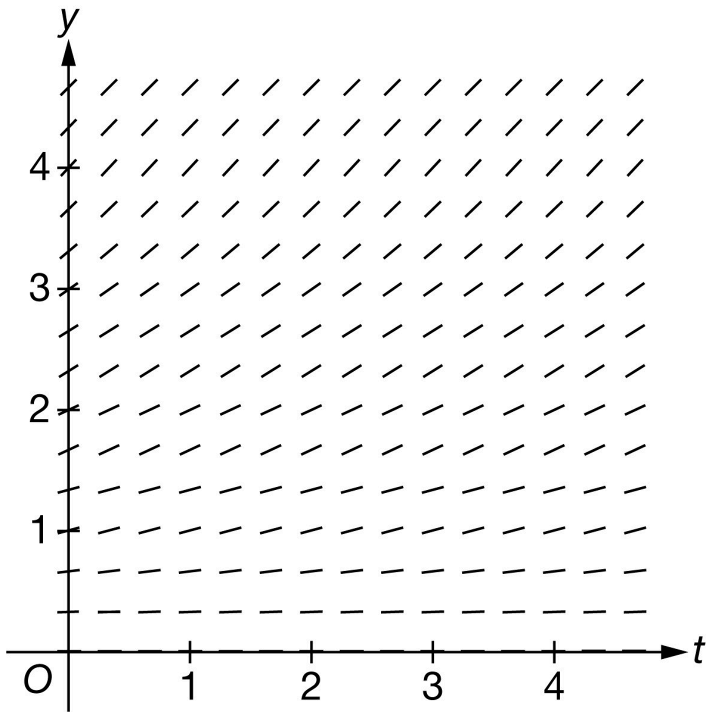
(e) It is known that $S'(6) = -40$. Write an equation for the line tangent to the graph of $y = S(t)$ at $t = 6$.

(f) It is known that $S''(t) < 0$ for $6 \le t \le 9$. If $S(8)$ is approximated using the equation from part (e), would the approximation be an overestimate or underestimate for the actual value of $S(8)$? Give a reason for your answer.

(g) For $0 \le t \le 12$, the total number of boxes of snacks remaining can be modeled by the differentiable function Q, where Q is a function of t. Assume the weight of each box is 8 ounces. Write an expression for the rate of change, in ounces per day, of the total weight of the boxes of snacks remaining at $t = 5$ days.

6-7

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.25y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Form of midpoint sum): Philosophy: The response should provide clear evidence adding the area of two rectangles whose heights are found using the midpoint of the interval.

Acceptable evidence:

$\cdot\, 6 \cdot (S(3) + S(9))$ or $6 \cdot S(3) + 6 \cdot S(9)$

$\cdot\, (6 - 0) \cdot 500 + (12 - 6) \cdot 240$

$\cdot\, 6 \cdot (500 + 240)$ or $6 \cdot 500 + 6 \cdot 240$

To earn this point the response can have at most one mistake in the expression or equation. So, expressions of the form $\frac{6}{b} \cdot (S(3) + S(9))$, for any nonzero value of b would earn the point. Note: The value $\frac{6}{b}$ might be distributed across the sum.

Expressions of the form $\frac{a}{b} \cdot (S(3) + S(9))$, for $a \neq 6$ will not earn this point.

Once the first point is earned, it cannot be lost.

Second Point (Answer)

To earn the point, the answer must be correct. It does not need to be simplified, but if it is, the simplification must be correct.

6-7

The answer point cannot be earned unless the first (midpoint sum) point was earned.

Minimal complete answers that earn both points:

· $6 \cdot (500 + 240)$

· $6 \cdot 500 + 6 \cdot 240$

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Form of midpoint sum
- [ ] Answer

Solution:

$$\int_0^{12} S(t)\,dt \approx S(3)(6-0) + S(9)(12-6)$$

$$= 500 \cdot 6 + 240 \cdot 6 = 4440$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Substitution) Philosophy: This point is earned when the response has all three components of the substitution done correctly.

Uses $u = 3t$, perhaps implicitly (any variable could be used for u)

Uses the correct constant of $\frac{1}{3}$

Uses the correct limits for their substitution.

· The limits might not be seen until the final evaluation, and at that point, response may have returned to the variable t.

· Limits for t attached to an integral involving u or vice versa will not earn this substitution point but could still earn the answer point, if the numerical answer is correct.

Second point: (Fundamental Theorem of Calculus) Philosophy: This point is earned when the FTC has been applied correctly to either $\int S'(3t)\,dt$ or $\int S'(u)\,du$.

The point is earned for any of the following:

6-7

- The equation $\int S'(3t)\,dt = S(3t)$ or $\int S'(u)\,du = S(u)$
- $\frac{1}{3}S(3t)\Big|_0^3$ or $\frac{1}{3}S(u)\Big|_0^9$
- $S(9) - S(0)$, $S(9)$, or $S(0)$

This point can be earned if the substitution point is not.

Note: $\int_a^b S'(u)\,du = S'(b) - S'(a)$ does not earn the point. The FTC point is for the response communicating that the antiderivative of S' is S.

Third Point (Answer)

Must be the correct numerical answer.

Must have earned the second point for the FTC.

Might have not earned the first point for substitution because of an incorrect use of limits but will still earn the third point for the correct answer.

$S(9)$ and $S(0)$ must be evaluated; the expression $\frac{1}{3}(S(9) - S(0))$ by itself does not earn the third point.

The answer may be unsimplified, but if simplified it must be done correctly.

Scoring common responses:

The minimal work required to earn all three points is $\frac{1}{3}(240 - 600)$.

The following responses would earn the first two points and be eligible for the answer point if the response continues:

- $\frac{1}{3}S(3t)\Big|_0^3$
- $\frac{1}{3}(S(9) - S(0))$

Responses of only the following would earn only the second point:

- $S(9) - S(0)$
- $S(3) - S(0)$
- $\int_0^9 S'(u)\,du = S(9) - S(0)$

6-7

· $S\left(3t\right)\Big|_0^3$ or $S\left(u\right)\Big|_0^9$

The following response would earn no points: -120 in the absence of one of the first two points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u=3t$. Then $du=3\ dt$.

When $t=0$, $u=0$.

When $t=3$, $u=9$.

$$\int_0^3 S'\left(3t\right)\ dt=\frac{1}{3}\int_0^9 S'\left(u\right)\ du$$

$$=\frac{1}{3}\left(S\left(9\right)-S\left(0\right)\right)$$

$$=\frac{1}{3}\left(240-600\right)=-120$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Approximation)

The response must be a difference and a quotient.

The answer does not need to be simplified, but if it is simplified it must be done correctly.

The minimal work that must be shown to earn this point is either $\frac{20-440}{6}$ or $\frac{-420}{12-6}$.

Second point (Units)

The units must be attached to a correct or incorrect value.

6-7

The response will not be penalized for any verbal attempt to discuss the answer after the correct approximation with units is presented.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$S'(9) \approx \frac{S(12)-S(6)}{12-6} = \frac{20-440}{6} = -70 \text{ boxes per day}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The interpretation must include three components:

· indicate the rate of change

· the specific context of the number of boxes remaining

· the specific point in time, $t = 9$.

Responses do not need to include correct units or any units.

The word "decreasing" may be used in place of the word "changing".

The response is not earned for:

· "the rate of boxes sold on day 9" as this does not indicate a change in the rate of boxes remaining.

· "decreasing at a rate of -70 boxes per day"

"the rate of change of S" without specifying that S is the number of boxes remaining, does not earn the point.

Acceptable ways to provide the specific point in time:

· On day 9 or after 9 days.

· At $t = 9$ (missing the unit of time)

· On week 9 (incorrect unit of time)

6-7

A response that clearly indicates an interval of time (e.g. “during the first 9 days” or “from day 9 on”) does not earn the point.

0	1

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$S'(9)$ is the rate at which the number of boxes of snacks remaining is changing, in boxes per day, on day 9.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of $S(t)$ in place of y earns the point.

The use of x in place of t earns the point.

Tangent lines do not need to be simplified, but any simplification must be done correctly.

Any correct form of the tangent line earns the point. For example:

· $y = 440 - 40(t - 6)$

· $y - 440 = -40(t - 6)$

· $y = -40t + 240 + 440$

· $y = -40t + 680$

0	1

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$y = S'(6)(t - 6) + S(6)$

6-7

$y = -40\,(t - 6) + 440$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the point the response must include $S''\,(t) < 0$ and overestimate.

"$S\,(t)$ is concave down" can be used instead of $S''\,(t) < 0$.

Information about $S'\,(t)$ by itself without reference to $S''\,(t) < 0$ is not sufficient and does not earn this point.

"Overestimate because it is concave down" is not sufficient by itself because of the use of "it". The response must clearly, directly or indirectly, indicate to what "it" is referring.

If a response makes the correct conclusion and states, $S''\,(t) < 0$, and then goes on and makes an incorrect statement or a statement that cannot be verified will not earn this point.

For example: "$S'\,(t)$ is concave down, overestimate" does not earn this point.

0	1 ✓

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because $S''\,(t) < 0$, the tangent line approximation would produce an overestimate.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression of only $8 \cdot Q'\,(5)$ earns this point.

Only expression $8 \cdot S'\,(5)$ also earns this point.

No justifications or explanations are required.

Any incorrect statement such as, $Q'\,(5) = 8 \cdot S'\,(5)$, does not earn this point.

0	1 ✓

6-7

The student response accurately includes the criteria below.

- [ ] Expression

Solution:

The rate of change, in ounces per day, of the total weight of the boxes of snacks remaining at $t = 5$ days is $8Q'(5)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Response clearly references what part of the differential equation is different from the slope field and the response needs only to clearly and explicitly reference one of them.

The response does not need to explicitly reference both the differential equation and the given slope field.

The distinction can be made with explicit reference to one of the two things and an implicit reference to the other

For example:

· "$\frac{dy}{dt}$ is negative, not positive". We will infer that negative is referring to the slopes shown in the slope field.

· "This would be correct for a positive derivative, but $\frac{dy}{dt}$ is negative." We will read "This" as "the slope field" in this sentence.

· "All of the slopes in the slope field are positive, not negative." We will infer that the values of $\frac{dy}{dt}$ at points in the first quadrant are negative.

A response that lacks even an implicit reference to either the differential equation or the slope field will not earn the point. For example:

· "$\frac{dy}{dt}$ is negative" lacks any reference to the slope field and so will not earn the point.

· "All the slopes in the graph are positive" lacks any reference to the differential equation, and so will not earn the point.

· Vague answers such as "The slope field has positive points" or "The field is increasing" will not earn the point.

0	1 ✓

The student response accurately includes the criteria below.

6-7

☐ Explanation

Solution:

When $y > 0$, $\frac{dy}{dt} < 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are positive.

6-7

57. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	2	5	10
$S(t)$ (boxes per day)	-60	-40	-30	-5

A student organization is selling boxes of snacks to raise money for a school trip. For time t days, $0 \le t \le 10$, the rate of change in the number of boxes of snacks remaining to sell is modeled by an increasing, twice-differentiable function S. Values of $S(t)$ at various times t are shown in the table above.

(a) Use a right Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{10} S(t)\,dt$. Show the calculations that lead to your answer.

(b) Is your answer from part (a) an overestimate or an underestimate for $\int_0^{10} S(t)\,dt$? Give a reason for your answer.

(c) Using correct units, approximate $S'(2)$ using the average rate of change of S over the interval $0 \le t \le 5$.

(d) Explain the meaning of $S'(2)$ in the context of the problem.

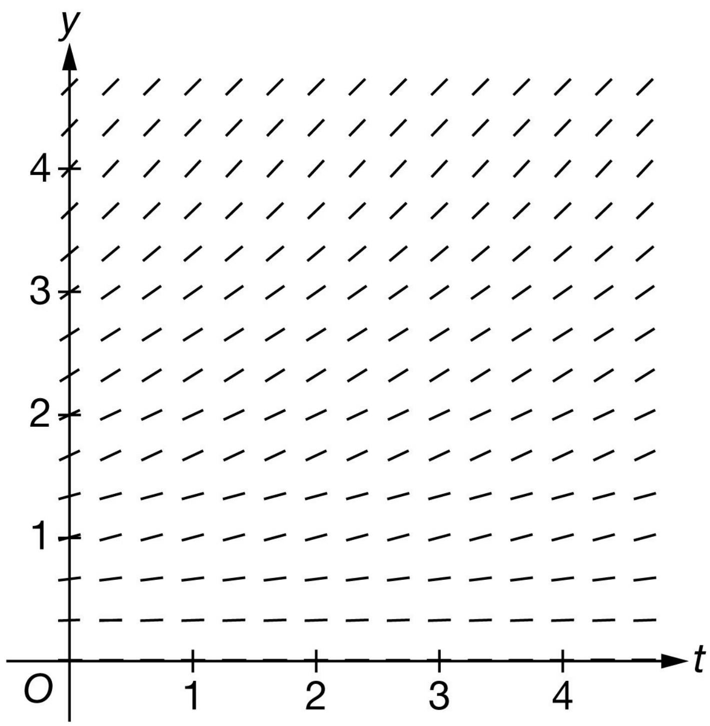
(e) It is known that $S'(5) = 6$. Write an equation for the line tangent to the graph of $y = S(t)$ at $t = 5$.

(f) Find the value of $\int_0^{10} S'\left(\frac{t}{2}\right)\,dt$. Show the calculations that lead to your answer.

(g) Another student organization is also selling boxes of snacks. For $0 \le t \le 10$, the total number of boxes of snacks remaining for the second student organization can be modeled by the decreasing and differentiable function M, where M is a function of t. Assume the weight of each box is 7.5 ounces. Write an expression for the rate of change, in ounces per day, of the total weight of the boxes of snacks remaining for the second student organization at $t = 6$ days.

6-7

(h) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = M'(t)$ for the function M defined in part (g).



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns, like so.

-40	2	-80
-30	3	-90
-5	5	$+(-25)$
		-195

or

$(2-0)(-40)$	$=-80$
$(5-2)(-30)$	$=-90$
$(10-5)(-5)$	$=-25$
	-195

Implicit responses such as those above are eligible for both points.

A response presenting a left Riemann sum (instead of right), will not earn any points for part (a).

The second point is for the correct numerical result.

6-7

In order to earn the second point, there must be supporting calculations.

The numerical values for $S(2)$, $S(5)$, and $S(10)$ must be pulled from the table

Any simplification shown must be correct.

Use the following as an illustration of how to score both points. (In most cases, you see scores for which the expression represents the entirety of relevant work shown for the response.)

· A response of $(-40)\cdot(2-0)+(-30)\cdot(5-2)+(-5)\cdot(10-5)$ earns both points.

· A response of $2\cdot S(2)+3\cdot S(5)+5\cdot S(10)$ earns the first point. Eligible for second point if more work is on the page.

· A response of $2\cdot S(2)+3\cdot S(5)+5\cdot S(10)=-195$ earns both points.

· A response of $2\cdot(-40)+3\cdot(-30)+5\cdot(-5)$ earns both points.

· A response of $2\cdot(-40)+3\cdot(-30)+4\cdot(-5)$ contains one numerical error. Earns the first point. Not eligible for the second point.

· A response of $2\cdot(-40)+3\cdot(-30)+5\cdot(-5)=-190$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.

· A response of $(-80)+(-90)+(-25)$ or $(-80)+(-90)+(-25)=-195$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we are not seeing an unsupported answer. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$2\cdot(-40)+3\cdot(-30)+5\cdot(-5)$

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Right Riemann sum
- [ ] Answer

Solution:

$$\int_0^{10} S(t)\,dt \approx S(2)(2-0)+S(5)(5-2)+S(10)(10-5)$$

6-7

$= (-40) \cdot 2 + (-30) \cdot 3 + (-5) \cdot 5 = -195$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an overestimate because S is increasing.

The response does not need an explicit reference to part (a) or to a right Riemann sum.

A response that appeals to concavity as a reason will NOT earn this point.

The following table contains some examples of what WILL or will NOT earn the point.

The following examples earn the point:

· Overestimate because S is increasing

· Overestimate because $S'(t) > 0$

· Overestimate because the graph of S is increasing (we know they are talking about the given function S)

· Overestimate because the function is increasing. (There is only one function in the problem at this point.)

The following examples do not earn the point:

· Overestimate because S is concave down/up

· Overestimate because the graph/curve is increasing. Since no graph/curve is provided, we do not know what graph/curve they are talking about.

· Overestimate because the table [or table values] is/are increasing. The behavior of the values shown in the table is not the same as the behavior of the function S. The table does not tell us what is happening to S between the given points.

0	1

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because S is an increasing function, and this is a right Riemann sum, this is an overestimate.

Part C

6-7

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

We need to see both a difference and a quotient. They don't have to be together.

The expression does not need to be simplified, but values for $S(5)$ and $S(0)$ must be pulled from the table.

The response does not need to include "approximately" in any way. We will read $=$ and $\approx$ as the same thing.

The second point is for correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols. For example:

· Boxes per day per day or Boxes/day/day

· Boxes per day^2 or Boxes/day^2

· Boxes per day squared

We do not read work shown in part (d) for this point. The units must appear in part (c).

The letter " B " is not a standard abbreviation for "boxes." It will not be interpreted as such.

The response will not be penalized for any attempt to verbally describe their answer after a correct answer has been presented.

Minimalist response: The following expression and units represent the minimum work that must be seen in a response in order to earn both points.

$\frac{-30+60}{5}\ \text{Boxes/day}^2$ or $\frac{-30-(-60)}{5}\ \text{Boxes/day}^2$

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$S'(2) \approx \frac{S(5)-S(0)}{5-0} = \frac{-30-(-60)}{5} = 6 \text{ boxes per day per day}$$

6-7

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

· Indicating a rate of a rate

· Including the context of boxes [remaining to be sold]

· Explicitly referring to the specified point in time $t = 2$

Incorrect or missing units will not be penalized as long as there is a clear connection to the context of the problem.

The words “increasing” and “decreasing” are interpreted as modifiers describing one of the specified rates.

If the numerical value from part (c) is used in this interpretation, then we expect that the choice of “increasing” or “decreasing” is consistent with that value.

The following examples use 6 as the imported value from part (c). If a different [incorrect] value is imported from part (c), then these statements should be consistent with that value. (Remember that we are reading $=$ and $\approx$ as the same thing.)

· “… increasing at a rate of 6 ” is OK [Units not required in part (d)].(the point is not yet earned with this phrase alone)

· “… decreasing at a rate of 6 ” is OK [Units not required in part (d)]. (the point is not yet earned with this phrase alone)

Note: the context is complicated. $S\ (t)$ is describing the rate of change of the remaining inventory of boxes. The values in the table indicate that the inventory is decreasing more slowly as time passes. Common sense might tell us that this equates to the idea that the rate at which boxes are sold is decreasing. Explaining the meaning using “boxes sold” rather than “boxes remaining” will not prevent the response from earning this point.

Rate of a rate

To earn this point, the response must explicitly discuss the rate of change of a rate of change. For example:

· The rate of change in the rate at which boxes are changing

· The rate of change in boxes is changing (or decreasing) at a rate of ...

· The rate at which the rate of change of boxes is changing

The words “increasing,” “decreasing,” or “changing” are not interpreted as rates.

Units are not interpreted as rates. (The phrase “boxes per day” is not enough to indicate a rate.)

Slope of tangent line, velocity, and acceleration are not appropriate interpretations for this context.

6-7

“Rate of change of S ” is not enough without describing what S represents in this context.

Context

A key component of the context for this explanation is “boxes.”

Any misstatement about boxes sold vs. boxes remaining to be sold will not prevent the response from earning this point.

Explicit reference to $t = 2$

Acceptable references to a specific point in time include (but are not limited to):

· On day 2 (where day could be replaced with any other unit of time)

· At $t = 2$ (with or without units of time)

· After 2 days (where days could be replaced with any other unit of time).

Note: we are choosing to assume that “after” is being used in lieu of “at” unless we have additional evidence to the contrary.

If the response is explicitly indicating an interval of time (e.g., “from day 2 on” or “during the first 2 days”), this point will not be earned.

0	1

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$S'(2)$ is the rate at which the rate of change in the number of boxes of snacks remaining is changing, in boxes per day per day, on day 2.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. Any of these equations can be written as is without any explanation or additional supporting work. If rewriting of equations is done (transforming to slope-intercept form, perhaps), it must be done correctly.

$y = 6(t - 5) - 30$

6-7

$y + 30 = 6\left(t - 5\right)$

$y = 6t - 60$

The key calculus concept being assessed in this part of the problem is that the value of $S'\left(5\right)$ is the slope of the tangent line. In that spirit…

· The response will not be penalized for using x instead of t.

We are accepting $S\left(t\right)$ in lieu of y in all forms of the tangent line equation.

0	1

The student response accurately includes the criteria below.

- [ ] Tangent line equation

Solution:

$y = S'\left(5\right)\left(t - 5\right) + S\left(5\right)$

$y = 6\left(t - 5\right) - 30$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are three components of substitution that we need to see:

· Uses $u = \frac{t}{2}$, which may be done implicitly

· Uses the correct constant of 2

· Uses the correct limits for the substitution (0 and 5). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

Either of the expressions $2S\left(\frac{t}{2}\right)\Big|_0^{10}$ or $2S\left(u\right)\Big|_0^{5}$ will earn the substitution point without any additional supporting work.

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

· Limits for t attached to an integral with u as the variable of integration (or vice versa),

6-7

· Improper equals signs, such as an indefinite integral equal to a definite integral

· Reversed limits of integration, for example: $\int_5^0 S'(u)\ du$

The second point is for the Fundamental Theorem of Calculus.

This point can be earned with any of the following, where k is a nonzero constant and $b = 5$ or $b = 10$. No explanation or additional supporting work is necessary.

· $kS\left(\frac{t}{2}\right)\Big|_0^b$

· $k(S(b) - S(0))$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is 60.

Values from the table must be used. $2(S(5) - S(0))$ is not enough to earn the answer point.

The expression $2(-30 - (-60))$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown they do not need to be correct to earn this point.

The response cannot earn this point without also earning the first two points.

Minimalist response: The expression $2(-30 - (-60))$ represents the minimum work that must been seen in a response in order to earn all 3 points.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{2}$. Then $du = \frac{dt}{2}$.

When $t = 0$, $u = 0$.

When $t = 10$, $u = 5$.

6-7

$$\int_0^{10} S'\left(\frac{t}{2}\right)\,dt = 2\int_0^5 S'\left(u\right)\,du$$

$$= 2\left(S\left(5\right) - S\left(0\right)\right)$$

$$= 2\left(-30 - \left(-60\right)\right) = 60$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression at $t = 6$ earns the point. It may be written as $\left(7.5\right) \cdot M'\left(6\right)$ or $\left(7.5\right) \cdot \left.\frac{dM}{dt}\right|_{t=6}$

The expression $\left(7.5\right) \cdot M'\left(t\right)$ is not sufficient because it does not use $t = 6$. .

The response must use M (not S or any other function name) in order to earn the point.

0	1

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in ounces per day, of the total weight of the boxes of snacks remaining for the second student organization at $t = 6$ days is $7.5M'\left(6\right)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between the differential equation $\frac{dy}{dt} = M'\left(t\right)$ and the given slope field.

In order to earn the point, the response must state that the values of the differential equation are negative while the slopes in the given slope field are positive or 0 (nonnegative).

Some ways to say that the values of differential equation are negative:

· M is decreasing

· $M'\left(t\right) < 0$

· M has negative slopes

6-7

When discussing the slopes in the slope field, the response must explicitly refer to the signs of the slopes.

The following examples could earn the point:

· The slope field has nonnegative slopes.

· The slopes shown are positive or 0.

· The slope field contains positive slopes.

· "The graph" or "The field" could be used in lieu of "The slope field."

The following examples will not earn the point:

· The slope field is increasing/positive.

· The slope field looks like it represents an increasing function.

· The slope field represents a quadratic [or exponential] function.

· "It" (rather than "the slope field")

Statements like " M is a decreasing function, so the signs of the slopes in the slope field should be negative" can earn this point because "should be" is indicating contrast with what is known.

The response will not be penalized for saying that the slopes in the slope field are "always" positive or "all" positive.

0	1 ✓

The student response accurately includes the criteria below.

☐ Explanation

Solution:

M is a decreasing function, so $\frac{dy}{dt} = M'(t) \leq 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are positive.

58. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	8	24	35
$P(t)$ (number of postcards remaining per minute)	-10	-8	-5	-4

A student is delivering postcards to teachers from students on a school trip. For time t minutes, $0 \leq t \leq 35$, the rate at which the total number of postcards remaining to deliver is changing, in number of postcards remaining per minute, is modeled by an increasing, twice-differentiable function P. Values of $P(t)$ at various times t are shown in the table above.

(a) Use a left Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{35} P(t)\,dt$. Show the calculations that lead to your answer.

(b) Is your answer from part (a) an overestimate or an underestimate for $\int_0^{35} P(t)\,dt$? Give a reason for your answer.

(c) Using correct units, approximate $P'(24)$ using the average rate of change of P over the interval $8 \leq t \leq 35$.

(d) Explain the meaning of $P'(24)$ in the context of the problem.

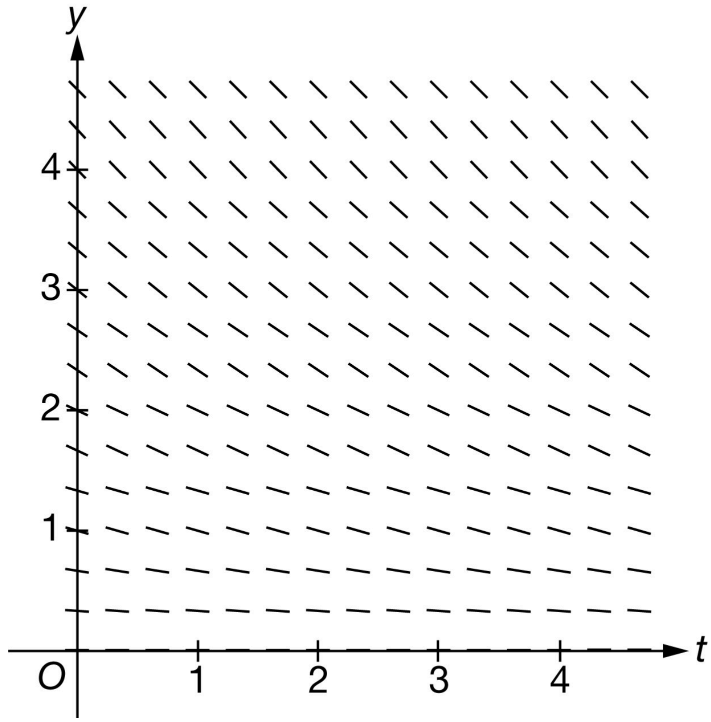
(e) It is known that $P'(8) = 0.2$. Write an equation for the line tangent to the graph of $y = P(t)$ at $t = 8$.

(f) Find the value of $\int_0^{24} P'\left(\frac{t}{3}\right)\,dt$. Show the calculations that lead to your answer.

(g) Another student brings a bag full of postcards to deliver as well. For $0 \leq t \leq 35$, the total number of postcards delivered to teachers by this student can be modeled by the increasing, differentiable function R, where R is a function of t. Assume the weight of each postcard is 4 grams. Write an expression for the rate of change, in grams per minute, of the total weight of postcards delivered by the second student at $t = 11$ minutes.

6-7

(h) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = R'(t)$ for the function R defined in part (g).



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a left Riemann sum. The setup must be a sum of three products.

A response may still earn this point if there is only 1 numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns, like so.

8	(-10)	-80
16	(-8)	-128
11	(-5)	$+(-55)$
		-263

or

$(8-0)(-10)$	$=-80$
$(24-8)(-8)$	$=-128$
$(35-24)(-5)$	$=-55$
	-263

Implicit responses such as those above are eligible for both points.

A response presenting a right Riemann sum (instead of left), will not earn any points for part (a).

The second point is for the correct numerical result.

6-7

In order to earn the second point, there must be supporting calculations.

The numerical values for $P(0)$, $P(8)$, and $P(24)$ must be pulled from the table

Any simplification shown must be correct.

Use the following as an illustration how to score of both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

· A response of $(-10) \cdot (8-0) + (-8) \cdot (24-8) + (-5) \cdot (35-24)$ earns both points.

· A response of $8 \cdot P(0) + 16 \cdot P(8) + 11 \cdot P(24)$ earns the first point. Eligible for second point if more work is on the page.

· A response of $8 \cdot P(0) + 16 \cdot P(8) + 11 \cdot P(24) = -263$ earns both points.

· A response of $8 \cdot (-10) + 16 \cdot (-8) + 11 \cdot (-5)$ earns both points.

· A response of $8 \cdot (-10) + 18 \cdot (-8) + 11 \cdot (-5)$ contains one numerical error. Earns the first point. Not eligible for the second point.

· A response of $8 \cdot (-10) + 16 \cdot (-8) + 11 \cdot (-5) = -200$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.

· A response of $(-80) + (-128) + (-55)$ or $(-80) + (-128) + (-55) = -263$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we are not seeing an unsupported answer. The second point will be earned.

The response does not need to include "approximately" in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points. $8 \cdot (-10) + 16 \cdot (-8) + 11 \cdot (-5)$

✓

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Left Riemann sum
- ☐ Answer

Solution:

$$\int_0^{35} P(t)\, dt \approx P(0)(8-0) + P(8)(24-8) + P(24)(35-24)$$

$$= (-10) \cdot 8 + (-8) \cdot 16 + (-5) \cdot 11 = -263$$

6-7

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an underestimate because P is increasing.

The response does not need an explicit reference to part (a) or to a left Riemann sum.

A response that appeals to concavity as a reason will NOT earn this point.

The following examples earn the point:

· Underestimate because P is increasing

· Underestimate because $P'(t) > 0$

· Underestimate because the graph of P is increasing (we know they are talking about the given function P)

· Underestimate because the function is increasing. (There is only one function in the problem at this point.)

The following examples do not earn the point:

· Underestimate because P is concave down/up

· Underestimate because the graph/curve is increasing. Since no graph/curve is provided, we do not know what graph/curve they are talking about.

· Underestimate because the table [or table values] is/are increasing. The behavior of the values shown in the table is not the same as the behavior of the function P. The table does not tell us what is happening to P between the given points.

✓

0	1

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because P is an increasing function, and this is a left Riemann sum, this is an underestimate.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

6-7

The first point is earned for the correct approximation shown with supporting work.

We need to see both a difference and a quotient. They don't have to be together.

The expression does not need to be simplified, but values for $P\left(35\right)$ and $P\left(8\right)$ must be pulled from the table.

Decimal approximation is not expected, but if presented, then the decimal value must be correct to three decimal places, either rounded or truncated. We do not read anything past the third decimal place.

· If the first three places match 0.148 then the point is earned

· It does not matter whether any subsequent decimal places match the correct value or not. (For example, the value 0.1487 will earn the point because the first three decimal places match the correct value.)

· Responses with fewer than three decimal places (0.14, 0.15, 0.1, etc.) will not earn the point.

The response does not need to include "approximately" in any way. We will read $=$ and $\approx$ as the same thing.

The second point is for correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols. For example:

· Postcards per minute per minute or Postcards/min/min

· $\text{Postcards per min}^2$ or Postcards/min^2

· Postcards per minute squared

We do not read work shown in part (d) for this point. The units must appear in part (c).

The letter "P" is not a standard abbreviation for "Postcards." It will not be interpreted as such.

The response will not be penalized for any attempt to verbally describe their answer after a correct answer has been presented.

Minimalist response: The following expression and units represent the minimum work that must be seen in a response in order to earn both points. $\frac{-4+8}{27}\text{ postcards/min}^2$ or $\frac{-4-(-8)}{27}\text{ postcards/min}^2$

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

6-7

Solution:

$$P'(24) \approx \frac{P(35)-P(8)}{35-8} = \frac{-4-(-8)}{27} = \frac{4}{27} \text{ postcard per minute per minute}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

· Indicating a rate of a rate

· Including the context of postcard delivery (postcards remaining)

· Explicitly referring to the specified point in time $t = 24$

Incorrect or missing units will not be penalized as long as there is a clear connection to the context of the problem.

The words “increasing” and “decreasing” are interpreted as modifiers describing one of the specified rates.

If the numerical value from part (c) is used in this interpretation, then we expect that the choice of “increasing” or “decreasing” is consistent with that value.

The following examples use $\frac{4}{27}$ as the imported value from part (c). If a different [incorrect] value is imported from part (c), then these statements should be consistent with that value. (Remember that we are reading $=$ and $\approx$ as the same thing.)

· “… increasing at a rate of $\frac{4}{27}$” is OK [Units not required in part (d)] (the point is not earned with this phrase alone)

· “… decreasing at a rate of $\frac{4}{27}$” is OK [Units not required in part (d)] (the point is not earned with this phrase alone)

Note: the context is complicated. $P(t)$ is describing the rate of change of the remaining undelivered postcards. The values in the table indicate that the number of undelivered postcards is decreasing more slowly as time passes. Common sense might tell us that this equates to the idea that the rate at which postcards are delivered is decreasing. Explaining the meaning using “postcards delivered” rather than “postcards remaining” will not prevent the response from earning this point.

Rate of a rate

To earn this point, the response must explicitly discuss the rate of change of a rate of change. For example:

· The rate of change in the rate at which postcards are changing

· The rate of change in postcards is changing/increasing/decreasing at a rate of . . .

6-7

The rate at which the rate of change of postcards is changing.

The words "increasing," "decreasing," or "changing" are not interpreted as rates.

Units are not interpreted as rates. (The phrase "postcards per minute" is not enough to indicate a rate.)

Slope of tangent line, velocity, and acceleration are not appropriate interpretations for this context.

"Rate of change of P" is not enough without describing what P represents in this context.

Context

A key component of the context for this explanation is "postcards."

Any misstatement about postcards delivered vs. postcards remaining to be delivered will not prevent the response from earning this point.

Explicit reference to $t = 24$

Acceptable references to a specific point in time include (but are not limited to):

· At minute 24 (where minute could be replaced with any other unit of time)

· At $t = 24$ (with or without units of time)

· After 24 minutes (where minutes could be replaced with any other unit of time).

Note: we are choosing to assume that "after" is being used in lieu of "at" unless we have additional evidence to the contrary.

If the response is explicitly indicating an interval of time (e.g., "from minute 24 on" or "during the first 24 minutes"), this point will not be earned.

0	1

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$P'(24)$ is the rate at which the rate of change in the number of postcards remaining is changing, in postcards per minute per minute, at minute 24.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-7

This point is earned with a correct tangent line equation in any form. Any of these equations can be written as is without any explanation or additional supporting work. If rewriting of equations is done (transforming to slope-intercept form, perhaps), it must be done correctly.

· $y = 0.2\,(t - 8) - 8$

· $y + 8 = 0.2\,(t - 8)$

· $y = 0.2t - 9.6$

The key calculus concept being assessed in this part of the problem is that the value of $P'(8)$ is the slope of the tangent line. In that spirit:

· The response will not be penalized for using x instead of t.

· We are accepting $P(t)$ in lieu of y in all forms of the tangent line equation.

0	1

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$y = P'\,(8)\,(t - 8) + P\,(8)$

$y = 0.2\,(t - 8) - 8$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are 3 components of substitution that we need to see:

· Uses $u = \frac{t}{3}$, which may be done implicitly

· Uses the correct constant of 3

· Uses the correct limits for the substitution (0 and 8). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

Either of the expressions $3P\,\left(\frac{t}{3}\right)\Big|_0^{24}$ or $3P\,(u)\,\Big|_0^{8}$ will earn the substitution point without any additional supporting work.

Special Handling: Notation Errors

6-7

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

· Limits for t attached to an integral with u as the variable of integration (or vice versa),

· Improper equals signs, such as an indefinite integral equal to a definite integral

· Reversed limits of integration, for example: $\int_8^0 P'(u)\ du$

The second point is for the Fundamental Theorem of Calculus.

This point can be earned with any of the following, where k is a nonzero constant and $b = 8$ or $b = 24$. No explanation or additional supporting work is necessary.

· $kP\left(\frac{t}{3}\right)\Big|_0^b$

· $k\left(P(b) - P(0)\right)$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is 6.

Values from the table must be used. $3\left(P(8) - P(0)\right)$ is not enough to earn the answer point.

The expression $3\left(-8 - (-10)\right)$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they do not need to be correct to earn this point.

The response cannot earn this point without also earning first two points.

Minimalist response: The expression $3\left(-8 - (-10)\right)$ represents the minimum work that must been seen in a response in order to earn all three points.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

6-7

Let $u = \frac{t}{3}$. Then $du = \frac{dt}{3}$.

When $t = 0$, $u = 0$.

When $t = 24$, $u = 8$.

$$\int_0^{24} P'\left(\frac{t}{3}\right)\, dt = 3\int_0^{8} P'(u)\, du$$

$= 3\left(P(8) - P(0)\right)$

$= 3\left(-8 - (-10)\right) = 6$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression at $t = 11$ earns the point. It may be written as $4 \cdot R'(11)$ or $4 \cdot \left.\frac{dR}{dt}\right|_{t=11}$

The expression $4 \cdot R'(t)$ is not sufficient because it does not use $t = 11$.

The response must use R (not M or G or any other function name) in order to earn the point.

0	1

The student response accurately includes the criteria below.

- [ ] Expression

Solution:

The rate of change, in grams per minute, of the total weight of postcards delivered by the second student at $t = 11$ minutes is $4R'(11)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between the differential equation $\frac{dy}{dt} = R'(t)$ and the given slope field.

In order to earn the point, the response must state that the values of the differential equation are positive while the slopes in the given slope field are negative or 0 (non-positive).

6-7

Some ways to say that the values of differential equation are positive:

· R is increasing

· $R'(t) > 0$

· R has positive slopes

When discussing the slopes in the slope field, the response must explicitly refer to the signs of the slopes.

The following examples could earn the point:

· The slope field has nonpositive slopes.

· The slopes shown are negative or 0.

· The slope field contains negative slopes.

· "The graph" or "The field" could be used in lieu of "The slope field."

The following examples will not earn the point:

· The slope field is decreasing/negative.

· The slope field looks like it represents a decreasing function.

· The slope field represents a quadratic [or exponential] function.

· "It" (rather than "the slope field")

Statements like "R is an increasing function, so the signs of the slopes in the slope field should be positive" can earn this point because "should be" is indicating contrast with what is known.

The response will not be penalized for saying that the slopes in the slope field are "always" negative or "all" negative.

0	1 ✓

The student response accurately includes the criteria below.

☐ Explanation

Solution:

R is an increasing function, so $\frac{dy}{dt} = R'(t) \geq 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are negative.

6-7

59. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	8	24
$P(t)$ (postcards per minute)	-6	-10	-12

A student is delivering postcards to teachers from students on a school trip. For time t minutes, $0 \le t \le 24$, the rate of change in the number of postcards remaining for the student to deliver is modeled by a twice-differentiable function P, in postcards remaining per minute. Values of $P(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $P'(8)$ using the average rate of change of P over the interval $0 \le t \le 24$.

(b) Explain the meaning of $P'(8)$ in the context of the problem.

(c) The function $N(t)$ models the number of postcards remaining for the student to deliver at time t minutes, where $N'(t) = P(t)$. It is known that $N(8) = 185$. Write an equation for the line tangent to the graph of $y = N(t)$ at $t = 8$.

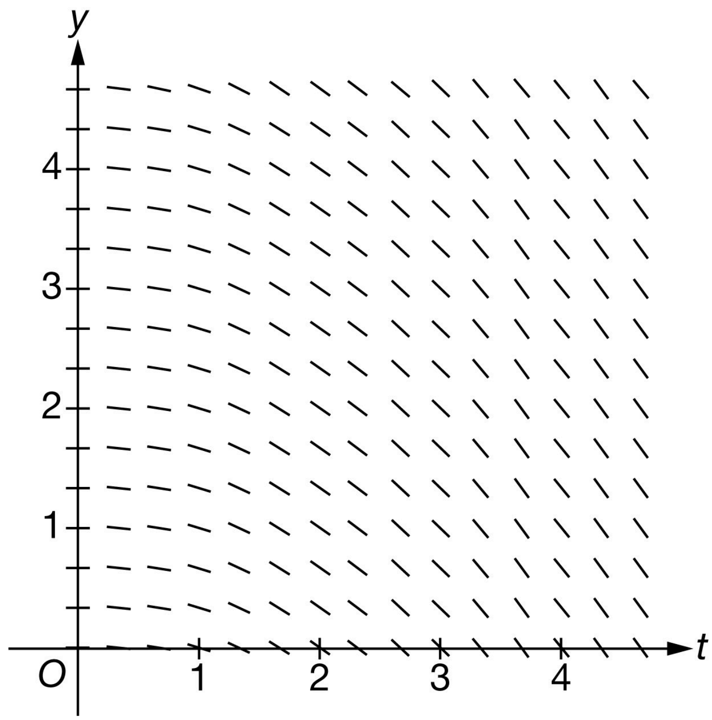
(d) Find the value of $\int_0^8 P'(3t)\,dt$. Show the calculations that lead to your answer.

(e) Use a trapezoidal sum with the two subintervals indicated by the table to approximate $\int_0^{24} P(t)\,dt$. Show the calculations that lead to your answer.

(f) Another student is also delivering postcards. For $0 \le t \le 24$, the rate of change in the number of postcards remaining for the second student to deliver, in postcards per minute, can be modeled by the differentiable function R. Assume the weight of each postcard is 4 grams. Write an expression in terms of P and R for the rate of change, in grams per minute, of the combined weight of the postcards remaining for both students at $t = 13$ minutes.

6-7

(g) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = -0.25y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

Answers without supporting work do not earn this point. Supporting work must be at least a difference and a quotient.

Response does not need to be simplified, but any simplification must be done correctly.

If a decimal is presented, it must be correct to three decimal places. We do not read anything past the third decimal place.

The second point is earned for the correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols.

The response will not be penalized for any verbal attempt to discuss the answer after the correct presentation of the approximation.

The minimum response needed to earn both points is: $\frac{-12+6}{24}$ postcards per minute per minute

6-7

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Approximation
- [ ] Units

Solution:

$$P'(8) \approx \frac{P(24)-P(0)}{24-0} = \frac{-12-(-6)}{24} = \frac{-6}{24}$$

$= -\frac{1}{4}$ postcard per minute per minute

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This part is intended to be independent from part (a), so we do not want or need to see any reference to the value obtained in (a). If that value is referenced, conclusion must be consistent.

Three components of a correct answer:

· Indicating a rate of a rate (look for the word rate to be used correctly twice)

· Including the context of postcards

· Referring to the specific point in time

Units will not be considered for this point. They do not need to be present, and if they are, they do not need to be correct.

Any incorrect statements make the response ineligible for the point. For example:

· $P'(8)$ is the average (or an approximation, or an estimate of) rate of change of . . . .

· $P'(8)$ is the acceleration (or rate of change of velocity) of . . . .

“Increasing” or “decreasing” may be used in place of “changing” as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Beware of phrases such as “decreasing at a negative rate.”

· Example: $P'(8)$ is the rate of decrease of the rate of change of the number of postcards at 8 minutes. (earns the point)

6-7

· Example: $P'(8)$ means the number of postcards per minute is decreasing by $\frac{1}{4}$ postcards/minute/minute. (does not earn the point)

Acceptable references to time include (but are not limited to):

· On (or at) minute 8

· At $t = 8$ (with correct units of time, without correct units of time, or even with incorrect units of time)

· After $t = 8$ (with correct units of time, without correct units of time, or even with incorrect units of time)

If the response clearly indicates an interval of time (e.g. "from minute 0 to minute 8 ") the response will not earn the point.

Examples that earn the point include:

· $P'(8)$ represents the rate at which the rate of change of the number of postcards is changing on minute 8.

· $P'(8)$ is the rate of change in the rate of change of the number of postcards at 8 minutes.

· $P'(8)$ represents the rate at which the rate of change of the number of postcards is decreasing at $t = 8$.

· $P'(8)$ equals (answer from part (a)) means the rate of change in the number of postcards per minute is increasing at a rate of (answer from part (a)) at $t = 8$.

· $P'(8)$ represents the rate at which the rate of change of the number of postcards is increasing on the eighth minute.

Examples that do not earn the point include:

· $P'(8)$ represents the rate at which the number of postcards is changing on minute 8.

· $P'(8)$ is the rate of change in the number of postcards per minute at $t = 8$ minutes

· The rate of change in the number of postcards per minute is decreasing at $t = 8$.

· The rate of change in the rate of change of the number of postcards per day is decreasing at $t = 8$.

· $P'(8) = -\frac{1}{4}$ means that the rate of change in the number of postcards is decreasing at a rate of $-\frac{1}{4}$ $t = 8$. (double negative)

0	1 ✓

The student response accurately includes the criteria below.

☐ Explanation

6-7

Solution:

$P'(8)$ is the rate at which the rate of change in the number of postcards remaining is changing, in postcards per minute per minute, at minute 8.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned with a correct tangent line equation in any form. Equivalent forms include (but are not limited to):

· $y = -10(t - 8) + 185$

· $y - 185 = -10(t - 8)$

· $y = -10t + 265$

The student may write x in place of t and still earn the point.

No work is required to earn the point.

We will accept $N(t)$ in lieu of y in the equation of the tangent line.

0	1 ✓

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$y = N'(8)(t - 8) + N(8)$

$y = -10(t - 8) + 185$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution.

Two things are required as evidence of correct substitution:

· Coefficient of $\frac{1}{3}$ correctly applied to an integral or an antiderivative

6-7

· Evidence that the correct limits of 0 to 24 will be applied

Examples that will earn the first point:

· $\frac{1}{3}\int_{0}^{124} P'(u)\,du$

· $\frac{1}{3}P(u)\Big|_0^{24}$ or $\frac{1}{3}P(3t)\Big|_0^{8}$

Responses that include misstatements such as using t -values for limits with u as the variable of integration in an intermediate step which is later corrected may still earn this point. Intermediate errors in bounds that are subsequently corrected will disqualify the student from earning the third point. If the bounds are never appropriately handled, the response cannot earn the first point. Scratch work that is not connected to the solution will not be read.

Examples:

· $\int_0^8 P'(3t)\,dt = \frac{1}{3}\int_0^8 P'(u)\,du = \frac{1}{3}P(3t)\Big|_0^8$ earns the first point, but will not earn the third point.

· $\int_0^8 P'(3t)\,dt = \frac{1}{3}\int_0^8 P'(u)\,du = \frac{1}{3}[P(8) - P(0)]$ does not earn the first point.

The second point is for the Fundamental Theorem of Calculus, demonstrating the knowledge that
$\int_a^b P'(u)\,du = P(u)|_a^b = P(b) - P(a)$

This point can be earned with either $kP(3t)\,|_a^b$, $kP(t)\,|_a^b$, or $k\,(P(b) - P(a))$ where k is a nonzero constant [not necessarily correct], $a = 0$, and $b = 8$ or $b = 24$.

The response can earn this point without earning the first point.

$\frac{1}{3}[P(24) - P(0)]$ earns the first two points.

The third point is for the correct numerical value. Our answer only.

Values from the table must be used (i.e., $\frac{1}{3}[P(24) - P(0)]$ is not sufficient to earn the third point).

If the response includes any linkage error, (i.e., the use of equal signs to equate two expressions that are not equal), the third point will not be earned. If an indefinite integral is linked with a definite integral the response will not be penalized.

· $\int_0^8 P'(3t)\,dt = \frac{1}{3}\int P'(u)\,du = \frac{1}{3}P(3t)\Big|_0^8 = \ldots$ is eligible for the third point.

Eligibility: Must have earned the first two points.

The minimum response needed to earn all three points is: $\frac{1}{3}(-12 + 6)$.

6-7

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3\,dt$.

When $t = 0$, $u = 0$.

When $t = 8$, $u = 24$.

$$\int_0^8 P'(3t)\,dt = \frac{1}{3}\int_0^{24} P'(u)\,du$$

$$= \frac{1}{3}(P(24) - P(0))$$

$$= \frac{1}{3}(-12 - (-6)) = -\frac{6}{3} = -2$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the form of a trapezoidal sum

Point is earned for the setup of the trapezoidal sum. Numerical values do not need to be pulled from the table for this point, but Δt values must be presented. Responses that contain one incorrect numerical value in the trapezoidal sum setup are still eligible to earn the first point, but will not earn the second.

Response may present a left Riemann sum and a right Riemann sum, then average the two. Again, one incorrect numerical value will not disqualify the response from earning this first point.

Response may compute trapezoids as areas of rectangles and triangles.

Trapezoidal areas may be given in a graph. In order to earn this point, the response must clearly show the computations leading to the values presented in the graph, and must show that these areas are added to obtain the final approximation value.

6-7

A response using subtraction instead of addition will not earn the point (e,g, $\frac{1}{2}[8(P(0)-P(8))+16(P(8)-M(24))]$ earns no points).

Response utilizing "Trapezoidal Rule" will not earn this point, unless this represents only one error in the numerical values.

$\cdot\frac{1}{2}(8)\left[P(0)+2P(8)+P(24)\right]$ is equivalent to $\frac{1}{2}[8(P(0)+P(8))+8(P(8)+P(24))]$, thus has only one incorrect numerical value (the second interval width is 8 instead of 16), so can earn the point.

$\cdot\frac{1}{2}(12)\left[P(0)+2P(8)+P(24)\right]$ has two incorrect numerical values (both interval widths are incorrect) so will not earn the point.

The second point is for the correct numerical result.

Our answer only. Response does not need to be simplified, but if it is, it must be simplified correctly.

The answer must be accompanied with supporting work. Supporting work may appear in a graph, with clearly labeled areas. It is possible to earn the second point without earning the first if the supporting work is not sufficient to earn the first point.

Pictures without clear area computations and a clear summation ($+$) of quantities can earn at most the second point.

Special case: A positive constant multiple of a complete and correctly expressed trapezoidal sum (with the correct values pulled from the table) will only the first point:

$\cdot k\left[\left(\frac{-6-10}{2}\right)\cdot 8+\left(\frac{-10-12}{2}\cdot 16\right)\right]$ for $k>0$ earns just the first point.

The minimum response needed to earn both points: $\left(\frac{-16}{2}\right)\cdot 8+\left(\frac{-22}{2}\right)\cdot 16$

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Form of trapezoidal sum
- ☐ Answer

Solution:

$$\int_0^{24} P(t)\,dt \approx \left(\frac{P(0)+P(8)}{2}\right)(8-0)+\left(\frac{P(8)+P(24)}{2}\right)(24-8)$$

$$=\left(\frac{-6-10}{2}\right)\cdot 8+\left(\frac{-10-12}{2}\right)\cdot 16=-240$$

6-7

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a correct constant multiple or correct sum.

Response must contain the constant 4 as a factor of the expression, and/or the correct sum $P(13) + R(13)$.

If the sum is incorrect, then the constant factor of 4 must be applied either to a sum that involves only $P(t)$ and $R(t)$, $P'(t)$ and $R'(t)$, or integrals of $P(t)$ and $R(t)$, or to one of the correct terms of the sum $P(13)$ or $R(13)$, with no additional incorrect terms.

Examples that earn the point include:

$\cdot\, 4\,[P(t) + R(t)]$

$\cdot\, 4\displaystyle\int [P(t) + R(t)]dt$

$\cdot\, 4\,[P' + R']$

$\cdot\, 4P(13)$

Examples that do not earn the point include:

$\cdot\, 4\,[P(t) + R'(t)]$

$\cdot\, 4\displaystyle\int P(t)dt$

$\cdot\, 4P' + R'$

$\cdot\, 4R(13) + t$

Special case**:** $4P(13) + R(13)$ is an error in parentheses, so will earn the first point but is not eligible to earn the second point.

If the constant multiple is missing or incorrect, the sum must be correct. $k[P(13) + R(13)]$ for any nonzero constant k will earn the first point.

The second point is for the correct expression.

$4P(13) + 4R(13)$ (or an equivalent expression) earns both points.

Inclusion of units (correct or incorrect) will not affect the score.

A completely correct expression with one additional constant term added/subtracted would earn only the first point. For example, $4P(13) + 4R(13) + 4R(2)$ earns the first point only.

6-7

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Constant multiple or sum
- [ ] Expression

Solution:

The rate of change, in grams per minute, of the combined weight of the postcards remaining for both students at $t = 13$ minutes is $4P(13) + 4R(13)$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for a correct statement identifying specific inconsistencies between the given differential equation and the given slope field. There are many such correct statements, such as:

· The response references a specific point on the y-axis (such as $(0, 4)$), and indicates that the given slope field shows a slope of zero, yet the differential equation would yield a positive slope.

· The response references a pattern of slopes, and explains why that pattern does not fit our differential equation (e.g. "the slope field shows slopes that change as t changes, but our differential equation depends only on y, so slopes should remain constant for a given value of y as t changes.")

Identifying a mismatch between the differential equation and the slope field at one point is sufficient to earn the point.

Response is not penalized for referencing "x" instead of "t".

If the response contains any incorrect statement, the response will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point, nor will responses that present information beyond what we are given. For example:

· Responses that specify that the slope field shows a particular nonzero slope at a particular point will not earn the point, as this cannot actually be determined visually (i.e., the response should not indicate that at $(2, 2)$ the slope shown on the slope field is -1)

· Responses that state a specific slope value on the slope field for a general value of x or y that are not correct for all points with that x or y value do not earn the point. (e.g. "the slope field shows a slope of 0 when $y = 4$" is an incorrect statement)

6-7

A response may involve a solution to the differential equation, with an argument based on the resulting function. If done correctly, and presented with a correct argument, this will earn the point.

0	1

The student response accurately includes the criteria below.

- [ ] Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

60. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	8	24	35
$P(t)$ (number of postcards remaining)	316	260	100	0

A principal is delivering postcards to teachers from students on a school trip. The principal begins with 316 postcards and delivers them over 35 minutes. For time t minutes, $0 \leq t \leq 35$, the total number of postcards remaining to deliver is modeled by a decreasing, twice-differentiable function P. Values of $P(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $P'(8)$ using the average rate of change of P over the interval $0 \leq t \leq 24$.

(b) Explain the meaning of $P'(8)$ in the context of the problem.

(c) It is known that $P'(24) = -11$. Write an equation for the line tangent to the graph of $y = P(t)$ at $t = 24$.

(d) Use a right Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{35} P(t)\,dt$. Show the computations that lead to your answer.

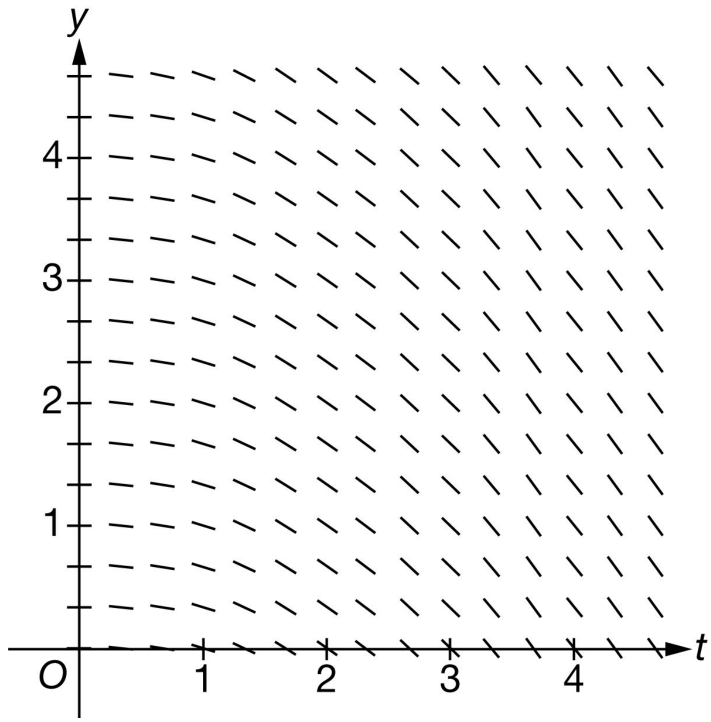
(e) Is the approximation from part (d) an overestimate or an underestimate for $\int_0^{35} P(t)\,dt$? Give a reason for your answer.

(f) Find the value of $\int_0^{8} P'(3t)\,dt$. Show the computations that lead to your answer.

(g) For $0 \leq t \leq 35$, the total number of postcards remaining can also be modeled by the differentiable function R, where R is a function of t. Assume the weight of each postcard is 4 grams. Write an expression for the rate of change, in grams per minute, of the total weight of the postcards remaining at $t = 13$ minutes.

6-7

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.4y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work. The supporting work must be at least a difference and a quotient. Correct values must be taken from the table to earn the point. The difference quotient does not need to be simplified.

Examples that earn the first point include:

- $\frac{100-316}{24-0}$
- $\frac{P(24)-P(0)}{24} = \frac{-216}{24}$
- $\frac{P(24)-P(0)}{24} = -9$

An example that does not earn the first point is $\frac{P(24)-P(0)}{24}$.

Minimal response that earns the first point: $\frac{100-316}{24-0}$ or $\frac{100-316}{24}$

Neither $\frac{-216}{24}$ nor -9, with no supporting work, will earn this point

The second point is for correct units

6-7

The correct units must be attached to a numerical value: either correct or incorrect.

The units may be written using words or a combination of words and symbols. For example:

· Postcards per minute

· Cards per min

· Postcards/min

The response will not be penalized for any verbal attempt to discuss the answer after the correct answer has been presented.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$P'\,(8) \approx \frac{P(24)-P(0)}{24-0} = \frac{100-316}{24} = -9 \text{ postcards per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

· Indicating a rate of change

· Including the context of postcards

· Referring to the specific point in time, $t = 8$

The response is not penalized for incorrect units in this part. (Whether correct or incorrect, units have no effect on the score in this part.)

Responses may use “changing” or “decreasing” as a modifier as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Example: “the number of postcards is decreasing at 9 postcards per minute.” is okay, but “the number of postcards is decreasing at a rate of -9 postcards per day...”would not earn the point.

Acceptable reference to a specific point in time includes (but is not limited to):

6-7

· On minute 8 and after 8 minutes

· At $t = 8$

· At $t = 8$ hours or any other incorrect unit of time

· If the response clearly indicates an interval of time (e.g., “from minute 8 on,” “during the first 8 minutes”, or “for $t = 0$ to $t = 8$ ”), the point is not earned.

Examples that earn the point include:

· $P'(8)$ is the rate of change of the number of postcards per minute at time $t = 8$

· The rate of change of number of postcards is -9 postcards per minute at minute 8

· $P'(8)$ is the rate of change in the number of postcards at time $t = 8$ in postcards per hour

· -9 is the rate of decrease in the number of postcards per minute at minute 8

Examples that do not earn the point include:

· $P'(8)$ is the average rate of change of the total number of postcards /minute at minute 8

· $P'(8)$ is the number of postcards per minute at minute 8

· The rate of change is -9 postcards per minute (Missing “at $t = 8$”)

· $P'(8)$ is the slope of the tangent line at $t = 8$

· Rate of change of P at time $t = 8$

· Postcards per minute decreases on minute 8

0	1 ✓

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$P'(8)$ is the rate at which the number of postcards remaining is changing, in postcards per minute, at time $t = 8$ minutes.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-7

This point is earned with a correct tangent line equation in any form. The use of x in place of t is acceptable.

Some examples include:

· $y - 100 = -11\,(t - 24)$, $y - 100 = -11\,(x - 24)$

· $y = -11\,(t - 24) + 100$, $y = -11\,(x - 24) + 100$

· $y = -11t + 364$, $y = -11x + 364$

The key calculus concept being assessed in this part of the problem is that the value of $P'\,(24)$ is the slope of the tangent line. In that spirit, we are accepting $P'\,(t) = \ldots$ in lieu of $y = \ldots$, but we are not accepting $P'\,(24) = \ldots$

Note that $P\,(t) = -11\,(t - 24) + 100$ earns the point, but $P\,(24) = -11\,(t - 24) + 100$ does not earn the point.

0	1

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$y = P'\,(24)\,(t - 24) + P\,(24)$

$y = -11\,(t - 24) + 100$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

· A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

· Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns. Implicit responses are eligible for both points.

A response presenting a left Riemann sum (instead of right), will not earn any points for part (d).

The second point is for the correct numerical result.

· In order to earn the second point, there must be supporting calculations.

6-7

· The numerical values for $P(8)$, $P(24)$, and $P(35)$ must be pulled from the table

· Any simplification shown must be correct.

Use the following as an illustration how to score both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

· A response of $260 \cdot (8-0) + 100 \cdot (24-8) + 0 \cdot (35-24)$ earns both points.

· A response of $8 \cdot P(8) + 16 \cdot P(24) + 11 \cdot P(35)$ earns the first point. Eligible for second point if more work is on the page.

· A response of $8 \cdot P(8) + 16 \cdot P(24) + 11 \cdot P(35) = 3680$ earns both points.

· A response of $8 \cdot 260 + 16 \cdot 100 + 11 \cdot 0$ earns both points.

· A response of $8 \cdot 260 + 12 \cdot 100 + 11 \cdot 0$ contains one numerical error. Earns the first point. Not eligible for the second point.

· A response of $8 \cdot 260 + 16 \cdot 100 + 11 \cdot 0 = 3860$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.

· A response of $2080 + 1600 + 0 = 3680$ or $2080 + 1600 = 3680$ does not earn the first point because we cannot see the sum of 3 products. However, this is the correct sum, so we see an answer with supporting work. The second point will be earned.

The response does not need to include "approximately" in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$8 \cdot 260 + 16 \cdot 100 + 11 \cdot 0$

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{35} P(t)\,dt \approx P(8)(8-0) + P(24)(24-8) + P(35)(35-24)$$

$$= 260 \cdot 8 + 100 \cdot 16 + 0 \cdot 11 = 3680$$

6-7

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an underestimate because P is decreasing.

The response does not need an explicit reference to the approximation in part (d), nor does there need to be an answer given in part (d). The following statements are sufficient for earning this point:

· Underestimate because P is decreasing

· Underestimate because the function is decreasing

Note: At this point, there is only one function being discussed in this problem, so we will assume "the function" is P.

When a response includes additional information about concavity, we interpret that as a lack of understanding about the question, so this will not earn the point. When the response references a graph or the given table rather than the function P, the point is not earned.

Examples that earn the point:

· The right Riemann sum underestimates because the function is decreasing.

· The sum is an underestimate because $P\ (t)$ is decreasing.

· The sum is an underestimate because $P'\ (t) < 0$.

Examples that do not earn the point:

· The sum is an underestimate because $P'\ (t)$ is decreasing. ($P'\ (t)$ decreasing is not known)

· The right Riemann sum underestimates because the graph is decreasing.

· The right Riemann sum underestimates because the function is decreasing and concave up.

0	1 ✓

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because P is a decreasing function, and this is a right Riemann sum, the answer from part (d) is an underestimate.

6-7

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are 3 components of substitution that we need to see:

· Uses $u = 3t$, which may be done implicitly

· Uses the correct constant of $\frac{1}{3}$

· Uses the correct limits for the substitution (0 and 24). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

· Either of the expressions below will earn the substitution point without any additional supporting work:
$\frac{1}{3} P\,(3t)\Big|_0^8$ or $\frac{1}{3} P\,(u)\Big|_0^{24}$

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point

· Limits for t attached to an integral with u as the variable of integration (or vice versa),

· Improper equals signs, such as an indefinite integral equal to a definite integral

· Reversed limits of integration, for example: $\int_{24}^{0} P'\,(u)du$

The second point is for the Fundamental Theorem of Calculus. This point can be earned with any of the following, where k is a nonzero constant and $b = 8$ or $b = 24$. No explanation or additional supporting work is necessary.

· $kP\,(3t)\Big|_0^b$

· $k\,(P\,(b) - P\,(0))$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is -72.

Values from the table must be used. $\frac{1}{3}[P\,(24) - P\,(0)]$ is not enough to earn the answer point.

The expression $\frac{1}{3}(100 - 316)$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

6-7

The response cannot earn this point without also earning the FTC point.

Minimalist response: The expression $\frac{1}{3}(-216)$ represents the minimum work that must been seen in a response in order to earn all three points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3\ dt$.

When $t = 0$, $u = 0$.

When $t = 8$, $u = 24$.

$$\int_0^8 P'(3t)dt = \frac{1}{3}\int_0^{24} P'(u)du$$

$$= \frac{1}{3}(P(24) - P(0))$$

$$= \frac{1}{3}(100 - 316) = -\frac{216}{3} = -72$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct expression at $t = 13$. No supporting work is required.

The correct expression, $4R'(13)$ or $4P'(13)$, earns this point.

No justification or explanations are required.

Any incorrect statement such as, $P'(13) = 4R'(13)$, does not earn this point.

0	1

6-7

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in grams per minute, of the total weight of the postcards remaining at $t = 13$ minutes is $4R'(13)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between two things: the differential equation, $\frac{dy}{dt} = -0.4y$, and the given slope field.

The response does not need to explicitly reference both the differential equation and the given slope field., but it must explicitly reference one and implicitly reference the other.

Acceptable responses include

· Identifying an inconsistency between the differential equation and the slope field at one point , e.g., at the point $(4,\ 0)$, the slope of the segment should be 0, but it is negative.

· Referencing a pattern of slopes, and explaining why that pattern does not fit the given differential equation.

A response that lacks even an implicit reference to the one of two things being compared will not earn the point. For example, merely stating either $\frac{dy}{dt}$ is negative, or that the slope field is negative discretely does not earn the point.

· "The values for y are less than or equal to zero so $\frac{dy}{dt} \leq 0$" lacks any reference to the slope field and will not earn the point.

· "All the slopes in the slope field are negative" lacks any reference to the differential equation and will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point. Some examples are:

· "The field is decreasing"

· Responses that state a specific slope value on the slope field for a general value of x or y that is not correct for all points with that x or y value do not earn the point. (e.g. "the slope field shows a slope of 0 when $y = 4$" is an incorrect statement)

A response may correctly solve or attempt to solve the differential equation and then provide a correct argument without reference to their solution. Such a response would earn the point.

6-7

A response might actually solve the differential equation and provide a correct argument on the resulting function.

0	1

The student response accurately includes the criteria below.

- [ ] Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

6-7

61. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	9	18	27	36
$P(t)$ (number of postcards remaining)	320	256	100	40	0

A principal is delivering postcards to teachers from students on a school trip. The principal begins with 320 postcards and delivers them over 36 minutes. For time t minutes, $0 \le t \le 36$, the total number of postcards remaining to deliver is modeled by a twice-differentiable function P. Values of $P(t)$ at various times t are shown in the table above.

(a) Use a midpoint Riemann sum with the two subintervals indicated by the table to approximate $\int_0^{36} P(t)\,dt$. Show the computations that lead to your answer.

(b) Find the value of $\int_0^{9} P'(4t)\,dt$. Show the computations that lead to your answer.

(c) Using correct units, approximate $P'(18)$ using the average rate of change of P over the interval $9 \le t \le 27$.

(d) Explain the meaning of $P'(18)$ in the context of the problem.

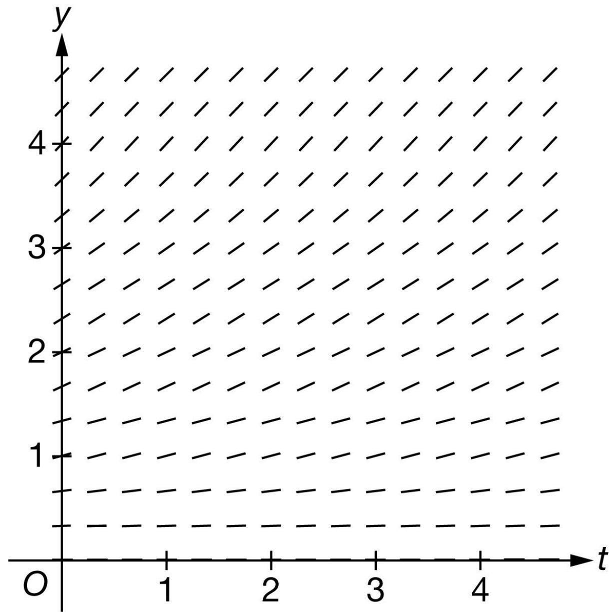
(e) It is known that $P'(9) = -12$. Write an equation for the line tangent to the graph of $y = P(t)$ at $t = 9$.

(f) It is known that $P''(t) < 0$ for $9 \le t \le 18$. If $P(15)$ is approximated using the equation from part (e), would the approximation be an overestimate or underestimate for the actual value of $P(15)$? Give a reason for your answer.

(g) For $0 \le t \le 36$, the total number of postcards remaining can be modeled by the differentiable function Z, where Z is a function of t. Assume the weight of each postcard is 4.5 grams. Write an expression for the rate of change, in grams per minute, of the total weight of the postcards remaining at $t = 19$ minutes.

6-7

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.3y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Form of midpoint sum): Philosophy: The response should provide clear evidence adding the area of two rectangles whose heights are found using the midpoint of the interval.

Acceptable evidence:

· $18 \cdot (P(9) + P(27))$ or $18 \cdot P(9) + 18 \cdot P(27)$

· $(18 - 0) \cdot 256 + (36 - 18) \cdot 40$

· $18 \cdot (256 + 40)$ or $18 \cdot 256 + 18 \cdot 40$

To earn this point the response can have at most one mistake in the expression or equation. So, expressions of the form $\frac{18}{b} \cdot (P(9) + P(27))$, for any nonzero value of b would earn the point. Note: The value $\frac{18}{b}$ might be distributed across the sum.

Expressions of the form $\frac{a}{b} \cdot (P(9) + P(27))$ for $a \neq 18$ will not earn this point.

Once the first point is earned, it cannot be lost.

Second Point (Answer)

To earn the point, the answer must be correct. It does not need to be simplified, but if it is, the simplification must be correct.

The answer point cannot be earned unless the first (midpoint sum) point was earned.

6-7

Minimal complete answers that earn both points:

· $18 \cdot (256 + 40)$

· $18 \cdot 256 + 18 \cdot 40$

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Form of midpoint sum
- [ ] Answer

Solution:

$$\int_0^{36} P(t)\ dt \approx P(9)(18-0) + P(27)(36-18)$$

$$= 256 \cdot 18 + 40 \cdot 18 = 5328$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Substitution) Philosophy: This point is earned when the response has all three components of the substitution done correctly.

Uses $u = 4t$, perhaps implicitly (any variable could be used for u)

Uses the correct constant of $\frac{1}{4}$

Uses the correct limits for their substitution.

· The limits might not be seen until the final evaluation, and at that point, response may have returned to the variable t.

· Limits for t attached to an integral involving u or vice versa will not earn this substitution point but could still earn the answer point, if the numerical answer is correct.

Second point: (Fundamental Theorem of Calculus) Philosophy: This point is earned when the FTC has been applied correctly to either $\int P'(4t)\ dt$ or $\int P'(u)\ du$.

The point is earned for any of the following:

6-7

· The equation $\int P'(4t)\ dt = P(4t)$ or $\int P'(u)\ du = P(u)$

· $\frac{1}{4}P(4t)\Big|_0^9$ or $\frac{1}{4}P(u)\Big|_0^{36}$

· $P(36) - P(0)$, $P(36)$, or $P(0)$

This point can be earned if the substitution point is not.

A response that applies the FTC to $P'(t)$ does not earn this point.

Note: $\int_a^b P'(u)\ du = P'(b) - P'(a)$ does not earn the point. The FTC point is for the response communicating that the antiderivative of P' is P.

Third Point (Answer)

Must be the correct numerical answer.

Must have earned the second point for the FTC.

Might have not earned the first point for substitution because of an incorrect use of limits, but will still earn the third point for the correct answer.

$P(36)$ and $P(0)$ must be evaluated; the expression $\frac{1}{4}(P(36) - P(0))$ by itself does not earn the third point.

The answer may be unsimplified, but if simplified it must be done correctly.

Scoring common responses:

The minimal work required to earn all three points is $\frac{1}{4}(0 - 320)$ or $-\frac{320}{4}$.

The following responses would earn the first two points and be eligible for the answer point if the response continues:

· $\frac{1}{4}P(4t)\Big|_0^9$

· $\frac{1}{4}(P(36) - P(0))$

Responses of only the following would earn only the second point:

· $P(36) - P(0)$

· $P(9) - P(0)$

· $P(36)$

6-7

· $\int_0^9 P'(u)\ du = P(9)$

· $P(4t)\Big|_0^9$ or $P(u)\Big|_0^{36}$

The following responses would earn no points:

· -80 in the absence of one of the first two points.

· $\int_0^9 P'(4t)\ dt = P(9)$

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 4t$. Then $du = 4\ dt$.

When $t = 0$, $u = 0$.

When $t = 9$, $u = 36$.

$$\int_0^9 P'(4t)\ dt = \frac{1}{4}\int_0^{36} P'(u)\ du$$

$$= \frac{1}{4}(P(36) - P(0))$$

$$= \frac{1}{4}(0 - 320) = -80$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Approximation)

The response must be a difference and a quotient.

The answer does not need to be simplified, but if it is simplified it must be done correctly.

6-7

The minimal work that must be shown to earn this point is either $\frac{40-256}{18}$ or $\frac{-216}{27-9}$.

Second point (Units)

The units must be attached to a value (either correct or incorrect).

The response will not be penalized for any verbal attempt to discuss the answer after the correct approximation with units is presented.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Approximation
- [ ] Units

Solution:

$$P'(18) \approx \frac{P(27)-P(9)}{27-9} = \frac{40-256}{18} = -12 \text{ postcards per minute}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The interpretation must include three components:

· indicate the rate of change

· the specific context of the number of postcards remaining

· the specific point in time, $t = 18$.

Responses do not need to include correct units or any units.

The word "decreasing" may be used in place of the word "changing".

The response is not earned for:

· "the rate of postcards delivered in minute 18 " as this does not indicate a change in the rate of delivery.

· "decreasing at a rate of -12 postcards per minute"

"the rate of change of P " without specifying that P is the number of postcards delivered, does not earn the point.

Acceptable ways to provide the specific point in time:

6-7

· In minute 18 or after 18 minutes.

· At $t = 18$ (missing the unit of time)

· After 18 days (incorrect unit of time)

A response that clearly indicates an interval of time (e.g. "during the first 18 minutes" or "from minute 18 on") does not earn the point.

0	1

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$P'(18)$ is the rate at which the number of postcards remaining is changing, in postcards per minute, at time $t = 18$ minutes

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of $P(t)$ in place of y earns the point.

The use of x in place of t earns the point.

Tangent lines do not need to be simplified, but any simplification must be done correctly.

Any correct form of the tangent line earns the point. For example:

o $y = 256 - 12(t - 9)$

o $y - 256 = 12(t - 9)$

o $y = -12t + 108 + 256$

o $y = -12t + 364$

0	1

The student response accurately includes the criteria below.

6-7

☐ Tangent line equation

Solution:

$y = P'(9)(t-9) + P(9)$

$y = -12(t-9) + 256$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point the response must include $P''(t) < 0$ and overestimate.

"$P(t)$ is concave down" can be used instead of $P''(t) < 0$.

Information about $P'(t)$ by itself without reference to $P''(t) < 0$ is not sufficient and does not earn this point.

"Overestimate because it is concave down" is not sufficient by itself because of the use of "it". The response must clearly, directly or indirectly, indicate to what "it" is referring.

If a response makes the correct conclusion and states $P''(t) < 0$ and then goes on and makes an incorrect statement or a statement that cannot be verified will not earn this point.

For example: " $P'(t)$ is concave down, overestimate" does not earn this point.

✓

0	1

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because $P''(t) < 0$, the tangent line approximation would produce an overestimate.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression of only $4.5 \cdot Z'(19)$ earns this point.

Only expression $4.5 \cdot P'(19)$ also earns this point.

No justifications or explanations are required.

6-7

Any incorrect statement such as, $Z'(19) = 4.5 \cdot P'(19)$, does not earn this point.

0	1

The student response accurately includes the criteria below.

- [ ] Expression

Solution:

The rate of change, in grams per minute, of the total weight of the postcards remaining at $t = 19$ minutes is $4.5Z'(19)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Response clearly references what part of the differential equation is different from the slope field and the response needs only to clearly and explicitly reference one of them.

The response does not need to explicitly reference both the differential equation and the given slope field.

The distinction can be made with explicit reference to one of the two things and an implicit reference to the other

For example:

· "$\frac{dy}{dt}$ is negative, not positive". We will infer that negative is referring to the slopes shown in the slope field.

· "This would be correct for a positive derivative, but $\frac{dy}{dt}$ is negative." We will read "This" as "the slope field" in this sentence.

· "All of the slopes in the slope field are positive, not negative." We will infer that the values of $\frac{dy}{dt}$ at points in the first quadrant are negative.

A response that lacks even an implicit reference to either the differential equation or the slope field will not earn the point. For example:

· "$\frac{dy}{dt}$ is negative" lacks any reference to the slope field and so will not earn the point.

· "All the slopes in the graph are positive" lacks any reference to the differential equation, and so will not earn the point.

· Vague answers such as "The slope field has positive points" or "The field is increasing" will not earn the point.

6-7

0	1

The student response accurately includes the criteria below.

- [ ] Explanation

Solution:

When $y > 0$, $\frac{dy}{dt} < 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are positive.

6-7

62. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	5	20
$R(t)$ (meters per minute)	40	200	160

A student on the school track and field team is running on the track. For time t minutes, $0 \leq t \leq 20$, the rate at which the student runs, in meters per minute, is modeled by a twice-differentiable function, $R(t)$. Values of $R(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $R'(5)$ using the average rate of change of R over the interval $0 \leq t \leq 20$.

(b) Explain the meaning of $R'(5)$ in the context of the problem.

(c) The function $D(t)$ models the number of meters run by the student at time t minutes, where $D'(t) = R(t)$. It is known that $D(5) = 800$. Write an equation for the line tangent to the graph of $y = D(t)$ at $t = 5$.

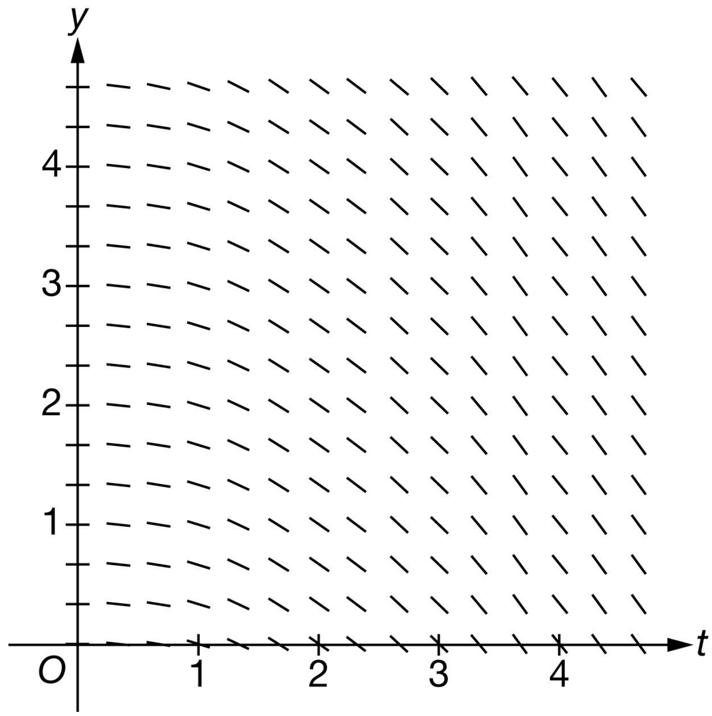
(d) Find the value of $\int_0^5 R'(4t)\,dt$. Show the calculations that lead to your answer.

(e) Use a trapezoidal sum with the two subintervals indicated by the table to approximate $\int_0^{20} R(t)\,dt$. Show the calculations that lead to your answer.

(f) The student and a friend team up to run on the school track for a fundraiser for 20 minutes. For $0 \leq t \leq 20$, the total number of laps run by the student is modeled by $M(t)$ and the total number of laps run by the friend is modeled by $L(t)$. Both M and L are differentiable functions of t. Each person raises 5 dollars for each lap run. Write an expression for the rate of change, in dollars per minute, of the total money raised at time $t = 18$ minutes.

6-7

(g) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = -0.35y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

Answers without supporting work do not earn this point. Supporting work must be at least a difference and a quotient.

Response does not need to be simplified, but any simplification must be done correctly.

If a decimal is presented, it must be correct to three decimal places. We do not read anything past the third decimal place.

The second point is earned for the correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols.

The response will not be penalized for any verbal attempt to discuss the answer after the correct presentation of the approximation.

The minimum response needed to earn both points is: $\frac{160-40}{20}$ meters per minute per minute.

6-7

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$R'(5) \approx \frac{R(20)-R(0)}{20-0} = \frac{160-40}{20} = 6 \text{ meters per minute per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This part is intended to be independent from part (a), so we do not want or need to see any reference to the value obtained in (a). If that value is referenced, conclusion must be consistent.

Three components of a correct answer:

· Indicating a rate of a rate (look for the word rate to be used correctly twice)

· Including the context of the student running

· Referring to the specific point in time

Because the context is one of motion, a description of $R'(5)$ as the acceleration of the runner at time $t = 5$ earns the point.

Units will not be considered for this point. They do not need to be present, and if they are, they do not need to be correct.

Any incorrect statement makes the response ineligible for the point. For example:

$R'(5)$ is the average (or an approximation of, or an estimate of) rate of change of ....

"Increasing" or "decreasing" may be used in place of "changing" as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Beware of phrases such as "decreasing at a negative rate."

· Example: $R'(5)$ is the rate of increase of the rate at which the student runs at 5 minutes. (earns the point)

· Example: $R'(5)$ means the rate at which the student runs is increasing by 6 meters / min / min. at time $t = 5$ (does not earn the point)

6-7

Acceptable references to time include (but are not limited to):

· At the fifth minute

· At $t = 5$ (with correct units of time, without correct units of time, or even with incorrect units of time)

· After $t = 5$ (with correct units of time, without correct units of time, or even with incorrect units of time)

If the response clearly indicates an interval of time (e.g. "during the first 5 minutes") the response will not earn the point.

Examples that earn the point include:

· $R'(5)$ represents the rate at which the rate at which the student runs is changing on day 5.

· $R'(5)$ is the rate of change in the velocity of the runner at 5 days.

· $R'(5)$ represents the rate at which the velocity of the runner is increasing at $t = 5$.

· $R'(5)$ equals (answer from part (a)) means the velocity of the runner is increasing at a rate of (answer from part (a)) at $t = 5$.

Examples that do not earn the point include:

· $R'(5)$ represents the rate at which the meters per minute is changing on day 5.

· $R'(5)$ is the rate of change in the meters run at $t = 5$ days.

· The rate at which the velocity changes is increasing at $t = 5$.

· The rate of change in the rate of change of the runner is increasing at $t = 5$.

0	1

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$R'(5)$ is the rate of change in the rate at which the student runs, in meters per minute per minute, at time $t = 5$ minutes.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-7

The point is earned with a correct tangent line equation in any form. Equivalent forms include (but are not limited to):

o $y = 200\,(t - 5) + 800$

o $y - 800 = 200\,(t - 5)$

o $y = 200t - 200$

The student may write x in place of t and still earn the point.

No work is required to earn the point.

We will accept $D\,(t)$ in lieu of y in the equation of the tangent line.

0	1

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$y = D'\,(5)\,(t - 5) + D\,(5)$

$y = 200\,(t - 5) + 800$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution.

Two things are required to award this point:

· Coefficient of $\frac{1}{4}$ correctly applied to an integral or an antiderivative

· Evidence that the correct limits of 0 to 20 will be or have been applied

Examples that will earn the first point (these are just a few):

· $\frac{1}{4}\int_0^{20} R'\,(u)\;du$

· $\frac{1}{4}R\,(u)\big|_0^{20}$ or $\frac{1}{4}R\,(4t)\big|_0^{5}$

6-7

Responses that include misstatements such as using t -values for limits with u as the variable of integration in an intermediate step which is later corrected may still earn this point. Intermediate errors in bounds that are subsequently corrected will disqualify the student from earning the third point. If the bounds are never appropriately handled, the response cannot earn the first point. Scratch work that is not connected to the solution will not be read.

Examples:

- $\int_0^5 R'(4t)\ dt = \frac{1}{4}\int_0^5 R'(u)\ du = \frac{1}{4}R(4t)\big|_0^5$ earns the first point, but will not earn the third point.
- $\int_0^5 R'(4t)\ dt = \frac{1}{4}\int_0^5 R'(u)\ du = \frac{1}{4}[R(5) - R(0)]$ does not earn the first point.

The second point is for the Fundamental Theorem of Calculus, demonstrating the knowledge that

$\int_a^b R'(u)\ du = R(u)\big|_a^b = R(b) - R(a)$

This point can be earned with either $kR(4t)\big|_a^b$, $kR(t)\big|_a^b$, or $k(R(b) - R(a))$ where k is a

nonzero constant [not necessarily correct], $a = 0$, and $b = 5$ or $b = 20$.

A response can earn this point without earning the first point.

$\frac{1}{4}[R(20) - M(0)]$ earns the first two points.

The third point is for the correct numerical value. Our answer only.

Values from the table must be used (i.e., $\frac{1}{4}[R(20) - M(0)]$ is not sufficient to earn the third point).

If the response includes any linkage error, (i.e., the use of equal signs to equate two expressions that are not equal), the third point will not be earned. If an indefinite integral is linked with a definite integral the response will not be penalized.

$\int_0^5 R'(4t)\ dt = \frac{1}{4}\int R'(u)\ du = \frac{1}{4}R(4t)\big|_0^5 = \ldots$ is eligible for the third point.

Eligibility: Must have earned the first two points.

The minimum response needed to earn all three points is: $\frac{1}{4}(160 - 40)$.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

Substitution

6-7

- ☐
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 4t$. Then $du = 4\ dt$.

When $t = 0$, $u = 0$.

When $t = 5$, $u = 20$.

$$\int_0^5 R'(4t)\ dt = \frac{1}{4}\int_0^{20} R'(u)\ du$$

$$= \frac{1}{4}(R(20) - R(0))$$

$$= \frac{1}{4}(160 - 40) = 30$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The **first point** is for the form of a trapezoidal sum

Point is earned for the setup of the trapezoidal sum. Numerical values do not need to be pulled from the table for this point, but Δt values must be presented. Responses that contain one incorrect numerical value in the trapezoidal sum setup are still eligible to earn the first point, but will not earn the second.

Response may present a left Riemann sum and a right Riemann sum, then average the two. Again, one incorrect numerical value will not disqualify the response from earning this first point.

Response may compute trapezoids as areas of rectangles and triangles.

Trapezoidal areas may be given in a graph. In order to earn this point, the response must clearly show the computations leading to the values presented in the graph, and must show that these areas are added to obtain the final approximation value.

A response using subtraction instead of addition will not earn the point

(e.g. $\frac{1}{2}[5(R(0) - R(5)) + 15(R(5) - R(20))]$ earns no points).

Response utilizing “Trapezoidal Rule” will not earn this point, unless this represents only one incorrect numerical value.

6-7

- $\frac{1}{2}(5)\left[R(0)+2R(5)+R(20)\right]$ is equivalent to $\frac{1}{2}[5(R(0)+R(5))+5(M(5)+R(20))]$, thus has only one incorrect numerical value (the second interval width is 5 instead of 15), so can earn the point.

- $\frac{1}{2}(10)\left[R(0)+2R(5)+R(20)\right]$ has two incorrect numerical values (both interval widths are incorrect) so will not earn the point.

The second point is for the correct numerical result.

Our answer only. Response does not need to be simplified, but if it is, it must be simplified correctly.

The answer must be accompanied with supporting work. Supporting work may appear in a graph, with clearly labeled areas. It is possible to earn the second point without earning the first point if the supporting work is not sufficient to earn the first point.

Pictures without clear area computations and a clear summation (+) of quantities can earn at most the second point.

Special case: A positive constant multiple of a complete and correctly expressed trapezoidal sum (with the correct values pulled from the table) will earn the first but not the second point:

- $k\left[\left(\frac{40+200}{2}\right)\cdot 5+\left(\frac{200+160}{2}\cdot 15\right)\right]$ for $k>0$ earns only the first point.

The minimum response needed to earn both points: $\left(\frac{240}{2}\right)\cdot 5+\left(\frac{360}{2}\right)\cdot 15$

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Form of trapezoidal sum
- ☐ Answer

Solution:

$$\int_0^{20} R(t)\,dt \approx \left(\frac{R(0)+R(5)}{2}\right)(5-0)+\left(\frac{R(5)+R(20)}{2}\right)(20-5)$$

$$=\left(\frac{40+200}{2}\right)\cdot 5+\left(\frac{200+160}{2}\right)\cdot 15=3300$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a correct constant multiple or correct sum.

6-7

Response must contain the constant 5 as a factor of the expression, and/or the correct sum $L'(18) + M'(18)$.

If the sum is incorrect, then the constant factor of 5 must be applied either to a sum that involves only $L(t)$ and $M(t)$, $L'(t)$ and $M'(t)$, or integrals of $L(t)$ and $M(t)$, or to one of the correct terms of the sum $L'(18)$ or $M'(18)$, with no additional incorrect terms.

Examples that earn the point include:

· $5[M'(t) + L'(t)]$

· $5\int [M(t) + L(t)]\, dt$

· $5[M + L]$

· $5M'(18)$

Examples that do not earn the point include:

· $5[M(t) + L'(t)]$

· $5\int M(t)\, dt$

· $5M' + L$

· $5L'(18) + t$

Special case**:** $5L'(18) + M'(18)$ is an error in parentheses, so will earn the first point but is not eligible to earn the second point.

If the constant multiple is missing or incorrect, the sum must be correct. $k[L'(18) + M'(18)]$ for any nonzero constant k will earn the first point.

The second point is for the correct expression.

$5L'(18) + 5M'(18)$ (or an equivalent expression) earns both points.

Inclusion of units (correct or incorrect) will not affect the score.

A completely correct expression with one additional constant term added/subtracted would earn a just the first point. For example, $5L'(18) + 5M'(18) + 5(2)$ earns only the first point.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

Constant multiple or sum

6-7

- [ ]
- [ ] Expression

Solution:

The rate of change, in dollars per minute, of the total money raised at time $t = 18$ minutes is $5M'(18) + 5L'(18)$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for a correct statement identifying specific inconsistencies between the given differential equation and the given slope field. There are many such correct statements, such as:

· The response references a specific point on the y -axis (such as $(0,\ 4)$), and indicates that the given slope field shows a slope of zero, yet the differential equation would yield a positive slope.

· The response references a pattern of slopes, and explains why that pattern does not fit our differential equation (e.g. “the slope field shows slopes that change as t changes, but our differential equation depends only on y, so slopes should remain constant for a given value of y as t changes.”)

Identifying a mismatch between the differential equation and the slope field at one point is sufficient to earn the point.

Response is not penalized for referencing “x ” instead of “t ”.

If the response contains any incorrect statement, the response will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point, nor will responses that present information beyond what we are given. For example:

· Responses that specify that the slope field shows a particular nonzero slope at a particular point will not earn the point, as this cannot actually be determined visually (i.e., the response should not indicate that at $(2,\ 2)$ the slope shown on the slope field is -1)

· Responses that state a specific slope value on the slope field for a general value of x or y that are not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may involve a solution to the differential equation, with an argument based on the resulting function. If done correctly, and presented with a correct argument, this will earn the point.

0	1 ✓

6-7

The student response accurately includes the criteria below.

- [ ] Explanation

Solution:

When $y=0$, $\dfrac{dy}{dt}=0$. However, in the given slope field, for $t>0$, the slopes of the line segments for $y=0$ are nonzero.

6-7

63. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	5	20	30
$D(t)$ (meters)	0	1000	4500	6500

A student on the school track and field team is running laps around a track. For time t minutes, $0 \le t \le 30$, the total distance run, in meters, is modeled by an increasing, twice-differentiable function D. Values of $D(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $D'(20)$ using the average rate of change of D over the interval $5 \le t \le 30$.

(b) Explain the meaning of $D'(20)$ in the context of the problem.

(c) It is known that $D'(5) = 200$. Write an equation for the line tangent to the graph of $y = D(t)$ at $t = 5$.

(d) Use a left Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{30} D(t)\,dt$. Show the computations that lead to your answer.

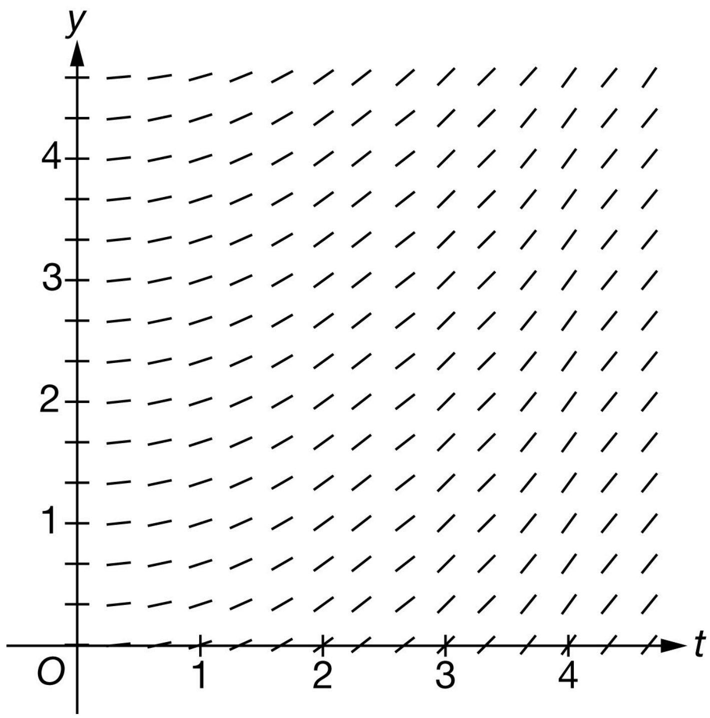
(e) Is the approximation from part (d) an overestimate or an underestimate for $\int_0^{30} D(t)\,dt$? Give a reason for your answer.

(f) Find the value of $\int_0^{5} D'(4t)\,dt$. Show the computations that lead to your answer.

(g) The student is using their running practice as a fundraiser. For $0 \le t \le 30$, the total number of laps run can be modeled by the differentiable function L, where L is a function of t. For every lap run, the student raises 5 dollars. Write an expression for the rate of change, in dollars per minute, of the total money raised at $t = 22$ minutes.

6-7

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = 0.5y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work. The supporting work must be at least a difference and a quotient. Correct values must be taken from the table to earn the point. The difference quotient does not need to be simplified.

Examples that earn the first point include:

- $\frac{6500-1000}{30-5}$
- $\frac{D(30)-D(5)}{25} = \frac{5500}{25}$
- $\frac{D(30)-D(5)}{25} = 220$

An example that does not earn the first point is $\frac{D(30)-D(5)}{30-5}$.

Minimal response that earns the point: $\frac{5500-1000}{30-5}$ or $\frac{6500-1000}{25}$

Neither $\frac{5500}{25}$ nor 220, with no supporting work, will earn this point

The second point is for correct units

6-7

The correct units must be attached to a numerical value: either correct or incorrect.

The units may be written using words or a combination of words and symbols. For example:

· Meters per minute

· Meters/minute

· m/min

The response will not be penalized for any verbal attempt to discuss the answer after the correct answer has been presented.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Approximation
- [ ] Units

Solution:

$$D'(20) \approx \frac{D(30)-D(5)}{30-5} = \frac{6500-1000}{25} = 220 \text{ meters per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

· Indicating a rate of change

· Including the context of student running

· Referring to the specific point in time $t = 20$

The response is not penalized for incorrect units in this part. (Whether correct or incorrect, units have no effect on the score in this part.)

Responses may use “changing” or “increasing” as a modifier as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Example: “The number of meters is increasing at a rate of 220 meters per minute” is okay.

Acceptable reference to a specific point in time includes (but is not limited to):

6-7

· On minute 20 and after 20 minutes

· At $t = 20$

· At $t = 20$ hours or any incorrect unit of time

If the response clearly indicates an interval of time (e.g., "from minute 20 on," "during the first 20 minutes," or "for $t = 0$ to $t = 20$ "), the point is not earned.

Examples that earn the point include:

· $D'(20)$ is the rate of change of the distance of meters per minute at time $t = 20$

· The rate of change (or speed/velocity/ how fast) is 220 meters per minute at 20 minutes

· $D'(20)$ is the rate of change in the distance ran at time $t = 20$ in meters per *second*

Examples that do not earn the point include:

· 220 is the rate of increase in the distance of meters per minute at 20 minutes

· $D'(20)$ is the average rate of change of the total meters ran per minute at minute 20

· $D'(20)$ is the number of meters per minute at minute 20

· The rate of change is 220 meters per minute (Missing "at $t = 20$ ")

· $D'(20)$ is the slope of the tangent line at $t = 20$

· Rate of change of D at time $t = 20$

· Increase in number of meters at $t = 20$

0	1

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$D'(20)$ is the rate at which the total distance run is changing, in meters per minute, at time $t = 20$ minutes.

– OR –

$D'(20)$ is the speed at which the student is running, in meters per minute, at time $t = 20$ minutes.

6-7

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. The use of x in place of t is acceptable.

Some examples include

· $y = 200\,(t - 5) + 1000$, $y = 200\,(x - 5) + 1000$

· $y - 1000 = 200\,(t - 5)$,

· $y - 1000 = 200\,(x - 5)$

· $y = 200t$, $y = 200x$

The key calculus concept being assessed in this part of the problem is that the value of $D'(5)$ is the slope of the tangent line. In that spirit, we are accepting $D'(t) = \ldots$ in lieu of $y = \ldots$, but we are not accepting $D'(5) = \ldots$

Note that $D\,(t) = 200\,(t - 5) + 1000$ earns the point, but $D\,(5) = 200\,(t - 5) + 1000$ does not earn the point.

0	1 ✓

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$y = D'(5)\,(t - 5) + D\,(5)$

$y = 200\,(t - 5) + 1000$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a left Riemann sum. The setup must be a sum of three products.

· A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

· Responses with the numerical error described above are not eligible for the second point.

6-7

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns. Implicit responses are eligible for both points.

A response presenting a right Riemann sum (instead of left), will not earn any points for part (a).

The second point is for the correct numerical result.

· In order to earn the second point, there must be supporting calculations.

· The numerical values for $D(0)$, $D(5)$, and $D(20)$ must be pulled from the table

· Any simplification shown must be correct.

Use the following as an illustration how to score both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

· A response of $0 \cdot (5-0)+1000 \cdot (20-5)+4500 \cdot (30-20)$ earns both points.

· A response of $5 \cdot D(0)+15 \cdot D(5)+10 \cdot D(20)$ earns the first point. Eligible for second point if more work is on the page.

· A response of $5 \cdot D(0)+15 \cdot D(5)+10 \cdot D(20)=6000$ earns both points.

· A response of $5 \cdot 0+15 \cdot 1000+10 \cdot 4500$ earns both points.

· A response of $5 \cdot 0+10 \cdot 1000+10 \cdot 4500$ contains one numerical error. Earns the first point. Not eligible for the second point.

· A response of $5 \cdot 0+15 \cdot 1000+10 \cdot 4500=50,000$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.

· A response of $0+15,000+45,000$ or $0+15,000+45,000=60,000$ does not earn the first point because we cannot see the sum of 3 products. However, this is the correct sum, so we see an answer with supporting work. The second point will be earned.

The response does not need to include "approximately" in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$5 \cdot 0+15 \cdot 1000+10 \cdot 4500$

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

☐ Left Riemann sum

6-7

☐ Answer

Solution:

$$\int_0^{30} D(t)\, dt \approx D(0)(5-0) + D(5)(20-5) + D(20)(30-20)$$
$$= 0 \cdot 5 + 1000 \cdot 15 + 4500 \cdot 10 = 60{,}000$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an underestimate because D is increasing.

The response does not need an explicit reference in part (d). The following statements are sufficient for earning this point:

· Underestimate because D is increasing

· Underestimate because the function is increasing

Note: At this point, there is only one function being discussed in this problem, so we will assume "the function" is D.

When a response includes additional information about concavity, we interpret that as a lack of understanding about the question, so this will not earn the point. When the response references a graph or the given table rather than the function D, the point is not earned.

Examples that earn the point:

· The left Riemann sum underestimates because the function is increasing.

· The sum is an underestimate because $D(t)$ is increasing.

· The sum is an underestimate $D'(t) > 0$.

Examples that do not earn the point:

· The sum is an underestimate because $D'(t)$ is increasing. ($D'(t)$ increasing is not known)

· The left Riemann sum underestimates because the graph is increasing.

· The left Riemann sum underestimates because the function is increasing and concave up.

✓

0	1

6-7

The student response accurately includes the criteria below.

- [ ] Answer with reason

Solution:

Because D is an increasing function, and this is a left Riemann sum, the answer from part (d) is an underestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are 3 components of substitution that we need to see:

· Uses $u = 4t$, which may be done implicitly

· Uses the correct constant of $\frac{1}{4}$

· Uses the correct limits for the substitution (0 and 20). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

· Either of the expressions below will earn the substitution point without any additional supporting work:
$\frac{1}{4} D(4t)\Big|_0^5$ or $\frac{1}{4} D(u)\Big|_0^{20}$

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point

· Limits for t attached to an integral with u as the variable of integration (or vice versa),

· Improper equals signs, such as an indefinite integral equal to a definite integral

· Reversed limits of integration, for example: $\int_{20}^{0} D'(u)\, du$

The second point is for the Fundamental Theorem of Calculus. This point can be earned with any of the following, where k is a nonzero constant and $b = 5$ or $b = 20$. No explanation or additional supporting work is necessary.

· $kD(4t)\Big|_0^b$

· $k(D(b) - D(0))$

The response cannot earn this point without also earning the FTC point.

The third point is for the correct numerical value, which is 1125.

6-7

Values from the table must be used. $\frac{1}{4}[D(20) - D(0)]$ is not enough to earn the answer point.

The expression $\frac{1}{4}(4500 - 0)$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

The response cannot earn this point without also earning first two points.

Minimalist response: The expression $\frac{1}{4}(4500)$ represents the minimum work that must been seen in a response in order to earn all three points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 4t$. Then $du = 4\ dt$.

When $t = 0$, $u = 0$.

When $t = 5$, $u = 20$.

$$\int_0^5 D'(4t)\ dt = \frac{1}{4}\int_0^{20} D'(u)\ du$$

$$= \frac{1}{4}(D(20) - D(0))$$

$$= \frac{1}{4}(4500 - 0) = 1125$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct expression at $t = 22$. No supporting work is required.

The correct expression of $5 \cdot L'(22)$ earns this point.

No justifications or explanations are required.

6-7

Any incorrect statement such as, $D'(22) = 5 \cdot L'(22)$, does not earn this point.

0	1

The student response accurately includes the criteria below.

- [ ] Expression

Solution:

The rate of change, in dollars per minute, of the total money raised at $t = 22$ minutes is $5L'(22)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between two things: the differential equation, $\frac{dy}{dt} = 0.5y$, and the given slope field.

The response does not need to explicitly reference both the differential equation and the given slope field, but it must explicitly reference one and implicitly reference the other.

Acceptable responses include

· Identifying an inconsistency between the differential equation and the slope field at one point , e.g., at the point $(4,\ 0)$, the slope of the segment should be 0, but it is positive.

· Referencing a pattern of slopes, and explaining why that pattern does not fit the given differential equation.

A response that lacks even an implicit reference to the one of two things being compared will not earn the point. For example, merely stating either $\frac{dy}{dt}$ is positive, or that the slope field is positive does not earn the point.

· "The values for are greater than or equal to zero so $\frac{dy}{dt} \geq 0$ " lacks any reference to the slope field and will not earn the point.

· "All the slopes in the slope field are positive" lacks any reference to the differential equation and will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point. Some examples are:

· "The field is increasing"

· Responses that state a specific slope value on the slope field for a general value of x or y that is not correct for all points with that x or y value do not earn the point. (e.g. "the slope field shows a slope of 0 when $y = 4$ " is an

6-7

incorrect statement)

A response may correctly solve or attempt to solve the differential equation and then provide a correct argument without reference to their solution. Such a response would earn the point.

A response might actually solve the differential equation and provide a correct argument on the resulting function.

0	1

The student response accurately includes the criteria below.

- [ ] Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

6-7

64. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	8	16	24	32
$D(t)$ (meters)	0	1500	2800	3900	5000

A student on the school track and field team is running laps around a track. For time t minutes, $0 \leq t \leq 32$, the total distance run, in meters, is modeled by a twice-differentiable function D. Values of $D(t)$ at various times t are shown in the table above.

(a) Use a midpoint Riemann sum with the two subintervals indicated by the table to approximate $\int_0^{32} D(t)\,dt$. Show the computations that lead to your answer.

(b) Find the value of $\int_0^{16} D'(2t)\,dt$. Show the computations that lead to your answer.

(c) Using correct units, approximate $D'(16)$ using the average rate of change of D over the interval $8 \leq t \leq 24$.

(d) Explain the meaning of $D'(16)$ in the context of the problem.

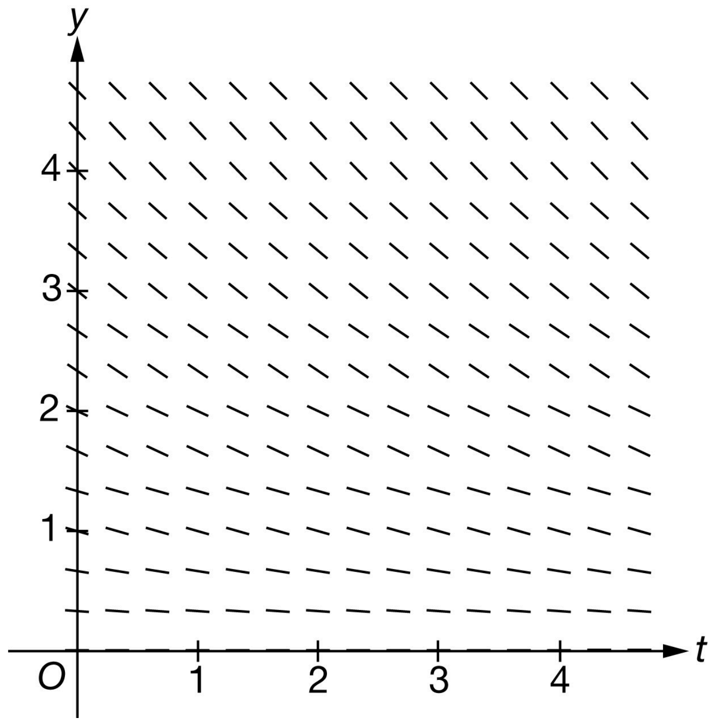
(e) It is known that $D'(24) = 140$. Write an equation for the line tangent to the graph of $y = D(t)$ at $t = 24$.

(f) It is known that $D''(t) < 0$ for $24 \leq t \leq 31$. If $D(30)$ is approximated using the equation from part (e), would the approximation be an overestimate or underestimate for the actual value of $D(30)$? Give a reason for your answer.

(g) A second student is raising funds by running laps for 60 minutes. For $0 \leq t \leq 60$, the total number of laps run by this student is modeled by $L(t)$, a differentiable function of t. For each lap run, the student raises 4 dollars. Write an expression for the rate of change, in dollars per minute, of the total money raised at time $t = 45$ minutes.

6-7

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = 0.25y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Form of midpoint sum): Philosophy: The response should provide clear evidence adding the area of two rectangles whose heights are found using the midpoint of the interval.

Acceptable evidence:

$\cdot\, 16 \cdot (D(8) + D(24))$ or $16 \cdot D(8) + 16 \cdot D(24)$

$\cdot\, (16 - 0) \cdot 1500 + (32 - 16) \cdot 3900$

$\cdot\, 16 \cdot (1500 + 3900)$ or $16 \cdot 1500 + 16 \cdot 3900$

To earn the point the response can have at most one mistake in the expression or equation. So, expressions of the form $\frac{16}{b} \cdot (D(8) + D(24))$, for any nonzero value of b would earn the point. Note: The value $\frac{16}{b}$ might be distributed across the sum.

Expressions of the form $\frac{a}{b} \cdot (D(8) + D(24))$, for $a \neq 16$ will not earn this point.

Once the first point is earned, it cannot be lost.

Second Point (Answer)

6-7

To earn the point, the answer must be correct. It does not need to be simplified, but if it is, the simplification must be correct.

The answer point cannot be earned unless the first (midpoint sum) point was earned.

Minimal complete answers that earn both points:

· $16 \cdot (1500 + 3900)$

· $16 \cdot 1500 + 16 \cdot 3900$

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Form of midpoint sum
- [ ] Answer

Solution:

$$\int_0^{32} D(t)\ dt \approx D(8)(16-0) + D(24)(32-16)$$

$$= 1500 \cdot 16 + 3900 \cdot 16 = 86{,}400$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Substitution) Philosophy: This point is earned when the response has all 3 components of the substitution done correctly.

Uses $u = 2t$, perhaps implicitly (any variable could be used for u)

Uses the correct constant of $\frac{1}{2}$

Uses the correct limits for their substitution.

· The limits might not be seen until the final evaluation, and at that point, response may have returned to the variable t.

· Limits for t attached to an integral involving u or vice versa will not earn this substitution point but could still earn the answer point, if the numerical answer is correct.

6-7

Second point: (Fundamental Theorem of Calculus) Philosophy: This point is earned when the FTC has been applied correctly to either $\int D'(2t)\ dt$ or $\int D'(u)\ du$.

The point is earned for any of the following:

· The equation $\int D'(2t)\ dt = D\ (2t)$ or $\int D'(u)\ du = D(u)$

· $\frac{1}{2}D\ (2t)\Big|_0^{16}$ or $\frac{1}{2}D\ (u)\Big|_0^{32}$

· $D\ (32) - D\ (0)$, $D\ (32)$, or $D\ (0)$

This point can be earned if the substitution point is not.

Note: $\int_a^b D'(u)\ du = D'(b) - D'(a)$ does not earn the point. The FTC point is for the response communicating that the antiderivative of D' is D.

Third Point (Answer)

Must be the correct numerical answer.

Must have earned the second point for the FTC.

Might have not earned the first point for substitution because of an incorrect use of limits, but will still earn the third point for the correct answer.

$D\ (32)$ and $D\ (0)$ must be evaluated; the expression $\frac{1}{2}(D\ (32) - D\ (0))$ by itself does not earn the third point.

The answer may be unsimplified, but if simplified it must be done correctly.

Scoring common responses:

The minimal work required to earn all three points is $\frac{1}{2}(5000 - 0)$ or $\frac{1}{2}(5000)$.

The following responses would earn the first two points and be eligible for the answer point if the response continues:

· $\frac{1}{2}D\ (2t)\Big|_0^{16}$

· $\frac{1}{2}(D\ (32) - D\ (0))$

Responses of only the following would earn only the second point:

· $D\ (32) - D\ (0)$

· $D\ (16) - D\ (0)$

6-7

· $D(32)$

· $\int_0^{16} D'(u)\ du = D(16)$

· $D\ (2t)|_0^{16}$ or $D\ (u)|_0^{32}$

The following responses would earn no points:

· 2500 in the absence of one of the first two points.

· $\int_0^{16} D'(2t)\ dt = D(16)$

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- [ ] Substitution
- [ ] Fundamental Theorem of Calculus
- [ ] Answer

Solution:

Let $u = 2t$. Then $du = 2\ dt$.

When $t = 0$, $u = 0$.

When $t = 16$, $u = 32$.

$$\int_0^{16} D'\ (2t)\ dt = \frac{1}{2}\int_0^{32} D'\ (u)\ du$$

$$= \frac{1}{2}(D\ (32) - D\ (0))$$

$$= \frac{1}{2}(5000 - 0) = 2500$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Approximation)

The response must be a difference and a quotient.

6-7

The answer does not need to be simplified, but if it is simplified it must be done correctly.

The minimal work that must be shown to earn this point is either $\frac{3900-1500}{16}$ or $\frac{2400}{24-8}$.

Second point (Units)

The units must be attached to a value (either correct or incorrect).

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Approximation
- [ ] Units

Solution:

$$D'(16) \approx \frac{D(24)-D(8)}{24-8} = \frac{3900-1500}{16}$$

$= 150$ meters per minute

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The interpretation must include three components

· indicate the rate of change, the velocity, or the speed

· the specific context of the total distance run

· the specific point in time, $t = 16$.

Responses do not need to include correct units or any units.

The phrase “the student is running” may be used in place of "the total distance run is changing”.

The phrase “the student is running” is not enough. The response must attach “running” to the word “rate”, as in “the student is running at a rate of”.

The response is not earned for “the rate of total distance run” as this does not indicate a change in the rate of distance.

“the rate of change of D” without specifying that D is the total distance run, does not earn the point.

6-7

Acceptable ways to provide the specific point in time:

· At minute 16 or after 16 minutes.

· At $t = 16$ (missing the unit of time)

· At 16 seconds (incorrect unit of time)

A response that clearly indicates an interval of time (e.g. "during the first 16 minutes" or "from minute 16 on") does not earn the point.

0	1 ✓

The student response accurately includes the criteria below.

- [ ] Explanation

Solution:

$D'(16)$ is the rate at which the total distance run is changing, in meters per minute, at time $t = 16$ minutes.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of $D(t)$ in place of y earns the point.

The use of x in place of t earns the point.

Tangent lines do not need to be simplified, but any simplification must be done correctly.

Any correct form of the tangent line earns the point. For example:

· $y = 3900 + 140(t - 24)$

· $y - 3900 = 140(t - 24)$

· $y = 140t - 3360 + 3900$

· $y = 140t + 540$

0	1 ✓

The student response accurately includes the criteria below.

6-7

☐ Tangent line equation

Solution:

$y = D'(24)(t - 24) + D(24)$

$y = 140(t - 24) + 3900$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point the response must include $D''(t) < 0$ and overestimate.

“$D(t)$ is concave down” can be used instead of $D''(t) < 0$.

Information about $D'(t)$ by itself without reference to $D''(t) < 0$ is not sufficient and does not earn this point.

“Overestimate because it is concave down” is not sufficient by itself because of the use of "it". The response must clearly, directly or indirectly, indicate to what "it" is referring.

If a response makes the correct conclusion and states $D''(t) < 0$ and then goes on and makes an incorrect statement or a statement that cannot be verified will not earn this point.

For example: “$D'(t)$ is concave down, overestimate” does not earn this point.

0	1 ✓

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because $D''(t) < 0$, the tangent line approximation would produce an overestimate.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression of only $4 \cdot L'(45)$ earns this point.

No justifications or explanations are required.

The expression $4 \cdot D'(45)$ does not earn this point.

6-7

Any incorrect statement such as, $L'(45) = 4 \cdot D'(45)$, does not earn this point.

0	1

The student response accurately includes the criteria below.

- [ ] Expression

Solution:

The rate of change, in dollars per minute, of the total money raised at time $t = 45$ minutes is $4L'(45)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Response clearly references what part of the differential equation is different from the slope field and the response needs only to clearly and explicitly reference one of them.

The response does not need to explicitly reference both the differential equation and the given slope field.

The distinction can be made with explicit reference to one of the two things and an implicit reference to the other

For example:

· "$\frac{dy}{dt}$ is positive, not negative". We will infer that negative is referring to the slopes shown in the slope field.

· "This would be correct for a negative derivative, but $\frac{dy}{dt}$ is positive." We will read "This" as "the slope field" in this sentence.

· "All of the slopes in the slope field are negative, not positive." We will infer that the values of $\frac{dy}{dt}$ at points in the first quadrant are positive.

A response that lacks even an implicit reference to either the differential equation or the slope field will not earn the point. For example:

· "$\frac{dy}{dt}$ is positive" lacks any reference to the slope field and so will not earn the point.

· "All the slopes in the graph are negative" lacks any reference to the differential equation, and so will not earn the point.

· Vague answers such as "The slope field has negative points" or "The field is decreasing" will not earn the point.

6-7

0	1

The student response accurately includes the criteria below.

Solution:

When $y > 0$, $\frac{dy}{dt} > 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are negative.

6-7

65. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	5	20	30
$D(t)$ (meters per minute)	0	100	120	150

A student on the school track and field team is running laps around a track. For time t minutes, $0 \leq t \leq 30$, the rate at which the student runs, in meters per minute, is modeled by an increasing, twice-differentiable function D. Values of $D(t)$ at various times t are shown in the table above.

(a) Use a right Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{30} D(t)\,dt$. Show the calculations that lead to your answer.

(b) Is your answer from part (a) an overestimate or an underestimate for $\int_0^{30} D(t)\,dt$? Give a reason for your answer.

(c) Using correct units, approximate $D'(5)$ using the average rate of change of D over the interval $0 \leq t \leq 20$.

(d) Explain the meaning of $D'(5)$ in the context of the problem.

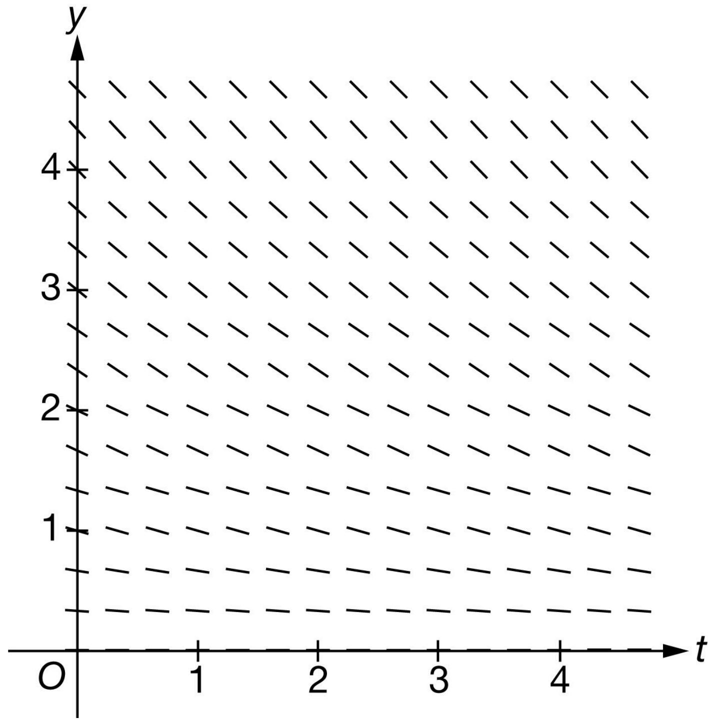
(e) It is known that $D'(20) = 2.5$. Write an equation for the line tangent to the graph of $y = D(t)$ at $t = 20$.

(f) Find the value of $\int_0^{20} D'\left(\frac{t}{4}\right)\,dt$. Show the calculations that lead to your answer.

(g) A friend on the track and field team is using her running practice as a fundraiser. For $0 \leq t \leq 30$, the total number of laps run by the friend can be modeled by the increasing, differentiable function L, where L is a function of t. For every lap run, the friend raises 4 dollars. Write an expression for the rate of change of the money, in dollars per minute, raised by the friend at $t = 19$ minutes.

6-7

(h) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = L'(t)$ for the function L defined in part (g).



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns, like so.

100	5	500
120	15	1800
150	10	+1500
		3800

or

$(5-0)(100)$	$= 500$
$(20-5)(120)$	$= 1800$
$(30-20)(150)$	$= 1500$
	3800

Implicit responses such as those above are eligible for both points.

6-7

A response presenting a left Riemann sum (instead of right), will not earn any points for part (a).

The second point is for the correct numerical result.

In order to earn the second point, there must be supporting calculations.

The numerical values for $D(5)$, $D(20)$, and $D(30)$ must be pulled from the table.

Any simplification shown must be correct.

Use the following as an illustration of how to score both points. (In most cases, you see scores for which the expression represents the entirety of relevant work shown for the response.)

· A response of $100 \cdot (5-0) + 120 \cdot (20-5) + 150 \cdot (30-20)$ earns both points.

· A response of $5 \cdot D(5) + 15 \cdot D(20) + 10 \cdot D(30)$ earns the first point. Eligible for second point if more work is on the page.

· A response of $5 \cdot D(5) + 15 \cdot D(20) + 10 \cdot D(30) = 3800$ earns both points.

· A response of $5 \cdot 100 + 15 \cdot 120 + 10 \cdot 150$ earns both points.

· A response of $5 \cdot 100 + 25 \cdot 120 + 10 \cdot 150$ contains one numerical error. Earns the first point. Not eligible for the second point.

· A response of $5 \cdot 100 + 15 \cdot 120 + 10 \cdot 150 = 3300$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.

· A response of $500 + 1800 + 1500$ or $500 + 1800 + 1500 = 3800$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we are not seeing an unsupported answer. The second point will be earned.

The response does not need to include "approximately" in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points. $5 \cdot 100 + 15 \cdot 120 + 10 \cdot 150$

✓

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Right Riemann sum
- ☐ Answer

Solution:

6-7

$$\int_0^{30} D(t)\,dt \approx D(5)(5-0) + D(20)(20-5) + D(30)(30-20)$$

$$= 100 \cdot 5 + 120 \cdot 15 + 150 \cdot 10 = 3800$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an overestimate because D is increasing.

The response does not need an explicit reference to part (a) or to a right Riemann sum.

A response that appeals to concavity as a reason will NOT earn this point.

The following table contains some examples of what WILL or will NOT earn the point.

The following examples earn the point:

· Overestimate because D is increasing

· Overestimate because $D'(t) > 0$

· Overestimate because the graph of D is increasing (we know they are talking about the given function D)

· Overestimate because the function is increasing. (There is only one function in the problem at this point.)

The following examples do not earn the point:

· Overestimate because D is concave down/up

· Overestimate because the graph/curve is increasing. Since no graph/curve is provided, we do not know what graph/curve they are talking about.

· Overestimate because the table [or table values] is/are increasing. The behavior of the values shown in the table is not the same as the behavior of the function D. The table does not tell us what is happening to D between the given points.

0	1

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because D is an increasing function, and this is a right Riemann sum, this is an overestimate.

6-7

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

We need to see both a difference and a quotient. They don't have to be together.

The expression does not need to be simplified, but values for $D\left(20\right)$ and $D\left(0\right)$ must be pulled from the table.

The response does not need to include "approximately" in any way. We will read $=$ and $\approx$ as the same thing.

The second point is for correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols. For example:

· Meters per minute per minute or Meters/min/min

· Meters per min^2 or Meters/min^2

· Meters per minute squared

We do not read work shown in part (d) for this point. The units must appear in part (c).

Use caution when interpreting abbreviations: m should be meters and min should be minutes, but responses may vary. For example,

· if a response includes units "Meters/m/m" we read "m" as minutes; or

· if a response includes units "m/min/min" we read "m" as meters; or

· if a response includes units "m/m/m" we cannot interpret the units and the response will not earn the second point.

The response will not be penalized for any attempt to verbally describe their answer after a correct answer has been presented.

Minimalist response: The following expression and units represent the minimum work that must be seen in a response in order to earn both points.

$\frac{120-0}{20}\ \text{meters/min}^2$ or $\frac{120}{20-0}\ \text{meters/min}^2$

In the minimalist response, we still need to see subtraction (a difference), so subtracting 0 must be explicit either in the numerator or in the denominator.

6-7

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$D'(5) \approx \frac{D(20)-D(0)}{20-0} = \frac{120-0}{20} = 6$ meters per minute per minute

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

· Three components of a correct answer:

· Indicating a rate of a rate

· Including the context of running

· Explicitly referring to the specified point in time ($t = 5$)

Incorrect or missing units will not be penalized as long as there is a clear connection to the context of the problem.

The words "increasing" and "decreasing" are interpreted as modifiers describing one of the specified rates.

If the numerical value from part (c) is used in this interpretation, then we expect that the choice of "increasing" or "decreasing" is consistent with that value.

The following examples use 6 as the imported value from part (c). If a different [incorrect] value is imported from part (c), then these statements should be consistent with that value. (Remember that we are reading $=$ and $\approx$ as the same thing.)

· "… increasing at a rate of 6 " is OK [Units not required in part (d)] (the point is not earned with this phrase alone)

· "… decreasing at a rate of 6 " is not OK and the point is not earned.

Rate of a rate

To earn this point, the response must explicitly discuss the rate of change of a rate of change in the context of motion [running]. For example:

6-7

· The rate of change in the rate at which students are running

· The rate of change in speed is changing (or increasing) at a rate of . . .

Since the runners are unlikely to be running backwards, we interpret speed as being the same as velocity.

· The rate at which the velocity of the runner is changing

· Acceleration of the runner (this is correct for the context)

The words "increasing," "decreasing," or "changing" are not interpreted as rates

Units are not interpreted as rates. (The phrase "meters per minute" is not enough to indicate a rate.)

Slope of tangent line is not an appropriate interpretation for this context.

"Rate of change of D " is not enough without describing what D represents in this context.

Context

The context is running, which could be indicated by words such as (but not limited to): meters, minutes, laps, speed, velocity, runner

Explicit reference to $t = 5$

Acceptable references to a specific point in time include (but are not limited to):

· At minute 5 (where minute could be replaced with any other unit of time)

· At $t = 5$ (with or without units of time)

· After 5 minutes (where minutes could be replaced with any other unit of time).

Note: we are choosing to assume that "after" is being used in lieu of "at" unless we have additional evidence to the contrary.

If the response is explicitly indicating an interval of time (e.g., "from minute 5 on" or "during the first 5 minutes"), this point will not be earned.

0	1 ✓

The student response accurately includes the criteria below.

☐ Explanation

Solution:

6-7

$D'(5)$ is the rate at which the rate at which the student runs is changing, in meters per minute per minute, at minute 5.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. Any of these equations can be written as is without any explanation or additional supporting work. If rewriting of equations is done (transforming to slope-intercept form, perhaps), it must be done correctly.

· $y = 2.5(t - 20) + 120$

· $y - 120 = 2.5(t - 20)$

· $y = 2.5t + 70$

The key calculus concept being assessed in this part of the problem is that the value of $D'(20)$ is the slope of the tangent line. In that spirit:

· The response will not be penalized for using x instead of t.

· We are accepting $D(t)$ in lieu of y in all forms of the tangent line equation.

0	1

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = D'(20)(t - 20) + D(20)$$

$$y = 2.5(t - 20) + 120$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are three components of substitution that we need to see:

· Uses $u = \frac{t}{4}$, which may be done implicitly

· Uses the correct constant of 4

6-7

· Uses the correct limits for the substitution (0 and 5). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

Either of the expressions $4D\left(\frac{t}{4}\right)\Big|_0^{20}$ or $4D\left(u\right)\Big|_0^{5}$ will earn the substitution point without any additional supporting work.

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

· Limits for t attached to an integral with u as the variable of integration (or vice versa),

· Improper equals signs, such as an indefinite integral equal to a definite integral

· Reversed limits of integration, for example: $\displaystyle\int_5^0 D'\left(u\right)\, du$

The second point is for the Fundamental Theorem of Calculus.

This point can be earned with any of the following, where k is a nonzero constant and $b = 5$ or $b = 20.$ No explanation or additional supporting work is necessary.

· $kD\left(\frac{t}{4}\right)\Big|_0^{b}$

· $k\left(D(b) - D(0)\right)$ or $kD(b)$ (because $D\left(0\right) = 0$)

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is $400.$

Values from the table must be used. $4\left(D\left(5\right) - D\left(0\right)\right)$ is not enough to earn the answer point.

The expressions $4\left(100 - 0\right)$ or $4\left(100\right)$ are sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they do not need to be correct to earn this point.

The response cannot earn this point without also earning the first two points.

Minimalist response: The expression $4\left(100\right)$ represents the minimum work that must been seen in a response in order to earn all three points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

6-7

- [ ] Substitution
- [ ] Fundamental Theorem of Calculus
- [ ] Answer

Solution:

Let $u = \frac{t}{4}$. Then $du = \frac{dt}{4}$.

When $t = 0$, $u = 0$.

When $t = 20$, $u = 5$.

$$\int_0^{20} D'\left(\frac{t}{4}\right)\, dt = 4\int_0^5 D'(u)\, du$$

$$= 4\left(D(5) - D(0)\right)$$

$$= 4\left(100 - 0\right) = 400$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression at $t = 19$ earns the point. It may be written as $4 \cdot L'(19)$ or $4 \cdot \frac{dL}{dt}\Big|_{t=19}$

The expression $4 \cdot L'(t)$ is not sufficient because it does not use $t = 19$.

The response must use L (not M or D or any other function name) in order to earn the point.

0	1 ✓

The student response accurately includes the criteria below.

- [ ] Expression

Solution:

The rate of change of the money, in dollars per minute, raised by the friend at $t = 19$ minutes is $4L'(19)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-7

This point is earned for a correct statement identifying the distinction between the differential equation $\frac{dy}{dt} = L'(t)$ and the given slope field.

In order to earn the point, the response must:

State that the values of the differential equation are positive while the slopes in the given slope field are negative or 0.

Some ways to say that the values of differential equation are positive:

· L is increasing

· $L'(t) > 0$

· L has positive slopes

When discussing the slopes in the slope field, the response must explicitly refer to the signs of the slopes.

The following examples could earn the point:

· The slope field has negative (non-positive) slopes.

· The slopes shown are negative or 0.

· "The graph" or "The field" could be used in lieu of "The slope field."

The following examples will not earn the point:

· The slope field is decreasing/negative.

· The slope field looks like it represents a decreasing function.

· The slope field represents a quadratic [or exponential] function.

· "It" (rather than "the slope field")

Statements like "L is an increasing function, so the signs of the slopes in the slope field should be positive" can earn this point because "should be" is indicating contrast with what is known.

The response will not be penalized for saying that the slopes in the slope field are "always" negative or "all" negative.

0	1 ✓

The student response accurately includes the criteria below.

☐ Explanation

6-7

Solution:

L is an increasing function, so $\frac{dy}{dt} = L'(t) \geq 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are negative.

6-7

66. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	4	8	12	16
$p(t)$ (feet)	3	9	16.5	26	36

The position of a particle, A, moving along the x-axis is modeled by a twice-differentiable function p, where time t is measured in seconds and $p(t)$ is measured in feet. Selected values of $p(t)$ are shown in the table above.

(a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate the value of $\int_0^{16} p(t)\,dt$. Show the computations that lead to your answer.

(b) Use the data in the table to approximate $p'(11)$ using the average rate of change of $p(t)$ over the interval $8 \leq t \leq 12$. Show the computations that lead to your answer. Indicate units of measure.

(c) Interpret the meaning of $p'(11)$ in the context of the problem.

(d) Must there exist a value of k, for $0 < k < 16$, such that $p'(k)$ is equal to the average rate of change of p over the interval $0 \leq x \leq 16$? Justify your answer.

(e) Find $\int_0^4 p'(2t+4)\,dt$. Show the computations that lead to your answer.

(f) Let $h(x) = \int_1^{4x} p(2t)\,dt$. Find $h'(1)$. Show the computations that lead to your answer.

(g) The velocity of a second particle, Q, can be modeled by a twice-differentiable function g. It is known that $g(3) = 2$, $g'(3) = -5$, and $g''(3) = 11$. Is the speed of particle Q increasing or decreasing at time $t = 3$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -xy^2 + 3$ with initial condition $f(5) = 2$. Write an equation for the line tangent to the graph of f at the point $(5, 2)$.

Part A

6-7

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Midpoint Riemann sum, with appropriate values taken from the table.

Need to see sum of two terms, each as a product of two factors, i.e. base$_1$ $\cdot$ height$_1$ + base$_2$ $\cdot$ height$_2$

$p(4) \cdot (8-0)+p(12) \cdot (16-8)$ or $p(3) \cdot (6)+p(9) \cdot (6)$ earns first point.

If at least three of the four base/height values are correct in the sum of products, the expression earns the first point.

Second Point

280 or equivalent

Can be earned without first point with Riemann sum setup demonstration. With credible evidence that a midpoint Riemann sum attempt was made without not enough work shown, this point can be earned in the presence of the correct answer. For example, a sum of two terms with no evidence of products $(72+208)$ or a graph with areas labeled and summed to 280 or equivalent, can earn this point.

Examples:

- $280=72+208$ by itself earns second point, not first point
- 280 declared, with $9 \cdot 8$ and $26 \cdot 8$ identified (perhaps graphically) earns second point, not first point
- $k(9 \cdot 8+26 \cdot 8)$, nonzero constant $k \neq 1$ by itself earns first point, not second point
- $p(4) \cdot (8)+p(12) \cdot (16-8)$ by itself earns first point, not second point

✓

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Form of midpoint Riemann sum
- [ ] Answer

Solution:

$$\int_0^{16} p(t)\, dt \approx p(4)(8-0)+p(12)(16-8)$$

$$=9 \cdot 8+26 \cdot 8=280$$

6-7

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct difference quotient, with appropriate values from the table.

Need to see both a difference and a quotient

$\frac{p(12)-p(8)}{12-8}$ by itself doesn't earn first point; however $\frac{26-16.5}{12-8}$ or $\frac{26-16.5}{4}$ by itself earns first point

Second Point

Earned for correct units of ft per sec or ft/sec or f/s.

Correct units must be tied to a number to earn second point.

Correct units must appear in part (b); correct units in part (c) only doesn't earn second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$p'(11) \approx \frac{26-16.5}{12-8} = \frac{9.5}{4} = 2.375 \text{ feet per second}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct $p'(11)$ interpretation.

Need two components in response:

· "rate of change of position" or "velocity" of the particle

· at $t = 11$

Correct ways to describe rate at time at $t = 11$ include:

6-7

· Rate at which position is changing/increasing

· Rate at which particle is moving right

· Instantaneous velocity or velocity of particle

Ambiguous/incorrect ways to describe rate include:

· Rate at which particle is changing/increasing

· Rate at which particle is moving

Units are not required in part (c). If units are presented for t, they must be in seconds.

0	1

The student response accurately includes the criteria below.

- [ ] Interpretation

Solution:

$p'(11)$ is the rate at which the position of the particle is changing, in feet per second, at time $t = 11$ seconds.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for both Mean Value Theorem (MVT) conditions with the appropriate implication that p is continuous because we know that p is differentiable.

Ways to earn first point include:

· p differentiable $\Rightarrow$ p continuous

· p differentiable, hence p continuous

· p differentiable, so p continuous

p differentiable and p continuous does not earn first point.

Second Point

Philosophy: Earned for concluding “yes” and invoking Mean Value Theorem or its equivalent formulation, with consideration of the continuity of p .

6-7

MVT title does not have to be stated to earn second point.

This point is not earned if other theorems (IVT, etc.) are invoked.

If presented, the value for $p'\,(k) = \frac{33.5}{16}$ must be correct to earn second point.

Examples:

· “By MVT, yes since p differentiable” earns neither point

· “By MVT, yes since p continuous” earns second point only

· “By MVT, yes since p differentiable and continuous” earns second point only

· “By MVT, yes since p differentiable hence continuous” earns both points

· “By MVT, yes since p twice differentiable hence continuous” earns both points

· “By MVT, yes since function differentiable hence continuous” earns both points

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Conditions
- [ ] Conclusion using Mean Value Theorem

Solution:

p is differentiable. $\Rightarrow p$ is continuous.

Therefore, the Mean Value Theorem can be applied to p on the interval $0 \le x \le 16$ to guarantee that there exists a value of k, for $0 < k < 16$, such that $p'\,(k)$ equals the average rate of change of p over the interval $0 \le x \le 16$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct substitution application.

Need three elements in response (with consistent use of u or another substitution variable):

· $u = 2t + 4$ (can be explicit or implicit)

6-7

· $du = 2\ dt$ (yielding integral coefficient of $\frac{1}{2}$)

· Correct integral limits throughout, with t from 0 to 4 and u from 4 to 12

Omitting the coefficient of $\frac{1}{2}$ or using some other value will not earn first point; response could earn second point, but not third point.

In the third element above, if response shows an otherwise correct substitution with unchanged limits of integration on a definite integral from 0 to 4, then point is not earned.

Alternately, the integral can be worked with as an indefinite integral with u returned to $2t+4$ in the antiderivative before evaluating as a means to eliminate the need to change limits. Such a response can earn the first point if handled correctly.

For example: $\int_0^4 p'\ (2t+4)\ dt = \frac{1}{2}p\ (2t+4)\ \Big|_0^4$ earns the substitution point.

Second Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, with evidence of connection between $\int p'(u)$ and $p(u)$.

Ways to earn second point include:

· $k\,(p\ (12) - p\ (4))$ or $k\,(p\ (2\ \cdot\ 4+4) - p\ (2\ \cdot\ 0+4))$ for k a nonzero constant

· $kp\ (2t+4)\,\big|_0^3$ or $kp\ (u)\ \big|_4^{12}$ for k a nonzero constant

· $\int p'\ (2t+4)\ dt = kp\ (2t+4)$ or $\int p'\ (u)\ du = kp\,(u)$ for any nonzero constant k, with or without appropriate limits of integration on either integral

First and second points can be earned simultaneously by responses such as $\int_0^4 p'\ (2t+4)\ dt = \frac{1}{2}p\ (2t+4)\ \Big|_0^4$

Third Point

Earned only for an answer of 8.5 or equivalent value.

Value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring Examples:

· $\frac{1}{2}\int p'\,(u)\ du = \frac{1}{2}p(u)$ earns $0-1-0$

· $\frac{1}{2}\int p'\ (u)\ du = \frac{1}{2}p\ (u)\big|_0^4 = \frac{1}{2}(p\ (4) - p\ (0))$ earns $0-1-0$

6-7

- $\int p'(u)\ du = p(u)\,|_4^{12} = (p(12) - p(4)) = 17$ earns $0 - 1 - 0$
- $2\int p'(u)\ du = 2p(u)|_0^4 = 2\,[26 - 9]$ earns $0 - 1 - 0$
- $\int_0^4 p'(2t+4)\ dt = p\left(t^2 + 4t\right)\big|_0^4 = p(32) - p(0)$ earns $0 - 0 - 0$
- $\frac{1}{2}\int_0^4 p'(u)\ du = \frac{1}{2}(p(12) - p(4)) = 8.5$ earns $0 - 1 - 1$
- $\int_0^4 p'(2t+4)\ dt = \frac{1}{2}(p(12) - p(4)) = 8.5$ earns $1 - 1 - 1$
- $\int_0^4 p'(2t+4)\ dt = \frac{1}{2}p(u)\big|_4^{12} = \frac{1}{2}(p(12) - p(4)) = 8.5$ earns $1 - 1 - 1$
- $\int_0^4 p'(2t+4)\ dt = \frac{1}{2}p(2t+4)\big|_0^4 = \frac{1}{2}(p(12) - p(4))$ earns $1 - 1 - 0$
- $\frac{1}{2}\int p'(u) = \frac{1}{2}p(u)$; and $\frac{1}{2}p(2t+4)\big|_0^4 = \frac{1}{2}(26 - 9)$ earns $1 - 1 - 1$

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t + 4$. Then $du = 2\ dt$.

When $t = 0$, $u = 4$.

When $t = 4$, $u = 12$.

$$\int_0^4 p'(2t+4)\ dt = \frac{1}{2}\int_4^{12} p'(u)\ du$$

$$= \frac{1}{2}(p(12) - p(4))$$

6-7

$= \frac{1}{2}(26 - 9) = 8.5$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, resulting in $a \cdot p\,(8x)$ for some nonzero constant a.

Ways to earn first point include responses of $p\,(2 \cdot 4x)$ or $p\,(8x)$ (Note $p\,(8t)$ would also earn this point.)

The expression $p\,(2 \cdot 4x) - p\,(2 \cdot 1)$ or $(p\,(2 \cdot 4x) - p\,(2 \cdot 1)) \cdot 4$ earns this point. The $p\,(2 \cdot 1)$ is viewed as a chain rule error.

Any response correctly presenting and evaluating the derivative of the integral in terms of the substitution $u = 2t$ earns first point. Part of this response may include $\int_1^{4x} p\,(2t)\,dt = \frac{1}{3}\int_2^{8x} p\,(u)\,du.$

Second Point

Philosophy: This point is earned for correctly applying the chain rule to find $h'\,(x)$ or $h'\,(1)$. That is accomplished by multiplying the term involving $p\,(x)$ by 4.

Can only be earned if response earned first (FTC) point.

The first two points can be, and often are, earned simultaneously.

The point is not earned if the presented derivative is $a \cdot p\,(8x)$ for any $a \neq 4$.

The point is not earned for responses of $4p\,(2t)\,|_1^{4x}$ or $4\,(p\,(8x) - p\,(b))$, for any value of b.

Third Point

Can only be earned if response earned first (FTC) and second (chain rule) points.

Earned only for answer of 66 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring examples:

$\cdot\, p\,(2 \cdot 4x) - p\,(2);\;\; p\,(8) - p\,(2)$ earns $1 - 0 - 0$

$\cdot\, p\,(2 \cdot 4x) \cdot 4 - p\,(2) \cdot 0;\;\; 66$ earns $1 - 1 - 1$

$\cdot\, 4p\,(2t)\,|_1^{4x} = 4\,(p\,(8) - p\,(2))$ earns $1 - 0 - 0$

$\cdot\, p\,(2x) \cdot 2;\;\; p\,(2) \cdot 2$ earns $0 - 0 - 0$

6-7

$\cdot\, p\,(2\,\cdot\,4x)\,\cdot\,4;\;\; p\,(8)\,\cdot\,4$ earns $1-1-0$

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Fundamental Theorem of Calculus
- [ ] Chain rule
- [ ] Answer

Solution:

$$h'\,(x)=\frac{d}{dx}\int_{1}^{4x}p\,(2t)\;dt=p\,(2\,\cdot\,4x)\,\cdot\,4$$

$$h'\,(1)=p\,(8)\,\cdot\,4=16.5\,\cdot\,4=66$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for narrowing consideration of given information to g and g'.

g and g' need not be evaluated to earn the first point.

Considering velocity and acceleration not sufficient to earn first point, unless acceleration is linked to g'.

If response considers g and g' and g'', then eventually g'' must be eliminated to earn first point.

Second Point

Philosophy: Earned for "decreasing" with reason, using the opposite signs of g and g'.

Ways to earn second point include:

· "decreasing since $g>0$ and "$g'<0$

· "particle slowing down since g and g' have opposite signs"

Can only be earned if response earned first point.

Examples:

· "decreasing since velocity and acceleration have opposite signs" earns neither point

6-7

· “decreasing since velocity > 0 and acceleration < 0 ”earns neither point

· “decreasing since $(2) \cdot (-5) = -10 < 0$ ” earns both points

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g and g'
- ☐ Answer with reason

Solution:

Because $g(3) = 2 > 0$, the velocity of the particle is positive. Because $g'(3) = -5 < 0$, the acceleration of the particle is negative. Because the velocity and acceleration have opposite signs, the speed of the particle is decreasing at time $t = 3$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct tangent line equation.

Correct responses include:

· $y = -17(x-5) + 2$ or $y - 2 = -17(x-5)$

· $y = -17x + 87$

· $y = \left(-5 \cdot 2^2 + 3\right)(x-5) + 2$

Incorrect responses include:

· $y = \left(-xy^2 + 3\right)(x-5) + 2$

· $y = 17x + 87$

· $f(x) = -17x + 87$

0	1

The student response accurately includes the criteria below.

6-7

- [ ] Tangent line equation

Solution:

$$\left.\frac{dy}{dx}\right|_{(5,\,2)} = -5 \cdot 2^2 + 3 = -17$$

$$y = -17\,(x - 5) + 2$$

6-7

67. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	3	6	9	12
$p(t)$ (feet)	-5	-8.5	-12.5	-15	-20

The position of a particle, A, moving along the x-axis is modeled by a twice-differentiable function p, where time t is measured in seconds and $p(t)$ is measured in feet. Selected values of $p(t)$ are shown in the table above.

(a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate the value of $\int_0^{12} p(t)\,dt$. Show the computations that lead to your answer.

(b) Use the data in the table to approximate $p'(5)$ using the average rate of change of $p(t)$ over the interval $3 \le t \le 6$. Show the computations that lead to your answer. Indicate units of measure.

(c) Interpret the meaning of $p'(5)$ in the context of the problem.

(d) Must there exist a value of k, for $0 < k < 12$, such that $p'(k)$ is equal to the average rate of change of p over the interval $0 \le x \le 12$? Justify your answer.

(e) Find $\int_0^{3} p'(2t+6)\,dt$. Show the computations that lead to your answer.

(f) Let $h(x) = \int_1^{2x} p(3t)\,dt$. Find $h'(2)$. Show the computations that lead to your answer.

(g) The velocity of a second particle, Q, can be modeled by a twice-differentiable function g. It is known that $g(3) = -1$, $g'(3) = -5$, and $g''(3) = 7$. Is the speed of particle Q increasing or decreasing at time $t = 3$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = 12x^2 - y^3$ with initial condition $f\left(\frac{1}{2}\right) = -2$. Write an equation for the line tangent to the graph of f at the point $\left(\frac{1}{2}, -2\right)$.

Part A

6-7

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct midpoint Riemann sum, with appropriate values taken from the table.

Need to see sum of two terms, each as a product of two factors, i.e. $\text{base}_1 \cdot \text{height}_1 + \text{base}_2 \cdot \text{height}_2$

$p(3) \cdot (6-0) + p(9) \cdot (12-6)$ or $p(3) \cdot (6) + p(9) \cdot (6)$

If at least three of the four base/height values are correct in the sum of products, the expression earns the first point. However, if one of these values is incorrect, the response does not earn the second point.

Second Point

Earned for an answer of -141 or equivalent.

Can be earned without the first point for the Riemann sum setup demonstration. With credible evidence that a midpoint Riemann sum attempt was made, but not enough work shown, this point can be earned in the presence of the correct answer. For example, a sum of two terms with no evidence of products $(-51-90)$, or a graph with areas labeled and summed to -141, can earn this point.

Examples:

· $-141 = -51 - 90$ by itself earns second point, not first point

· -141 declared, with $-8.5 \cdot 6$ and $-15 \cdot 6$ identified earns second point, not first point (perhaps graphically)

· $k(-8.5 \cdot 6 - 15 \cdot 6)$, nonzero constant $k \neq 1$ by itself earns first point, not second point

· $p(3) \cdot (6-0) + p(9) \cdot (12-6)$ by itself earns first point, not second point

✓

0	1	2

The student response accurately includes **both** of the criteria below.

☐ Form of midpoint Riemann sum

☐ Answer

Solution:

$$\int_0^{12} p(t)\, dt \approx p(3)(6-0) + p(9)(12-6)$$

$$= -8.5 \cdot 6 + (-15) \cdot 6 = -141$$

6-7

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for a correct difference quotient, with appropriate values from the table.

The response must show both a difference and a quotient.

$\frac{p(6)-p(3)}{6-3}$ by itself doesn't earn first point; however $\frac{-12.5-(-8.5)}{6-3}$ or A $\frac{-12.5-(-8.5)}{3}$ by itself earns the first point

Second Point

Earned for correct units of ft per sec or ft/sec or f/s.

Correct units must be tied to a number to earn second point.

Correct units must appear in part (b); correct units in part (c) only do not earn the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$p'(5) \approx \frac{-12.5-(-8.5)}{6-3} = \frac{-4}{3}$ feet per second

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct $p'(5)$ interpretation.

Need two components in the response:

· "rate of change of position" or "velocity" of the particle

· at $t = 5$

6-7

Correct ways to describe the rate of change include:

· rate at which position (of the particle) is changing/decreasing

· rate at which particle is moving left

· instantaneous velocity or velocity of the particle

Ambiguous/incorrect ways to describe rate include:

· rate at which particle is changing/decreasing

· rate at which particle is moving

The response does not need to include the value of the approximation found in part (b), but will not be penalized for doing so (e.g. "the rate at which the particle's position is decreasing is $\frac{4}{3}$ at time $t = 5$" earns the point).

If the response includes the approximation value from part (b) and makes an incorrect statement such as "the rate at which the position of the particle is decreasing is $-\frac{4}{3}$ at time $t = 5$", the point is not earned.

Units are not required in part (c). If units are presented for t, they must be in seconds.

0	1 ✓

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$p'(5)$ is the rate at which the position of the particle is changing, in feet per second, at time $t = 5$ seconds.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for both Mean Value Theorem (MVT) conditions with the appropriate implication that p is continuous because we know that p is differentiable.

Ways to earn first point include:

· p differentiable $\Rightarrow$ p continuous

· p differentiable, hence p continuous

6-7

· p differentiable, so p continuous

p differentiable and p continuous does not earn first point.

Second Point

Philosophy: Earned for concluding “yes” and invoking Mean Value Theorem or its equivalent formulation, with consideration of the continuity of p.

MVT title does not have to be stated to earn second point.

This point is not earned if other theorems (IVT, etc.) are invoked.

If presented, the value for $p'(c) = -\frac{5}{4}$ must be correct to earn second point.

Examples:

· “By MVT, yes since p differentiable” earns neither point

· “By MVT, yes since p continuous” earns second point only

· “By MVT, yes since p differentiable and continuous” earns second point only

· “By MVT, yes since p differentiable hence continuous” earns both points

· “By MVT, yes since p twice differentiable hence continuous” earns both points

· “By MVT, yes since function differentiable hence continuous” earns both points

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Conditions
- [ ] Conclusion using Mean Value Theorem

Solution:

p is differentiable. $\Rightarrow p$ is continuous.

Therefore, the Mean Value Theorem can be applied to p on the interval $0 \leq x \leq 12$ to guarantee that there exists a value of k, for $0 < k < 12$, such that $p'(k)$ equals the average rate of change of p over the interval $0 \leq x \leq 12$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

6-7

First Point

Philosophy: Earned for correct substitution.

A correct substitution requires three elements (with consistent use of u or another substitution variable):

· $u = 2t + 6$ (can be explicit or implicit)

· $du = 2\ dt$ (yielding integral coefficient of $\frac{1}{2}$)

· Correct integral limits throughout, with t from 0 to 3 and u from 6 to 12

The substitution could be implicit: $\int_0^3 p'\left(2t+6\right)\ dt = \frac{1}{2}p\left(2t+6\right)\Big|_0^3$

Omitting the coefficient of $\frac{1}{2}$ or using some other constant will not earn the substitution or answer points; the response could still earn the second point.

In the third element above, if response shows an otherwise correct substitution with unchanged limits of integration on a definite integral from 0 to 3, then point is not earned.

Alternately, the integral can be worked with as an indefinite integral with u returned to $2t + 6$ in the antiderivative before evaluating as a means to eliminate the need to change limits. Such a response can earn the first point, if handled correctly.

Second Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, with evidence of connection between $\int p'\left(u\right)$ and $p\left(u\right)$.

Ways to earn second point include:

· $k\left(p\left(12\right) - p\left(6\right)\right)$ or $k\left(p\left(2\cdot 3+6\right) - p\left(2\cdot 0+6\right)\right)$ for k a nonzero constant

· $kp\left(2t+6\right)\Big|_0^3$ or $kp\left(u\right)\Big|_6^{12}$ for k a nonzero constant

· $\int p'\left(2t+6\right)\ dt = kp\left(2t+6\right)$ or $\int p'\left(u\right)\ du = kp\left(u\right)$ for any nonzero constant k, with or without appropriate limits of integration on either integral

The first two points can be earned simultaneously by responses such as $\int_0^3 p'\left(2t+6\right)\ dt = \frac{1}{2}p\left(2t+6\right)\Big|_0^3$

Third Point

Earned only for an answer of -3.75 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

6-7

Scoring Examples:

- $\frac{1}{2}\int p'(u)\ du = \frac{1}{2}p(u)$ earns $0-1-0$
- $\frac{1}{2}\int p'(u)\ du = \frac{1}{2}p(u)\big|_0^3 = \frac{1}{2}(p(3)-p(0))$ earns $0-1-0$
- $\int p'(u)\ du = p(u)\big|_6^{12} = (p(12)-p(6)) = -7.5$ earns $0-1-0$
- $2\int p'(u)\ du = 2p(u)\big|_0^3 = 2[-8.5+5]$ earns $0-1-0$
- $\int_0^3 p'(2t+6)\ dt = p\left(t^2+6t\right)\big|_0^3 = p(27)-p(0)$ earns $0-0-0$
- $\frac{1}{2}\int_0^3 p'(u)\ du = \frac{1}{2}(p(12)-p(6)) = -3.75$ earns $0-1-1$
- $\int_0^3 p'(2t+6)\ dt = \frac{1}{2}(p(12)-p(6)) = -3.75$ earns $1-1-1$
- $\int_0^3 p'(2t+6)\ dt = \frac{1}{2}p(u)\big|_6^{12} = \frac{1}{2}(p(12)-p(6)) = -3.75$ earns $1-1-1$
- $\int_0^3 p'(2t+6)\ dt = \frac{1}{2}p(2t+6)\big|_0^3 = \frac{1}{2}(p(12)-p(6))$ earns $1-1-0$
- $\frac{1}{2}\int p'(u) = \frac{1}{2}p(u)$; and $\frac{1}{2}p(2t+6)\big|_0^3 = \frac{1}{2}(-20-(12.5))$ earns $1-1-1$

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- [ ] Substitution
- [ ] Fundamental Theorem of Calculus
- [ ] Answer

Solution:

Let $u = 2t+6$. Then $du = 2\ dt$.

When $t=0$, $u=6$.

6-7

When $t = 3$, $u = 12$.

$$\int_0^3 p'\left(2t+6\right)\,dt = \frac{1}{2}\int_6^{12} p'\left(u\right)\,du$$

$$= \frac{1}{2}\left(p\left(12\right) - p\left(6\right)\right)$$

$$= \frac{1}{2}\left(-20 - \left(-12.5\right)\right) = -3.75$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, resulting in $a \cdot p\left(6x\right)$ for some nonzero constant a.

Ways to earn first point include responses of $p\left(3 \cdot 2x\right)$ or $p\left(6x\right)$ (Note $p\left(6t\right)$ would also earn this point.)

The expression $p\left(3 \cdot 2x\right) - p\left(3 \cdot 1\right)$ or $\left(p\left(3 \cdot 2x\right) - p\left(3 \cdot 1\right)\right) \cdot 3$ earns this point. The $p\left(3 \cdot 1\right)$ is viewed as a chain rule error.

Any response correctly presenting and evaluating the derivative of the integral in terms of the substitution $u = 3t$ earns first point. Part of this response may include $\int_1^{2x} p\left(3t\right)\,dt = \frac{1}{3}\int_3^{6x} p\left(u\right)\,du$.

Second Point

Philosophy: This point is earned for correctly applying the chain rule to find $h'\left(x\right)$ or $h'\left(2\right)$. That is accomplished by multiplying the term involving $p\left(x\right)$ by 2.

Can only be earned if response earned first (FTC) point.

The first two points can be, and often are, earned simultaneously.

The point is not earned if the presented derivative is $a \cdot p\left(6x\right)$ for any $a \neq 2$.

The point is not earned for responses of $2p\left(3t\right)\big|_1^{2x}$ or $2\left(p\left(6x\right) - p\left(b\right)\right)$, for any value of b.

Third Point

Can only be earned if response earned first (FTC) and second (chain rule) points.

Earned only for answer of -40 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring examples:

6-7

$\cdot\, p\,(3\cdot 2x) - p\,(3);\ p\,(12) - p\,(3) = -20 + 8.5$ earns $1-0-0$

$\cdot\, p\,(3\cdot 2x)\cdot 2 - p\,(3)\cdot 0;\ \ -40$ earns $1-1-1$

$\cdot\, 2p\,(3t)\,|_1^{2x} = 2\,(p\,(12) - p\,(3))$ earns $1-0-0$

$\cdot\, p\,(3x)\cdot 2;\ p\,(6)\cdot 2 = -12.5\cdot 2$ earns $0-0-0$

$\cdot\, p\,(6x)\cdot 2;\ p\,(12)\cdot 2$ earns $1-1-0$

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Fundamental Theorem of Calculus
- [ ] Chain rule
- [ ] Answer

Solution:

$$h'\,(x) = \frac{d}{dx}\int_1^{2x} p\,(3t)\ dt = p\,(3\cdot 2x)\cdot 2$$

$$h'\,(2) = p\,(12)\cdot 2 = -20\cdot 2 = -40$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for narrowing consideration of given information to g and g'.

g and g' need not be evaluated to earn the first point.

Considering velocity and acceleration not sufficient to earn first point, unless acceleration is linked to g'.

If response considers g and g' and g'', then eventually g'' must be eliminated to earn first point.

Second Point

Philosophy: Earned for “increasing” with reason, using the opposite signs of g and g'.

Ways to earn second point include:

$\cdot$ “increasing since $g < 0$ and $g' < 0$ ”

6-7

· "particle speeding up since g and g' have the same signs"

Can only be earned if response earned first point.

A response that calls out all three given values ($g\ (3)$, $g'\ (3)$, and $g''\ (3)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g and g' only, both points can be earned.

Examples:

· "increasing since velocity and acceleration have the same signs" earns neither point

· "increasing since velocity < 0 and acceleration < 0 " earns neither point

· "increasing since $(-1)\ \cdot\ (-5) = 5 > 0$ " earns both points

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Narrows consideration to g and g'
- [ ] Answer with reason

Solution:

Because $g\ (3) = -1 < 0$, the velocity of the particle is negative. Because $g'\ (3) = -5 < 0$, the acceleration of the particle is negative. Because the velocity and acceleration have the same signs, the speed of the particle is increasing at time $t = 3$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct tangent line equation, written in any form.

Correct responses include:

· $y = 11\left(x - \frac{1}{2}\right) - 2$ or $y + 2 = 11\left(x - \frac{1}{2}\right)$

· $y = 11x - \dfrac{15}{2}$

· $y = \left(12 \cdot \left(\frac{1}{2}\right)^2 - (-2)^3\right)\left(x - \frac{1}{2}\right) - 2$

Incorrect responses include:

· $y = \left(12x^2 - y^3\right)\left(x - \frac{1}{2}\right) - 2$

6-7

· $y = 11x - 2$

· $f(x) = 11x - \frac{15}{2}$. ($f(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

0	1

The student response accurately includes the criteria below.

- [ ] Tangent line equation

Solution:

$$\left.\frac{dy}{dx}\right|_{\left(\frac{1}{2},-2\right)} = 12 \cdot \left(\frac{1}{2}\right)^2 - (-2)^3 = 11$$

$$y = 11\left(x - \frac{1}{2}\right) - 2$$

6-7

68. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	5	10	15	20
$p(t)$ (feet)	-25	-20	-14	-8	-2.5

The position of a particle, A, moving along the x-axis is modeled by a twice-differentiable function p, where time t is measured in seconds and $p(t)$ is measured in feet. Selected values of $p(t)$ are shown in the table above.

(a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate the value of $\int_0^{20} p(t)\,dt$. Show the computations that lead to your answer.

(b) Use the data in the table to approximate $p'(7)$ using the average rate of change of $p(t)$ over the interval $5 \le t \le 10$. Show the computations that lead to your answer. Indicate units of measure.

(c) Interpret the meaning of $p'(7)$ in the context of the problem.

(d) Must there exist a value of b, for $0 < b < 20$, such that $p'(b)$ is equal to the average rate of change of p over the interval $0 \le x \le 20$? Justify your answer.

(e) Find $\int_0^{5} p'(2t+5)\,dt$. Show the computations that lead to your answer.

(f) Let $h(x) = \int_3^{5x} p(4t)\,dt$. Find $h'(1)$. Show the computations that lead to your answer.

(g) The velocity of a second particle, Q, can be modeled by a twice-differentiable function g. It is known that $g(1) = -2$, $g'(1) = 4$, and $g''(1) = 7$. Is the speed of particle Q increasing or decreasing at time $t = 1$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -2x^3y^2 - 5$ with initial condition $f(-1) = 2$. Write an equation for the line tangent to the graph of f at the point $(-1, 2)$.

Part A

6-7

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Midpoint Riemann sum, with appropriate values taken from the table.

Need to see sum of two terms, each as a product of two factors, i.e. $\text{base}_1 \cdot \text{height}_1 + \text{base}_2 \cdot \text{height}_2$

$p(5) \cdot (10-0) + p(15) \cdot (20-10)$ or $p(5) \cdot (10) + p(15) \cdot (10)$ earns first point.

If at least three of the four base/height values are correct in the sum of products, the expression earns the first point.

Second Point

-280 or equivalent

Can be earned without first point with Riemann sum setup demonstration. With credible evidence that a midpoint Riemann sum attempt was made without not enough work shown, this point can be earned in the presence of the correct answer. For example, a sum of two terms with no evidence of products $(-200-80)$ or a graph with areas labeled and summed to -280 or equivalent, can earn this point.

Examples:

· $-280 = -200 - 80$ by itself earns second point, not first point

· -280 declared, with $-20 \cdot 10$ and $-8 \cdot 10$ identified (perhaps graphically) earns second point, not first point

· $k(-20 \cdot 10 - 8 \cdot 10)$, nonzero constant $k \neq 1$ by itself earns first point, not second point

· $p(3) \cdot (10) + p(15) \cdot (20-10)$ by itself earns first point, not second point

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint Riemann sum
- ☐ Answer

Solution:

$$\int_0^{20} p(t)\,dt \approx p(5)(10-0) + p(15)(20-10)$$

$$= -20 \cdot 10 + (-8) \cdot 10 = -280$$

6-7

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct difference quotient, with appropriate values from the table.

Need to see both a difference and a quotient

$\frac{p(10)-p(5)}{10-5}$ by itself doesn't earn first point; however $\frac{-14-(-20)}{10-5}$ or $\frac{-14-(-20)}{5}$ by itself earns first point

Second Point

Earned for correct units of ft per sec or ft/sec or f/s.

Correct units must be tied to a number to earn second point.

Correct units must appear in part (b); correct units in part (c) only doesn't earn second point.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$p'(7) \approx \frac{-14-(-20)}{10-5} = \frac{6}{5} \text{ feet per second}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct $p'(7)$ interpretation.

Need two components in response:

· "rate of change of position" or "velocity" of the particle

· at $t = 7$

Correct ways to describe rate at time at $t = 7$ include:

6-7

· Rate at which position is changing/increasing

· Rate at which particle is moving right

· Instantaneous velocity or velocity of particle

Ambiguous/incorrect ways to describe rate include:

· Rate at which particle is changing/increasing

· Rate at which particle is moving

Units are not required in part (c). If units are presented for t, they must be in seconds.

0	1

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$p'(7)$ is the rate at which the position of the particle is changing, in feet per second, at time $t = 7$ seconds.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for both Mean Value Theorem (MVT) conditions with the appropriate implication that p is continuous because we know that p is differentiable.

Ways to earn first point include:

· p differentiable $\Rightarrow$ p continuous

· p differentiable, hence p continuous

· p differentiable, so p continuous

p differentiable and p continuous does not earn first point.

Second Point

Philosophy: Earned for concluding "yes" and invoking Mean Value Theorem or its equivalent formulation, with consideration of the continuity of p.

6-7

MVT title does not have to be stated to earn second point.

This point is not earned if other theorems (IVT , etc.) are invoked.

If presented, the value for $p'(b) = \frac{22.5}{20}$ must be correct to earn second point.

Examples:

· “By MVT , yes since p differentiable” earns neither point

· “By MVT , yes since p continuous” earns second point only

· “By MVT , yes since p differentiable and continuous” earns second point only

· “By MVT , yes since p differentiable hence continuous” earns both points

· “By MVT , yes since p twice differentiable hence continuous” earns both points

· “By MVT , yes since function differentiable hence continuous” earns both points

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Conditions
- [ ] Conclusion using Mean Value Theorem

Solution:

p is differentiable. $\Rightarrow p$ is continuous.

Therefore, the Mean Value Theorem can be applied to p on the interval $0 \leq x \leq 20$ to guarantee that there exists a value of b, for $0 < b < 20$, such that $p'(b)$ equals the average rate of change of p over the interval $0 \leq x \leq 20$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct substitution application.

Need three elements in response (with consistent use of u or another substitution variable):

· $u = 2t + 5$ (can be explicit or implicit)

6-7

· $du = 2dt$ (yielding integral coefficient of $\frac{1}{2}$)

· Correct integral limits throughout, with t from 0 to 5 and u from 5 to 15

Omitting the coefficient of $\frac{1}{2}$ or using some other value will not earn first point; response could earn second point, but not third point.

In the third element above, if response shows an otherwise correct substitution with unchanged limits of integration on a definite integral from 0 to 5, then point is not earned.

Alternately, the integral can be worked with as an indefinite integral with u returned to $2t+5$ in the antiderivative before evaluating as a means to eliminate the need to change limits. Such a response can earn the first point if handled correctly.

For example: $\int_0^5 p'(2t+5)dt = \frac{1}{2}p(2t+5)\Big|_0^5$ earns the substitution point.

Second Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, with evidence of connection between $\int p'(u)$ and $p(u)$.

Ways to earn second point include:

· $k(p(15)-p(5))$ or $k(p(2\cdot 5+5)-p(2\cdot 0+5))$ for k a nonzero constant

· $kp(2t+5)\Big|_0^5$ or $kp(u)\Big|_5^{15}$ for k a nonzero constant

· $\int p'(2t+5)dt = kp(2t+5)$ or $\int p'(u)du = kp(u)$ for any nonzero constant k, with or without appropriate limits of integration on either integral

First and second points can be earned simultaneously by responses such as $\int_0^5 p'(2t+5)dt = \frac{1}{2}p(2t+5)\Big|_0^5$

Third Point

Earned only for an answer of 6 or equivalent value.

Value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring Examples:

· $\frac{1}{2}\int p'(u)du = \frac{1}{2}p(u)$ earns $0-1-0$

· $\frac{1}{2}\int p'(u)du = \frac{1}{2}p(u)\Big|_0^5 = \frac{1}{2}(p(5)-p(0))$ earns $0-1-0$

6-7

- $\int p'(u)du = p(u)|_5^{15} = (p(15) - p(5)) = 12$ earns $0 - 1 - 0$
- $2\int p'(u)du = 2p(u)|_0^5 = 2[-8 - (-20)]$ earns $0 - 1 - 0$
- $\int_0^5 p'(2t+5)dt = p\left(t^2 + 5t\right)\Big|_0^5 = p(50) - p(0)$ earns $0 - 0 - 0$
- $\frac{1}{2}\int_0^5 p'(u)du = \frac{1}{2}(p(15) - p(5)) = 6$ earns $0 - 1 - 1$
- $\int_0^5 p'(2t+5)dt = \frac{1}{2}(p(15) - p(5)) = 6$ earns $1 - 1 - 1$
- $\int_0^5 p'(2t+5)dt = \frac{1}{2}p(u)\Big|_5^{15} = \frac{1}{2}(p(15) - p(5)) = 6$ earns $1 - 1 - 1$
- $\int_0^5 p'(2t+5)dt = \frac{1}{2}p(2t+5)\Big|_0^5 = \frac{1}{2}(p(15) - p(5))$ earns $1 - 1 - 0$
- $\frac{1}{2}\int p'(u) = \frac{1}{2}p(u)$; and $\frac{1}{2}p(2t+5)\Big|_0^5 = \frac{1}{2}(-8 - (-20))$ earns $1 - 1 - 1$

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t + 5$. Then $du = 2\ dt$.

When $t = 0$, $u = 5$.

When $t = 5$, $u = 15$.

$$\int_0^5 p'(2t+5)\ dt = \frac{1}{2}\int_5^{15} p'(u)\ du$$

$$= \frac{1}{2}(p(15) - p(5))$$

$$= \frac{1}{2}(-8 - (-20)) = 6$$

6-7

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, resulting in $a \cdot p(20x)$ for some nonzero constant a .

Ways to earn first point include responses of $p(4 \cdot 5x)$ or $p(20x)$ (Note $p(20t)$ would also earn this point.)

The expression $p(4 \cdot 5x) - p(4 \cdot 3)$ or $(p(4 \cdot 5x) - p(4 \cdot 3)) \cdot 5$ earns this point. The $p(4 \cdot 3)$ is viewed as a chain rule error.

Any response correctly presenting and evaluating the derivative of the integral in terms of the substitution $u = 4t$ earns first point. Part of this response may include $\int_3^{5x} p(4t)\ dt = \frac{1}{4} \int_{12}^{20x} p(u)\ du$.

Second Point

Philosophy: This point is earned for correctly applying the chain rule to find $h'(x)$ or $h'(1)$. That is accomplished by multiplying the term involving $p(x)$ by 5.

Can only be earned if response earned first (FTC) point.

The first two points can be, and often are, earned simultaneously.

The point is not earned if the presented derivative is $a \cdot p(20x)$ for any $a \neq 5$.

The point is not earned for responses of $5p(4t)\,|_3^{5x}$ or $5(p(20x) - p(b))$, for any value of b.

Third Point

Can only be earned if response earned first (FTC) and second (chain rule) points.

Earned only for answer of -12.5 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring examples:

- $p(4 \cdot 5x) - p(12);\ p(20) - p(12)$ earns $1 - 0 - 0$

- $p(4 \cdot 5x) \cdot 5 - p(12) \cdot 0;\ \ -12.5$ earns $1 - 1 - 1$

- $5p(4t)\,|_3^{5x} = 5(p(20) - p(12))$ earns $1 - 0 - 0$

- $p(4x) \cdot 5;\ p(4) \cdot 5$ earns $0 - 0 - 0$

6-7

· $p\left(4 \cdot 5x\right) \cdot 5;\ p\left(20\right) \cdot 5$ earns $1 - 1 - 0$

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Fundamental Theorem of Calculus
- [ ] Chain rule
- [ ] Answer

Solution:

$$h'\left(x\right) = \frac{d}{dx}\int_{3}^{5x} p\left(4t\right)\,dt = p\left(4 \cdot 5x\right) \cdot 5$$

$$h'\left(1\right) = p\left(20\right) \cdot 5 = -2.5 \cdot 5 = -12.5$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for narrowing consideration of given information to g and g'.

g and g' need not be evaluated to earn the first point.

Considering velocity and acceleration not sufficient to earn first point, unless acceleration is linked to g'.

If response considers g and g' and g'', then eventually g'' must be eliminated to earn first point.

Second Point

Philosophy: Earned for "decreasing" with reason, using the opposite signs of g and g'.

Ways to earn second point include:

· "decreasing since $g < 0$ and $g' > 0$ "

· "particle slowing down since g and g' have opposite signs"

Can only be earned if response earned first point.

Examples:

· "decreasing since velocity and acceleration have opposite signs" earns neither point

6-7

· "decreasing since velocity < 0 and acceleration > 0 " earns neither point

"decreasing since $(-2) \cdot 4 = -8 < 0$ " earns both points

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g and g'
- ☐ Answer with reason

Solution:

Because $g(1) = -2 < 0$, the velocity of the particle is negative. Because $g'(1) = 4 > 0$, the acceleration of the particle is positive. Because the velocity and acceleration have opposite signs, the speed of the particle is decreasing at time $t = 1$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct tangent line equation.

Correct responses include:

· $y = 3(x+1) + 2$ or $y - 2 = 3(x+1)$

· $y = 3x + 5$

· $y = \left(-2 \cdot (-1)^3 \cdot 2^2 - 5\right)(x+1) + 2$

Incorrect responses include:

· $y = \left(-2x^3y^2 - 5\right)(x+1) + 2$

· $y = -3x + 5$

· $f(x) = 3x + 5$

0	1

The student response accurately includes the criteria below.

6-7

- [ ] Tangent line equation

Solution:

$$\left.\frac{dy}{dx}\right|_{(-1,\,2)} = -2(-1)^3 \cdot 2^2 - 5 = 3$$

$$y = 3\,(x+1) + 2$$

6-7

69. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	4	8	12	16
$p(t)$ (feet)	25	21	16	12	7.5

The position of a particle, A, moving along the x-axis is modeled by a twice-differentiable function p, where time t is measured in seconds and $p(t)$ is measured in feet. Selected values of $p(t)$ are shown in the table above.

(a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate the value of $\int_0^{16} p(t)\,dt$. Show the computations that lead to your answer.

(b) Use the data in the table to approximate $p'(2)$ using the average rate of change of $p(t)$ over the interval $0 \le t \le 4$. Show the computations that lead to your answer. Indicate units of measure.

(c) Interpret the meaning of $p'(2)$ in the context of the problem.

(d) Must there exist a value of d, for $0 < d < 16$, such that $p'(d)$ is equal to the average rate of change of p over the interval $0 \le x \le 16$? Justify your answer.

(e) Find $\int_0^4 p'(3t+4)\,dt$. Show the computations that lead to your answer.

(f) Let $h(x) = \int_1^{2x} p(3t)\,dt$. Find $h'(2)$. Show the computations that lead to your answer.

(g) The velocity of a second particle, Q, can be modeled by a twice-differentiable function g. It is known that $g(3) = 2$, $g'(3) = -4$, and $g''(3) = -8$. Is the speed of particle Q increasing or decreasing at time $t = 3$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -2x^2 + y^2$ with initial condition $f(3) = 5$. Write an equation for the line tangent to the graph of f at the point $(3, 5)$.

Part A

6-7

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Midpoint Riemann sum, with appropriate values taken from the table.

Need to see sum of two terms, each as a product of two factors, i.e. $\text{base}_1 \cdot \text{height}_1 + \text{base}_2 \cdot \text{height}_2$

$p(4) \cdot (8-0) + p(12) \cdot (16-8)$ or $p(4) \cdot (8) + p(12) \cdot (8)$ earns first point.

If at least three of the four base/height values are correct in the sum of products, the expression earns the first point.

Second Point

264 or equivalent

Can be earned without first point with Riemann sum setup demonstration. With credible evidence that a midpoint Riemann sum attempt was made without not enough work shown, this point can be earned in the presence of the correct answer. For example, a sum of two terms with no evidence of products $(168+96)$ or a graph with areas labeled and summed to 264 or equivalent, can earn this point.

Examples:

· $264 = 168 + 96$ by itself earns second point, not first point

· 264 declared, with $21 \cdot 8$ and $12 \cdot 8$ identified (perhaps graphically) earns second point, not first point

· $k(21 \cdot 8 + 21 \cdot 8)$, nonzero constant $k \neq 1$ by itself earns first point, not second point

· $p(4) \cdot (8) + p(12) \cdot (16-8)$ by itself earns first point, not second point

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint Riemann sum
- ☐ Answer

Solution:

$$\int_0^{16} p(t)\,dt \approx p(4)(8-0) + p(12)(16-8)$$

$$= 21 \cdot 8 + 12 \cdot 8 = 264$$

6-7

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct difference quotient, with appropriate values from the table.

Need to see both a difference and a quotient

$\frac{p(4)-p(0)}{4-0}$ by itself doesn't earn first point; however $\frac{21-25}{4-0}$ or $\frac{21-25}{4}$ by itself earns first point

Second Point

Earned for correct units of ft per sec or ft/sec or f/s.

Correct units must be tied to a number to earn second point.

Correct units must appear in part (b); correct units in part (c) only doesn't earn second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$p'(2) \approx \frac{21-25}{4-0} = \frac{-4}{4} = -1 \text{ foot per second}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct $p'(2)$ interpretation.

Need two components in response:

· "rate of change of position" or "velocity" of the particle

· at $t = 2$

Correct ways to describe rate at time at $t = 2$ include:

6-7

· Rate at which position is changing/decreasing

· Rate at which particle is moving left

· Instantaneous velocity or velocity of particle

Ambiguous/incorrect ways to describe rate include:

· Rate at which particle is changing/decreasing

· Rate at which particle is moving

Units are not required in part (c). If units are presented for t, they must be in seconds.

0	1

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$p'(2)$ is the rate at which the position of the particle is changing, in feet per second, at time $t = 2$ seconds.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for both Mean Value Theorem (MVT) conditions with the appropriate implication that p is continuous because we know that p is differentiable.

Ways to earn first point include:

· p differentiable $\Rightarrow$ p continuous

· p differentiable, hence p continuous

· p differentiable, so p continuous

p differentiable and p continuous does not earn first point.

Second Point

Philosophy: Earned for concluding “yes” and invoking Mean Value Theorem or its equivalent formulation, with consideration of the continuity of p.

6-7

MVT title does not have to be stated to earn second point.

This point is not earned if other theorems (IVT, etc.) are invoked.

If presented, the value for $p'(d) = -\frac{17.5}{16}$ must be correct to earn second point.

Examples:

· "By MVT, yes since p differentiable" earns neither point

· "By MVT, yes since p continuous" earns second point only

· "By MVT, yes since p differentiable and continuous" earns second point only

· "By MVT, yes since p differentiable hence continuous" earns both points

· "By MVT, yes since p twice differentiable hence continuous" earns both points

· "By MVT, yes since function differentiable hence continuous" earns both points

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Conditions
- [ ] Conclusion using Mean Value Theorem

Solution:

p is differentiable. $\Rightarrow p$ is continuous.

Therefore, the Mean Value Theorem can be applied to p on the interval $0 \leq x \leq 16$ to guarantee that there exists a value of d, for $0 < d < 16$, such that $p'(d)$ equals the average rate of change of p over the interval $0 \leq x \leq 16$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct substitution application.

Need three elements in response (with consistent use of u or another substitution variable):

· $u = 3t + 4$ (can be explicit or implicit)

6-7

· $du = 3\ dt$ (yielding integral coefficient of $\frac{1}{3}$)

· Correct integral limits throughout, with t from 0 to 4 and u from 4 to 16

Omitting the coefficient of $\frac{1}{3}$ or using some other value will not earn first point; response could earn second point, but not third point.

In the third element above, if response shows an otherwise correct substitution with unchanged limits of integration on a definite integral from 0 to 4, then point is not earned.

Alternately, the integral can be worked with as an indefinite integral with u returned to $3t + 4$ in the antiderivative before evaluating as a means to eliminate the need to change limits. Such a response can earn the first point if handled correctly.

For example: $\int_0^4 p'\ (3t+4)\ dt = \frac{1}{3} p\ (3t+4)\ \Big|_0^4$ earns the substitution point.

Second Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, with evidence of connection between $\int p'\ (u)$ and $p\ (u)$.

Ways to earn second point include:

· $k\,(p\ (16) - p\ (4))$ or $k\,(p\ (3 \cdot 4 + 4) - p\ (3 \cdot 0 + 4))$ for k a nonzero constant

· $kp\ (3t+4)\ \Big|_0^4$ or $kp\ (u)\ \Big|_4^{16}$ for k a nonzero constant

· $\int p'\ (3t+4)\ dt = kp\ (3t+4)$ or $\int p'\ (u)\ du = kp\,(u)$ for any nonzero constant k, with or without appropriate limits of integration on either integral

First and second points can be earned simultaneously by responses such as $\int_0^4 p'\ (3t+4)\ dt = \frac{1}{2} p\ (3t+4)\ \Big|_0^4$

Third Point

Earned only for an answer of -4.5 or equivalent value.

Value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring Examples:

· $\frac{1}{3} \int p'(u)\ du = \frac{1}{3} p(u)$ earns $0 - 1 - 0$

· $\frac{1}{3} \int p'\ (u)\ du = \frac{1}{3} p\ (u)\Big|_0^4 = \frac{1}{3}(p\ (4) - p\ (0))$ earns $0 - 1 - 0$

6-7

- $\int p'(u)\ du = p(u)\,\big|_4^{16} = (p(16) - p(4)) = -13.5$ earns $0-1-0$
- $3\int p'(u)\ du = 3p(u)\big|_4^{16} = 3[7.5 - 21]$ earns $0-1-0$
- $\int_0^4 p'(3t+4)\ dt = p\left(1.5t^2 + 4t\right)\big|_0^4 = p(40) - p(0)$ earns $0-0-0$
- $\frac{1}{3}\int_0^4 p'(u)\ du = \frac{1}{3}(p(16) - p(4)) = -4.5$ earns $0-1-1$
- $\int_0^4 p'(3t+4)\ dt = \frac{1}{3}(p(16) - p(4)) = -4.5$ earns $1-1-1$
- $\int_0^4 p'(3t+4)\ dt = \frac{1}{3}p(u)\big|_4^{16} = \frac{1}{3}(p(16) - p(4)) = -4.5$ earns $1-1-1$
- $\int_0^4 p'(3t+4)\ dt = \frac{1}{3}p(3t+4)\big|_0^4 = \frac{1}{3}(p(16) - p(4))$ earns $1-1-0$
- $\frac{1}{3}\int p'(u) = \frac{1}{3}p(u)$; and $\frac{1}{3}p(3t+4)\big|_0^4 = \frac{1}{3}(7.5 - 21)$ earns $1-1-1$

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- [ ] Substitution
- [ ] Fundamental Theorem of Calculus
- [ ] Answer

Solution:

Let $u = 3t + 4$. Then $du = 3\ dt$.

When $t = 0$, $u = 4$.

When $t = 4$, $u = 16$.

$$\int_0^4 p'(3t+4)\ dt = \frac{1}{3}\int_4^{16} p'(u)\ du$$

$$= \frac{1}{3}(p(16) - p(4))$$

6-7

$= \frac{1}{3}(7.5 - 21) = -4.5$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, resulting in $a \cdot p(6x)$ for some nonzero constant a.

Ways to earn first point include responses of $p(3 \cdot 2x)$ or $p(6x)$ (Note $p(6t)$ would also earn this point.)

The expression $p(3 \cdot 2x) - p(3 \cdot 1)$ or $(p(3 \cdot 2x) - p(3 \cdot 1)) \cdot 2$ earns this point. The $p(3 \cdot 1)$ is viewed as a chain rule error.

Any response correctly presenting and evaluating the derivative of the integral in terms of the substitution $u = 3t$ earns first point. Part of this response may include $\int_1^{2x} p(3t)\,dt = \frac{1}{3}\int_3^{6x} p(u)\,du$.

Second Point

Philosophy: This point is earned for correctly applying the chain rule to find $h'(x)$ or $h'(2)$. That is accomplished by multiplying the term involving $p(x)$ by 2.

Can only be earned if response earned first (FTC) point.

The first two points can be, and often are, earned simultaneously.

The point is not earned if the presented derivative is $a \cdot p(6x)$ for any $a \neq 2$.

The point is not earned for responses of $2p(3t)\,|_1^{2x}$ or $2(p(6x) - p(b))$, for any value of b.

Third Point

Can only be earned if response earned first (FTC) and second (chain rule) points.

Earned only for answer of 24 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring examples:

- $p(3 \cdot 2x) - p(3);\ p(12) - p(3)$ earns $1 - 0 - 0$
- $p(3 \cdot 2x) \cdot 2 - p(3) \cdot 0;\ 24$ earns $1 - 1 - 1$
- $2p(3t)\,|_1^{2x} = 2(p(12) - p(3))$ earns $1 - 0 - 0$
- $p(3x) \cdot 2;\ p(6) \cdot 2$ earns $0 - 0 - 0$

6-7

· $p\left(6x\right)\ \cdot\ 2;\ p\left(12\right)\ \cdot\ 2$ earns $1-1-0$

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'\left(x\right)=\frac{d}{dx}\int_{1}^{2x}p\left(3t\right)\,dt=p\left(3\ \cdot\ 2x\right)\ \cdot\ 2$$

$$h'\left(2\right)=p\left(12\right)\ \cdot\ 2=12\ \cdot\ 2=24$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for narrowing consideration of given information to g and g'.

g and g' need not be evaluated to earn the first point.

Considering velocity and acceleration not sufficient to earn first point, unless acceleration is linked to g'.

If response considers g and g' and g'', then eventually g'' must be eliminated to earn first point.

Second Point

Philosophy: Earned for “decreasing” with reason, using the opposite signs of g and g'.

Ways to earn second point include:

· “decreasing since $g<0$ and $g'>0$”

· “particle slowing down since g and g' have opposite signs”

Can only be earned if response earned first point.

Examples:

· “decreasing since velocity and acceleration have opposite signs” earns neither point

6-7

· "decreasing since velocity < 0 and acceleration > 0" earns neither point

"decreasing since $(2) \cdot (-4) = -8 < 0$" earns both points

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Narrows consideration to g and g'
- [ ] Answer with reason

Solution:

Because $g(3) = 2 > 0$, the velocity of the particle is positive. Because $g'(3) = -4 < 0$, the acceleration of the particle is negative. Because the velocity and acceleration have opposite signs, the speed of the particle is decreasing at time $t = 3$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct tangent line equation.

Correct responses include:

$y = 7(x-3) + 5$ or $y - 5 = 7(x-3)$

$y = 7x - 16$

$y = \left(-2 \cdot 3^2 + 5^2\right)(x-3) + 5$

Incorrect responses include:

$y = \left(-2x^2 + y^2\right)(x-3) + 5$

$y = -7x - 16$

$f(x) = 7x - 16$

0	1

The student response accurately includes the criteria below.

6-7

- [ ] Tangent line equation

Solution:

$$\left.\frac{dy}{dx}\right|_{(3,\,5)} = -2 \cdot 3^2 + 5^2 = 7$$

$$y = 7\,(x - 3) + 5$$

6-7

70. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	3	6	9	12
$p(t)$ (feet)	3	5.5	7	13	17

The position of a particle, A, moving along the x-axis is modeled by a twice-differentiable function p, where time t is measured in seconds and $p(t)$ is measured in feet. Selected values of $p(t)$ are shown in the table above.

(a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate the value of $\int_0^{12} p(t)\,dt$. Show the computations that lead to your answer.

(b) Use the data in the table to approximate $p'(10)$ using the average rate of change of $p(t)$ over the interval $9 \leq t \leq 12$. Show the computations that lead to your answer. Indicate units of measure.

(c) Interpret the meaning of $p'(10)$ in the context of the problem.

(d) Must there exist a value of c, for $0 < c < 12$, such that $p'(c)$ is equal to the average rate of change of p over the interval $0 \leq x \leq 12$? Justify your answer.

(e) Find $\int_0^3 p'(2t+3)\,dt$. Show the computations that lead to your answer.

(f) Let $h(x) = \int_2^{3x} p(3t)\,dt$. Find $h'(1)$. Show the computations that lead to your answer.

(g) The velocity of a second particle, Q, can be modeled by a twice-differentiable function g. It is known that $g(2) = -6$, $g'(2) = 4$, and $g''(2) = -3$. Is the speed of particle Q increasing or decreasing at time $t = 2$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = 4x^2y^3$ with initial condition $f\left(\frac{1}{2}\right) = -2$. Write an equation for the line tangent to the graph of f at the point $\left(\frac{1}{2}, -2\right)$.

Part A

6-7

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct midpoint Riemann sum, with appropriate values taken from the table.

Need to see sum of two terms, each as a product of two factors, i.e. $\text{base}_1 \cdot \text{height}_1 + \text{base}_2 \cdot \text{height}_2$

$p(3) \cdot (6-0) + p(9) \cdot (12-6)$ or $p(3) \cdot (6) + p(9) \cdot (6)$ earns first point.

If at least three of the four base/height values are correct in the sum of products, the expression earns the first point.

Second Point

111 or equivalent

Can be earned without first point with Riemann sum setup demonstration. With credible evidence that a midpoint Riemann sum attempt was made without not enough work shown, this point can be earned in the presence of the correct answer. For example, a sum of two terms with no evidence of products $(33+78)$ or a graph with areas labeled and summed to 111 or equivalent, can earn this point.

Examples:

· $111 = 33 + 78$ by itself earns second point, not first point

· 111 declared, with $5.5 \cdot 6$ and $13 \cdot 6$ identified (perhaps graphically) earns second point, not first point

· $k(5.5 \cdot 6 + 13 \cdot 6)$, nonzero constant $k \neq 1$ by itself earns first point, not second point

· $p(3) \cdot (6) + p(9) \cdot (12-6)$ by itself earns first point, not second point

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Form of midpoint Riemann sum
- [ ] Answer

Solution:

$$\int_0^{12} p(t)\,dt \approx p(3)(6-0) + p(9)(12-6)$$

$$= 5.5 \cdot 6 + 13 \cdot 6 = 111$$

6-7

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct difference quotient, with appropriate values from the table.

Need to see both a difference and a quotient

$\frac{p(12)-p(9)}{12-9}$ by itself doesn't earn first point; however $\frac{17-13}{12-9}$ or $\frac{17-13}{3}$ by itself earns first point

Second Point

Earned for correct units of ft per sec or ft/sec or f/s.

Correct units must be tied to a number to earn second point.

Correct units must appear in part (b); correct units in part (c) only doesn't earn second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$p'(10) \approx \frac{17-13}{12-9} = \frac{4}{3}$ feet per second

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct $p'(10)$ interpretation.

Need two components in response:

· "rate of change of position" or "velocity" of the particle

· at $t = 10$

Correct ways to describe rate at time at $t = 10$ include:

6-7

· Rate at which position is changing/increasing

· Rate at which particle is moving right

· Instantaneous velocity or velocity of particle

Ambiguous/incorrect ways to describe rate include:

· Rate at which particle is changing/increasing

· Rate at which particle is moving

Units are not required in part (c). If units are presented for t, they must be in seconds.

0	1

The student response accurately includes the criteria below.

- [ ] ·Interpretation

Solution:

$p'(10)$ is the rate at which the position of the particle is changing, in feet per second, at time $t = 10$ seconds.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for both Mean Value Theorem (MVT) conditions with the appropriate implication that p is continuous because we know that p is differentiable.

Ways to earn first point include:

· p differentiable $\Rightarrow$ p continuous

· p differentiable, hence p continuous

· p differentiable, so p continuous

p differentiable and p continuous does not earn first point.

Second Point

Philosophy: Earned for concluding "yes" and invoking Mean Value Theorem or its equivalent formulation, with consideration of the continuity of p.

6-7

MVT title does not have to be stated to earn second point.

This point is not earned if other theorems (IVT, etc.) are invoked.

If presented, the value for $p'(c) = \frac{7}{6}$ must be correct to earn second point.

Examples:

· “By MVT, yes since p differentiable” earns neither point

· “By MVT, yes since p continuous” earns second point only

· “By MVT, yes since p differentiable and continuous” earns second point only

· “By MVT, yes since p differentiable hence continuous” earns both points

· “By MVT, yes since p twice differentiable hence continuous” earns both points

· “By MVT, yes since function differentiable hence continuous” earns both points

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Conditions
- [ ] Conclusion using Mean Value Theorem

Solution:

p is differentiable. $\Rightarrow p$ is continuous.

Therefore, the Mean Value Theorem can be applied to p on the interval $0 \leq x \leq 12$ to guarantee that there exists a value of c, for $0 < c < 12$, such that $p'(c)$ equals the average rate of change of p over the interval $0 \leq x \leq 12$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct substitution application.

Need three elements in response (with consistent use of u or another substitution variable):

· $u = 2t + 3$ (can be explicit or implicit)

6-7

· $du = 2\ dt$ (yielding integral coefficient of $\frac{1}{2}$)

· Correct integral limits throughout, with t from 0 to 3 and u from 3 to 9

Omitting the coefficient of $\frac{1}{2}$ or using some other value will not earn first point; response could earn second point, but not third point.

In the third element above, if response shows an otherwise correct substitution with unchanged limits of integration on a definite integral from 0 to 3, then point is not earned.

Alternately, the integral can be worked with as an indefinite integral with u returned to $2t + 3$ in the antiderivative before evaluating as a means to eliminate the need to change limits. Such a response can earn the first point if handled correctly.

For example: $\int_0^3 p'\ (2t+3)\ dt = \frac{1}{2}p\ (2t+3)\ \big|_0^3$ earns the substitution point.

Second Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, with evidence of connection between $\int p'\ (u)$ and $p\ (u)$.

Ways to earn second point include:

· $k\,(p\ (9) - p\ (3))$ or $k\,(p\ (2\ \cdot\ 3+3) - p\ (2\ \cdot\ 0+3))$ for k a nonzero constant

· $kp\ (2t+3)\ |_0^3$ or $kp\ (u)\ |_3^9$ for k a nonzero constant

· $\int p'\ (2t+3)\ dt = kp\ (2t+3)$ or $\int p'\ (u)\ du = kp\,(u)$ for any nonzero constant k, with or without appropriate limits of integration on either integral

First and second points can be earned simultaneously by responses such as $\int_0^3 p'\ (2t+3)\ dt = \frac{1}{2}p\ (2t+3)\ \big|_0^3$

Third Point

Earned only for an answer of 3.75 or equivalent value.

Value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring Examples:

· $\frac{1}{2}\int p'(u)\ du = \frac{1}{2}p(u)$ earns $0-1-0$

· $\frac{1}{2}\int p'\ (u)\ du = \frac{1}{2}p\ (u)\big|_0^3 = \frac{1}{2}(p\ (3) - p\ (0))$ earns $0-1-0$

6-7

- $\int p'(u)\ du = p(u)|_3^9 = (p(9) - p(3)) = 7.5$ earns $0 - 1 - 0$
- $2\int p'(u)\ du = 2p(u)|_0^3 = 2[13 - 5.5]$ earns $0 - 1 - 0$
- $\int_0^3 p'(2t+3)\ dt = p\left(t^2 + 3t\right)\big|_0^3 = p(18) - p(0)$ earns $0 - 0 - 0$
- $\frac{1}{2}\int_0^3 p'(u)\ du = \frac{1}{2}(p(9) - p(3)) = 3.75$ earns $0 - 1 - 1$
- $\int_0^3 p'(2t+3)\ dt = \frac{1}{2}(p(9) - p(3)) = 3.75$ earns $1 - 1 - 1$
- $\int_0^3 p'(2t+3)\ dt = \frac{1}{2}p(u)\big|_3^9 = \frac{1}{2}(p(9) - p(3)) = 3.75$ earns $1 - 1 - 1$
- $\int_0^3 p'(2t+3)\ dt = \frac{1}{2}p(2t+3)\big|_0^3 = \frac{1}{2}(p(9) - p(3))$ earns $1 - 1 - 0$
- $\frac{1}{2}\int p'(u) = \frac{1}{2}p(u); \frac{1}{2}p(2t+3)\big|_0^3 = \frac{1}{2}(13 - 5.5)$ earns $1 - 1 - 1$

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t + 3$. Then $du = 2\ dt$.

When $t = 0$, $u = 3$.

When $t = 3$, $u = 9$.

$$\int_0^3 p'(2t+3)\ dt = \frac{1}{2}\int_3^9 p'(u)\ du$$

$$= \frac{1}{2}(p(9) - p(3))$$

$$= \frac{1}{2}(13 - 5.5) = 3.75$$

6-7

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, resulting in $a \cdot p(9x)$ for some nonzero constant a.

Ways to earn first point include responses of $p(3 \cdot 3x)$ or $p(9x)$ (Note $p(9t)$ would also earn this point.)

The expression $p(3 \cdot 3x) - p(3 \cdot 2)$ or $(p(3 \cdot 3x) - p(3 \cdot 2)) \cdot 3$ earns this point. The $p(3 \cdot 2)$ is viewed as a chain rule error.

Any response correctly presenting and evaluating the derivative of the integral in terms of the substitution $u = 3t$ earns first point. Part of this response may include $\int_2^{3x} p(3t)\, dt = \frac{1}{3}\int_6^{9x} p(u)\, du$.

Second Point

Philosophy: This point is earned for correctly applying the chain rule to find $h'(x)$ or $h'(1)$. That is accomplished by multiplying the term involving $p(x)$ by 3.

Can only be earned if response earned first (FTC) point.

The first two points can be, and often are, earned simultaneously.

The point is not earned if the presented derivative is $a \cdot p(9x)$ for any $a \neq 3$.

The point is not earned for responses of $3p(3t)\,|_2^{3x}$ or $3(p(9x) - p(b))$, for any value of b.

Third Point

Can only be earned if response earned first (FTC) and second (chain rule) points.

Earned only for answer of 39 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring examples:

- $p(3 \cdot 3x) - p(6);\ p(9) - p(6) = 13 - 7$ earns $1 - 0 - 0$
- $p(3 \cdot 3x) \cdot 3 - p(6) \cdot 0;\ 39$ earns $1 - 1 - 1$
- $3p(3t)\,|_2^{3x} = 3(p(9) - p(6))$ earns $1 - 0 - 0$
- $p(3x) \cdot 3;\ p(3) \cdot 3 = 5.5 \cdot 3$ earns $0 - 0 - 0$

6-7

· $p\left(9x\right)\ \cdot\ 3;\ \ p\left(9\right)\ \cdot\ 3$ earns $1-1-0$

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'\left(x\right)=\frac{d}{dx}\int_{2}^{3x}p\left(3t\right)\,dt=p\left(3\ \cdot\ 3x\right)\ \cdot\ 3$$

$$h'\left(1\right)=p\left(9\right)\ \cdot\ 3=13\ \cdot\ 3=39$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for narrowing consideration of given information to g and g'.

g and g' need not be evaluated to earn the first point.

Considering velocity and acceleration not sufficient to earn first point, unless acceleration is linked to g'.

If response considers g and g' and g'', then eventually g'' must be eliminated to earn first point.

Second Point

Philosophy: Earned for “decreasing” with reason, using the opposite signs of g and g'.

Ways to earn second point include:

· “decreasing since $g<0$ and $g'>0$ ”

· “particle slowing down since g and g' have opposite signs”

Can only be earned if response earned first point.

Examples:

· “decreasing since velocity and acceleration have opposite signs” earns neither point

6-7

· "decreasing since velocity < 0 and acceleration > 0 " earns neither point

· "decreasing since $(-6) \cdot 4 = -24 < 0$ " earns both points

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Narrows consideration to g and g'
- [ ] Answer with reason

Solution:

Because $g(2) = -6 < 0$, the velocity of the particle is negative. Because $g'(2) = 4 > 0$, the acceleration of the particle is positive. Because the velocity and acceleration have opposite signs, the speed of the particle is decreasing at time $t = 2$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct tangent line equation.

Correct responses include:

· $y = -8\left(x - \frac{1}{2}\right) - 2$ or $y + 2 = -8\left(x - \frac{1}{2}\right)$

· $y = -8x + 2$

· $y = \left(4 \cdot \left(\frac{1}{2}\right)^2 \cdot (-2)^3\right)\left(x - \frac{1}{2}\right) - 2$

Incorrect responses include:

· $y = \left(4x^2y^3\right)\left(x - \frac{1}{2}\right) - 2$

· $y = 8x + 2$

· $f(x) = -8x + 2$

0	1

The student response accurately includes the criteria below.

6-7

- [ ] Tangent line equation

Solution:

$$\left.\frac{dy}{dx}\right|_{\left(\frac{1}{2},\,-2\right)} = 4 \cdot \left(\frac{1}{2}\right)^2 \cdot (-2)^3 = -8$$

$$y = -8\left(x - \frac{1}{2}\right) - 2$$

6-7

71. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	4	10	12
$v(t)$ (meters per second)	2	6	15	20

The velocity of a particle, P, moving along the x-axis is modeled by a differentiable function v, where time t is measured in seconds and $v(t)$ is measured in meters per second. Selected values of $v(t)$ are shown in the table above.

(a) Use the data in the table to approximate $v'(11)$ using the average rate of change of $v(t)$ over the interval $10 \leq t \leq 12$. Show the computations that lead to your answer. Indicate units of measure.

(b) Interpret the meaning of $v'(11)$ in the context of the problem.

(c) Justify why there must be a time $t = c$, for $4 \leq c \leq 10$, when the velocity of the particle is 8 meters per second.

(d) Use a right Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^{12} v(t)\,dt$. Show the computations that lead to your answer.

(e) Find $\int_4^{12} v'\left(\frac{t}{2} - 2\right) dt$. Show the computations that lead to your answer.

(f) Let $h(x) = \int_2^{4x} v(3t)\,dt$. Find $h'(1)$. Show the computations that lead to your answer.

(g) The position of a second particle, Q, can be modeled by a twice-differentiable function g. It is known that $g(4) = 7$, $g'(4) = -3$, and $g''(4) = -2$. Is the speed of particle Q increasing or decreasing at time $t = 4$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = 3x^2 - y^3$ with initial condition $f(2) = 1$. Write an equation for the line tangent to the graph of f at the point $(2, 1)$.

Part A

6-7

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Approximation

Need to see both a difference and a quotient as the supporting work. Minimal answers that earn the point include $\frac{20-15}{12-10}$, $\frac{20-15}{2}$, and $\frac{5}{12-10}$.

The answer does not need to be simplified, but any attempt at simplification must be correct.

Numerical values must be pulled from the table. $\frac{v(12)-v(10)}{12-10}$ alone is not sufficient as it does not pull values from the table (yet).

Second point: Units

$\frac{m}{s^2}$ or meters per second per second will earn the point.

The units must be tied to a number. Units alone do not earn this point.

Beware of $\left(\frac{m}{s}\right)^2$ or $\frac{m^2}{s}$ (which do not earn the point).

This point must be earned in part (a). If no units are present in part (a) and correct units are shown in part (b), this point is not earned.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$v'(11) \approx \frac{20-15}{12-10} = \frac{5}{2} \text{ meters per second per second}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Interpretation

There are two components to look for:

· “rate of change of velocity” or “acceleration” of the particle

6-7

· at $t = 11$

"The rate at which the velocity is increasing at $t = 11$" is acceptable. If the answer to part (a) was reported to be negative, "the rate at which the velocity is decreasing at $t = 11$" is acceptable. We will not penalize for the same error twice.

Units are not required in part (b). If units are presented for t, they must be s (seconds).

0	1 ✓

The student response accurately includes the criteria below.

- [ ] Interpretation

Solution:

$v'(11)$ is the rate at which the velocity of the particle is changing, in meters per second per second, at time $t = 11$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: v is continuous

It must be clear that the reason given as to why v is continuous is that v is differentiable.

Neither "v is differentiable and continuous" nor "v differentiable, v continuous" are sufficient to earn this point.

"v is differentiable hence continuous," by contrast, earns the point.

"v differentiable $\therefore$ v continuous" earns the point.

Second point: Bounding with table values

Must see the values 6 and 15 pulled from the table and correctly compared with 8, in words or using inequality symbols.

The Intermediate Value Theorem (IVT) does not need to be referenced by name.

That said, this point will not be earned if other theorems (MVT, etc.) are invoked.

Responses often "say too much"; especially common are responses that indicate that v is always increasing on the interval $[6, \ 15]$. This point will not be earned if the response contains such errors in reasoning.

6-7

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] v continuous
- [ ] Bounding 8 with table values

Solution:

v is differentiable. $\Rightarrow v$ is continuous.

$v(4) = 6 < 8 < 15 = v(10)$

By the Intermediate Value Theorem, there must be a time $t = c$, for $4 < c < 10$, such that the velocity of the particle is 8 meters per second.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Form of right Riemann sum

Must see a sum of three terms, each of which is a product of two factors: $(b_1 \cdot h_1) + (b_2 \cdot h_2) + (b_3 \cdot h_3)$.

If exactly 5 out of 6 of these numbers $(b_1, \ b_2, \ b_3, \ h_1, \ h_2, \ h_3)$ are correctly placed in the above expression, we are convinced that the response carries the form of a right Riemann sum. In this case, award the first point. Such a response cannot earn the second point. If fewer than 5 out of 6 values are correct, no points are earned in this problem.

To earn this point, values do not need yet to be imported from the table: the expression $v(4) \cdot (4 - 0) + v(10) \cdot (10 - 4) + v(12) \cdot (12 - 10)$ is sufficient.

Once the first point is earned, it cannot be lost. Subsequent errors result in the second point not being earned.

Second point: Answer

154 or equivalent.

It is possible to earn this point without earning the first point. If there is credible evidence that an attempt was made to find a right Riemann sum but not enough work is shown, this point can be earned in the presence of the correct answer. For example, a sum of three terms with no evidence of products $(24 + 90 + 40)$, or a graph with areas labeled and summed to 154 or equivalent, can earn this point.

The response of only $6 \cdot 4 + 15 \cdot 6 + 20 \cdot 2$ earns both points.

6-7

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Form of right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{12} v(t)\, dt \approx v(4)(4-0) + v(10)(10-4) + v(12)(12-10)$$

$$= 6 \cdot 4 + 15 \cdot 6 + 20 \cdot 2 = 154$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Substitution

There are three elements to look for in a correct substitution. (another variable may be chosen instead of u.)

$\cdot\, u = \frac{t}{2} - 2$

$\cdot\, du = \dfrac{dt}{2}$ or equivalent

$\cdot$ Appropriate limits of integration throughout (0 and 4 for u, 4 and 12 for t).

Substitution can be implicit. A coefficient of 2 accompanying a correct antiderivative convinces us of this:

$$\int_4^{12} v'\left(\frac{t}{2} - 2\right) dt = 2v\left(\tfrac{t}{2} - 2\right)\Big|_4^{12}$$

To elaborate on the third element above: if a response shows an otherwise correct substitution but the limits of integration on a definite integral are unchanged (4 and 12), this point is not earned.

Alternately, the integral can be worked with as an indefinite integral with the variable returned to $\frac{t}{2} - 2$ before evaluating as a means of eliminating the need to change limits. Such a response can earn the first point, if handled correctly.

Second point: Fundamental Theorem of Calculus

We must be convinced that the response has indicated an appropriate connection between $\displaystyle\int v'$ and v.

6-7

Expressions/equations that earn this point are:

· $k\left(v\left(4\right)-v\left(0\right)\right)$ or $k\left(v\left(\frac{12}{2}-2\right)-v\left(\frac{4}{2}-2\right)\right)$, where k is a nonzero constant.

· $kv\left(\frac{t}{2}-2\right)\Big|_{4}^{12}$ or $k\,v\left(u\right)\Big|_{0}^{4}$, where k is a nonzero constant.

· $\int v'\left(\frac{t}{2}-2\right)\ dt=kv\left(\frac{t}{2}-2\right)$ or $k\int v'\left(u\right)\ \ du=kv\left(u\right)$, where k is a nonzero constant, with or without appropriate limits of integration displayed on either integral.

· $\int_{4}^{12} v'\left(\frac{t}{2}-2\right)\ dt=k\,v\left(\frac{t^2}{4}-2t\right)$ where k is a nonzero constant is a common response. This does not earn the point. Likewise, similar mishandled antiderivatives do not earn the point.

Third point: Answer

8 or equivalent, with credible supporting work, earns this point.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u=\frac{t}{2}-2$. Then $du=\frac{dt}{2}$.

When $t=4$, $u=0$.

When $t=12$, $u=4$.

$$\int_{4}^{12} v'\left(\frac{t}{2}-2\right)\ dt=2\int_{0}^{4} v'\left(u\right)\ du$$

$$=2\left(v\left(4\right)-v\left(0\right)\right)$$

$$=2(6-2)=8$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

6-7

First point: Fundamental Theorem of Calculus

Substitutes $4x$ into $v(3t)$. We must see either $v(3 \cdot 4x)$ or $v(12x)$.

The expression $v(3 \cdot 4x) - v(3(2))$ or $v(3 \cdot 4x) \cdot 4 - v(3(2))$ will earn this point. The error of subtracting $v(3(2))$ will be viewed as an error in applying the chain rule.

Beware of possible substitutions prior to using FTC:

· For example, $u = 3t$, $\frac{1}{3}du = dt$ and rewrites: $\int_2^{4x} v(3t)\ dt = \frac{1}{3}\int_6^{12x} v(u)\ du.$

· Or $u = 4x$, $\frac{du}{dx} = 4$ and rewrites: $\frac{d}{dx}[h(x)] = \frac{d}{du}[h(u)] \cdot \frac{du}{dx}$

· Such approaches can earn this point if done properly.

Second point: Chain rule

Multiplies by 4

It is not possible to earn this point without having earned the first point.

Depending on substitutions, this may look more like $\frac{1}{3}$ and 12 multiplied separately.

Third point: Answer

80 or equivalent

It is not possible to earn this point without having earned both the first and second points.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx}\int_2^{4x} v(3t)\ dt = v(3 \cdot 4x) \cdot 4$$

$$h'(1) = v(12) \cdot 4 = 20 \cdot 4 = 80$$

6-7

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Narrows consideration to g' and g''

You must be convinced the response considers only the functions g' and g'' as being critical to the reasoning.

Second point: Answer with reason

A correct response must:

· Conclude "increasing"

· Appeal to the signs of g' and g''

It is not possible to earn this point without having earned the first point.

A response that mentions velocity and acceleration must tie velocity to g' and acceleration to g''. A response that says only "increasing because velocity and acceleration are both negative" earns neither of the two points. It is not clear that the response is considering g' and g''. A response that calls out all three given values ($g\ (4)$, $g'\ (4)$, and $g''\ (4)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g' and g'' only, both points can be earned.

The phrase "speeding up" will be interpreted as "increasing."

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g' and g''
- ☐ Answer with reason

Solution:

Because $g'\ (4) = -3 < 0$, the velocity of the particle is negative. Because $g''(4) = -2 < 0$, the acceleration of the particle is negative. Therefore, because both the velocity and acceleration are negative, the speed of the particle is increasing at time $t = 4$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Tangent line equation

6-7

This point is earned only for the correct tangent line equation written in any form.

Some common forms:

- $y - 1 = \left(3(2)^2 - (1)^3\right)(x - 2)$
- $y - 1 = 11\,(x - 2)$
- $y = 11x - 21$

f does not earn the point. This is not the equation of a line. The response can go on present one of the correct linear forms, and then would earn the point.

$f\,(x) = 11\,(x - 2) + 1$ or equivalent does not earn the point. $f\,(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

Some responses have the correct tangent line equation written initially, with subsequent errors in simplification. Such responses do not earn the point.

The point can be earned after pursuing a less fruitful approach (such as solving the differential equation) and then restarting.

0	1 ✓

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left.\frac{dy}{dx}\right|_{(2,\,1)} = 3 \cdot 2^2 - 1^3 = 11$$

$$y = 11\,(x - 2) + 1$$

6-7

72. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	3	8	11
$v(t)$ (meters per second)	-15	-12	-8	-3

The velocity of a particle, P, moving along the x-axis is modeled by a differentiable function v, where time t is measured in seconds and $v(t)$ is measured in meters per second. Selected values of $v(t)$ are shown in the table above.

(a) Use the data in the table to approximate $v'(10)$ using the average rate of change of $v(t)$ over the interval $8 \le t \le 11$. Show the computations that lead to your answer. Indicate units of measure.

(b) Interpret the meaning of $v'(10)$ in the context of the problem.

(c) Justify why there must be a time $t = k$, for $0 \le k \le 3$, when the velocity of the particle is -13 meters per second.

(d) Use a right Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^{11} v(t)\, dt$. Show the computations that lead to your answer.

(e) Find $\int_4^{14} v'\left(\frac{t}{2}+1\right) dt$. Show the computations that lead to your answer.

(f) Let $h(x) = \int_3^{\frac{11}{2}x} v(2t)\, dt$. Find $h'(1)$. Show the computations that lead to your answer.

(g) The position of a second particle, Q, can be modeled by a twice-differentiable function g. It is known that $g(2) = -5.5$, $g'(2) = 4$, and $g''(2) = 3$. Is the speed of particle Q increasing or decreasing at time $t = 2$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = y^2 - xy$ with initial condition $f(3) = 2$. Write an equation for the line tangent to the graph of f at the point $(3, 2)$.

Part A

6-7

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Approximation.

Need to see both a difference and a quotient as the supporting work. Minimal answers that earn the point include $\frac{-3+8}{11-8}$, $\frac{-3-(-8)}{3}$, and $\frac{5}{11-8}$.

The answer does not need to be simplified, but any attempt at simplification must be correct.

Numerical values must be pulled from the table. $\frac{v(11)-v(8)}{11-8}$ alone will not earn the point as it does not pull values from the table (yet).

Second point: Units

$\frac{m}{s^2}$ end fraction or meters per second per second will earn the point.

The units must be tied to a number. Units alone do not earn this point.

Beware of $\left(\frac{m}{s}\right)^2$ or $\frac{m^2}{s}$ (which do not earn the point).

This point must be earned in part (a). If no units are present in part (a) and correct units are shown in part (b), this point is not earned.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$v'(10) \approx \frac{-3-(-8)}{11-8} = \frac{5}{3} \text{ meters per second per second}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Interpretation

There are two components to look for:

6-7

· "rate of change of velocity" or "acceleration" of the particle

· at $t = 10$

"The rate at which the velocity is increasing at $t = 10$" is acceptable. If the answer to part (a) was reported to be negative, "the rate at which the velocity is decreasing at $t = 10$" is acceptable. We will not penalize for the same error twice.

Units are not required in part (b). If units are presented for t, they must be s (seconds).

0	1 ✓

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$v'(10)$ is the rate at which the velocity of the particle is changing, in meters per second per second, at time $t = 10$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: v is continuous

It must be clear that the reason given as to why v is continuous is that v is differentiable.

Neither "v is differentiable and continuous" nor "v differentiable, v continuous" are sufficient to earn this point.

"v is differentiable hence continuous," by contrast, earns the point.

"v differentiable $\therefore$ v continuous" earns the point.

Second point: Bounding with table values

Must see the values -15, and -12 pulled from the table and correctly compared to -13, in words or using inequality symbols.

The Intermediate Value Theorem (IVT) does not need to be referenced by name.

That said, this point will not be earned if other theorems (MVT, etc.) are invoked.

Responses often "say too much"; especially common are responses that indicate that v is always increasing on the interval $[0, 3]$. This point will not be earned if the response contains such errors in reasoning.

6-7

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] v continuous
- [ ] Bounding -13 with table values

Solution:

v is differentiable. $\Rightarrow v$ is continuous.

$v(0) = -15 < -13 < -12 = v(3)$

By the Intermediate Value Theorem, there must be a time $t = k$, for $0 \leq k \leq 3$, such that the velocity of the particle is -13 meters per second.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Form of right Riemann sum

Must see a sum of three terms, each of which is a product of two factors: $(b_1 \cdot h_1) + (b_2 \cdot h_2) + (b_3 \cdot h_3)$.

If exactly 5 out of 6 of these numbers $(b_1, b_2, b_3, h_1, h_2, h_3)$ are correctly placed in the above expression, we are convinced that the response carries the form of a right Riemann sum. In this case, award the first point. Such a response cannot earn the second point. If fewer than 5 out of 6 values are correct, no points are earned in this problem.

To earn this point, values do not need yet to be imported from the table: the expression $v(3) \cdot (3-0) + v(8) \cdot (8-3) + v(11) \cdot (11-8)$ is sufficient.

Once the first point is earned, it cannot be lost. Subsequent errors result in the second point not being earned.

second point: Answer

-85 or equivalent.

It is possible to earn this point without earning the first point. If there is credible evidence that an attempt was made to find a right Riemann sum but not enough work is shown, this point can be earned in the presence of the correct answer. For example, a sum of three terms with no evidence of products $(-36 + (-40) + (-9))$, or a graph with areas labeled and summed to -85 or equivalent, can earn this point.

The response of only $(-12) \cdot 3 + (-8) \cdot 5 + (-3) \cdot 3$ earns both points.

6-7

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Form of right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{11} v\,(t)dt \approx v\,(3)\,(3-0) + v\,(8)\,(8-3) + v\,(11)\,(11-8)$$

$$= -12 \cdot 3 + (-8) \cdot 5 + (-3) \cdot 3 = -85$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Substitution

There are three elements to look for in a correct substitution. (another variable may be chosen instead of u.)

$\cdot\, u = \frac{t}{2} + 1$

$\cdot\, du = \dfrac{dt}{2}$ or equivalent

$\cdot$ Appropriate limits of integration throughout (3 and 8 for u, 4 and 14 for t).

Substitution can be implicit. A coefficient of 2 accompanying a correct antiderivative convinces us of this:

$$\int_4^{14} v'\left(\frac{t}{2}+1\right)dt = 2v\left(\tfrac{t}{2}+1\right)\Big|_4^{14}$$

To elaborate on the third element above: if a response shows an otherwise correct substitution but the limits of integration on a definite integral are unchanged (4 and 14), this point is not earned.

Alternately, the integral can be worked with as an indefinite integral with the variable returned to $\frac{t}{2}+1$ before evaluating as a means of eliminating the need to change limits. Such a response can earn the first point, if handled correctly.

Second point: Fundamental Theorem of Calculus

We must be convinced that the response has indicated an appropriate connection between $\displaystyle\int v'$ and v.

6-7

Expressions/equations that earn this point are:

· $k\left(v\left(8\right)-v\left(3\right)\right)$ or $k\left(v\left(\frac{14}{2}+1\right)-v\left(\frac{4}{2}+1\right)\right)$, where k is a nonzero constant.

· $kv\left(\frac{t}{2}+1\right)\Big|_4^{14}$ or $k\,v\left(u\right)\Big|_3^8$, where k is a nonzero constant.

· $\displaystyle\int v'\left(\frac{t}{2}+1\right)dt = kv\left(\frac{t}{2}+1\right)$ or $\displaystyle k\int v'\left(u\right)du = kv\left(u\right)$, where k is a nonzero constant, with or without appropriate limits of integration displayed on either integral.

· $\displaystyle\int_4^{14} v'\left(\frac{t}{2}+1\right)dt = k\,v\left(\frac{t^2}{4}+t\right)$ where k is a nonzero constant is a common response. This does not earn the point. Likewise, similar mishandled antiderivatives do not earn the point.

Third point: Answer

8 or equivalent, with credible supporting work, earns this point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Substitution
- [ ] Fundamental Theorem of Calculus
- [ ] Answer

Solution:

Let $u=\frac{t}{2}+1$. Then $du=\frac{dt}{2}$.

When $t=4$, $u=3$.

When $t=14$, $u=8$.

$$\int_4^{14} v'\left(\frac{t}{2}+1\right)dt = 2\int_3^8 v'\left(u\right)du$$

$$=2\left(v\left(8\right)-v\left(3\right)\right)$$

$$=2\left(-8-\left(-12\right)\right)=8$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

6-7

First point: Fundamental Theorem of Calculus (FTC)

Substitutes $\frac{11}{2}x$ into $v(2t)$. We must see either $v\left(2\cdot\frac{11}{2}x\right)$ or $v(11x)$.

The expression $v\left(2\cdot\frac{11}{2}x\right) - v\left(2\left(3\right)\right)$ or $v\left(2\cdot\frac{11}{2}x\right)\cdot\frac{11}{2} - v\left(2\left(3\right)\right)$ will earn this point. The error of subtracting $v\left(2\left(3\right)\right)$ will be viewed as an error in applying the chain rule.

Beware of possible substitutions prior to using FTC:

· For example, $u = 2t$, $\frac{1}{2}du = dt$ and rewrites: $\int_3^{\frac{11}{2}x} v\left(2t\right)dt = \frac{1}{2}\int_6^{11x} v\left(u\right)du.$

· Or $u = \frac{11}{2}x$, $\frac{du}{dx} = \frac{11}{2}$ and rewrites: $\frac{d}{dx}\left[h\left(x\right)\right] = \frac{d}{du}\left[h\left(u\right)\right]\cdot\frac{du}{dx}$

· Such approaches can earn this point if done properly.

Second point: Chain rule

Multiplies by $\frac{11}{2}$

It is not possible to earn this point without having earned the first point.

Depending on substitutions, this may look more like $\frac{1}{2}$ and 11 multiplied separately.

Third point: Answer

$-\frac{33}{2}$ or equivalent

It is not possible to earn this point without having earned both the first and second points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'\left(x\right) = \frac{d}{dx}\int_3^{\frac{11}{2}x} v\left(2t\right)\,dt = v\left(2\cdot\frac{11}{2}x\right)\cdot\frac{11}{2}$$

6-7

$h'(1) = v(11) \cdot \frac{11}{2} = -3 \cdot \frac{11}{2} = -\frac{33}{2}$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Narrows consideration to g' and g''

You must be convinced the response considers only the functions g' and g'' as being critical to the reasoning.

Second point: Answer with reason

A correct response must:

· Conclude "increasing"

· Appeal to the signs of g' and g''

It is not possible to earn this point without having earned the first point.

A response that mentions velocity and acceleration must tie velocity to g' and acceleration to g''. A response that says only "increasing because velocity and acceleration are both negative" earns neither of the two points. It is not clear that the response is considering g' and g''. A response that calls out all three given values ($g(2)$, $g'(2)$, and $g''(2)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g' and g'' only, both points can be earned.

The phrase "speeding up" will be interpreted as "increasing."

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g' and g''
- ☐ Answer with reason

Solution:

Because $g'(2) = 4 > 0$, the velocity of the particle is positive. Because $g''(2) = 3 > 0$, the acceleration of the particle is positive. Therefore, because both the velocity and acceleration are positive, the speed of the particle is increasing at time $t = 2$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-7

First point: Tangent line equation

This point is earned only for the correct tangent line equation written in any form.

Some common forms:

- $y - 2 = \left((2)^2 - 3 \cdot 2\right)(x - 3)$
- $y - 2 = -2(x - 3)$
- $y = -2x + 8$

$y - 2 = \left(y^2 - xy\right)(x - 3)$ does not earn the point. This is not the equation of a line. The response can go on to present one of the correct linear forms, and then would earn the point.

$f(x) = -2(x - 3) + 2$ or equivalent does not earn the point. $f(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

Some responses have the correct tangent line equation written initially, with subsequent errors in simplification. Such responses do not earn the point.

The point can be earned after pursuing a less fruitful approach (such as solving the differential equation) and then restarting.

0	1 ✓

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left.\frac{dy}{dx}\right|_{(3,2)} = 2^2 - 3 \cdot 2 = -2$$

$$y = -2(x - 3) + 2$$

6-7

73. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	3	8	11
$v(t)$ (meters per second)	-2	-5	-11	-12

The velocity of a particle, P, moving along the x-axis is modeled by a differentiable function v, where time t is measured in seconds and $v(t)$ is measured in meters per second. Selected values of $v(t)$ are shown in the table above.

(a) Use the data in the table to approximate $v'(5)$ using the average rate of change of $v(t)$ over the interval $3 \le t \le 8$. Show the computations that lead to your answer. Indicate units of measure.

(b) Interpret the meaning of $v'(5)$ in the context of the problem.

(c) Justify why there must be a time $t = d$, for $0 \le d \le 3$, when the velocity of the particle is -4 meters per second.

(d) Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^{11} v(t)\, dt$. Show the computations that lead to your answer.

(e) Find $\int_0^{10} v'\left(\frac{t}{2} + 3\right) dt$. Show the computations that lead to your answer.

(f) Let $r(x) = \int_0^{\frac{3}{2}x} v(2t)\, dt$. Find $r'(1)$. Show the computations that lead to your answer.

(g) The position of a second particle, Q, can be modeled by a twice-differentiable function g. It is known that $g(3) = -4$, $g'(3) = -2$, and $g''(3) = 6$. Is the speed of particle Q increasing or decreasing at time $t = 3$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = 3x^2 - y^3$ with initial condition $f(-1) = 2$. Write an equation for the line tangent to the graph of f at the point $(-1, 2)$.

Part A

6-7

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Approximation.

Need to see both a difference and a quotient as the supporting work. Minimal answers that earn the point include $\frac{-11-(-5)}{8-3}$, $\frac{-11-(-5)}{5}$, and $\frac{-6}{8-3}$.

The answer does not need to be simplified, but any attempt at simplification must be correct.

Numerical values must be pulled from the table. $\frac{v(8)-v(3)}{8-3}$ alone will not earn the point as it does not pull values from the table (yet).

Second point: Units

$\frac{m}{s^2}$ or meters per second per second will earn the point.

The units must be tied to a number. Units alone do not earn this point.

Beware of $\left(\frac{m}{s}\right)^2$ or $\frac{m^2}{s}$ (which do not earn the point).

This point must be earned in part (a). If no units are present in part (a) and correct units are shown in part (b), this point is not earned.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Approximation
- [ ] Units

Solution:

$$v'(5) \approx \frac{-11-(-5)}{8-3} = -\frac{6}{5} \text{ meters per second per second}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Interpretation

There are two components to look for:

6-7

· “rate of change of velocity” or “acceleration” of the particle

· at $t = 5$

“The rate at which the velocity is decreasing at $t = 5$” is acceptable. If the answer to part (a) was reported to be positive, “the rate at which the velocity is increasing at $t = 5$” is acceptable. We will not penalize for the same error twice.

Responses that say “velocity is decreasing at a rate of $-\frac{6}{5}\,\frac{m}{s^2}$” or similar do not earn the point. This is interpreted to mean that the velocity is increasing.

Units are not required in part (b). If units are presented for t, they must be s (seconds).

0	1

The student response accurately includes the criteria below.

- [ ] Interpretation

Solution:

$v'\,(5)$ is the rate at which the velocity of the particle is changing, in meters per second per second, at time $t = 5$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: v is continuous

It must be clear that the reason given as to why v is continuous is that v is differentiable.

Neither “ v is differentiable and continuous” nor “v differentiable, v continuous” are sufficient to earn this point.

“v is differentiable hence continuous,” by contrast, earns the point.

"v differentiable $\therefore$ v continuous” earns the point.

Second point: Bounding with table values

Must see the values -5 and -2 pulled from the table and correctly compared with -4, in words or using inequality symbols.

The Intermediate Value Theorem (IVT) does not need to be referenced by name.

That said, this point will not be earned if other theorems (MVT, etc.) are invoked.

6-7

Responses often “say too much”; especially common are responses that indicate that v is always decreasing on the interval $[0, \ 3]$. This point will not be earned if the response contains such errors in reasoning.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] v continuous
- [ ] Bounding -4 with table values

Solution:

v is differentiable. $\Rightarrow$ v is continuous.

$v(3) = -5 < -4 < -2 = v(0)$

By the Intermediate Value Theorem, there must be a time $t = d$, $0 < d < 3$, such that the velocity of the particle is -4 meters per second.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Form of left Riemann sum

Must see a sum of three terms, each of which is a product of two factors: $(b_1 \cdot h_1) + (b_2 \cdot h_2) + (b_3 \cdot h_3)$

If exactly 5 out of 6 of these numbers $(b_1, \ b_2, \ b_3, \ h_1, \ h_2, \ h_3)$ are correctly placed in the above expression, we are convinced that the response carries the form of a left Riemann sum. In this case, award the first point. Such a response cannot earn the second point. If fewer than 5 out of 6 values are correct, no points are earned in this problem.

To earn this point, values do not need yet to be imported from the table: the expression $v(0) \cdot (3-0) + v(3) \cdot (8-3) + v(8) \cdot (11-8)$ is sufficient.

Once the first point is earned, it cannot be lost. Subsequent errors result in the second point not being earned.

Second point: Answer

-64 or equivalent.

It is possible to earn this point without earning the first point. If there is credible evidence that an attempt was made to find a left Riemann sum but not enough work is shown, this point can be earned in the presence of the correct answer. For example, a sum of three terms with no evidence of products $(-6 + -25 + -33)$ or a graph with areas labeled and summed to -64 or equivalent, can earn this point.

6-7

The response of only $(-2 \cdot 3) + (-5 \cdot 5) + (-11 \cdot 3)$ earns both points.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Form of left Riemann sum
- [ ] Answer

Solution:

$$\int_0^{11} v(t)\ dt \approx v(0)(3-0) + v(3)(8-3) + v(8)(11-8)$$

$$= -2 \cdot 3 + (-5) \cdot 5 + (-11) \cdot 3 = -64$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Substitution

There are three elements to look for in a correct substitution. (another variable may be chosen instead of u.)

$\cdot\, u = \frac{t}{2} + 3$

$\cdot\, du = \frac{dt}{2}$ or equivalent

$\cdot$ Appropriate limits of integration throughout (3 and 8 for u, 0 and 10 for t).

Substitution can be implicit. A coefficient of 2 accompanying a correct antiderivative convinces us of this:

$$\int_0^{10} v'\left(\frac{t}{2}+3\right)\ dt = 2v\left(\tfrac{t}{2}+3\right)\Big|_0^{10}$$

To elaborate on the third element above: if a response shows an otherwise correct substitution but the limits of integration on a definite integral are unchanged (0 and 10), this point is not earned.

Alternately, the integral can be worked with as an indefinite integral with the variable returned to $\frac{t}{2} + 3$ before evaluating as a means of eliminating the need to change limits. Such a response can earn the first point, if handled correctly.

Second point: Fundamental Theorem of Calculus

We must be convinced that the response has indicated an appropriate connection between $\int v'$ and v.

6-7

Expressions/equations that earn this point are:

$k\left(v\left(8\right)-v\left(3\right)\right)$ or $k\left(v\left(\frac{10}{2}+3\right)-v\left(\frac{0}{2}+3\right)\right)$,

· where k is a nonzero constant.

· $kv\left(\frac{t}{2}+3\right)\Big|_0^{10}$ or $k\,v\left(u\right)\Big|_3^8$, where k is a nonzero constant.

· $\int v'\left(\frac{t}{2}+3\right)dt=kv\left(\frac{t}{2}+3\right)$ or $k\int v'\left(u\right)\ du=kv\left(u\right)$, where k is a nonzero constant, with or without appropriate limits of integration displayed on either integral.

· $\int_0^{10} v'\left(\frac{t}{2}+3\right)dt=k\,v\left(\frac{t^2}{4}+3t\right)$ where k is a nonzero constant is a common response. This does not earn the point. Likewise, similar mishandled antiderivatives do not earn the point.

Third point: Answer

-12 or equivalent, with credible supporting work, earns this point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Substitution
- [ ] Fundamental Theorem of Calculus
- [ ] Answer

Solution:

Let $u=\frac{t}{2}+3$. Then $du=\frac{dt}{2}$.

When $t=0$, $u=3$.

When $t=10$, $u=8$.

$$\int_0^{10} v'\left(\frac{t}{2}+3\right)\ dt=2\int_3^8 v'\left(u\right)\ du$$

$=2\left(v\left(8\right)-v\left(3\right)\right)$

$=2\left(-11-\left(-5\right)\right)=-12$

Part F

6-7

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Fundamental Theorem of Calculus

Substitutes $\frac{3}{2}x$ into $v\left(2t\right)$. We must see either $v\left(2\cdot\frac{3}{2}x\right)$ or $v\left(3x\right)$.

The expression $v\left(2\cdot\frac{3}{2}x\right)-v\left(2\left(0\right)\right)$ or $v\left(2\cdot\frac{3}{2}x\right)\cdot\frac{3}{2}-v\left(2\left(0\right)\right)$ will earn this point. The error of subtracting $v\left(2\left(0\right)\right)$ will be viewed as an error in applying the chain rule.

Beware of possible substitutions prior to using FTC:

· For example, $u=2t$, $\frac{1}{2}du=dt$ and rewrites: $\int_0^{\frac{3}{2}x} v\left(2t\right)\ dt=\frac{1}{2}\int_0^{3x} v\left(u\right)\ du.$

· Or $u=\frac{3}{2}x$, $\frac{du}{dx}=\frac{3}{2}$ and rewrites: $\frac{d}{dx}[h\left(x\right)]=\frac{d}{du}[h\left(u\right)]\cdot\frac{du}{dx}$

· Such approaches can earn this point if done properly.

Second point: Chain rule

Multiplies by $\frac{3}{2}$.

It is not possible to earn this point without having earned the first point.

Depending on substitutions, this may look more like $\frac{1}{2}$ and 3 multiplied separately.

Third point: Answer

$-\frac{15}{2}$ or equivalent

It is not possible to earn this point without having earned both the first and second points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

6-7

$$r'(x) = \frac{d}{dx}\int_0^{\frac{3}{2}x} v(2t)\,dt = v\left(2 \cdot \tfrac{3}{2}x\right) \cdot \tfrac{3}{2}$$

$$r'(1) = v(3) \cdot \frac{3}{2} = -5 \cdot \frac{3}{2} = -\frac{15}{2}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Narrows consideration to g' and g''

You must be convinced the response considers only the functions g' and g'' as being critical to the reasoning.

Second point: Answer with reason

A correct response must:

· Conclude “decreasing”

· Appeal to the signs of g' and g''

It is not possible to earn this point without having earned the first point.

A response that mentions velocity and acceleration must tie velocity to g' and acceleration to g'' . A response that says only “decreasing because velocity and acceleration have different signs” earns neither of the two points. It is not clear that the response is considering g' and g''

A response that calls out all three given values ($g(3)$, $g'(3)$, and $g''(3)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g' and g'' only, both points can be earned.

The phrase “slowing down” will be interpreted as “decreasing.”

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g' and g''
- ☐ Answer with reason

Solution:

Because $g'(3) = -2 < 0$, the velocity of the particle is negative. Because $g''(3) = 6 > 0$, the acceleration of the particle is positive. Because the signs of the velocity and acceleration are different, the speed of the particle is decreasing at time $t = 3$.

6-7

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Tangent line equation

This point is earned only for the correct tangent line equation written in any form.

Some common forms:

· $y-2=\left(3(-1)^2-2^3\right)(x+1)$

· $y-2=-5(x+1)$

· $y=-5x-3$

$y-2=\left(3x^2-y^3\right)(x+1)$ does not earn the point. This is not the equation of a line. The response can go on to present one of the correct linear forms, and then would earn the point.

$f(x)=-5(x+1)+2$ or equivalent does not earn the point. $f(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

Some responses have the correct tangent line equation written initially, with subsequent errors in simplification. Such responses do not earn the point.

The point can be earned after pursuing a less fruitful approach (such as solving the differential equation) and then restarting.

0	1 ✓

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left.\frac{dy}{dx}\right|_{(-1,2)}=3\cdot(-1)^2-2^3=-5$$

$$y=2-5(x+1)$$

6-7

74. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	2	8	14
$v(t)$ (meters per second)	15	12	6	0

The velocity of a particle, P, moving along the x-axis is modeled by a differentiable function v, where time t is measured in seconds and $v(t)$ is measured in meters per second. Selected values of $v(t)$ are shown in the table above.

(a) Use the data in the table to approximate $v'(6)$ using the average rate of change of $v(t)$ over the interval $2 \leq t \leq 8$. Show the computations that lead to your answer. Indicate units of measure.

(b) Interpret the meaning of $v'(6)$ in the context of the problem.

(c) Justify why there must be a time $t = k$, for $8 \leq k \leq 14$, when the velocity of the particle is 3 meters per second.

(d) Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^{14} v(t)\,dt$ Show the computations that lead to your answer.

(e) Find $\int_3^{21} v'\left(\frac{t}{3}+7\right)dt$. Show the computations that lead to your answer.

(f) Let $p(x) = \int_0^{4x} v(2t)\,dt$. Find $p'(1)$. Show the computations that lead to your answer.

(g) The position of a second particle, Q, can be modeled by a twice-differentiable function g. It is known that $g(4) = 5$, $g'(4) = 2$, and $g''(4) = -6$. Is the speed of particle Q increasing or decreasing at time $t = 4$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -2x + 3y^2$ with initial condition $f(3) = 2$. Write an equation for the line tangent to the graph of f at the point $(3, 2)$.

Part A

6-7

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Approximation.

Need to see both a difference and a quotient as the supporting work. Minimal answers that earn the point include $\frac{6-12}{8-2}$, $\frac{6-12}{6}$, and $-\frac{6}{8-2}$.

The answer does not need to be simplified, but any attempt at simplification must be correct.

Numerical values must be pulled from the table. $\frac{v(8)-v(2)}{8-2}$ alone will not earn the point as it does not pull values from the table (yet).

Note: the reciprocal of the quotient gives the same numerical answer. This does not earn the point.

Also: $\frac{v(14)-v(8)}{14-8}$ happens to equal -1 as well. A response using an incorrect pair of points does not earn the point.

Second point: Units

$\frac{m}{s^2}$ or meters per second per second will earn the point.

The units must be tied to a number. Units alone do not earn this point.

Beware of $\left(\frac{m}{s}\right)^2$ or $\frac{m^2}{s}$ (which do not earn the point).

This point must be earned in part (a). If no units are present in part (a) and correct units are shown in part (b), this point is not earned.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$v'(6) \approx \frac{6-12}{8-2} = -1 \text{ meter per second per second}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-7

First point: Interpretation

There are two components to look for:

· "rate of change of velocity" or "acceleration" of the particle

· at $t = 6$

"The rate at which the velocity is decreasing at $t = 6$" is acceptable. If the answer to part (a) was reported to be positive, "the rate at which the velocity is increasing at $t = 6$ is acceptable. We will not penalize for the same error twice.

Responses that say "velocity is decreasing at a rate of $-1\,\frac{m}{s^2}$" or similar do not earn the point. This is interpreted to mean that the velocity is increasing.

Units are not required in part (b). If units are presented for t, they must be s (seconds).

0	1 ✓

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$v'(6)$ is the rate at which the velocity of the particle is changing, in meters per second per second, at time $t = 6$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: v is continuous

It must be clear that the reason given as to why v is continuous is that v is differentiable.

Neither "v is differentiable and continuous" nor "v differentiable, v continuous" are sufficient to earn this point.

"v is differentiable hence continuous," by contrast, earns the point.

v differentiable $\therefore$ v continuous" earns the point.

Second point: Bounding with table values

Must see the values 0 and 6 pulled from the table and correctly compared with 3, in words or using inequality symbols.

6-7

The Intermediate Value Theorem (IVT) does not need to be referenced by name.

That said, this point will not be earned if other theorems (MVT, etc.) are invoked.

Responses often "say too much"; especially common are responses that indicate that v is always decreasing on the interval $[8,\ \ 14]$. This point will not be earned if the response contains such errors in reasoning.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ v continuous
- ☐ Bounding 3 with table values

Solution:

v is differentiable. $\Rightarrow v$ is continuous.

$v\ (14) = 0 < 3 < 6 = v\ (8)$

By the Intermediate Value Theorem, there must be a time $t = k$, for $8 < k < 14$, such that the velocity of the particle is 3 meters per second.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Form of left Riemann sum

Must see a sum of three terms, each of which is a product of two factors:

$(b_1\ \ \cdot h_1) + (b_2\ \cdot\ h_2) + (b_3\ \cdot\ h_3)$

If exactly 5 out of 6 of these numbers $(b_1,\ \ b_2,\ \ b_3,\ \ h_1,\ \ h_2,\ \ h_3)$ are correctly placed in the above expression, we are convinced that the response carries the form of a left Riemann sum. In this case, award the first point. Such a response cannot earn the second point. If fewer than 5 out of 6 values are correct, no points are earned in this problem.

To earn this point, values do not need yet to be imported from the table: the expression $v\ (0)\ \ \cdot\ \ (2-0) + v\ (2)\ \cdot\ \ (8-2) + v\ (8)\ \cdot\ \ (14-8)$ is sufficient.

Once the first point is earned, it cannot be lost. Subsequent errors result in the second point not being earned.

Second point: Answer

138 or equivalent.

6-7

It is possible to earn this point without earning the first point. If there is credible evidence that an attempt was made to find a left Riemann sum but not enough work is shown, this point can be earned in the presence of the correct answer. For example, a sum of three terms with no evidence of products $(30 + 72 + 36)$, or a graph with areas labeled and summed to 138 or equivalent, can earn this point.

The response of only $15 \cdot 2 + 12 \cdot 6 + 6 \cdot 6$ earns both points.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Form of left Riemann sum
- [ ] Answer

Solution:

$$\int_0^{14} v\,(t)\;dt \approx v\,(0)\,(2-0) + v\,(2)\,(8-2) + v\,(8)\,(14-8)$$

$$= 15 \cdot 2 + 12 \cdot 6 + 6 \cdot 6 = 138$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

1st point: Substitution

There are three elements to look for in a correct substitution. (another variable may be chosen instead of u.)

$\cdot\, u = \frac{t}{3} + 7$

$\cdot\, du = \frac{dt}{3}$ or equivalent

$\cdot$ Appropriate limits of integration throughout (8 and 14 for u, 3 and 21 for t).

Substitution can be implicit. A coefficient of 3 accompanying a correct antiderivative convinces us of this:

$$\int_3^{21} v'\left(\frac{t}{3} + 7\right) dt = 3v\left(\frac{t}{3} + 7\right)\Big|_3^{21}$$

To elaborate on the third element above: if a response shows an otherwise correct substitution but the limits of integration on a definite integral are unchanged (3 and 21), this point is not earned.

Alternately, the integral can be worked with as an indefinite integral with the variable returned to $\frac{t}{3} + 7$ before evaluating as a means of eliminating the need to change limits. Such a response can earn the first point, if handled

6-7

correctly.

Second point: Fundamental Theorem of Calculus

We must be convinced that the response has indicated an appropriate connection between $\int v'$ and v.

Expressions/equations that earn this point are:

· $k\left(v\left(14\right)-v\left(8\right)\right)$ or $k\left(v\left(\frac{21}{3}+7\right)-v\left(\frac{3}{3}+7\right)\right)$, where k is a nonzero constant

· $kv\left(\frac{t}{3}+7\right)\Big|_3^{21}$ or $k\,v\left(u\right)\Big|_8^{14}$, where k is a nonzero constant

· $\int v'\left(\frac{t}{3}+7\right)\ dt=kv\left(\frac{t}{3}+7\right)$ or $k\int v'\left(u\right)\ \ du=kv\left(u\right)$, where k is a nonzero constant, with or without appropriate limits of integration displayed on either integral

· $\int_3^{21}v'\left(\frac{t}{3}+7\right)\ dt=k\,v\left(\frac{t^2}{6}+7t\right)$ where k is a nonzero constant is a common response. This does not earn the point. Likewise, similar mishandled antiderivatives do not earn the point.

Third point: Answer

-18 or equivalent, with credible supporting work, earns this point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u=\frac{t}{3}+7$.

Then $du=\frac{dt}{3}$.

When $t=3$, $u=8$.

When $t=21$, $u=14$.

$$\int_3^{21}v'\left(\frac{t}{3}+7\right)\ dt=3\int_8^{14}v'\left(u\right)\ du$$

6-7

$= 3\left(v\left(14\right) - v\left(8\right)\right)$

$= 3(0 - 6) = -18$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Fundamental Theorem of Calculus (FTC)

Substitutes $4x$ into $v\left(2t\right)$. We must see either $v\left(2 \cdot 4x\right)$ or $v\left(8x\right)$.

The expression $v\left(2 \cdot 4x\right) - v\left(2\left(0\right)\right)$ or $v\left(2 \cdot 4x\right) \cdot 4 - v\left(2\left(0\right)\right)$ will earn this point. The error of subtracting $v\left(2\left(0\right)\right)$ will be viewed as an error in applying the chain rule.

Beware of possible substitutions prior to using FTC:

· For example, $u = 2t$, $\frac{1}{2}du = dt$ and rewrites: $\int_0^{4x} v\left(2t\right)\ dt = \frac{1}{2}\int_0^{8x} v\left(u\right)\ du$.

· Or Alt text: $u = 4x$, $\frac{du}{dx} = 4$ and rewrites: $\frac{d}{dx}[h\left(x\right)] = \frac{d}{du}[h\left(u\right)] \cdot \frac{du}{dx}$

· Such approaches can earn this point if done properly.

Second point: Chain rule

Multiplies by 4

It is not possible to earn this point without having earned the first point.

Depending on substitutions, this may look more like $\frac{1}{2}$ and 8 multiplied separately.

Third point: Answer

24 or equivalent

It is not possible to earn this point without having earned both the first and second points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

6-7

Solution:

$$p'(x) = \frac{d}{dx}\int_0^{4x} v(2t)\,dt = v(2 \cdot 4x) \cdot 4$$

$$p'(1) = v(8) \cdot 4 = 6 \cdot 4 = 24$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Narrows consideration to g' and g''

You must be convinced the response considers only the functions g' and g'' as being critical to the reasoning.

Second point: Answer with reason

A correct response must:

· Conclude "decreasing"

· Appeal to the signs of g' and g''

It is not possible to earn this point without having earned the first point.

A response that mentions velocity and acceleration must tie velocity to g' and acceleration to g''. A response that says only "decreasing because velocity and acceleration have different signs" earns neither of the two points. It is not clear that the response is considering g' and g''.

A response that calls out all three given values ($g(4)$, $g'(4)$, and $g''(4)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g' and g'' only, both points can be earned.

The phrase "slowing down" will be interpreted as "decreasing."

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g' and g''
- ☐ Answer with reason

Solution:

Because $g'(4) = 2 > 0$, the velocity of the particle is positive. Because $g''(4) = -6 < 0$, the acceleration of the particle is negative. Therefore, because the signs of the velocity and acceleration are different, the speed of the

6-7

particle is decreasing at time $t = 4$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Tangent line equation

This point is earned only for the correct tangent line equation written in any form.

Some common forms:

- $y - 2 = \left(-2\,(3) + 3(2)^2\right)(x - 3)$
- $y - 2 = 6\,(x - 3)$
- $y = 6x - 16$

$y - 2 = \left(-2x + 3y^2\right)(x - 3)$ does not earn the point. This is not the equation of a line. The response can go on to present one of the correct linear forms, and then would earn the point.

$f\,(x) = 6\,(x - 3) + 2$ or equivalent does not earn the point. $f\,(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

Some responses have the correct tangent line equation written initially, with subsequent errors in simplification. Such responses do not earn the point.

The point can be earned after pursuing a less fruitful approach (such as solving the differential equation) and then restarting.

0	1 ✓

The student response accurately includes the criteria below.

- [ ] Tangent line equation

Solution:

$$\left.\frac{dy}{dx}\right|_{(3,\,2)} = -2 \cdot 3 + 3 \cdot 2^2 = 6$$

$$y = 6(x - 3) + 2$$

6-7

75. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	2	5	11
$v(t)$ (meters per second)	3	5.5	7	13

The velocity of a particle, P, moving along the x-axis is modeled by a differentiable function v, where time t is measured in seconds and $v(t)$ is measured in meters per second. Selected values of $v(t)$ are shown in the table above.

(a) Use the data in the table to approximate $v'(8)$ using the average rate of change of $v(t)$ over the interval $5 \leq t \leq 11$. Show the computations that lead to your answer. Indicate units of measure.

(b) Interpret the meaning of $v'(8)$ in the context of the problem.

(c) Justify why there must be a time $t = b$, for $0 \leq b \leq 2$, when the velocity of the particle is 5 meters per second.

(d) Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^{11} v(t)\,dt$. Show the computations that lead to your answer.

(e) Find $\int_0^9 v'\left(\frac{t}{3}+2\right)dt$. Show the computations that lead to your answer.

(f) Let $h(x) = \int_1^{\frac{5}{2}x} v(2t)\,dt$. Find $h'(1)$. Show the computations that lead to your answer.

(g) The position of a second particle, Q, can be modeled by a twice-differentiable function g. It is known that $g(2) = 5.5$, $g'(2) = -4$, and $g''(2) = -3$. Is the speed of particle Q increasing or decreasing at time $t = 2$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -3x^2y^3$ with initial condition $f(3) = 2$. Write an equation for the line tangent to the graph of f at the point $(3, 2)$.

Part A

6-7

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Approximation

Need to see both a difference and a quotient as the supporting work. Minimal answers that earn the point include $\frac{13-7}{11-5}$, $\frac{13-7}{6}$, and $\frac{6}{11-5}$.

The answer does not need to be simplified, but any attempt at simplification must be correct.

Numerical values must be pulled from the table. $\frac{v(11)-v(5)}{11-5}$ alone will not earn the point as it does not pull values from the table (yet).

Note: the reciprocal of this quotient gives the same numerical value. This does not earn the point.

Second point: Units

$\frac{m}{s^2}$ or meters per second per second will earn the point.

The units must be tied to a number. Units alone do not earn this point.

Beware of $\left(\frac{m}{s}\right)^2$ or $\frac{m^2}{s}$ (which do not earn the point).

This point must be earned in part (a). If no units are present in part (a) and correct units are shown in part (b), this point is not earned.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$v'(8) \approx \frac{13-7}{11-5} = 1 \text{ meter per second per second}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Interpretation

There are two components to look for:

6-7

· "rate of change of velocity" or "acceleration" of the particle

· at $t = 8$

"The rate at which the velocity is increasing at $t = 8$" is acceptable. If the answer to part (a) was reported to be negative, "the rate at which the velocity is decreasing at $t = 8$" is acceptable. We will not penalize for the same error twice.

Units are not required in part (b). If units are presented for t, they must be s (seconds).

0	1

The student response accurately includes the criteria below.

- [ ] Interpretation

Solution:

$v'(8)$ is the rate at which the velocity of the particle is changing, in meters per second per second, at time $t = 8$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: v is continuous

It must be clear that the reason given as to why v is continuous is that v is differentiable.

Neither "v is differentiable and continuous" nor "v differentiable, v continuous" are sufficient to earn this point.

"v is differentiable hence continuous," by contrast, earns the point.

"v differentiable $\therefore$ v continuous" earns the point.

Second point: Bounding with table values

Must see the values 3 and 5.5 pulled from the table and correctly compared to 5, in words or using inequality symbols.

The Intermediate Value Theorem (IVT) does not need to be referenced by name.

That said, this point will not be earned if other theorems (MVT, etc.) are invoked.

Responses often "say too much"; especially common are responses that indicate that v is always increasing on the interval $[0, \ 2]$. This point will not be earned if the response contains such errors in reasoning.

6-7

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] v continuous
- [ ] Bounding 5 with table values

Solution:

v is differentiable. $\Rightarrow v$ is continuous.

$v(0) = 3 < 5 < 5.5 = v(2)$

By the Intermediate Value Theorem, there must be a time $t = b$, for $0 \leq b \leq 2$, such that the velocity of the particle is 5 meters per second.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Form of left Riemann sum

Must see a sum of three terms, each of which is a product of two factors: $(b_1 \cdot h_1) + (b_2 \cdot h_2) + (b_3 \cdot h_3)$.

If exactly 5 out of 6 of these numbers $(b_1, \ b_2, \ b_3, \ h_1, \ h_2, \ h_3)$ are correctly placed in the above expression, we are convinced that the response carries the form of a left Riemann sum. In this case, award the first point. Such a response cannot earn the second point. If fewer than 5 out of 6 values are correct, no points are earned in this problem.

To earn this point, values do not need yet to be imported from the table: the expression $v(0) \cdot (2-0) + v(2) \cdot (5-2) + v(5) \cdot (11-5)$ is sufficient.

Once the first point is earned, it cannot be lost. Subsequent errors result in the second point not being earned.

Second point: Answer

64.5 or equivalent.

It is possible to earn this point without earning the first point. If there is credible evidence that an attempt was made to find a left Riemann sum but not enough work is shown, this point can be earned in the presence of the correct answer. For example, a sum of three terms with no evidence of products $(6 + 16.5 + 42)$, or a graph with areas labeled and summed to 64.5 or equivalent, can earn this point.

The response of only $3 \cdot 2 + 5.5 \cdot 3 + 7 \cdot 6$ earns both points.

6-7

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Form of left Riemann sum
- ☐ Answer

Solution:

$$\int_0^{11} v(t)\ dt \approx v(0)(2-0) + v(2)(5-2) + v(5)(11-5)$$

$$= 3 \cdot 2 + 5.5 \cdot 3 + 7 \cdot 6 = 64.5$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Substitution

There are three elements to look for in a correct substitution. (another variable may be chosen instead of u.)

$\cdot\, u = \frac{t}{3} + 2$

$\cdot\, du = \dfrac{dt}{3}$ or equivalent

$\cdot$ Appropriate limits of integration throughout (2 and 5 for u, 0 and 9 for t).

Substitution can be implicit. A coefficient of 3 accompanying a correct antiderivative convinces us of this:

$$\int_0^9 v'\left(\frac{t}{3} + 2\right)\ dt = 3v\left(\tfrac{t}{3} + 2\right)\Big|_0^9$$

To elaborate on the third element above: if a response shows an otherwise correct substitution but the limits of integration on a definite integral are unchanged (0 and 9), this point is not earned.

Alternately, the integral can be worked with as an indefinite integral with the variable returned to $\frac{t}{3} + 2$ before evaluating as a means of eliminating the need to change limits. Such a response can earn the first point, if handled correctly.

Second point: Fundamental Theorem of Calculus

We must be convinced that the response has indicated an appropriate connection between $\displaystyle\int v'$ and v.

6-7

Expressions/equations that earn this point are:

· $k\left(v\left(5\right)-v\left(2\right)\right)$ or $k\left(v\left(\frac{9}{3}+2\right)-v\left(\frac{0}{3}+2\right)\right)$, where k is a nonzero constant.

· $kv\left(\frac{t}{3}+2\right)\Big|_0^9$ or $k\,v\left(u\right)\Big|_2^5$, where k is a nonzero constant.

· $\int v'\left(\frac{t}{3}+2\right)\ dt=kv\left(\frac{t}{3}+2\right)$ or $k\int v'\left(u\right)\ \ du=kv\left(u\right)$, where k is a nonzero constant, with or without appropriate limits of integration displayed on either integral.

· $\int_0^9 v'\left(\frac{t}{3}+2\right)\ dt=k\,v\left(\frac{t^2}{6}+2t\right)$ where k is a nonzero constant is a common response. This does not earn the point. Likewise, similar mishandled antiderivatives do not earn the point.

Third point: Answer

4.5 or equivalent, with credible supporting work, earns this point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u=\frac{t}{3}+2$. Then $du=\dfrac{dt}{3}$.

When $t=0$, $u=2$.

When $t=9$, $u=5$.

$$\int_0^9 v'\left(\frac{t}{3}+2\right)\ dt=3\int_2^5 v'\left(u\right)\ du$$

$=3\left(v\left(5\right)-v\left(2\right)\right)$

$=3\left(7-5.5\right)=4.5$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

6-7

First point: Fundamental Theorem of Calculus (FTC)

Substitutes $\frac{5}{2}x$ into $v\left(2t\right)$. We must see either $v\left(2\cdot\frac{5}{2}x\right)$ or $v\left(5x\right)$.

The expression $v\left(2\cdot\frac{5}{2}x\right)-v\left(2\left(1\right)\right)$ or $v\left(2\cdot\frac{5}{2}x\right)\cdot\frac{5}{2}-v\left(2\left(1\right)\right)$ will earn this point. The error of subtracting $v\left(2\left(1\right)\right)$ will be viewed as an error in applying the chain rule.

Beware of possible substitutions prior to using FTC:

· For example, $u=2t$, $\frac{1}{2}\ du=dt$ and rewrite $\int_1^{\frac{5}{2}x}v\left(2t\right)\ dt=\frac{1}{2}\int_2^{5x}v\left(u\right)\ du.$

· Or $u=\frac{5}{2}x$, $\frac{du}{dx}=\frac{5}{2}$ and rewrites: $\frac{d}{dx}[h\left(x\right)]=\frac{d}{du}[h\left(u\right)]\cdot\frac{du}{dx}$

· Such approaches can earn this point if done properly.

Second point: Chain rule

Multiplies by $\frac{5}{2}$

It is not possible to earn this point without having earned the first point.

Depending on substitutions, this may look more like $\frac{1}{2}$ and 5 multiplied separately.

Third point: Answer

$\frac{35}{2}$ or equivalent

It is not possible to earn this point without having earned both the first and second points.

✓

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'\left(x\right)=\frac{d}{dx}\int_1^{\frac{5}{2}x}v\left(2t\right)\ dt=v\left(2\cdot\frac{5}{2}x\right)\cdot\frac{5}{2}$$

6-7

$h'(1) = v(5) \cdot \frac{5}{2} = 7 \cdot \frac{5}{2} = \frac{35}{2}$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Narrows consideration to g' and g''

You must be convinced the response considers only the functions g' and g'' as being critical to the reasoning.

Second point: Answer with reason

A correct response must:

· Conclude "increasing"

· Appeal to the signs of g' and g''

It is not possible to earn this point without having earned the first point.

A response that mentions velocity and acceleration must tie velocity to g' and acceleration to g''. A response that says only "increasing because velocity and acceleration are both negative" earns neither of the two points. It is not clear that the response is considering g' and g''. A response that calls out all three given values ($g(2)$, $g'(2)$, and $g''(2)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g' and g'' only, both points can be earned.

The phrase "speeding up" will be interpreted as "increasing."

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g' and g''
- ☐ Answer with reason

Solution:

Because $g'(2) = -4 < 0$, the velocity of the particle is negative. Because $g''(2) = -3 < 0$, the acceleration of the particle is negative. Therefore, because both the velocity and acceleration are negative, the speed of the particle is increasing at time $t = 2$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-7

First point: Tangent line equation

This point is earned only for the correct tangent line equation written in any form.

Some common forms:

- $y - 2 = -3(3)^2(2)^3\,(x-3)$
- $y - 2 = -216\,(x-3)$
- $y = -216x + 650$

$y - 2 = -3x^2y^3\,(x-3)$ does not earn the point. This is not the equation of a line. The response can go on to present one of the correct linear forms, and then would earn the point.

$f\,(x) = -216\,(x-3) + 2$ or equivalent does not earn the point. $f\,(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

Some responses have the correct tangent line equation written initially, with subsequent errors in simplification. Such responses do not earn the point.

The point can be earned after pursuing a less fruitful approach (such as solving the differential equation) and then restarting.

0	1 ✓

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left.\frac{dy}{dx}\right|_{(3,\,2)} = -3\cdot 3^2\cdot 2^3 = -216$$

$$y = -216(x-3) + 2$$

6-7

76.

x	2	3	4
$f(x)$	1	2	6
$f'(x)$	4	5	3

The table above gives values of the differentiable function f and its derivative at selected values of x.

If g is the inverse function of f, which of the following is an equation of the line tangent to the graph of g at the point where $x = 2$?

(A) $y = -\frac{1}{5}(x - 2) + 3$

(B) $y = -\frac{1}{4}(x - 2) + 1$

(C) $y = \frac{1}{5}(x - 2) + 3$ ✓

(D) $y = 4(x - 2) + 1$

Answer C

Correct. Since $f\left(g\left(x\right)\right) = x$, the chain rule can be used to determine that $f'\left(g\left(x\right)\right)g'\left(x\right) = 1$. Substituting $x = 2$ gives $1 = f'\left(g\left(2\right)\right)g'\left(2\right) = f'\left(3\right)g'\left(2\right) \Rightarrow g'\left(2\right) = \frac{1}{f'(3)} = \frac{1}{5}$. Since $g\left(2\right) = f^{-1}\left(2\right) = 3$, an equation of the line tangent to the graph of g at $x = 2$ is therefore $y = \frac{1}{5}(x - 2) + 3$.

77. Let f be the function with first derivative $f'\left(x\right) = -3 + \sqrt{x^4 + x^2}$. If $f\left(2\right) = 1.572$, what is the value of $f\left(0\right)$?

(A) -2.607

(B) -1.035

(C) 4.179 ✓

(D) 4.472

Answer C

Correct. By the Fundamental Theorem of Calculus, $\int_0^2 f'(x)\,dx = f(2) - f(0)$. Therefore

$$f(0) = f(2) - \int_0^2 f'(x)\,dx = 1.572 - \int_0^2 \left(-3 + \sqrt{x^4 + x^2}\right)dx = 4.179.$$

The calculator is used to evaluate the definite integral.

6-7

78. $\int_1^2 \left(4x^3 - x\right) dx =$

(A) $\frac{27}{2}$

(B) 27

(C) 36

(D) 57

Answer A

Correct. This question involves using the basic power rule for antidifferentiation and correctly substituting the endpoints and evaluating, as follows.

$$\int_1^2 \left(4x^3 - x\right) dx = x^4 - \tfrac{1}{2}x^2\Big|_1^2 = (16 - 2) - \left(1 - \tfrac{1}{2}\right) = 14 - \tfrac{1}{2} = \tfrac{27}{2}$$

79. $\int_0^{\frac{\pi}{4}} \sin x \, dx =$

(A) $-\frac{\sqrt{2}}{2}$

(B) $1 - \frac{\sqrt{2}}{2}$

(C) $\frac{\sqrt{2}}{2} - 1$

(D) $\frac{\sqrt{2}}{2}$

Answer B

Correct. An antiderivative of $\sin x$ is $-\cos x$. Therefore by the Fundamental Theorem of Calculus,

$$\int_0^{\frac{\pi}{4}} \sin x \, dx = -\cos x\Big|_0^{\frac{\pi}{4}} = -\cos\left(\frac{\pi}{4}\right) + \cos(0) = -\frac{\sqrt{2}}{2} + 1.$$

6-7

80. The derivative of a function f is given by $f'(x) = -1 + \sqrt{x + e^{x/3}}$. It is known that $f(0) = 3$. In which of the following intervals is there a value of x such that $f(x) = 6$?

(A) $3 < x < 4$ ✓

(B) $5 < x < 6$

(C) $6 < x < 7$

(D) $10 < x < 11$

Answer A

Correct. The Fundamental Theorem of Calculus is used to find values of f, as follows.

$$f(3) = f(0) + \int_0^3 f'(x)\,dx = 3 + \int_0^3 \left(-1 + \sqrt{x + e^{x/3}}\right)dx = 5.252$$

$$f(4) = f(0) + \int_0^4 f'(x)\,dx = 3 + \int_0^4 \left(-1 + \sqrt{x + e^{x/3}}\right)dx = 6.843$$

Because f is differentiable, it is continuous. Since $f(3) < 6 < f(4)$, the Intermediate Value Theorem guarantees that there is a value c in the interval $(3, 4)$ such that $f(c) = 6$. Because $f'(x) > 0$ for $x > 0$, there is no other interval in which $f(x) = 6$ could have a solution.

81. $\int_1^3 \left(4x^3 - 6x + 5\right)dx =$

(A) 66 ✓

(B) 92

(C) 96

(D) 282

Answer A

Correct. This question involves using the basic power rule for antidifferentiation and correctly substituting the endpoints and evaluating, as follows.

$$\int_1^3 \left(4x^3 - 6x + 5\right)dx = \left(x^4 - 3x^2 + 5x\right)\Big|_1^3 = (81 - 27 + 15) - (1 - 3 + 5) = 69 - 3 = 66$$

6-7

82. $\int_0^{\frac{\pi}{2}} \sin\left(\frac{x}{2}\right) dx =$

(A) $\sqrt{2} - 2$

(B) $\frac{2-\sqrt{2}}{4}$

(C) $\frac{2-\sqrt{2}}{2}$

(D) $2 - \sqrt{2}$ ✓

Answer D

Correct. The antiderivative of $\sin x$ is $-\cos x$. Therefore, the antiderivative of $\sin\left(\frac{x}{2}\right)$ is $-2\cos\left(\frac{x}{2}\right)$ where the cosine function has been multiplied by 2 to adjust for the chain rule. Therefore, by the Fundamental Theorem of Calculus, $\int_0^{\frac{\pi}{2}} \sin\left(\frac{x}{2}\right)dx = -2\cos\left(\frac{x}{2}\right)\Big|_0^{\frac{\pi}{2}} = -2\cos\left(\frac{\pi}{4}\right) + 2 = -\sqrt{2} + 2.$

83. For which of the following values of b does $\int_2^b (2x-3)\, dx = 0$?

(A) -2 only

(B) 2 only

(C) -2 and 2

(D) 1 and 2 ✓

Answer D

Correct. The values of b would be the solutions to

$$0 = \int_2^b (2x-3)dx = \left(x^2 - 3x\right)\Big|_2^b = \left(b^2 - 3b\right) - (4-6) = b^2 - 3b + 2 = (b-2)(b-1).$$

Therefore $b = 1$ and $b = 2$.

An alternative geometric method is to first recognize that $b = 2$ is always a solution, since a definite integral with the same lower and upper limits is equal to 0. Then observe that the line $y = 2x - 3$ crosses the x-axis at $x = \frac{3}{2}$ so that the region bounded by the graph below the x-axis between $x = 1$ and $x = \frac{3}{2}$ would have the same area as the region bounded by the graph above the x-axis between $x = \frac{3}{2}$ and $x = 2$, and hence

$$\int_2^1 (2x-3)dx = -\int_1^2 (2x-3)dx = 0.$$